

ACADEMY OF MARITIME EDUCATION AND TRAINING DEEMED TO BE UNIVERSITY

SELF ASSESSMENT REPORT(TIER - I) FOR NAVAL ARCHITECTURE AND OFFSHORE ENGINEERING

Part A : Institutional Information

1 Name and Address of the Institution

ACADEMY OF MARITIME EDUCATION AND TRAINING DEEMED TO BE UNIVERSITY,
NO,135, EAST COST ROAD, KANATHUR, REDDIKUPPAM CHENNAI. KANCHIPURAM

2 Name and Address of Affiliating University

3 Year of establishment of the Institution:

2007

4 Type of the Institution:

- | | |
|--|---|
| ☐ Institute of National Infornance | ☐ Autonomous |
| ☐ University | ☐ Any other(please specify) |
| ☒ Deemed University | |

5 Ownership Status:

- | | |
|--|--|
| ☐ Central Government | ☒ Trust |
| ☐ State Government | ☐ Society |
| ☐ Government Aided | ☐ Section 25 Company |
| ☐ Self financing | ☐ Any Other(Please Specify) |

6 Other Academic Institutions of the Trust/Society/Company etc., if any

Name of Institutions	Year of Establishment	Programs of Study	Location

7 Details of all the programs being offered by the Institution under consideration:

Name of Program	Program Applied level	Start of year	Year of AICTE approval	Initial Intake	Intake Increase	Current Intake	Accreditation status	From	To	Program for consideration	Program for Duration
B.E - Naval Architecture and Offshore Engineering	UG	2007	2007	90	Yes	30	Applying first time	--	--	Yes	4
Sanctioned Intake for Last Five Years for the B.E - Naval Architecture and Offshore Engineering											
Academic Year				Sanctioned Intake							
2022-23				30							
2021-22				30							
2020-21				54							
2019-20				60							
2018-19				120							
2017-18				65							
M.E - Naval Architecture and Offshore Engineering	PG	2018	2018	18	Yes	16	Eligible but not applied	--	--	No	2
Sanctioned Intake for Last Five Years for the M.E - Naval Architecture and Offshore Engineering											
Academic Year				Sanctioned Intake							
2022-23				16							
2021-22				16							
2020-21				16							
2019-20				18							
2018-19				18							
2017-18				0							
Electrical and Electronics Engineering - Marine	UG	2008	2008	60	Yes	30	Granted accreditation for 3 years for the period (specify period)	2021	2024	0	4
Sanctioned Intake for Last Five Years for the Electrical and Electronics Engineering - Marine											
Academic Year				Sanctioned Intake							
2022-23				30							
2021-22				30							
2020-21				54							
2019-20				60							
2018-19				60							
2017-18				45							
Mechanical Engineering	UG	2014	2014	60	Yes	108	Granted accreditation for 3 years for the period (specify period)	2021	2024	0	4

Name of Program	Program Applied level	Start of year	Year of AICTE approval	Initial Intake	Intake Increase	Current Intake	Accreditation status	From	To	Program for consideration	Program for Duration
Sanctioned Intake for Last Five Years for the Mechanical Engineering											
Academic Year				Sanctioned Intake							
2022-23				108							
2021-22				108							
2020-21				108							
2019-20				120							
2018-19				120							
2017-18				70							
Power Systems	PG	2018	2018	18	Yes	16	Eligible but not applied	--	--	0	2
Sanctioned Intake for Last Five Years for the Power Systems											
Academic Year				Sanctioned Intake							
2022-23				16							
2021-22				16							
2020-21				16							
2019-20				18							
2018-19				18							
2017-18				0							
Thermal & Fluid Engineering	PG	2018	2018	18	Yes	16	Eligible but not applied	--	--	0	2
Sanctioned Intake for Last Five Years for the Thermal & Fluid Engineering											
Academic Year				Sanctioned Intake							
2022-23				16							
2021-22				16							
2020-21				16							
2019-20				18							
2018-19				18							
2017-18				0							
Petroleum Engineering	PG	2018	2018	18	Yes	16	Eligible but not applied	--	--	0	2

Name of Program	Program Applied level	Start of year	Year of AICTE approval	Initial Intake	Intake Increase	Current Intake	Accreditation status	From	To	Program for consideration	Program for Duration
Sanctioned Intake for Last Five Years for the Petroleum Engineering											
Academic Year				Sanctioned Intake							
2022-23				16							
2021-22				16							
2020-21				16							
2019-20				18							
2018-19				18							
2017-18				0							
Marine Engineering	UG	2001	2001	40	Yes	300	Applying first time	--	--	No	4
Sanctioned Intake for Last Five Years for the Marine Engineering											
Academic Year				Sanctioned Intake							
2022-23				300							
2021-22				300							
2020-21				300							
2019-20				360							
2018-19				360							
2017-18				250							
COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)	UG	2022	2022	30	No	30	Not eligible for accreditation	--	--	0	4
COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)	UG	2022	2022	30	No	30	Not eligible for accreditation	--	--	0	4
Petroleum Engineering	UG	2008	2008	60	Yes	30	Not eligible for accreditation	--	--	No	4
Sanctioned Intake for Last Five Years for the Petroleum Engineering											
Academic Year				Sanctioned Intake							
2022-23				30							
2021-22				30							
2020-21				54							
2019-20				60							
2018-19				120							
2017-18				50							
Food Processing Technology	UG	2016	2016	60	Yes	30	Not eligible for accreditation	--	--	0	4

Name of Program	Program Applied level	Start of year	Year of AICTE approval	Initial Intake	Intake Increase	Current Intake	Accreditation status	From	To	Program for consideration	Program for Duration
Sanctioned Intake for Last Five Years for the Food Processing Technology											
Academic Year				Sanctioned Intake							
2022-23				30							
2021-22				30							
2020-21				54							
2019-20				60							
2018-19				60							
2017-18				25							
Mining Engineering	UG	2016	2016	60	Yes	30	Eligible but not applied	--	--	No	4
Sanctioned Intake for Last Five Years for the Mining Engineering											
Academic Year				Sanctioned Intake							
2022-23				30							
2021-22				30							
2020-21				54							
2019-20				60							
2018-19				60							
2017-18				25							
MBA(SHIPPING AND LOGISTICS MANAGEMENT)	PG	2007	2007	60	Yes	90	Granted accreditation for 3 years for the period (specify period)	2021	2024	No	2
Sanctioned Intake for Last Five Years for the MBA(SHIPPING AND LOGISTICS MANAGEMENT)											
Academic Year				Sanctioned Intake							
2022-23				90							
2021-22				90							
2020-21				90							
2019-20				120							
2018-19				120							
2017-18				75							

8 Programs to be considered for Accreditation vide this application:

S No	Level	Discipline	Program
1	Under Graduate	Engineering & Technology	Marine Engg.
2	Under Graduate	Engineering & Technology	NAVAL ARCHITECTURE AND OFFSHORE ENGINEERING

9 Total number of employees**A. Regular* Employees (Faculty and Staff):**

Items	2022-23		2021-22		2020-21	
	MIN	MAX	MIN	MAX	MIN	MAX
Faculty in Engineering (Male)	111	111	132	132	144	144
Faculty in Engineering (Female)	36	36	42	42	46	46
Faculty in Maths, Science & Humanities teaching in engineering program (Male)	23	23	24	24	28	28
Faculty in Maths, Science & Humanities teaching in engineering program (Female)	20	20	17	17	16	16
Non-teaching staff (Male)	226	226	329	329	71	71
Non-teaching staff (Female)	46	46	36	36	4	4

B. Contractual* Employees (Faculty and Staff):

Items	2022-23		2021-22		2020-21	
	MIN	MAX	MIN	MAX	MIN	MAX
Faculty in Engineering (Male)	0	0	0	0	0	0
Faculty in Engineering (Female)	0	0	0	0	0	0
Faculty in Maths, Science & Humanities teaching in engineering Programs (Male)	0	0	0	0	0	0
Faculty in Maths, Science & Humanities teaching in engineering Programs (Female)	0	0	0	0	0	0
Non-teaching staff (Male)	0	0	0	0	0	0
Non-teaching staff (Female)	0	0	0	0	0	0

10 Total number of Engineering students:

Engineering and Technology- UG	☒ Shift1	☐ Shift2
Engineering and Technology- PG	☒ Shift1	☐ Shift2
Engineering and Technology- Polytechnic	☐ Shift1	☐ Shift2
MBA	☒ Shift1	☐ Shift2
MCA	☐ Shift1	☐ Shift2

Engineering and Technology- UG Shift-1

Course Name	2022-23	2021-22	2020-21
Total no. of Boys	1456	1553	1773
Total no. of Girls	54	40	47
Total	1510	1593	1820

Engineering and Technology- PG Shift-1

Course Name	2022-23	2021-22	2020-21
Total no. of Boys	68	72	86
Total no. of Girls	6	8	6
Total	74	80	92

Engineering and Technology- MBA Shift-1

Course Name	2022-23	2021-22	2020-21
Total no. of Boys	153	132	125
Total no. of Girls	16	11	9
Total	169	143	134

11 Vision of the Institution:

To sustain identity as a World Class Leader in Maritime Education and empower learners with wholesome knowledge through progressive innovation in training, research and development which will render students a unique learning experience and a transformation impact on the Global Society.

12 Mission of the Institution:

AMET will strive continuously to

1. Impart value-based higher education and technical knowledge with uncompromising strides of an outstanding quality.
2. Emerge as a Centre of Excellence inculcating skill development in recent technologies in accordance with industrial trends.
3. Create world class research capabilities on par with the finest in the world and broaden student's horizons beyond classroom education.
4. Nurture talent and entrepreneurship and enable all round personality development in students.
5. Empower students from across socio economic strata.
6. Make a positive difference to society through technical education.

13 Contact Information of the Head of the Institution and NBA coordinator, if designated:

Head of the Institution	
Name	Dr. T. Sasilatha
Designation	Principal
Mobile No.	9444752994
Email ID	deaneeem@ametuniv.ac.in

☒ **NBA Coordinator, If Designated**

Name	Dr. T. Sasilatha
Designation	NBA Coordinator
Mobile No.	9444752994
Email ID	deaneeem@ametuniv.ac.in

PART B: Criteria Summary

Criteria No.	Criteria	Total Marks	Institute Marks
1	VISION, MISSION AND PROGRAM EDUCATIONAL OBJECTIVES	50	50.00
2	PROGRAM CURRICULUM AND TEACHING - LEARNING PROCESSES	100	100.00
3	COURSE OUTCOMES AND PROGRAM OUTCOMES	175	175.00
4	STUDENTS' PERFORMANCE	100	74.47
5	FACULTY INFORMATION AND CONTRIBUTIONS	200	198.00
6	FACILITIES AND TECHNICAL SUPPORT	80	80.00
7	CONTINUOUS IMPROVEMENT	75	75.00
8	FIRST YEAR ACADEMICS	50	47.38
9	STUDENT SUPPORT SYSTEMS	50	50.00
10	GOVERNANCE, INSTITUTIONAL SUPPORT AND FINANCIAL RESOURCES	120	120.00
	Total	1000	969

Part B : Criteria Summary

1 VISION, MISSION AND PROGRAM EDUCATIONAL OBJECTIVES (50)

Total Marks 50.00

1.1 State the Vision and Mission of the Department and Institute (5)

Total Marks 5.00

Institute Marks : 5.00

Vision of the institute	To sustain identity as a World Class Leader in Maritime Education and empower learners with wholesome knowledge through progressive innovation in training, research and development which will render students a unique learning experience and a transformation impact on the Global Society.													
Mission of the institute	AMET will strive continuously to 1. Impart value-based higher education and technical knowledge with uncompromising strides of an outstanding quality. 2. Emerge as a Centre of Excellence inculcating skill development in recent technologies in accordance with industrial trends. 3. Create world class research capabilities on par with the finest in the world and broaden student’s horizons beyond classroom education. 4. Nurture talent and entrepreneurship and enable all round personality development in students. 5. Empower students from across socio economic strata. 6. Make a positive difference to society through technical education.													
Vision of the Department	To educate and train students in Naval Architecture and Offshore Engineering who are academically bright, technically creative, ethically sound, emotionally strong, and valuable to society through innovative teaching, learning and research.													
Mission of the Department	<table><tr><th>Mission No.</th><th>Mission Statements</th></tr><tr><td>M1</td><td>1. To upgrade technical knowledge through industry institute interaction, to educate students in order to produce innovative and socio-economically conscious Naval Architects and Offshore Engineers.</td></tr><tr><td>M2</td><td>2. To inspire students to pursue higher education in Naval Architecture and Offshore Engineering and other allied fields in the maritime sector.</td></tr><tr><td>M3</td><td>3. To nurture innovativeness, creative, strong ethics, positive attitude and entrepreneurship skills in students to meet the needs of the industry and society.</td></tr><tr><td>M4</td><td>4. To establish high quality teaching and research environment to offer state-of-the-art undergraduate, graduate and doctoral programs.</td></tr><tr><td>M5</td><td>5. To create world-class research facilities and infrastructure in the field of Naval Architecture and Offshore Engineering.</td></tr></table>		Mission No.	Mission Statements	M1	1. To upgrade technical knowledge through industry institute interaction, to educate students in order to produce innovative and socio-economically conscious Naval Architects and Offshore Engineers.	M2	2. To inspire students to pursue higher education in Naval Architecture and Offshore Engineering and other allied fields in the maritime sector.	M3	3. To nurture innovativeness, creative, strong ethics, positive attitude and entrepreneurship skills in students to meet the needs of the industry and society.	M4	4. To establish high quality teaching and research environment to offer state-of-the-art undergraduate, graduate and doctoral programs.	M5	5. To create world-class research facilities and infrastructure in the field of Naval Architecture and Offshore Engineering.
Mission No.	Mission Statements													
M1	1. To upgrade technical knowledge through industry institute interaction, to educate students in order to produce innovative and socio-economically conscious Naval Architects and Offshore Engineers.													
M2	2. To inspire students to pursue higher education in Naval Architecture and Offshore Engineering and other allied fields in the maritime sector.													
M3	3. To nurture innovativeness, creative, strong ethics, positive attitude and entrepreneurship skills in students to meet the needs of the industry and society.													
M4	4. To establish high quality teaching and research environment to offer state-of-the-art undergraduate, graduate and doctoral programs.													
M5	5. To create world-class research facilities and infrastructure in the field of Naval Architecture and Offshore Engineering.													

1.2 State the Program Educational Objectives (PEOs) (5)

Total Marks 5.00

PEO No.	Program Educational Objectives Statements
PEO1	1. Become successful Naval Architects and Offshore Engineers with breadth and depth of knowledge who are competent, innovative and productive in addressing the needs of the global industry.
PEO2	2. Pursue higher education and research in Naval Architecture and Offshore Engineering, and other allied disciplines.
PEO3	3. Develop innovative thinking skills and pursue a life-long learning process to augment professional growth.
PEO4	4. Exhibit high standards of ethical conduct, positive attitude and societal responsibility.

1.3 Indicate where the Vision, Mission and PEOs are published and disseminated among stakeholders (15)

Total Marks 15.00

The information regarding the vision, mission and PEOs are made available in the

- University website (<https://www.ametuniv.ac.in/objectives.html> (<https://www.ametuniv.ac.in/objectives.html>)),
- Department notice boards
- Department level documents like course files and curriculum books, etc.
- Posters are displayed in classrooms, faculty rooms, HoD chamber, and laboratories, and also at the main entrance of the Department building.

The vision, mission and PEOs are disseminated among stakeholders in the following ways:

1. **Faculty meeting:** At the beginning of each semester a faculty meeting is arranged to discuss the vision, mission and PEOs of the program.
2. **Student induction program:** As a part of the orientation program of the first year students, a separate session is scheduled exclusively for discussing the vision, mission and PEOs of the Department. For other students, a senior faculty/ HoD discusses the vision, mission and PEO during the start of the academic year.
3. **Parent-teacher meeting:** During parent-teacher meeting, the vision, mission and PEOs are disseminated.
4. **Alumni meeting:** During alumni meeting, the vision, mission and PEOs are disseminated.
5. **Professional activities:** The vision, mission and PEOs are disseminated through conference brochures, seminar pamphlets, etc.

The Fig 1.3 below identifies the various internal and external stakeholders involved in formulating the vision, mission and PEOs of the Department.

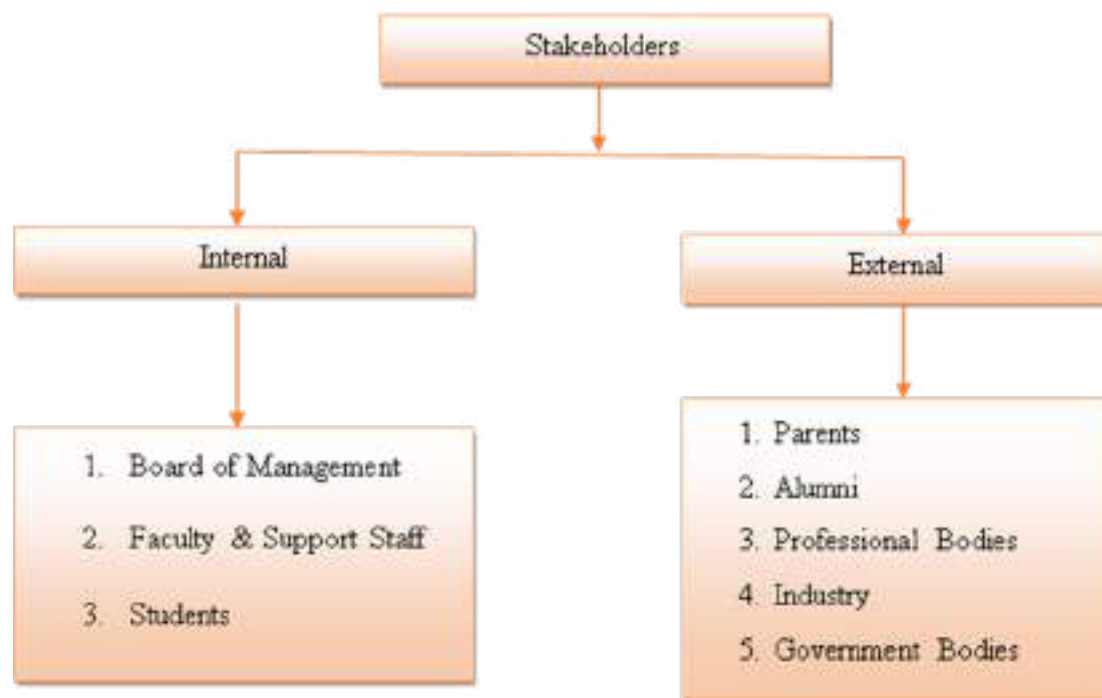


Fig 1.3: Internal and External Stakeholders

Table 1.3: Process of dissemination among the various stakeholders

Stakeholders	Process of Dissemination	Frequency of Meeting	Responsibility

Faculty	Department Faculty meetings	First meeting of every semester	HoD
Students	Orientation Program	Annual	HoD
Parents	Parent - Teacher Meetings	Biannual	Faculty Members
Alumni	Alumni Interaction	Annual	Alumni Co-Ordinator
Recruiters	Placement drives	Annual	Placement Director
University Management	Management review meeting with all HoDs	Annual	Vice Chancellor
Professional Bodies	Interaction in Workshop/Seminar/Conference	First meeting of every semester	President of the Student Chapter
Governing Bodies	Board of Management	Annual	Vice Chancellor

1.4 State the process for defining the Vision and Mission of the Department, and PEOs of the program (15)

Total Marks 15.00

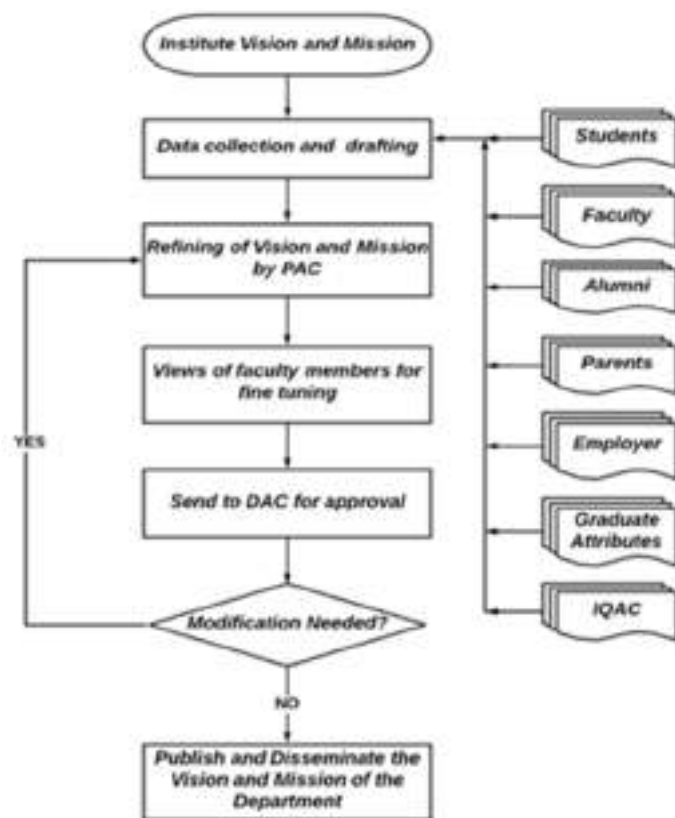


Fig 1.4(a): Process for defining Vision and Mission of the Department

Step 1: The Vision and Mission statements of the Institute are taken and inputs are gathered from the stakeholders (Students, Faculty, Parents, Employer, and Alumni). The views of the Internal Quality Assurance Cell (IQAC) and the Graduate Attributes (Program Outcomes or POs) are also considered and consolidated.

Step 2: The consolidated data are refined to formulate the Vision and Mission of the Department by the Program Assessment Committee (PAC).

Step 3: The formulated statements are circulated to the faculty members of the Department. After acquiring their views, the statements are fine-tuned in accordance with global trends.

Step 4: The fine-tuned statements are sent to the Department Advisory Committee (DAC) for approval. After reaching a consensus, the Vision and Mission statement of the Department are published and disseminated. If any modifications are suggested, the above process is repeated again.

Process for defining Program Educational Objectives (PEOs)

The Program Educational Objectives (PEOs) are established through a consultation process involving the core constituents such as students, alumni, faculty, employer and professional body members. The process for defining the PEOs is shown in the figure below.

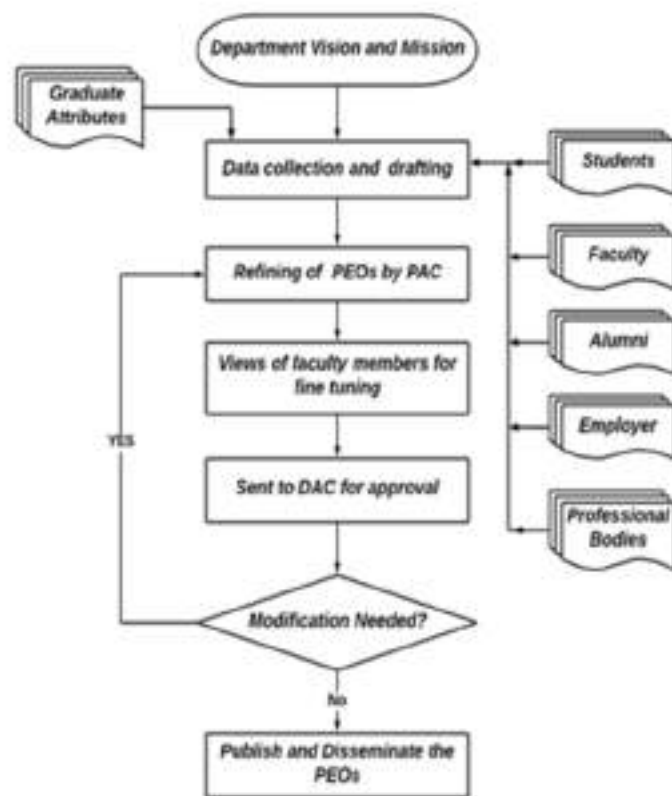


Fig 1.4(b): Process for formulating the PEOs of the Program

The PEOs are established through the following steps,

Step 1: Vision and Mission of the Department are considered along with the graduate attributes and inputs from various stakeholders.

Step 2: The collected inputs are reviewed by the Program Assessment Committee (PAC) for refining the PEOs.

Step 3: The refined PEOs are discussed with the faculty members of the Department. Their views are considered for fine tuning the objectives.

Step 4: The fine-tuned statements are scrutinized by the Department Advisory Committee (DAC) and a consensus is reached.

Step 5: After acquiring the approval of DAC, the framed PEOs are published and disseminated. If any modifications are suggested, the above process is repeated again.

1.5 Establish consistency of PEOs with Mission of the Department (10)

Total Marks 10.00

Table 1.5: Consistency of PEOs with Mission of the Department

PEO Statements	M1	M2	M3	M4	M5	Justification
PEO1: Become successful Naval Architects and Offshore Engineers with breadth and depth of knowledge who are competent, innovative and productive in addressing the needs of the industry.	3	3	3	3	2	M1: Strongly supports PEO1. Mission 1 focus on the upgradation of knowledge through industry institute interaction activities. Interaction with industries and real world helps the students to upgrade their knowledge and skills. M2: Strongly support PEO1. Mission 2 focus on pursuing higher education and motivate the students for their overall personality development in the field of Naval Architecture and Offshore Engineering. M3: Strongly support PEO1. Mission 3 focus on Innovativeness, Strong ethics and positive attitude which helps in making decisions in all critical situations. M4: Strongly support PEO1. Mission 4 focuses on establishing high quality teaching and research environment. The quality teaching and research environment is established to offer state-of-the-art in all programs. M5: Moderately support PEO1 because Mission 5 clearly focus on promoting world class research facilities to meet the challenging needs of society.
PEO2 Pursue higher education and research in Naval Architecture and Offshore Engineering, and other allied disciplines.	2	3	2	3	3	M1: Moderately support PEO2. Interaction with industries and real world helps students to become competent, productive and better successful professional. M2: Strongly support PEO2. By pursuing higher studies, students are able to excel in their career as per the industrial demands. M3: Moderately supports PEO2. Apart from skills and knowledge, ethics and positive attitude is also required to become an Engineering professional with a good career. M4 Strongly support PEO2. Quality teaching and research environment helps students to acquire technical knowledge and skills in order to produce highly innovative Engineers for the industry. M5: Strongly support PEO2. By creating world-class research facilities and infrastructure, students are able to develop their own enterprises or pursue higher education and productive in addressing the needs of the industry.
PEO3 Develop innovative thinking skills and pursue a life-long learning process to augment professional growth.	3	3	3	3	3	M1: Strongly supports PEO3. To practice the ethics of their professional career, updating the knowledge is essential through industry interaction activities. M2: Strongly support to achieve PEO3 because it focuses on higher education to meet the local and global demands in specialized areas. M3: Strongly support PEO3 since strong ethics and positive attitude is required to become a successful professional in the society. M4: Strongly supports PEO3 By acquiring quality education, students will be able to adapt any kind of environment and to have good ethics in their professional career. M5: Strongly support PEO3 because student can adapt to any environment and practice their profession facing the challenges.

PEO4 Exhibit high standards of ethical conduct, positive attitude and societal responsibility.						
	3	3	3	2	3	M1: Strongly support PEO4. Collaboration with industries helps students to update their knowledge to meet all the challenges. M2: Strongly supports PEO4 as Mission 2 focuses on knowledge in Naval Architecture and Offshore Engineering field. Updating knowledge and skills is always required to meet all the challenges at their work places. M3: Strongly supports PEO4. Since strong ethics and positive attitude of students, developing entrepreneurship skills will make a successful professional, who can face all challenges in the society. M4: Moderately supports PEO4. Since Mission 4 focus on quality teaching and research to produce innovative and socio-economically conscious Naval Architects and Offshore Engineers, to meet the challenges in maritime sector. M5: Strongly supports PEO4 as Mission 5 clearly focus on research and motivate the students to meet all challenges at their profession.

PEO Statements	M1	M2	M3	M4	M5
1. Become successful Naval Architects and Offshore Engineers with breadth and depth of knowledge who are competent, innovative and productive in addressing the needs of the global industry.	3 ▾	3 ▾	3 ▾	3 ▾	2 ▾
2. Pursue higher education and research in Naval Architecture and Offshore Engineering, and other allied disciplines.	2 ▾	3 ▾	2 ▾	3 ▾	3 ▾
3. Develop innovative thinking skills and pursue a life-long learning process to augment professional growth.	3 ▾	3 ▾	3 ▾	3 ▾	3 ▾
4. Exhibit high standards of ethical conduct, positive attitude and societal responsibility.	3 ▾	3 ▾	3 ▾	2 ▾	3 ▾

2 PROGRAM CURRICULUM AND TEACHING - LEARNING PROCESSES (100)

Total Marks 100.00

2.1 Program Curriculum (30)

Total Marks 30.00

An extensive multi-step process is in place for the design of program curriculum to ensure that it is continuously updated and stays tuned to the fast-paced changes in technology and industry requirements. Periodic changes are made to accommodate the current trends in the job market, change in technology, societal needs, and students' career. The process used to design program curriculum is shown in the figure below.

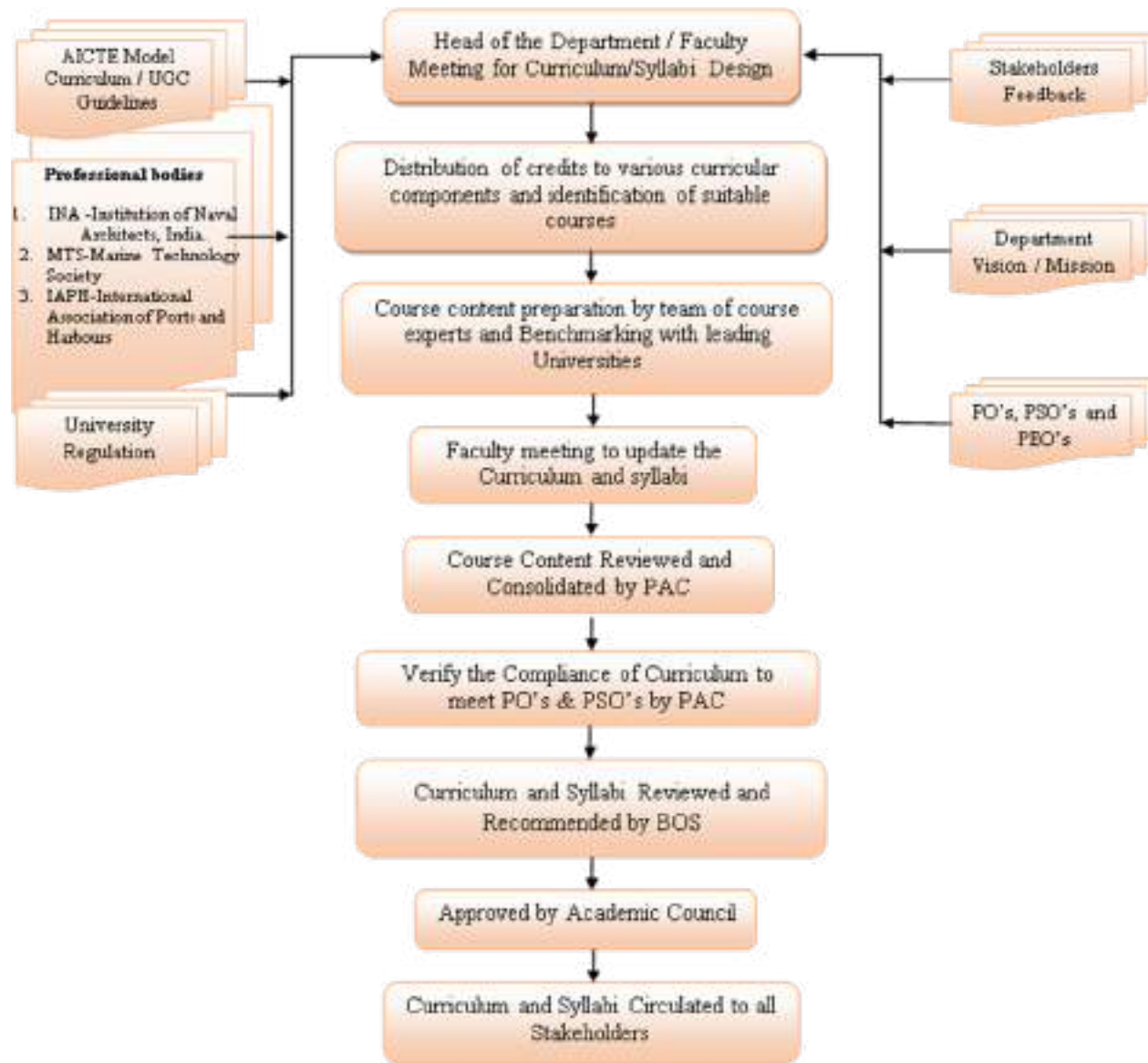


Fig 2.1.1 Process used to design the program curriculum

Department of Naval Architecture and Offshore Engineering follows a systematic process in the design and development of the curriculum. The steps involved in designing the program curriculum are given in the table below.

Table 2.1.1:Steps in designing the curriculum

Steps	Process
1	The design of the curriculum is done by keeping the stakeholders feedback, Department vision and mission, professional bodies' expectation, UGC regulations, university guidelines, NBA expectations as inputs and AICTE model regulations as guidelines for framing the curricula and syllabi. The consolidated inputs are presented in the faculty meeting to identify the gaps in the curriculum keeping the needs of the industry in view and other needs of the external stakeholders.
2	In the faculty meeting, the credits are distributed to various curricular components and suitable courses are identified. The course experts are assigned with the preparation of the content of the courses grouped under various curricular components.
3	The prepared curriculum is then compared with the curriculum and syllabi of the leading universities for benchmarking.
4	Faculty meeting is conducted to discuss the draft curriculum and the course contents with course objectives and Course Outcomes (COs). The domain experts verify the mapping of COs with POs and PSOs.
5	The course contents are reviewed by the Program Assessment Committee (PAC) of the Department. PAC also verifies the compliance of the draft curriculum to meet the POs and PSOs.
6	A program articulation matrix is prepared and reviewed by PAC to establish the extent of compliance. On satisfactory compliance of the COs with POs and PSOs, the draft curriculum is presented to Board of Studies (BoS).
7	The revised curriculum, approved by Board of Studies (BoS), is submitted for the approval of the Academic Council.
8	The approved curriculum is printed as per the university format and disseminated to stakeholders.

2.1.2 Structure of the Curriculum (5)

Institute Marks : 5.00

ID	Course Code	Course Title	Lecture (L)	Tutorial (T)	Practical (P)	Total Hours	Theory Credits	Practical Credits	Total Credits
1	C101	Technical English I	2	0	0	2	2	0	2
2	C102	Engineering Physics I	3	0	0	3	3	0	3
3	C103	Engineering Mathematics I	3	1	0	4	4	0	4
4	C104	Engineering Chemistry	4	0	0	4	4	0	4
5	C105	Induction program	0	0	0	0	0	0	0
6	C106	English Language Laboratory I	0	0	2	2	0	1	1
7	C107	Engineering Physics Laboratory	0	0	2	2	0	1	1
8	C108	Engineering Chemistry Laboratory	0	0	2	2	0	1	1
9	C109	Workshop Practices	0	0	4	4	0	2	2
10	C110	Engineering Physics II	2	0	0	2	2	0	2
11	C111	Engineering Mathematics II	3	1	0	4	4	0	4
12	C112	Basic Electrical Engineering	3	1	0	4	4	0	4
13	C113	Programming for Problem Solving	3	0	0	3	3	0	3
14	C114	Environmental Science	2	0	0	2	0	0	0
15	C115	English Laboratory II	0	0	2	2	0	1	1
16	C116	Basic Electrical Engineering Laboratory	0	0	2	2	0	1	1
17	C117	Programming for Problem Solving Laboratory	0	0	2	2	0	1	1
18	C118	Engineering Graphics and Design	1	0	4	5	0	3	3
19	C201	Engineering Mathematics – III	3	1	0	4	4	0	4
20	C202	Introduction to Naval Architecture	3	0	0	3	3	0	3
21	C203	Marine Materials and Welding Technology	3	0	0	3	3	0	3
22	C204	Marine Hydrodynamics	3	1	0	4	4	0	4
23	C205	Strength of Materials	3	0	0	3	3	0	3
24	C206	Engineering Mechanics	3	0	0	3	3	0	3
25	C207	Value Added Training Program - I	0	0	0	0	0	0	0

26	C208	Industrial Visit - I	0	0	0	0	0	0	0
27	C209	Interpersonal Communication	0	0	2	2	0	1	1
28	C210	Strength of Materials Lab	0	0	2	2	0	1	1
29	C211	Electrical Workshop	0	0	2	2	0	1	1
30	C212	Ship Drawing - Lines Plan	0	0	4	4	0	2	2
31	C213	Thermodynamics & Marine Machinery	3	0	0	3	3	0	3
32	C214	Marine Production Technology	3	0	0	3	3	0	3
33	C215	Elements of Offshore Engineering	3	1	0	4	4	0	4
34	C216	Theory of Ships	2	1	0	3	3	0	3
35	C217	Ethics and Values	3	0	0	3	3	0	3
36	C218	OEC - I	3	0	0	3	3	0	3
37	C219	Value Added Training Program - II	0	0	0	0	0	0	0
38	C220	Industrial Visit - II	0	0	0	0	0	0	0
39	C221	Professional Communication	0	0	2	2	0	1	1
40	C222	Hydrostatics & Stability Laboratory	0	0	4	4	0	2	2
41	C223	Surface Modelling and Analysis - Software Laboratory	0	0	2	2	0	1	1
42	C301	Biology for Engineers	2	0	0	2	2	0	2
43	C302	Strength of Ships	3	1	0	4	4	0	4
44	C303	Ship Resistance and Propulsion	3	1	0	4	4	0	4
45	C304	Object Oriented Programming	3	0	0	3	3	0	3
46	C305/ C306/	PEC - I	3	0	0	3	3	0	3
47	C305 C306	OEC - II	3	0	0	3	3	0	3
48	C309	Indian Constitution	2	0	0	2	0	0	0
49	C310	Value Added Training Program - III	0	0	0	0	0	0	0
50	C311	Industrial Visit - III	0	0	0	0	0	0	0
51	C312	Object Oriented Programming	0	0	2	2	0	1	1

52	C313	Ship Strength Laboratory	0	0	2	2	0	1	1
53	C314	Structural Modelling - Software Laboratory	0	0	2	2	0	1	1
54	C315	Internship – I	0	0	0	0	0	1	1
55	C316	Design of Offshore Structures	3	0	0	3	3	0	3
56	C317	Ship Motion and Control	3	1	0	4	4	0	4
57	C318	Ship Design	3	1	0	4	4	0	4
58	C319/ C320/	PEC - II	3	0	0	3	3	0	3
59	C321/ C322/	PEC - III	3	0	0	3	3	0	3
60	C323/ C324/	OEC - III	3	0	0	3	3	0	3
61	C327	Essence of Indian Traditional Knowledge	2	0	0	2	0	0	0
62	C328	Finishing School Training - I	0	0	0	0	0	0	0
63	C329	Value Added Training Program - IV	0	0	0	0	0	0	0
64	C330	Industrial Visit - IV	0	0	0	0	0	0	0
65	C331	Marine Hydrodynamics Laboratory	0	0	2	2	0	1	1
66	C332	Offshore Structure Design - Software Laboratory	0	0	2	2	0	1	1
67	C401	Advanced Ship Technology	3	0	0	3	3	0	3
68	C402/ C403/	PEC - IV	3	0	0	3	3	0	3
69	C404/ C405/	PEC - V	3	0	0	3	3	0	3
70	C406/ C407/	PEC - VI	3	0	0	3	3	0	3
71	C408/ C409/	OEC - IV	3	0	0	3	3	0	3
72	C414	Finishing School Training - II	0	0	0	0	0	0	0
73	C415	Value Added Training Program - V	0	0	0	0	0	0	0
74	C416	Industrial Visit - V	0	0	0	0	0	0	0
75	C417	Ship System Drawing & Launching Calculations Laboratory	0	0	2	2	0	1	1
76	C418	Numerical Ship Hydrodynamics – Software Laboratory	0	0	2	2	0	1	1
77	C419	Mini Project	0	0	4	4	0	2	2

78	C420	Internship - II	0	0	0	0	0	1	1
79	C421	Major Project / Industry Internship Project	0	0	22	22	0	11	11
		Total	115	11	78	204	119	42	161

2.1.3 State the components of the curriculum (5)

Institute Marks : 5.00

Course Components	Curriculum Content (% of total number of credits of the program)	Total number of contact hours	Total number of credits
Basic Sciences	15.5%	27.00	25
Engineering Sciences	16.1%	35.00	26
Humanities and Social Scie	5.9%	15.00	9
Program Core	34.7%	45.00	56
Program Electives	11.2%	18.00	18
Open Electives	7.5%	12.00	12
Project(s)	8.1%	22.00	13
Internships/Seminars	1.24%	0.00	2
Any other (Please specify)	0	6.00	0
Total number of Credits			161

2.1.4 State the process used to identify extent of compliance of the curriculum for attaining the Program Outcomes and Program Specific Outcomes as mentioned in Annexure I (10)

Institute Marks : 10.00

The curriculum for B.E – Naval Architecture and Offshore Engineering maintains a balance among various categories of courses from science, mathematics, engineering, humanities and social science including management, projects and internship components. The syllabus for each course has been designed to comply with POs and PSOs of the program.

The Program Outcomes (POs) and Program Specific Outcomes (PSOs) of B.E – Naval Architecture and Offshore Engineering are shown in the table below.

Table B.2.1.4(a) : Program Outcomes of the curriculum

Program Outcomes (POs)	
PO 1	Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO 2	Problem Analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO 3	Design/development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO 4	Conduct Investigations of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO 5	Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO 6	The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO 7	Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO 8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO 9	Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO 10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions

PO 11	Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments
PO 12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.
Program Specific Outcomes (PSOs)	
PSO1	Develop innovative and contemporary designs of ships and offshore structures through advanced analysis and experiments to meet the challenges of marine operations in the shipbuilding, offshore, and other maritime industries.
PSO2	Implement engineering design and construction concepts to produce solutions in the field of Naval Architecture and Offshore Engineering.
PSO3	Apply research based knowledge and research methods to analyse and interpret data using modern tools towards the solution of complex problems in the area of Naval Architecture and Offshore Engineering.

The flowchart shown below is used to identify the extent of compliance of the curriculum.

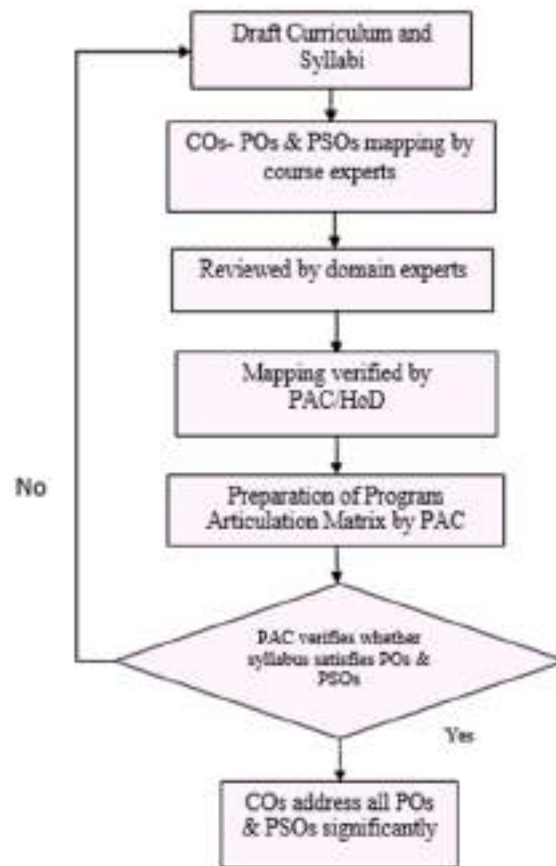


Fig 2.1.4 Process used to identify the extent of compliance of the curriculum

The following steps are adopted to identify the extent of compliance of the curriculum.

Step 1: Course content is prepared by course experts with COs-POs and PSOs mapping.

Step 2: Course content with COs-POs mapping is reviewed by the domain experts.

Step 3: The COs-POs and PSOs mapping is verified by the PAC/HoD.

Step 4: PAC prepares the Program Articulation Matrix and verifies whether all courses are significantly mapped with the POs and PSOs.

Step 5: It is ensured that all POs and PSOs are adequately covered by the courses being taught and each course is mapped to a high correlation level with at least one PO. It is also ensured that all POs and PSOs have high correlation with adequate number of courses.

Step 6: PAC verifies whether syllabus is able to contribute significantly the POs and PSOs. Otherwise suggestions are given to the course experts to enrich the course content to attain the POs and PSOs.

The courses that are highly correlated with each of the POs and PSOs of regulation D (2018-2022) are shown in the table below.

Table B.2.1.4(b) : Courses highly correlated with POs and PSOs

POs /PSOs	Courses Mapped Significantly
PO1	C202,C203,C204,C205,C206,C210, C212,C211,C213,C214,C215,C216, C306,C301,C307,C304,C312,C314,C313,C315,C305,C317,C317,C318,C319, C320,C321,C322,C323,C324,C325,C331,C332,C401,C402,C403,C404,C405, C406,C407,C408,C409,C410,C411,C412,C417,C418,C419,C420,C421
PO2	C112,C113,C114,C115,C117,C116,C118,C201,C202,C203,C204,C205,C206,C210, C209,C212,C211,C213,C214,C215,C216,C217,C221,C222,C223,C302,C303,C306 ,C301,C307,C304,C312,C314,C313,C315,C305,C317,C317,C318,C319,C320,C321, C322,C323,C324,C325,C331,C332,C401,C402,C403,C404,C405,C406,C407,C408, C409,C410,C411,C412,C417,C418,C419,C420,C421.
PO3	C103,C104,C106,C107,C108,C109,C110,C111,C112,C113,C114,C115,C117,C116, C118,C201,C202,C203,C204,C205,C206,C210,C209,C212,C211,C213,C214,C215, C216,C217,C221,C222,C223,C302,C303,C306,C301,C307,C304,C312,C314,C313, C315.C305.C317,C317,C318,C319,C320,C321,C322,C323,C324,C325,C331,C332, C401,C402,C403,C404,C405,C406,C407,C408,C409,C410,C411,C412
PO4	C107,C108,C112,C117,C116,C201,C202,C205,C210,C212,C215,C216,C303,C306, C307,C313,C317,C318,C323,C401,C402,C403,C404.
PO5	C107,C109,C117,C202,C212,C215,C401,C402,C404,C405,C407,C409,C410,C411,C418.
PO6	C101,C106,C114,C115,C211,C217,C322,C401,C419,C421.
PO7	C104,C108,C114,C203,C214,C216,C321,C322,C408,C412,C419,C421.
PO8	C114,C217,C315,C318,C321,C322,C323,C401,C402,C410,C412,C419,C420,C421.
PO9	C115,C117,C210,C209,C212,C211,C315,C331,C402,C420.
PO10	C115,C117,C118,C209,C211,C217,C221,C315,C402,C419,C420.
PO11	C209,C217,C318,C320,C325,C331,C402,C403,C404.
PO12	C101,C106,C111,C112,C115,C116,C118,C203,C205,C212,C211,C213,C214, C215,C222,C302,C303,C317,C317,C318,C322,C332,C401,C403,C404.
PSO1	C112,C116,C203,C204,C205,C206,C210,C213,C214,C215,C216,C223,C302,C303, C306,C307,C314,C313,C315,C317,C317,C318,C319,C322,C323,C324,C325,C331 C332,C401,C402,C403,C404.
PSO2	C112,C116,C201,C202,C203,C204,C205,C206,C210,C213,C214,C215,C216,C223, C302,C303,C306,C307,C314,C313,C315,C317,C317,C318,C321,C322,C323,C324,C332, C401,C402,C403,C404.
PSO3	C112,C116,C203,C204,C205,C210,C213,C214,C216,C223,C302,C303,C306,C307,C314,C313, C315,C317,C317,C318, C319,C321,C322,C323,C324,C332,C401,C403,C404.

2.2 Teaching-Learning Processes (70)

Total Marks 70.00

The course choices are collected from the faculty by the PAC. Based on the specialization of faculty and their experience in handling the course, the courses are allotted and time table is prepared. The lesson plans are prepared by the faculty with the careful selection of activity based learning techniques and the same is approved by PAC.

To strengthen the teaching-learning process, the following initiatives are taken:

- Adherence to Academic Calendar
- Instruction methods and pedagogical initiatives
- Continuous assessment in the laboratory
- Methodologies to support slow learners/advanced learners
- Student feedback and action taken

The teaching-learning process followed in the Department is described in the figure below.

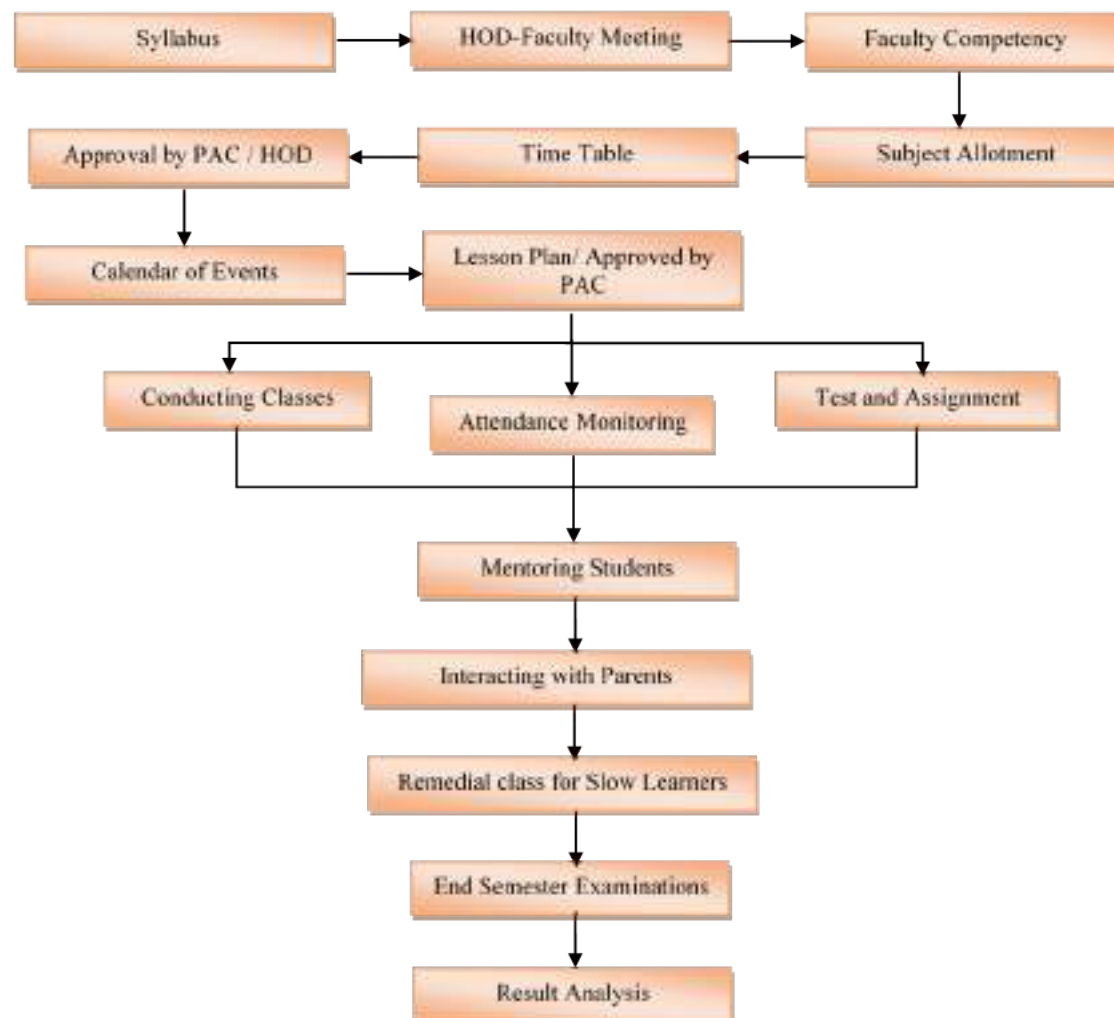


Fig 2.2.1(a)Teaching-Learning Process

To strengthen the teaching-learning process, following initiatives have been taken:

Adherence to Academic Calendar

Academic Calendar is prepared by the office of Registrar, prior to the commencement of the academic year. Important events and other academic activities such as university reopening, number of working days, induction program, holidays, commencement of Continuous Assessment Test (CAT), Model Examination, End Semester Examination, practical examination, etc. are scheduled and communicated to the students through Academic Calendar, which is also available in the university website. A sample Academic Calendar is shown in the table below.

Sample Academic Calendar for the academic year 2021-2022**Table B.2.2.1(a): July 2021**

Date	Day	Events	No. of Working Day
July 2021			
15/07/2021	Thursday	Commencement of online classes for Odd Semester	1
16/07/2021	Friday		2
17/07/2021	Saturday		3
18/07/2021	Sunday	Sunday	Holiday
19/07/2021	Monday		4
20/07/2021	Tuesday		5
21/07/2021	Wednesday	Bakrid	Holiday
22/07/2021	Thursday		6
23/07/2021	Friday		7
24/07/2021	Saturday		8
25/07/2021	Sunday	Sunday	Holiday
26/07/2021	Monday		9
27/07/2021	Tuesday		10
28/07/2021	Wednesday		11
29/07/2021	Thursday	Department Club Activities	
30/07/2021	Friday		12
31/07/2021	Saturday		13

Table B.2.2.1(b): August 2021

Date	Day	Events	No. of Working Day
August 2021			
01/08/2021	Sunday	Sunday	Holiday
02/08/2021	Monday		14
03/08/2021	Tuesday		15
04/08/2021	Wednesday		16

05/08/2021	Thursday		17
06/08/2021	Friday		18
07/08/2021	Saturday		19
08/08/2021	Sunday	Sunday	Holiday
09/08/2021	Monday	Industrial Visit	20
10/08/2021	Tuesday		21
11/08/2021	Wednesday		22
12/08/2021	Thursday		23
13/08/2021	Friday		24
14/08/2021	Saturday	Department Club Activities	
15/08/2021	Sunday	Independence day	Holiday
16/08/2021	Monday		25
17/08/2021	Tuesday		26
18/08/2021	Wednesday		27
19/08/2021	Thursday		28
20/08/2021	Friday	Moharam Festival	Holiday
21/08/2021	Saturday		29
22/08/2021	Sunday	Sunday	Holiday
23/08/2021	Monday		30
24/08/2021	Tuesday	Submission of CAT-I Question Papers	31
25/08/2021	Wednesday		32
26/08/2021	Thursday		33
27/08/2021	Friday		34
28/08/2021	Saturday		35
29/08/2021	Sunday	Sunday	Holiday
30/08/2021	Monday	Krishna Jayanthi	Holiday
31/08/2021	Tuesday	Commencement of CAT-I Examination	36

Table B.2.2.1(c): September 2021

Date	Day	Events	No. of Working Day
September 2021			
01/09/2021	Wednesday		37
02/09/2021	Thursday		38
03/09/2021	Friday		39
04/09/2021	Saturday		40
05/09/2021	Sunday	Sunday	Holiday

06/09/2021	Monday		41
07/09/2021	Tuesday		42
08/09/2021	Wednesday	Industrial Visit	43
09/09/2021	Thursday		44
10/09/2021	Friday	Vinayagar Chathurthi	Holiday
11/09/2021	Saturday		45
12/09/2021	Sunday	Sunday	Holiday
13/09/2021	Monday		46
14/09/2021	Tuesday		47
15/09/2021	Wednesday	Department Club Activities	
16/09/2021	Thursday	Submission of CAT-I marks	48
17/09/2021	Friday		49
18/09/2021	Saturday		50
19/09/2021	Sunday	Sunday	Holiday
20/09/2021	Monday		51
21/09/2021	Tuesday		52
22/09/2021	Wednesday		53
23/09/2021	Thursday		54
24/09/2021	Friday		55
25/09/2021	Saturday		56
26/09/2021	Sunday	Sunday	Holiday
27/09/2021	Monday		57
28/09/2021	Tuesday		58
29/09/2021	Wednesday		59
30/09/2021	Thursday		60

Table B.2.2.1(d): October 2021

Date	Day	Events	No. of Working Day
October 2021			
01/10/2021	Friday	Industrial visit	61
02/10/2021	Saturday	Gandhi Jayanthi	Holiday
03/10/2021	Sunday	Sunday	Holiday
04/10/2021	Monday		62
05/10/2021	Tuesday		63
06/10/2021	Wednesday		64
07/10/2021	Thursday		65
08/10/2021	Friday		66

09/10/2021	Saturday		67
10/10/2021	Sunday	Sunday	Holiday
11/10/2021	Monday	Submission of CAT-II Question Papers	68
12/10/2021	Tuesday		69
13/10/2021	Wednesday		70
14/10/2021	Thursday	Ayudha Pooja	Holiday
15/10/2021	Friday	Vijayadasami	Holiday
16/10/2021	Saturday	Department Club Activities	
17/10/2021	Sunday	Sunday	Holiday
18/10/2021	Monday	Commencement of CAT-II Examination Exams	71
19/10/2021	Tuesday	Miladinabi	Holiday
20/10/2021	Wednesday		72
21/10/2021	Thursday		73
22/10/2021	Friday		74
23/10/2021	Saturday		75
24/10/2021	Sunday	Sunday	Holiday
25/10/2021	Monday		76
26/10/2021	Tuesday		77
27/10/2021	Wednesday		78
28/10/2021	Thursday		79
29/10/2021	Friday		80
30/10/2021	Saturday		81
31/10/2021	Sunday	Sunday	Holiday

Table B.2.2.1(e): November 2021

Date	Day	Events	No. of Working Day
November 2021			
01/11/2021	Monday	Submission of CAT-II Marks	82
02/11/2021	Tuesday		83
03/11/2021	Wednesday	Alumini Guest Lecture	84
04/11/2021	Thursday	Deepavali festival	Holiday
05/11/2021	Friday		85
06/11/2021	Saturday	Submission of Model Question Papers	86
07/11/2021	Sunday	Holiday	Holiday
08/11/2021	Monday		87

09/11/2021	Tuesday		88
10/11/2021	Wednesday		89
11/11/2021	Thursday		90
12/11/2021	Friday	Department Club Activities	
13/11/2021	Saturday	Commencement of Model Exams	
14/11/2021	Sunday	Holiday	Holiday
15/11/2021	Monday		
16/11/2021	Tuesday		
17/11/2021	Wednesday		
18/11/2021	Thursday		
19/11/2021	Friday		
20/11/2021	Saturday		
21/11/2021	Sunday	Holiday	Holiday
22/11/2021	Monday	Practical Examinations	
23/11/2021	Tuesday		
24/11/2021	Wednesday		
25/11/2021	Thursday		
26/11/2021	Friday	Submission of Model Exam Marks and Internal Marks	
27/11/2021	Saturday		
28/11/2021	Sunday	Holiday	Holiday
29/11/2021	Monday	Commencement of End Semester Examinations	
30/11/2021	Tuesday		

Table B.2.2.1(f): December 2021

Date	Day	Events	No. of Working Day
December 2021			
01/12/2021	Wednesday		
02/12/2021	Thursday	Work Load Distribution to the faculty for 2021-22 (Even Semester)	
03/12/2021	Friday		
04/12/2021	Saturday		
05/12/2021	Sunday	Holiday	Holiday
06/12/2021	Monday		
07/12/2021	Tuesday		
08/12/2021	Wednesday		

09/12/2021	Thursday		
10/12/2021	Friday		
11/12/2021	Saturday		
12/12/2021	Sunday	Holiday	Holiday
13/12/2021	Monday		
14/12/2021	Tuesday	Time Table uploading for 2021-22 (Even Semester)	
15/12/2021	Wednesday	University End Semester Examination Ends	
16/12/2021	Thursday	ODD Semester Vacation Commences	
17/12/2021	Friday	Course coordinator's Meeting	
18/12/2021	Saturday	Preparation of Academic Calendar for 2021-22 (Even Semester)	
19/12/2021	Sunday	Holiday	Holiday
20/12/2021	Monday		
21/12/2021	Tuesday	Lesson Plan Submission for 2021-22 (Even Semester)	
22/12/2021	Wednesday		
23/12/2021	Thursday	One day National Workshop	
24/12/2021	Friday		
25/12/2021	Saturday		
26/12/2021	Sunday	Holiday	Holiday
27/12/2021	Monday		
28/12/2021	Tuesday		
29/12/2021	Wednesday		
30/12/2021	Thursday		

Action plan includes major routine tasks for timely execution of the teaching-learning process. Based on this, the lesson plan and assessment plan are prepared for every course. Department also schedules conferences, seminars, guest lectures and industrial visits based on the department academic calendar. Sample departmental action plan for odd and even semesters are shown in the tables below.

Table B.2.2.1(f) Action plan for conducting important events of the Department during 2021-2022

Date	Events
15.07.2021	University reopens for ODD Semester
24.08.2021	CAT I Question Paper submission
09.08.2021	Industrial visit
31.08.2021	Commencement of CAT I Examination
16.09.2021	Submission of CAT I Marks

01.10.2021	Industrial visit
11.10.2021	Submission of CAT II Question Papers
18.10.2021	Commencement of CAT II Examination
01.11.2021	Submission of CAT II marks
03.11.2021	Alumini Guest Lecture
06.11.2021	Submission of Model Exam Question papers
13.11.2021	Commencement of Model Examination
06.11.2021	Commencement of University End Semester Practical Examinations
13.11.2021	Commencement of Model Examinations
22.11.2021	Commencement of Practical Examinations
26.11.2021	Submission of Model Exam Marks and Internal Marks
29.11.2021	Commencement of End Semester Examinations
1 st week of December	Work Load Distribution to the faculty for 2021-22 (Even Semester)
2 nd week of December	Time Table uploading for 2021-22 (Even Semester)
3 rd week of December	University End Semester Examination Ends
3 rd week of December	ODD Semester Vacation Commences
3 rd week of December	Course coordinator's Meeting
3 rd week of December	Preparation of Academic Calendar for 2021-22 (Even Semester)
3 rd week of December	Lesson Plan Submission for 2021-22 (Even Semester)
23.12.2021	One day National Workshop

Table B.2.2.1(g) Deviation in the Academic Calendar 2021-2022

S.No	Proposed Dates	Planned Tasks	Revised dates	Reason
1	1 st October 2021	Industrial Visit	6 th October 2021	Due to heavy rain

The main items in the action plan of the Department are shown in the table below

Table B.2.2.1(h) Department Action Plan

Department Action Plans	Period	Remarks
Course choices from Faculty	2-3 weeks prior to the commencement of semester	Course choices are requested from faculty for the forthcoming semester.

Work Load Distribution	Work load Distribution 2-3 weeks prior to the commencement of semester	Allotment of courses are done by the Timetable coordinator and HoD based on the specialization of faculty or based on the expertise on a particular course.
Time Table Uploading	2 weeks prior to the commencement of semester	Time table is prepared by the Time-table committee and the same is approved by PAC and HoD, is uploaded in MYCAMU.
Course/ Lesson Plan	2 weeks prior to the commencement of semester	Course plans are collected from the faculty for their respective courses and reviewed by PAC.
CAT, Model exam and ESE question papers	2 weeks for CAT and 4 weeks for Model and ESE prior to the commencement of examination	Question papers have been collected from the faculty and reviewed at the Department level by PAC .
Mini Project (IV year)	Within 1 week after commencement of semester	Students are allowed to choose the guide, identify the project and same is submitted to project coordinator for record.
Main Project (IV Year)	One month prior to the commencement of the VIII Semester.	Students are allowed to and choose the guide based on the area of interest, identify the project and same is submitted to project coordinator for record. Project coordinator prepares the schedules for review and presentation.

Pedagogical Initiatives

Pedagogy plays an important role in delivery of content and it varies with the audience. Course allocation is made based on the choice/expertise of the faculty members 2 -3 weeks before the commencement of course. Class time tables and lab manuals/study material are prepared and distributed to students. Since the teaching-learning is outcome based, various modes of course delivery are implemented in such a way to identify students' skill.

Various pedagogical initiatives used for course delivery are listed below:

- Ø ICT based learning: Use of LCD projectors, smart board and internet connection in class rooms.
- Ø Learning Management System (LMS) of AMET and Google class room for quiz, assignment, notes, etc.
- Ø Project based learning.
- Ø Flipped classrooms like product demo, quiz, MCQ, animation, simulation video, NPTEL, MOOC courses, etc.
- Ø Digital texts enable real time learning, monitoring, comprehension and online assessment.
- Ø Workshops/ Value Added Courses.
- Ø Expert talk/ Guest lectures.
- Ø Laboratory/ video based demonstration: Demonstration of system or parts of a real-world system using modern tools.
- Ø Group discussion/ presentation: Students learn through group discussion and are asked to deliver short presentation on a topic.

Faculty undergo Faculty Development Program (FDP) and learn various pedagogical methods/tools to improve teaching-learning process.

Sample list of Training Program attended by faculty is listed in table below.

Table B.2.2.1(i) List of Faculty who attended FDP, Workshops and Seminars

2022-2023

S.No	Faculty Name	Seminar / Workshop/FDP	Title of the program	Venue	Date & Duration
1	Mr.MSP Raju	Workshop	Professional development program on implementation of NEP 2020 for University and college teachers	UGC-IGNOU, Delhi	18/11/2022 to 26/11/2022 (9 days)
2	Mr.MSP Raju	FDP	Accreditation and outcome based pedagogic principle for effective teaching	AICTE-ATAL Academy	14/11/2022 to 25/11/2022 (11 days)
3	Mr.MSP Raju	FDP	Accreditation and outcome based learning	NPTEL- AICTE	Aug- Oct 2022 8 weeks
4	Mr.MSP Raju	FDP	Spoken Sanskrit: Basic and Intermediate level	NPTEL- AICTE	Jul- Oct 2022 12 weeks
5	Mr.Thalapathy Vigneshwar S	FDP	Design Thinking	ICT Academy,Adhiyaman College, Hosur, Tamilnadu	28/11/2022 to 29/11/2022 (2 days)
6	Mr.R. Jaya Surya	FDP	Universal Human Value	AICTE	15/8/22 - 20/8/22 (5days)
7	Dr.Sivabalan P	FDP	Accreditation and outcome based pedagogic principles for effective teaching	AICTE & AMET University (Online)	14/11/2022 to 25/11/2022 (11 days)
8	Mr.A.Prem Anandh	Workshop	Executive management for shipping industry	Business college of Athens, Greece & AMET University, Online	23/01/2023 to 30/01/2023 (7 days)
2021-2022					
1.	Mr.MSP Raju	FDP	Design Technology and Innovation –	NPTEL Course	Jan - April 2022 12 weeks
2.	Mr.MSP Raju	FDP	Applications of Artificial Intelligence in Digital Manufacturing –	NIT Warangal	21.02.2022to 03.03.2022 & 12 Days
3.	Mr.MSP Raju	FDP	UHV-II	AICTE, Delhi	20/06/2022 to 25/06/2022

4.	Dr.P.Sivabalan	FDP	Applications of Artificial Intelligence in Digital Manufacturing –	NIT Warangal	21.02.2022to 03.03.2022 & 12 Days
5.	Dr.P.Sivabalan	FDP	NPTEL – Air Pollution & Control	NPTEL	Jan - April 2022
6.	Dr.R.Lilly	FDP	Artificial Intelligence for IoT Services in Cloud: Techniques and Applications	NIT,Jalandar (Offline)	28.02.22 to 4.03.22. 5 Days
7.	Dr.R.Lilly	FDP	Offshore Structures under special Environmental loads including Fire	Jan-Apr 2022	4 Months (NPTEL) 3 Days
8.	Dr.R.Lilly	Seminar	Advanced Emerging Technologies, Research and Practical Applications	Institute for Engineering Research	18/05/2022 1 day
9.	Dr.R.Lilly	Seminar	How to face Interview?	Institute for Engineering Research	29/06/2022 1 day
10.	Dr.R.Lilly	Seminar	India's 5G future	Institute for Engineering Research	15/10/2022 1 day
11.	Dr.R.Lilly	Seminar	Evolution of Intelligent system	Institute for Engineering Research	11/05/2022 1 day
12.	Dr.R.Lilly	Seminar	Current Trends in Machine Learning	Institute for Engineering Research	04/05/2022
13.	Dr.R.Lilly	Seminar	Energy Conservation and Energy Efficiency for Domestic Sector	AMET University	09/01/2022
14.	Dr.R.Lilly	Seminar	Maintaining health at workplace	AMET University	10/02/2022
15.	Dr.R.Lilly	Seminar	Consumer Protection Laws	Consumer Club,Vaniyambadi	24/12/2021
16.	Mr. A .Prem Anandh	FDP	Safe campus Program	AIU & Maratha foundation , Online	30/07/2021- 31/07/2021 2 days
17.	Mr. A .Prem Anandh	FDP	Applications of Artificial Intelligence in Digital Manufacturing –	NIT Warangal	21.02.2022to 03.03.2022 & 12 Days
18.	Mr.R. Jaya Surya	FDP	Offshore Structures under special Environmental Loads	NPTEL	Jan-Apr 2022 12 weeks
2020-2021					

1	Dr.R.Lilly	FDP	Managing Infrastructure projects using Emerging Technologies	Thiyagarajar College of Engineering, Madurai	28/02/2022 to 04/03/2022. 5 Days
2	Dr.R.Lilly	FDP	An Interdisciplinary Research Metrics	SRM (Rampuram Cmapus)	Jan-Apr 2022 3 months
3	Dr.P.Sivabalan	FDP	Blended And Flipped Approach In Teaching And Learning	ATAL	24/08/2020 to 28/08/2020 5 Days
4	Dr.P.Sivabalan	FDP	Green Technology & Sustainability Engineering	ATAL	11/01/2021 to 15/01/2021 5 Days
5	Dr.P.Sivabalan	FDP	Block chain Technology	ATAL	18/01/2021 to 22/01/21 5 Days
6	Dr.P.Sivabalan	FDP	Waste Technology	ATAL	27/01/2021 to 31/01/2021 5 Days
7	Mr..MSP. Raju	FDP	Inculcating Human Values in Technical Education	AICTE	19/04/2021 to 23/04/2021 5 days
8	Mr.A.Prem Anandh	Workshop	Intellectual property rights	TNSCST >N Arts college, Dindigal (Online)	21/06/2021 to 22/06/2022 2 days
9	Mr.A.Prem Anandh	FDP	Innovation, IPR, Technology Development and Entrepreneurship	SCC- KSCST IP Cell ,Institution's Innovation Council (SCC-IIC) &St.Claret College Karnataka	14/06/2021 to 19/06/2021 6 days

10	Mr.R Jayasurya	FDP	An Interdisciplinary Research Metrics: Fostering Faculty Knowledge and Skills	SRM UNIVERSITY, Chennai	3/05/2021 to 08/05/2021 6 days
11	Mr.R Jayasurya	FDP	" Inculcating Universal Human Values in Technical Education	JNTU,AP,KALIKIRI	12/07/2021 to 16/07/2021 5 days
12	Dr.P.A.Prasob	FDP	Innovation, IPR, Technology Development and Entrepreneurship	SCC- KSCST IP Cell ,Institution's Innovation Council (SCC-IIC) &St.Claret College Karnataka	14/06/2021 to 19/06/2021 6 days
13	Mr.A.Prem Anandh	FDP	"Bioconversion Technologies and Start-up Opportunities in Biofuels Production	Silver Jubilee Govt College.(ATAL Academy)	14/06/2021 to 18/06/2021

In addition to the above, innovative teaching methods are adopted to deliver the course content more effectively, wherever appropriate. A few innovative teaching methods are given in the table below.

Table B.2.2.1(j) Innovative teaching methods for sample courses

S.No	Pedagogical methods	Innovative teaching practices	Course	Faculty Name
1	Video, Animations, assignments	Use of Model	Introduction to Naval Architecture	Dr.P.Sivabalan
2	Video, assignments	Lab demo	Marine materials and welding technology	Dr.P.A.Prasob
3	Video, Lab demo, assignments	Lab demo	Marine Hydrodynamics	Dr.B.M.Shameem
4	Video, Assignments, Lab demo	Lab demo	Strength of materials	Dr.R.Lilly
5	Video, assignments	Guest lecture / Expert talk	Marine production technology	Dr.P.A.Prasob
6	Video, assignments	Guest lecture / Expert talk	Elements of Offshore Engineering	Dr.R.Lilly
7	Video, assignments	Use of Model	Theory of ships	Dr.P.Sivabalan
8	Video, assignments	Guest lecture / Expert talk	Ship resistance and propulsion	Prof.MSP.Raju
9	Video, Lab demo, assignments	Guest lecture / Expert talk	Design of offshore structures	Dr.K.Thiruvengatasamy
10	Video, assignments	Lab demo	Ship motion and control	Dr.J.Suneela
11	Video, Lab demo, assignments	CAD model	Ship design	Prof.MSP.Raju

Table B.2.2.1(k) Sample Courses with Activity based Learning

S.No.	Course Name	Topic Covered	Activity	Conducted Date	Course Incharge	Year of Study	RELEVANCE TO POs, PSOs
-------	-------------	---------------	----------	----------------	-----------------	---------------	------------------------

1	Theory of ships	Geometry and main parts of ship	Demo	02.10.2021	Mr.R.Jayasurya	II Yr NAOE	PO1,PO2, PO3, PO4, PO7, PO12, PSO1,PSO2,PSO3
2	Marine materials and welding technology	Submerged arc welding	Video	11.10.2022	Dr.P.Sivabalan	II Yr NAOE	PO1, PO2, PO3, PO4, PO5, PO7,PO12, PSO1,PSO2,PSO3
3	Ship Resistance and propulsion	Estimation of resistance and design of propellers	MCQ	16.09.2021	Mr. MSP Raju	III Yr NAOE	PO1,PO2,PO3,PO4, PO5, PO7, PO12, PSO1, PSO2,PSO3
4	Marine hydrodynamics	Model making techniques	Demo	21.10.2021	Dr.B.M.Shameem	II Yr NAOE	PO1,PO2,PO3,PO4, PO12 PSO1, PSO2,PSO3

Methodologies to support weak students and encourage bright students

Based on the academic performance of the students, slow learners are identified and encouraged to attend additional classes to understand the subject in a better way. Mentors assist the slow learners in all aspects. Advance learners are encouraged to attend seminars, group discussions and present their ideas in Symposium, National/ International Conferences. High-talented students are encouraged to take case studies and industry related projects. The process followed to identify and support slow learners and bright students is shown in the figure below.

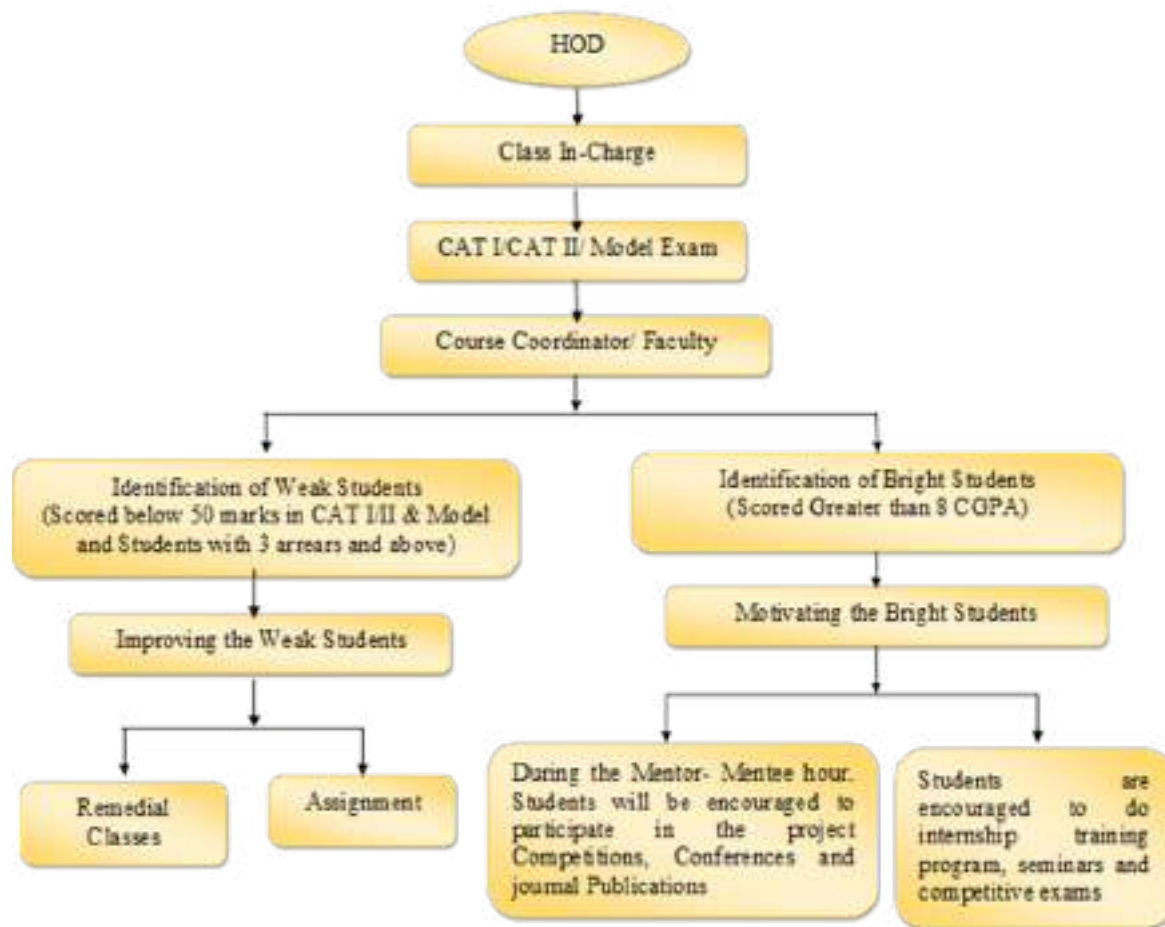


Fig 2.2.1(b) Process followed to identify and support slow learners and bright students

Weak students:

(i) Identification

Weak students are based on assessment that meets any one of the following criteria:

- Based on number of arrears – students with 3 or more than 3 arrears.
- Poor performance in the Continuous Assessment Test – If the student gets assessment marks < 50% in CAT 1 or CAT 2.
- Lack of ability to understand and solve problem during class hours.

(ii) Support system

- Counseling through mentors, and also by student counselor, if required.
- Provision for special classes to improve the academic performance of the students.

Bright students:

(i) Identification

Bright students are identified based on assessment through any one of the following criteria:

- a. Student previous academic year performance - If the student's CGPA > 8.
- b. Performance in the Continuous Assessment Test – If the student gets Assessment marks > 80% in CAT 1 or CAT 2 or in Assignment.
- c. Demonstration of superior problem solving and creative skills

(ii) Support System

1. Students are motivated to pursue online MOOC / Swayam – NPTEL courses.
2. Students are encouraged to take up innovative projects.
3. Special coaching classes are arranged for competitive exams like GATE, GRE, TOEFL, IELTS, etc.
4. Students are motivated to participate in inter-institute technical competitions.

Quality of classroom teaching

All class rooms are equipped with ICT based learning tools, namely, LCD projectors, smart board, desktops with internet connection. The classrooms are spacious and well maintained. A teacher, in addition to white board, can make use of power point presentations, animations, and other audio-visual resources in the classroom to enhance the quality of teaching.

Conduct of experiments

All software labs are well equipped with high end configuration computers and required software. All labs are well maintained and safety precautions are followed. Each student are individually provided with a computer for the laboratory experiments. Every laboratory exercise / experiment is categorized as

- v. Demo Exercise/Experiment
- v. Modelling Exercise/Experiment
- v. Viva / Oral exam
- v. Open-ended Exercise/Experiment

Continuous assessment in the laboratory

Student performance in the laboratory class is evaluated and records are maintained. Continuous evaluation of each laboratory experiment is done based on the following parameters shown in the table below.

Table B.2.2.1(l) Marks distribution for laboratory courses

Component of Evaluation	Internal/ external	Marks	Total Marks
Familiarization with the software/ equipment	Internal	15	40
Practical Performance during classes		15	
Record Note Book		10	
Practical Exam	External	30	60
Result		10	
Viva Voce		20	
Grand Total			100

Rubrics for continuous evaluation of laboratory courses is shown in the table below

Table B.2.2.1(m) Rubrics for continuous evaluation (Laboratory courses)

Assessment Parameter	Proficient (3 marks)	Acceptable (2 marks)	Needs Improvement (Less than 2)
Familiarization with the software/equipment	Listed the modelling tools required to execute the given task with their specifications and type	Listed the modelling tools required to execute the given task with incorrect specifications and type	Listed the modelling tools only partially.
Computer modelling of 3D geometry/use of preprocessor and post processor	Design the model with the appropriate tools and execute the task.	Design the model with the irrelevant tools and execute the task.	Improper Design the model with the irrelevant tools and could not execute the task.
Analysis Results	Obtained the correct results, which are matched with the master record results/pre-determined values	Output results are not matched with master record values or pre-determined values. Up to 5 % deviations are there	Output results are much deviated from expected or completely wrong results
Interpretation of results	Acquired meaningful and usable information and to make the most informed decisions	acquired meaningful information but not usable to make the most informed decisions	Not able to acquire meaningful and usable information
Viva-voce	Answers all questions with proper explanation	Answers all questions but with less relevance or explanation	Not able to answer most of the questions relevantly

Student feedback of teaching-learning process and actions taken

Student feedback is a mandatory procedure and the key elements of this procedure are detailed below.

Students feedback on curriculum, teaching-learning and resources through Class Committee Meeting (CCM):

Class Committee Meeting (CCM) is chaired by the HoD along with course faculty and student members. The queries related to the teaching-learning process, ambience of the class and other factors of concern are discussed. Two CCMs are conducted per semester. At the end of every semester, feedback is collected from students on curriculum, teaching-learning and resources provided to them. Analysis is done on the feedback received and suitable action taken.

Students' feedback on faculty:

Feedback from students are collected at the end of each semester for all courses with regard to content delivery, interaction, syllabus coverage in a 1-5 scale.

Sample feedback on faculty



Feedback on Faculty

Sl.No	Parameter	Feedback					Overall rating (1-5)
		Strongly Agree	Agree	Not sure	Disagree	Strongly Disagree	
Faculty							
1.	Punctuality of the faculty to the class.						
2.	Faculty communicates and interacts effectively with the students.						
3.	Faculty effectively addressing the Course outcomes during teaching and covering all CO's in internal and End Semester Examinations.						
4.	Faculty encourages the students to clarify their doubts.						
5.	Usage of modern teaching aids such as smart board, audio visuals, multimedia, LCD Projector, etc.,						
6.	Faculty is enthusiastic, confident and dynamic enough to handle classes without any hassle.						
7.	Faculty supports weak students and encourages bright students for presenting papers, projects etc and provides necessary counselling based on academic performance.						
8.	Faculty use student centric methods, such as experiential learning, participative learning and problem solving methodologies for enhancing learning experiences.						
Any other comments:							

2.2.2 Quality of end semester examination, internal semester question papers, assignments and evaluation (15)

Institute Marks : 15.00

Process for internal semester question paper setting and evaluation and effective process implementation.

The process for Continuous Internal Assessment question paper setting and evaluation is shown in the figure below

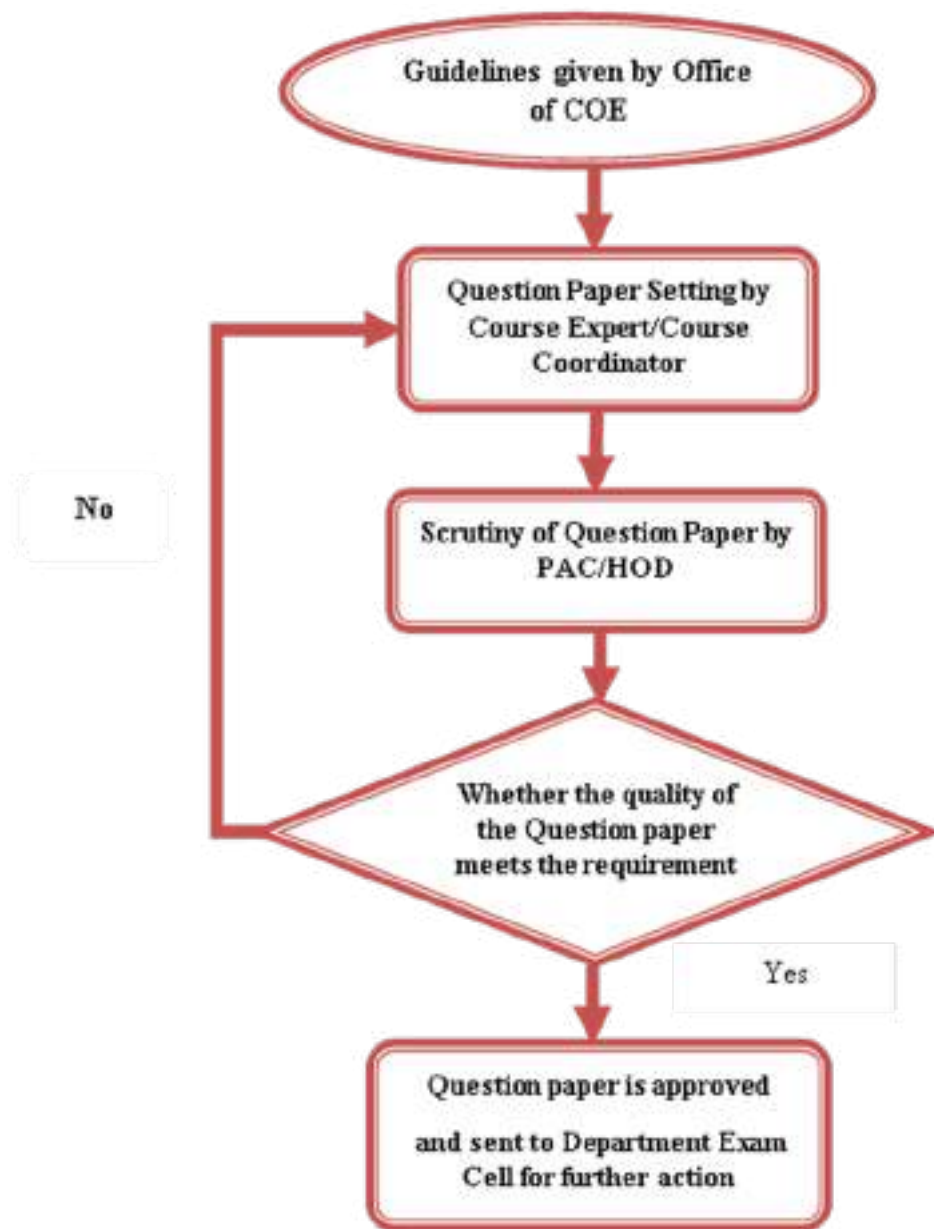


Fig 2.2.2 Process for Continuous Internal Assessment Question Paper Setting

Internal Question Paper Setting

- v. The question papers are prepared based on course outcomes. Each question is mapped with the corresponding course outcome.
- v. The question papers are also prepared based on Bloom's taxonomy framework which ensures quality in accordance with Program Outcomes and Program Specific Outcomes.
- v. Question papers for CAT 1, CAT 2 and Model Examination are set by the course faculty.
- v. The question papers are verified by PAC and HoD to ensure the compliance.

Continuous Assessment Test (CAT)

- v. Two Continuous Assessment Tests are conducted for each course per semester.

The question paper patterns are given below.

Internal Question paper pattern:

Continuous Assessment (CAT I & II)

Part A: 6 x 2 marks = 12 marks

Part B: 1 x 10 marks = 10 marks (compulsory question)

Part C: 2 x 14 marks = 28 marks (either/or choice type)

Total = 50 marks

Model Examination

- v. For each course of the program, a Model Examination is conducted prior to the end semester examination.
- v. This assessment process helps the students to prepare for the end semester examination and also gives insight into the overall perspective of the course.

The question paper pattern for conducting the Model Exam is mentioned below:

Model Examination

Part A: 10 x 2 marks = 20 marks

Part B: 1 x 10 marks = 10 marks (compulsory question)

Part C: 5 x 14 marks = 70 marks (either/or choice type)

Total = 100 marks

End Semester Examination

End Semester Examination is conducted for students who have 75% attendance. The answer scripts are evaluated by the Internal/ External evaluators to assess the outcomes.

The questions are of three categories:

- One third of the questions are direct questions, which could be answered by average students.
- One third of the questions is analytical type. Certain amount of thinking, analysis and mathematical knowledge are required to answer them.
- Remaining one third of the questions are indirect questions. Certain amount of critical thinking is needed.

Distribution of question & marks among the Units of syllabus is shown in table below.

Assessment Procedure – Tests and Examinations

For each theory course, the assessment pattern for CAT is illustrated in the following Table.

Table B.2.2.2(a) Theory Course Assessment pattern

S.No	Assessment	Syllabus Coverage	Marks	Weightage
1	CAT I	Unit I, II	50	10
2	CAT II	Unit III, IV	50	10
3	Model Examination	All 5 Units	100	10

4	Assignment/ Seminar	-	20	5
5	Attendance	-	-	5
Total Internal				40

For each practical course, the assessment pattern for ESE is illustrated in the following Table.

Table B.2.2.2(b) Practical Course Assessment pattern

S.No	Assessment	Marks	Weightage
1	Internal Assessment	40	40
2	ESE	100	60
Total Marks			100

Scheme of evaluation:

- Evaluation of Internal assessment test: Immediately after the completion of internal assessment test, the answer scripts are handed over to the respective course faculty/coordinator for evaluation. The answer scripts are evaluated as per the answer key. The evaluated answer scripts are shown to the student for verification. Marks obtained by the students are entered in the CO attainment sheet and the consolidated marks are entered in MYCAMU.
- Evaluation of End Semester Examination: After the completion of the end semester examination, a schedule for central evaluation is made by CoE. During the central evaluation, internal and external examiners evaluate their assigned answer scripts. During the evaluation, examiners are provided with the question paper, answer key and evaluation guidelines. Corrected answer scripts are verified by the chief examiner

Process to ensure questions from outcomes/learning levels perspective

The faculty who handles the course prepares the question papers for CAT and Model examination based on the Bloom's taxonomy action verbs and course outcomes.

- The question papers (CAT and Model Exam) are scrutinized by the Department exam cell coordinator and PAC.
- The HoD then approves the question papers. If any question paper is not up to the standard, it is sent back to the faculty for revision.
- For End Semester Examination, course experts are identified from the Department and from a premier institution outside the university, and one question papers is obtained from both of them. The CoE scrutinizes these two sets of question papers along with the HoD. The CoE selects one question paper from these two sets confidentially and uses it for the End Semester Exam.

Mapping of course outcomes and questions is shown in the table below.

Table B.2.2.2(c) Mapping between CO and questions

CO	Knowledge level (Blooms taxonomy, K1, K2, K3...	Question Numbers	Total Marks
CO1	K1 to K4	1, 2, 12	18
CO2		3, 4, 13	18
CO3		5, 6, 14	18
CO4		7, 8, 15	18
CO5		9, 10, 16	18
CO1 to CO6 (any one)		11	10

Evidence of COs coverage in class test/mid-term tests

Distribution of question & marks among the Units of syllabus is shown in the table below

Table B.2.2.2(d) Distribution of question & marks among the Units of syllabus

Unit	Question Numbers	Total Marks
1	1, 2, 12	18
2	3, 4, 13	18
3	5, 6, 14	18
4	7, 8, 15	18
5	9, 10, 16	18
Unit 1 to 5	11 (compulsory question)	10

Mapping of course outcomes is shown in the table below.

Table B.2.2.2(e) Mapping of course outcomes

CO	Question Numbers	Total Marks
CO1	1, 2, 12	18
CO2	3, 4, 13	18
CO3	5, 6, 14	18
CO4	7, 8, 15	18
CO5	9, 10, 16	18
CO1 to CO6 (any one)	11	10

Sample Model Exam Question Paper is Shown below



MODEL EXAMINATIONS - NOVEMBER 2022

Programme & Batch:

Semester:

Course Title:

Course Code:

Duration: 3 hours

Maximum Marks: 100 marks

Instructions

1. Before attempting any question paper, be sure that you got the correct question paper.
2. The marking data, if any, may be inserted suitably.
3. Use the slashes wherever necessary.

Question No.	Question	Marks	RTI (S1-S4)	CD
--------------	----------	-------	-------------	----

Part A (10x2 = 20 Marks) Answer all Questions

1		2		
2		3		
3		3		
4		2		
5		2		
6		3		
7		3		
8		3		
9		3		
10		2		

Part B (1x10 = 10 Marks) Compulsory Question

11		10		
----	--	----	--	--

Part C (2x10 = 20 Marks) Answer All Questions

10 (a)		10		
(OR)				
12 (a)		11		
13 (a)		11		
(OR)				
14 (a)		11		
15 (a)		11		
(OR)				
16 (a)		11		
17 (a)		11		
(OR)				
18 (a)		11		
(OR)				
19 (a)		11		

Knowledge Level as per Bloom Taxonomy
(E)- Remember (U)- Understand (A)- Apply (A)- Analyse

Quality of Assignment and its relevance to COs

Assignments are given to the students to achieve the outcomes of the courses to promote the self-learning and developing their analytical and problem-solving skills. The assignments are designed to assess the application-oriented knowledge gained by the students in the relevant course. Evaluation of the assignment is done based on the rubrics. Class assignments are designed to meet higher cognitive levels. In every semester, at least 3 assignments are given to students covering all the CO's. Assignments are given weightage of 5 marks out of 40 marks in internal assessment for theory courses. All assignment are framed and mapped with COs and Bloom's taxonomy action verbs. The relevance of COs in assignments is shown in the table below.

Table B.2.2.2(f) Relevance of COs in assignments

S.No	Assignment Details	COs covered
1	Assignment - 1	CO1, CO2
2	Assignment - 2	CO3, CO4
3	Assignment - 3	CO5, CO6

2.2.3 Quality of student projects (20)

Institute Marks : 20.00

Identification of projects and allocation methodology to faculty members

A brief idea of various fields for selecting project has been given to the students by the HoD and project coordinator. The projects are chosen by the students based on their interest. Two credits for mini project and 11 credits for major project are allotted. In both projects, a preliminary literature survey is done by students and later the objectives and scope are refined in the zeroth review. Project is evaluated based on the rubrics. Students are expected to clearly define the objectives, work plan and methodology with the support of literature survey. Students are assessed based on the evaluation by departmental project review committee. Assessment is done based on the quality, applicability, technical knowhow and innovative ideas. Marks are awarded based on internal reviews and end semester project viva- voce examination. The faculty encourages the students to carry out in house projects or industrial projects and all necessary support is provided. Students are encouraged to present their project work in conferences and journals.

The process followed in reviewing student projects is given in the figure below.

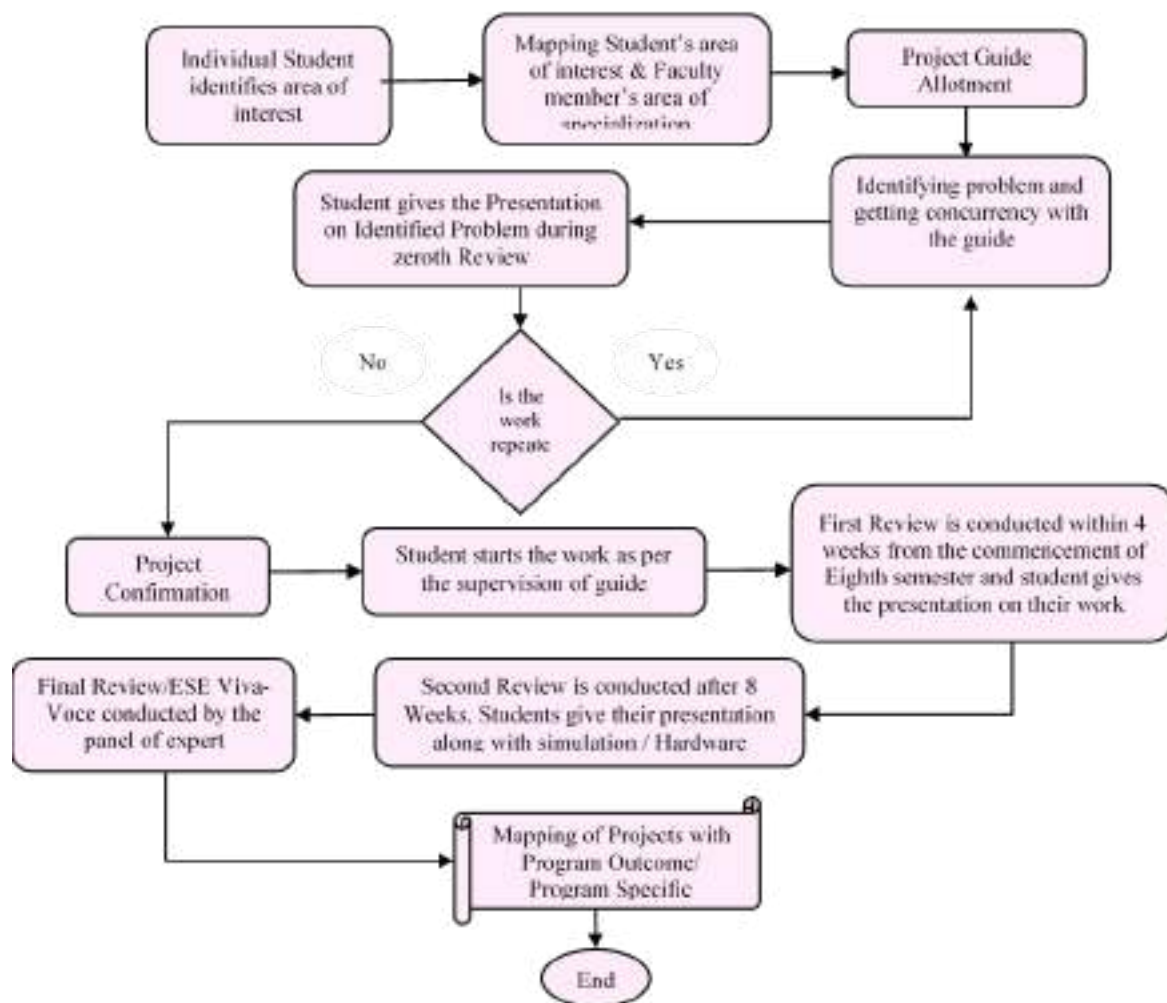


Fig 2.2.3 Process followed in reviewing student projects

Table B.2.2.3(a) List of Faculty and their specialization

S.No	Name of the Faculty	Specialization
1.	Dr.S. K. Bhattacharyya	Hydrodynamics and Ship structures
2.	Dr K.Thiruvengatasamy	Coastal structures and Dredging
3.	Mr M Surya Prakash Raju	Ship Hydrodynamics and Classification Society rules
4.	Dr.P.Sivabalan	Ship Motion and Control
5.	Dr.A.Rajesh Kanna	Enhanced Oil Recovery
6.	Dr.P.A Prasob	Design and Manufacturing
7.	Dr.R.Lilly	Fluid Mechanics
8.	Mr.A.PremAnandh	Ship Construction
9.	Mr.Kolli Sujit Kumar	Ship Stability
10.	Mr. R Jaya Surya	Ship Construction Engineering
11.	Mr.S.Thalapathy Vigneshwar	Hydrodynamics
12.	Dr.G.Venila	Information Technology
13.	Dr.A.Adinarayanan	Production engineering
14.	Dr.Suneela J	Hydrodynamics
15.	Dr.V.Arularasi	Advanced Construction Techniques
16.	Mr.S .Ramaswami	Computer Aided Design and Manufacturing
17.	Dr.Ponmani Swaminathan	Drilling Engineering

Types and relevance of the projects and their contribution towards attainment of POs and PSOs

A list of major projects for the academic year 2021-2022 is given in the table below. The projects are on ship design and they support the following POs and PSOs : PO1, PO2, PO3, PO5, PO6, PO7, PO8, PO9 ,PO12, PSO1 and PSO2..

Table B.2.2.3(b) Academic year 2021-2022

S.NO	REG. NO.	ROLL NO	NAME OF THE CANDIDATE	TITLE OF THE PROJECT
1	ANA18001	NA1134	BABITHA S	PRELIMINARY DESIGN OF 58000DWT BULK CARRIER
2	ANA18002	NA1135	CHRISTINAL B	DESIGN OF 50,000 TONS DWT OIL TANKER
3	ANA18003	NA1136	DHANUSHMA	DESIGN OF 200 PAX ELECTRIC ROPAX FERRY
4	ANA18004	NA1137	A K VISHNUNATH	DESIGN OF 1800 TEU CONTAINER SHIP OF 20 KNOTS SERVICE SPEED
5	ANA18005	NA1138	AKASH A	PRELIMINARY DESIGN OF BULK CARRIER , 40000 T
6	ANA18006	NA1139	AKKASH S	PRELIMINARY DESIGN OF AN OIL TANKER DWT 115000T

7	ANA18007	NA1140	ANDREWS JEBARAJ A	PRELIMINARY DESIGN OF BULK CARRIER , 55000T
8	ANA18008	NA1141	B ARAVIND	DESIGN OF 3000 m3 LNG BUNKERING VESSEL
9	ANA18009	NA1142	ASHWIN VINOD	DESIGN OF A CONTAINER SHIP OF 1600 TEU
10	ANA18010	NA1143	ASUTOSH PADHY	PRELIMINARY DESIGN OF BULK CARRIER, 44000t DWT
11	ANA18011	NA1144	ASWIN S NAIR	86-m HSLC ALLUMINIUM CATAMARAN ROPAX FERRY
12	ANA18012	NA1145	BALAJI P	[DESIGN OF BULKCARRIER VESSEL WITH 55000 DWT]
13	ANA18013	NA1146	BASKAR ANTONY PETLEES A	PRELIMINARY DESIGN OF CRUDE OIL CARRIER, 45000T
14	ANA18014	NA1147	BHUVANESWARAN S B	PRELIMINARY DESIGN OF BULK CARRIER, 45000T
15	ANA18015	NA1148	GOTTAM KARTHIK REDDY	DESIGN OF 120000 T DWT OF CRUDE OIL CARRIER
16	ANA18016	NA1149	JEGATHEESWARAN R	DESIGN OF 4000 m3 LNG BUNKERING VESSEL
17	ANA18017	NA1150	KISHORE V	PRELIMINARY DESIGN OF BULK CARRIER , 58000T
18	ANA18018	NA1151	KRISHNA PRASAD THILAKAN	PRELIMINARY DESIGN OF 1400 TEU CONTAINER VESSEL
19	ANA18019	NA1152	MOHAMMED THAMIMULLAH	PRELIMINARY DESIGN OF 45000T DWT LNG CARRIER
20	ANA18020	NA1153	MOHAMMAD ARSHAD P	DESIGN OF 30,000 TONS DWT OIL TANKER
21	ANA18021	NA1154	MURALI KRISHNAN G	PRELIMINARY DESIGN OF CONTAINER SHIP,1500 TEU
22	ANA18022	NA1155	NAVEEN PRASAD R	NAVAL ARCHITECTURE
23	ANA18023	NA1156	PRAVEENBALAJI S	DESIGN OF BULKCARRIER VESSEL WITH 46000 DWT
24	ANA18024	NA1157	PREMNATH O L	PRELIMINARY DESIGN OF BULK CARRIER , 43000T
25	ANA18026	NA1159	RAGESH R	PRELIMINARY DESIGN OF BULK CARRIER , 49000T
26	ANA18027	NA1160	REGIS BENNY AROCKIAM S	DESIGN OF 1500 TEU PANAMAX CONTAINER VESSEL
27	ANA18028	NA1161	RISABH BANTIYA S	DESIGN OF 4500 TEU CONTAINER
28	ANA18029	NA1162	V ROHAN KUMAR	PRELIMINARY DESIGN OF AN OIL TANKER DWT 110000T
29	ANA18030	NA1163	SANKAR LINGAM V	DESIGN OF 40000 DWT BULKCARRIER
30	ANA18031	NA1164	SARANG.S.A	DESIGN OF 1200 TEU CONTAINER SHIP

31	ANA18032	NA1165	SATHISH P	PRELIMINARY DESIGN OF BULK CARRIER 33000 T
32	ANA18033	NA1166	SHAIK SHAJAN	PRELIMINARY DESIGN OF CONTAINER SHIP,2500 TEU
33	ANA18034	NA1167	SHARUK BASHA S	PRELIMINARY DESIGN OF BULK CARRIER , 28000T
34	ANA18035	NA1168	SHOBHIT SHARMA	PRELIMINARY DESIGN OF BULK CARRIER, 41000T
35	ANA18036	NA1169	SREERAG.A.C	DESIGN OF A CONTAINER SHIP OF 1100 TEU
36	ANA18037	NA1170	SRI SANJAI V	PRELIMINARY DESIGN OF 70000T DWT BULKCARRIER
37	ANA18038	NA1171	SRIHARIRAM D	PRELIMINARY DESIGN OF BULK CARRIER , 50000T
38	ANA18039	NA1172	SRILESH M P	1300 TEU CONTAINER PROJECT
39	ANA18040	NA1173	VIGNESH L	DESIGN OF PLATFFORM SUPPLY VESSEL OF DWT 5000 TONNES
40	ANA18041	NA1174	VIKRAM V	DESIGN OF 150000 T DWT OF CRUDE OIL CARRIER
41	ANA18042	NA1175	VIVEK KUMAR CHAUHAN	DESIGN OF 47000 DWT BULK CARRIER
42	ANA18043	NA1176	VIVEK KUMAR SINGH	DESIGN OF 3500 TEU PANAMAX CONTAINER VESSEL
43	ANA18044	NA1177	VIVEK R	PRELIMINARY DESIGN OF 2800 TEU OF CONTAINER
44	ANA18045	NA1178	YOGI P	DESIGN OF 35 T BOLLARD PULL HARBOUR TUG VESSEL
45	ANA18046	NA1180	HARI ARJUN VIGNESH KUMAR	PRELIMINARY DESIGN OF 2500TEU CONTAINER VESSEL
46	ANA18047	NA1181	KOLA YASHO KRISHNA DEO	PRELIMINARY DESIGN OF BULK CARRIER , 55000T
47	ANA18048	NA1182	JEGANATHAN M	PRELIMINARY DESIGN OF BULK CARRIER,40000T
48	ANA18051	NA4183	ANUGRAH	PRELIMINARY DESIGN OF BULK CARRIER , 60000T
49	ANA18052	NA4184	BOOMINAGARAJ L	1400 TEU CONTAINER PROJECT
50	ANA18053	NA4185	CHANDU SHAJI	DESIGN OF 1400 TEU CONTAINER SHIP
51	ANA18054	NA4186	JISHNU S SARMA	PRELIMINARY DESIGN OF 6000 TEU CONTAINER VESSEL
52	ANA18055	NA4187	MANI KANDAN R	2500 TEU CONTAINER PROJECT
53	ANA18056	NA4188	MANOJ D	DESIGN OF PLATFFORM SUPPLY VESSEL OF DWT 4000 TONNES
54	ANA18057	NA4189	RAGHVENDRA PRATAP SINGH	DESIGN OF 3400 TEU CONTAINER SHIP

55	ANA18058	NA4190	SHETE RAHUL RATAN	DESIGN OF 200000 T DWT OF CRUDE OIL CARRIER
56	ANA18059	NA4191	TOSHID T R	4000 TEU CONTAINER PROJECT
57	ANA18060	NA4192	VAISHNAVI	PRELIMINARY DESIGN OF 4500 TEU CONTAINER VESSEL
58	ANA18061	NA4193	YUVRAJ SHAURYA SINGH RANA	2400 TEU CONTAINER PROJECT
59	ANA18049L	NA1219L	HEMACHANDRAN R	PRELIMINARY DESIGN OF AN OIL TANKER DWT 110000T
60	ANA18050L	NA1220L	RAMAIAH MUTHU.S	PRELIMINARY DESIGN OF 3000 TEU CONTAINER VESSEL
61	ANA18051MT		RAUSHAN KUMAR	DESIGN OF CONTAINER VESSEL 2700 TEU

Project related to Industry

The projects related to industry are shown in the table below.

Table B.2.2.3(c) Student projects related to industry (Mini, Major or Internship projects)

S.No.	Register no.	Name of the student	Name of the industry	Project title	Supported POs	Supported PSOs
1	ANA19030	SANJANA J T	PacMaK DB Naval Charterers & Consultants (Pte). Ltd, 3 Shenton Way, #16-02A Shenton House, Singapore 068805	DESIGN OF BULK CARRIER USING LIQUID HYDROGEN FUEL	PO1, PO2, PO3, PO4, PO5, PO7, PO9, PO12	PSO1, PSO2
2	ANA19009	CHANDANA SARAN VENKATA AMARNATH SWAMY	PacMaK DB Naval Charterers & Consultants (Pte). Ltd, 3 Shenton Way, #16-02A Shenton House, Singapore 068805	DESIGN OF CHEMICAL TANKER	PO1, PO2, PO3, PO4, PO5, PO7, PO9, PO12	PSO1
3	ANA19025	NILAPU KARAN RAO	PacMaK DB Naval Charterers & Consultants (Pte). Ltd, 3 Shenton Way, #16-02A Shenton House, Singapore 068805	DESIGN OF BULK CARRIER	PO1, PO2, PO3, PO4, PO5, PO7, PO9, PO12	PSO1, PSO2
4	ANA19035	VARUN BHARAT SINGH	PacMaK DB Naval Charterers & Consultants (Pte). Ltd, 3 Shenton Way, #16-02A Shenton House, Singapore 068805	DESIGN OF BULK CARRIER SHIP	PO1, PO2, PO3, PO4, PO5, PO7, PO9, PO12	PSO1
5	ANA19020	KUMTAKAR YASH SANJAY	PacMaK DB Naval Charterers & Consultants (Pte). Ltd, 3 Shenton Way, #16-02A Shenton House, Singapore 068805	DESIGN OF CONTAINER SHIP	PO1, PO2, PO3, PO4, PO5, PO7, PO9, PO12	PSO1, PSO2

6	ANA19002	AJITH	PacMaK DB Naval Charterers & Consultants (Pte). Ltd, 3 Shenton Way, #16-02A Shenton House, Singapore 068805	DESIGN OF OIL TANKER SHIP	PO1, PO2, PO3, PO4, PO5, PO7, PO9, PO12	PSO1
7	ANA19006	AVINES S	Accurant Engineers and Consultants, behind Indu Photos, Cochin University road, University, Kochi, Kerala 682520 KOCHI	DESIGN OF BALLAST PIPING AND CALCULATING EEDI AND EEXI	PO1, PO2, PO3, PO4, PO5, PO7, PO9, PO12	PSO1
8	ANA19042L	VETRI VEL MRA	Accurant Engineers and Consultants, behind Indu Photos, Cochin University road, University, Kochi, Kerala 682520 KOCHI	DESIGN OF BALLAST PIPING AND CALCULATING EEDI AND EEXI	PO1, PO2, PO3, PO4, PO5, PO7, PO9, PO12	PSO1, PSO2
9	ANA19029	RAJA GOPAL S	Accurant Engineers and Consultants, behind Indu Photos, Cochin University road, University, Kochi, Kerala 682520 KOCHI	DESIGN OF BALLAST PIPING AND CALCULATING EEDI AND EEXI	PO1, PO2, PO3, PO4, PO5, PO7, PO9, PO12	PSO1
10	ANA19038L	SACHIN GUPTA	Vedam Design & Technical Consultancy Pvt Ltd., 2nd floor, Kalpataru Prime, Nehru Nagar, Wagle Industrial Estate, Thane West, Thane, Maharashtra 400604	CONCEPT DESIGN OF 60TEU CONTAINER-BARGE, EEXI CALCULATION, SEEMP REPORT, 3D MODELLING	PO1, PO2, PO3, PO4, PO5, PO7, PO9, PO12	PSO1

Process for Monitoring and Evaluation

Mini Project and Major Project

Internal Review – 40 Marks

Based on the performance of the students, internal review (zeroth, first, second review) marks are awarded by the departmental project review committee.

External Review – 60 Marks

External examiner reviews the students projects and awards marks according to the rubrics followed in the Department.

The various stages of project monitoring and evaluation are described below.

1. The progress of a project is monitored by the guide on weekly basis.
2. The continuous progress is assessed through periodic review by the project review committee of the department. The purpose of the review is to track progress in the project work and to ensure that the student takes the correct direction to achieve the objectives of the project.
3. Projects are evaluated on the basis of rubrics.
4. Internal assessment marks (40%) for the project are awarded by the project review committee.
5. External assessment marks (60%) is based on the evaluation done by the external examiner.

The rubrics for project evaluation are shown in the tables below.

RUBRICS FOR MAJOR PROJECT

Table B.2.2.3(d) PROJECT VIVA VOCE & REPORT EVALUATION

Sl.No	CRITERIA	MARKS	EXCELLENT (100%)	GOOD (80%-90%)	AVERAGE (60%-70%)	BELOW AVERAGE (≤50%)
1	NOVELTY /OBJECTIVE	10	Novel idea, clearly articulated objective	Novel idea, objectives are not sufficiently focused	Reasonably good idea, objective are poorly chosen	No novelty, no clarity on objectives
2	LITERATURE SURVEY/ PARENT SHIP ANALYSIS	10	Extensive survey done with clarity and supporting evidences.	Survey not extensive, but covers essential details.	Survey covers the area partially	Survey inadequate
3	METHODOLOGY	10	Methods used are correct and well implemented, Student explains well the background of the procedures used.	Most of the methods used are appropriate and are well executed. Some of the methods may not be the best choice or minor details are incorrect.	Some of the methods used are appropriate but some mistakes are made in use of methodology or implementation.	Most of the methods used are inappropriate and many mistakes are made in use of methodology or implementation.
4	DESIGN	50	Design covers hull shape, general arrangement, propeller design, sea keeping, rudder design, hull scantling and economic analysis.	All aspects of design are not done with sufficient detail.	Some of the aspects of design are not covered, such as scantling calculations, cost analysis etc.	Some of the aspects of design are not covered. The ones covered are of poor quality.
5	RESULTS	10	Precise output and acceptable results.	Not precise but reasonably acceptable results.	Not precise but some of the results are acceptable.	Many of the results are unacceptable.
6	VIVA VOCE	10	Able to answer all questions with exceptional clarity.	Able to answer all questions with sufficient clarity.	Able to answer only some questions with clarity.	No clarity in answering questions.

Evidences of papers published/ awards received by projects

The students are encouraged to present their innovative work in the national and international conferences and journals. A few examples of students' participation in national and international conferences are given in table below.

Table B.2.2.3(e) Papers Published/ Events participated by Students

2020-2021					
S.No	Name of the Students	Title of the Paper	Name of the Conference/Event/ Symposium/Seminar	Venue	Date
1	Ardra M P	Power prediction of catamaran ferry	Emerging Technology on Energy System[NCETES 2020]	Prasad V. Potluri Siddhartha Institute of Technology, Kanuru, Vijayawada - 520 007	11 th September 2020.

2	Indumathi S	A study report on k-joint of jacket platform structures	Machining Behavior of Fly ash composites	The Quaide Milleth College for Men, Medavakkam, Chennai, Tamil Nadu, India	4 th and 5 th July 2020.
3	Deebak K	Corrosion of offshore structure	Machining Behavior of Fly ash composites	The Quaide Milleth College for Men, Medavakkam, Chennai, Tamil Nadu, India	4 th and 5 th July 2020.
4	Sharon shaju	Comparison of Aluminum and steel materials application of deck house for offshore support vessel	Machining Behavior of Fly ash composites	The Quaide Milleth College for Men, Medavakkam, Chennai, Tamil Nadu, India	4 th and 5 th July 2020.
5	Prithiv raj	Resistance prediction using CFD	Challenges and Opportunities in Renewable Energy, Smart Systems and E-Mobility[ICCORSE 2020]	Easwari Engineering College, Ramapuram, Chennai	30 th and 31 st July 2020.
6	Samuel	Resistance calculation of ice breaker ship	Challenges and Opportunities in Renewable Energy, Smart Systems and E-Mobility[ICCORSE 2020]	Easwari Engineering College, Ramapuram, Chennai	30 th and 31 st July 2020.
7	Venu Aravindh	Effect of bulbous bow type to ship resistance in catamaran	International Conference on Sustainable Innovation in Science and Technology[ICSIST 2021]	G.H.Raisoni University, Saikheda	26 th and 27 st February 2021
8	S.vikramselvagesh	Offshore wind mile economic feasibility	International Conference on Sustainable Innovation in Science and Technology[ICSIST 2021]	G.H.Raisoni University, Saikheda	26 th and 27 st February 2021
9	Ujjwal raj	Trim and bulbous bow optimization	International Conference on Sustainable Innovation in Science and Technology[ICSIST 2021]	G.H.Raisoni University, Saikheda	26 th and 27 st February 2021

10	Balachandar B	Analysis of ship resistance in CFD	International e-Conference on Recent Trends and Technologies in Soft Computing[ICRTSC 20]	St.Peters Institute of Higher Education and Research,Avadi,Chennai	28 th and 29 th September 2020
11	Gowtham satya kumar	Hull form optimization for bulk carrier using data compression and CFD	International e-Conference on Recent Trends and Technologies in Soft Computing[ICRTSC 20]	St.Peters Institute of Higher Education and Research,Avadi,Chennai	28 th and 29 th September 2020
12	Mohamed Nooh	A case study on alternative marine fuel	International Conference on Sustainable Innovation in Science and Technology[ICSIST 2021]	G.H.Raisoni University,Sai kheda	26 th and 27 st February 2021
13	Mohammed Binyamine	Fiber reinforced polymer composites in boat Manufacturing	International Conference on Sustainable Innovation in Science and Technology[ICSIST 2021]	G.H.Raisoni University,Sai kheda	26 th and 27 st February 2021
2021-2022					
S.No	Name of the Students	Title of the Paper	Name of the Conference/Event/Symposium/Seminar	Venue	Date
1	Aby Joseph	Proposal of system for High Efficiency Energy Extraction from LNG and study of use of EGR and SCR system in Ships	International conference on ship building and Offshore Engineering	AMET University	June 24th and 25th ,2021

2	Akkash S	Performance evaluation of coastal structures for erosion control at muttukadu (kovalam) along the east coast of chennai using DELFT3D	International Conference on Future Technologies in Manufacturing, Automation, Design and Energy[ICOFT 2021]	National Institute of Technology ,Puducherry,India	16 th and 18 th Decembe r 2021.
3	Balaji P	Performance analysis of aluminum and steel structures by using ABAQUS software.	Challenges and Oppurtunities in Renewable Energy, Smart Systems and E-Mobility[ICCORSE 2020]	Easwari Engineering College,Ramapuram,Chennai	7 th and 8 st October 2021.
4	Nithish Balaji	Engine derating with machine learning A solution to energy efficiency	International conference on ship building and Offshore Engineering	AMET University	June 24th and 25th ,2021
5	Rohan kumar V	Resistance analysis of crude carrier hull using STAR CCM+	Challenges and Oppurtunities in Renewable Energy, Smart Systems and E-Mobility[ICCORSE 2020]	Easwari Engineering College,Ramapuram,Chennai	7 th and 8 st October 2021.
6	Sreerag A C	Resistance prediction of a -container vessel using STARCCM+	Challenges and Oppurtunities in Renewable Energy, Smart Systems and E-Mobility[ICCORSE 2020]	Easwari Engineering College,Ramapuram,Chennai	7 th and 8 st October 2021.
7	Vivek kumar chauhan	Estimation of 32 passenger solar ferry	Challenges and Oppurtunities in Renewable Energy, Smart Systems and E-Mobility[ICCORSE 2020]	Easwari Engineering College,Ramapuram,Chennai	7 th and 8 st October 2021.

8	Aryan Bharmani Ayush Dave Nishil Gandhi Aadil Khan	Strength comparison of tubular joints using stiffener and collar plate	International conference on ship building and Offshore Engineering	AMET University	June 24th and 25th, 2021
9	Vikram.V	Wave force calculation of caisson breakwater using Godas formula	Challenges and Opportunities in Renewable Energy, Smart Systems and E-Mobility[ICCORSE 2020]	Easwari Engineering College, Ramapuram, Chennai	7 th and 8 th October 2021.
10	Sri hari ram D	Frequency and acceleration response of a base excited cylindrical column	International Conference on Future Technologies in Manufacturing, Automation, Design and Energy[ICOFT 2021]	National Institute of Technology, Puducherry, India	16 th and 18 th December 2021.
11	Vignesh.B	Motion analysis of kite assisted vessel using strip theory	International conference on ship building and Offshore Engineering	AMET University	June 24th and 25th, 2021
12	Vivek R	Hydrodynamic analysis of container vessel of 14000 TEU	2021 IEEE Conference	IEEE Conference on Norbert Wiener in the 21st Century (21CW)	22 nd and 25 th July 2021.
13	Asutosh padhy	Containerization and autonomous ships	2021 IEEE Conference	IEEE Conference on Norbert Wiener in the 21st Century (21CW)	22 nd and 25 th July 2021.
14	Dhanushma	Trim optimization on container using CFD	International Conference on Future Technologies in Manufacturing, Automation, Design and Energy[ICOFT 2021]	National Institute of Technology, Puducherry, India	16 th and 18 th December 2021.
2022-23					
S.No	Name of the Students	Title of the Paper	Name of the Conference/Event/Symposium/Seminar	Venue	Date

1	Abishek Nanda Kumar	Impact of sloshing on cargo tanks with and without baffles	International Conference on Sustainable Development, Innovation and Information Exchange	Savitha Institutions	29 th to 30 th July 2022
2	Chandana Saran	Drag prediction of submarines for different nose shapes	Role of Engineers for Creating a Sustainable & Self Reliant India	The Institution of Engineers (India)	16 th to 18 th December 2022
3	Dhanush S	Structural beam analysis using sacs	International Conference on Electrical engineering and Multidisciplinary Research [ICOFT 2021]	Chennai Institute of Technology ,Kundrathur ,Chennai India	23 rd and 24 th June 2022
4	Edwin E	Bulbous bow optimization using starcmm+	International Conference on Sustainable Development, Innovation and Information Exchange	Savitha Institutions	29 th to 30 th July 2022
5	Kumtakar yash sanjay	Computer vision for navigation using colregs	International Conference on Sustainable Development, Innovation and Information Exchange	Savitha Institutions	29 th to 30 th July 2022
6	Pratyush	Python based engine power estimate for bulk carrier using ship's displacement	International Conference on Electrical engineering and Multidisciplinary Research [ICOFT 2021]	Chennai Institute of Technology ,Kundrathur ,Chennai India	23 rd and 24 th June 2022
7	Nilapu karan rao	Object detection and localisation using python	International Conference on Sustainable Development, Innovation and Information Exchange	Savitha Institutions	29 th to 30 th July 2022
8	Yaswanth babu K	Multiphase flow analysis on ship hull using ansys	Role of Engineers for Creating a Sustainable & Self Reliant India	The Institution of Engineers (India)	16 th to 18 th December 2022

9	Jagadeeshvaran S	Motion analysis of ship in regular waves	Role of Engineers for Creating a Sustainable & Self Reliant, India	The Institution of Engineers (India)	16 th to 18 th December 2022
10	Sanjana J T	Performance prediction of patrol boat	International Conference on Electrical engineering and Multidisciplinary Research [ICOFT 2021]	Chennai Institute of Technology ,Kundrathur ,Chennai India	23 rd and 24 th June 2022
11	Judith eromy J	Analysis of mooring joint forces acting on fpso using hydrodynamic time response system	International Conference on Electrical engineering and Multidisciplinary Research [ICOFT 2021]	Chennai Institute of Technology ,Kundrathur ,Chennai India	23 rd and 24 th June 2022

Impact Analysis of Student Project:

1. Innovative solutions for various industrial/societal problems are developed.
2. Skill set of the students are improved.
3. Acquire knowledge on various aspects of project management.
4. Confidence level of the students is increased.
5. Document preparation and presentation.
6. More tendencies to showcase their project work in project exhibition were observed.

2.2.4 Initiatives related to industry interaction (10)

Institute Marks : 10.00

To promote Industry-Institute interactions, following initiatives are being taken by the Department.

Industry involvement in the program design and curriculum

Table B.2.2.4(a) Industry Supported Laboratories

S.No	Name of the Laboratory	Industry Collaboration	Activity	Title	Beneficiary
1	Non-destructive testing laboratory	Chennai Engineers Academy, 50/1, Girija Apartments MMDA, 100 Feet Rd, Vinayakapuram, Koyambedu, Chennai, Tamil Nadu 600106	Value Added Training Program	1.Non-destructive testing in shipbuilding	50
				2.Advanced Welding Technology in Shipbuilding	50
				3.Welding technologies in ship building industry	30

MoU with Industries / Organizations

MoUs of the Department and beneficiary activities are shown in the table below

Table B.2.2.4(b) MoUs of the Department

S.No	Name of the institution/ industry/ corporate house	Logo
1	L&T Shipbuilding, 887G+562, Kattupalli Road, Kattupalli Village, Ponneri taluk, Thiruvellur District, Thiruvallur, Tamil Nadu 600120	
2	Terran Engineering Solutions, Flat No-402, Balaji Elegance, ZADGAON Ratnagiri - 415612. Maharashtra - India	
3	Chennai Engineers Academy, 50/1, Girija Apartments MMDA, 100 Feet Rd, Vinayakapuram, Koyambedu, Chennai, Tamil Nadu 600106 & NA&OE, AMET	
4	FIETZ Global Engineering & Technology Pvt. Ltd, 2F, Ragamalika Phase II, Medavakkam Main Rd, Chennai, Tamil Nadu 600100	
5	AVEVA Information Technology and Pvt.Ltd,403-404,Alpha Hirandhani Gardens,Powai,Mumbai	
6	Tamil Nadu Dr.Jayalalitha Fisheries University,Nagapattinam	
7	EPR Labs, 88/234,2nd Floor, Rangarajapuram Main Rd, Kodambakkam, Chennai, Tamil Nadu 600024	
8	Chennai Port Trust, 1 Rajaji Salai Chennai - 600 001,Tamil Nadu,India	
9	TVS Marine Pvt Ltd, No 116/146, 4th Main Road, Ponnammanmedu, Chennai - 600110 (Prakash Nagar)	
10	Conceptia Software Technologies Private Limited, 22, Divya Shobha Building, 2nd Floor, 100 Feet Ring Road 6th Block, 3rd Phase, Kathreguppe, Banashankari 3rd Stage, Bengaluru, Karnataka 560085	

11	Ambition Marine Services Pvt Ltd 3086 B-Wing. Oberoi Garden Estate, Chandivali Farm Rd, Andheri East, Mumbai, Maharashtra 400072	
12	CADD Centre – Sholinganallur 275, First Floor OMR, Sholinganallur, Chennai, Tamil Nadu 600119	
13	CMCS, 6th Floor, Jayalakshmi Estate, Chennai, Tamil Nadu 600 006, IN	
14	Tamil Nadu Centre of excellence in Advance Manufacturing, 104 B South, First Floor, Tidel Park No 4, Rajiv Gandhi Salai, Taramani, Chennai – 600 113	
15	PacMak DB, Naval Charteres & Consultants Pvt.Ltd, Prince Infocity II, Unit No 1, 1st Floor, 283/3 & 283/4, Rajiv Gandhi Salai (OMR), Kandanchavady, Perungudi, Chennai 600096 India	

Table B.2.2.4(c) Beneficiary activities with MoUs

S. No	MoU partner	Activity under MoU	Topics	Year
1	Chennai Engineers Academy, 50/1, Girija Appartments MMDA, 100 Feet Rd, Vinayakapuram, Koyambedu, Chennai, Tamil Nadu 600106 & NA&OE, AMET	Value Added Training Program	Non-destructive testing in shipbuilding	September 2021
2	Chennai Engineers Academy, 50/1, Girija Appartments MMDA, 100 Feet Rd, Vinayakapuram, Koyambedu, Chennai, Tamil Nadu 600106 & NA&OE, AMET	Value Added Training Program	Advanced Welding Technology in Shipbuilding	July 2021
3	TVS Maarine Pvt Ltd, No 116/146, 4th Main Road, Ponniammanmedu, Chennai - 600110 (Prakash Nagar)	Value Added Training Program	Marine Materials	June 2021
4	EPR Labs, 88/234, 2nd Floor, Rangarajapuram Main Rd, Kodambakkam, Chennai, Tamil Nadu 600024	Value Added Training Program	Machine Learning	July 2021
5	Tamil Nadu Dr. Jayalalitha Fisheries University, Nagapattinam	National conference	Aquatic Foods & Fishing Vessel Technology	October 2022
6	Chennai Port Trust, 1 Rajaji Salai Chennai - 600 001, Tamil Nadu, India	Industrial visit	Second year students	2021-2022
7	AVEVA Information Technology and Pvt.Ltd, 403-404, Alpha Hirandhani Gardens, Powai, Mumbai	Workshop	Marine heating, Ventilation and Air Conditioning	October 2022
8	Chennai Port Trust, 1 Rajaji Salai Chennai - 600 001, Tamil Nadu, India	Industrial visit	Third year students	2021-2022

9	Tamil Nadu Dr.Jayalalitha Fisheries University,Nagapattinam	Workshop	Fishing Vessel Technology	September 2022
10	Chennai Port Trust, 1 Rajaji Salai Chennai - 600 001,Tamil Nadu,India	Internship	Third year students	2021-2022
11	CADD Centre – Sholinganallur 275, First Floor OMR,Sholinganallur, Chennai, Tamil Nadu 600119	Workshop	Application of CAD in Ship design	August 2022
12	Ambition Marine Services Pvt Ltd 3086 B-Wing. Oberoi Garden Estate, Chandivali Farm Rd, Andheri East, Mumbai, Maharashtra 400072	Placement	Final year students	2021-2022
13	TVS Maarine Pvt Ltd, No 116/146, 4th Main Road, Ponniammammedi, Chennai - 600110 (Prakash Nagar)	Value Added Training Program	Application of lightweight materials in ship construction	October 2022
14	CMCS, 6th Floor, Jayalakshmi Estate, Chennai, Tamil Nadu 600 006, IN	Workshop	Project Management using oracle primavera	January 2022
15	Conceptia Software Technologies Private Limited, 22, Divya Shobha Building, 2nd Floor, 100 Feet Ring Road 6th Block, 3rd Phase, Kathreguppe, Banashankari 3rd Stage, Bengaluru, Karnataka 560085	Placement	Final year students	2021-2022
16	Tamil Nadu Centre of Excellence in Advance Manufacturing,104 B South , First Floor, Tidel Park No 4, Rajiv Gandhi Salai, Taramani,Chennai – 600 113	Workshop	Hydrodynamic Analysis using STAR CCM+	December 2022
17	EPR Labs, 88/234,2nd Floor, Rangarajapuram Main Rd, Kodambakkam, Chennai, Tamil Nadu 600024	Value Added Training Program	Data Analytics applications in ships	October 2022
18	Conceptia Software Technologies Private Limited, 22, Divya Shobha Building, 2nd Floor, 100 Feet Ring Road 6th Block, 3rd Phase, Kathreguppe, Banashankari 3rd Stage, Bengaluru, Karnataka 560085	Workshop	Marine & Offshore Design	November 2022

19	L&T Shipbuilding, 887G+562, Kattupalli Road, Kattupalli Village, Ponneri taluk, Thiruvellur District, Thiruvallur, Tamil Nadu 600120	Industrial Visit	Second year students	2021-2022
20	Chennai Engineers Academy, 50/1, Girija Appartments MMDA, 100 Feet Rd, Vinayakupuram, Koyambedu, Chennai, Tamil Nadu 600106 & NA&OE, AMET	Value Added Training Program	Welding technologies in ship building industry	February 2023
21	AVEVA Information Technology and Pvt.Ltd,403-404,Alpha Hirandhani Gardens,Powai,Mumbai	Workshop	Ship piping & Instrumentation	January 2023
22	AVEVA Information Technology and Pvt.Ltd,403-404,Alpha Hirandhani Gardens,Powai,Mumbai	Seminar	Ship system design	January 2023
23	EPR Labs, 88/234,2nd Floor, Rangarajapuram Main Rd, Kodambakkam, Chennai, Tamil Nadu 600024	Value Added Training Program	Application of Python program	January 2023
24	Ambition Marine Services Pvt Ltd ,3086 B-Wing. Oberoi Garden Estate, Chandivali Farm Rd, Andheri East, Mumbai, Maharashtra 400072	Internship	Third year students	2022-2023

The participation of experts from Industry in curriculum development is shown in the table below.

Table B.2.2.4(d) Participation of experts from Industry in curriculum development

S.No	Name of the expert from industry/ corporate house	Nature of Association
1	Mr.Palaniyappan, General Manager, Vik Sandvik	Member of BOS
2	Cdr. Kamal Kanagat, General Manager, Larsen and Toubro, Ship Building, Chennai.	Member of BOS
3	Mr. G.Janakiraman, CEO, M/s FIETZ Global, Chennai	Member of BOS
4	Mr. Rakesh Roy, CEO, M/s Vedam and Design Technical consultancy Pvt.Ltd , Mumbai	Member of BOS

Industry involvement in partial delivery of any regular courses for students

The guest lectures and webinars conducted by the Department from 2020 onwards are given in the table below.

Table B.2.2.4(e) Guest lecturers / Invited lectures / partial delivery of course by industrial experts

S.No	Title	Activity	Relevant Course	Industrial expert	Beneficiary	No.of Beneficiary	No.of hours	Date
------	-------	----------	-----------------	-------------------	-------------	----------------------	----------------	------

1	Project Management using Primavera	Career development activity	Shipyard practices and project management	Mr.Vivek Infinity PMC Solution Pvt Ltd	Final year students	75	2	June 13th, 2020
2	Innovation and Intellectual Property Rights	Webinar	Major project	Dr.Deepa Bopanna, Indian Registered Patent Attorney	Final year students	73	2	June 25th, 2020
3	Non-destructive testing methods	Industrial Guest Lecture	Marine materials and welding technology	Mr.Mugunthan Chennai Engineers Academy	Second year students	28	2	May 6,2021
4	Ship General arrangement	Industrial Guest Lecture	Ship drawing lines plan	Mrs. K. Kusuma, Manager, VIKSANDVIK Visakhapatnam.	Second year students	28	2	May 7,2021
5	Design Parameters and Structural Design of Fishing Vessel	Industrial Guest Lecture	Fishing vessel technology	Mr. M. Jagannatham Raju DGM, VIKSANDVIK Visakhapatnam.	Third year students	34	2	May 8,2021
6	Role and Relevance of Numerical tools in the Maritime Industry	Industrial Guest Lecture	Numerical ship hydrodynamics software laboratory	Dr. D. Doreen Robin Founder and Director Computational Intelligence Research Foundation	Final year students	35	2	May 24,2021
7	Ship structural inspection and dry dock supervision	Alumni guest lecture	Hull inspection	Mr Naveen Shanmugan,Assistant Technical Superintendent,LDPL Middle East Shipping ,Dubai	Final year students	33	2	May 29th, 2021
8	Mind management	National webinar	Universal human values-II, Ethics and values	Swamini Vimalananda, Chinmaya Mission, Coimbatore	Second year students	34	2	June 5, 2021
9	Sustainable ship recycling	Alumni guest lecture	Quality, health, safety and environmental management	Mr.P.Sriram Krishnan, Recycling supervisor& IHM Surveyor, Sea Sentinels PTE Ltd., Singapore	Third year students	40	2	June 27th, 2021
10	Higher studies in university of Strathclyde	Promotion event		Professor Peilin Zhou, Dr Iraklis, Lazakis, University of Strathclyde	Third year students	35	2	July 15, 2021
11	Design and analysis of marine structures	Alumni guest lecture	Marine production technology	Mr. Rogero Fernanado, CEO, OPTIO Marine Pvt Ltd	Second year students	30	2	February 6th, 2022

12	Green ship recycling	Alumni guest lecture	Advanced ship design	Mr. P.Sriram Krishnan, Recycling supervisor & IHM Surveyor, Sea Sentinels Pte Ltd, Singapore	Third year students	27	2	February 25, 2022
13	International happiness day	National webinar	Universal human values-II, Ethics and values	Dr.S.Asokan, M S (General Surgery), Director CEO, GEM Hospital, Chennai.	All year students	66	2	March 20, 2022
14	Detailed ship design	Alumni guest lecture	CAD/CAM in ship building	Mrs. Shiva Shankari & Mr. Tamizharuvi, Maniyan, Cyber Marine, Mumbai.	Third year students	35	2	April 24th, 2022
15	Machine learning and fleet performance	Alumni guest lecture	Numerical analysis and programming	Mr. Abhiroop K, Naval Architect, Vedam Technologies Pvt Ltd.	Third year students	35	2	May 21st, 2022
16	Challenges in dry docking	Alumni guest lecture	Shipyard practices and project management	Mr. Vasantha Kumar Rajan (MRINA) Docking & Hasmat Manager 21 Knots Design & Consulting Pvt. Ltd	Final year students	36	2	June 6, 2022
17	Current Trend in Naval Architecture & Offshore Engineering	Alumni guest lecture	Advanced ship design	Mr. Sriram.S Naval Architect/Structural Engineer TUFF Offshore Energy & Engineering PVT Ltd	Third year students	34	2	November 19, 2022

The value added courses conducted by the Department from 2022 are shown in the table below.

Table B.2.2.4(f) Value Added Courses conducted

S.No	Name of the Institution/ Industry/ Corporate house	Name of the Activity	No. of beneficiaries	Duration (Hours)	Academic Year
1	Chennai Engineers Academy 50/1, Girija Apartments MMDA, 100 Feet Rd, Vinayapuram, Koyambedu, Chennai, Tamil Nadu 600106 & NA&OE, AMET	Advanced Welding Technology in Shipbuilding	50	30	2020-2021
2	EPR Labs, 88/234, 2nd Floor, Rangarajapuram Main Rd, Kodambakkam, Chennai, Tamil Nadu 600024	Machine Learning	16	10	2020-2021

3	EPR Labs, 88/234,2nd Floor, Rangarajapuram Main Rd, Kodambakkam, Chennai, Tamil Nadu 600024	Artificial Intelligence in Shipping	14	8	2020-2021
4	Chennai Engineers Academy 50/1, Girija Appartments MMDA, 100 Feet Rd, Vinayakapuram,Koyambedu, Chennai, Tamil Nadu 600106 & NA&OE, AMET	Non-destructive testing in shipbuilding	50	30	2021-2022
5	TVS Maarine Pvt Ltd, No 116/146, 4th Main Road, Ponnammanmedu, Chennai - 600110 (Prakash Nagar)	Marine Materials	35	30	2021-2022
6	Cohort Offshore and Marine Pvt. Ltd, No.5A/495, Caldwell colony taticorin Thoothukudi TN 628003 & NA&OE, AMET	Application of FRP in maritime industry	40	30	2021-2022
7	TVS Maarine Pvt Ltd, No 116/146, 4th Main Road, Ponnammanmedu, Chennai - 600110 (Prakash Nagar)	Corrosion and control in ships	45	30	2021-2022
8	EPR Labs, 88/234,2nd Floor, Rangarajapuram Main Rd, Kodambakkam, Chennai, Tamil Nadu 600024	Data Analytics applications in ships	32	25	2021-2022
9	EPR Labs, 88/234,2nd Floor, Rangarajapuram Main Rd, Kodambakkam, Chennai, Tamil Nadu 600024	Robotics and its applications	14	15	2021-2022
10	Chennai Engineers Academy 50/1, Girija Appartments MMDA, 100 Feet Rd, Vinayakapuram,Koyambedu, Chennai, Tamil Nadu 600106 & NA&OE, AMET	Welding technologies in ship building industry	40	30	2022-2023
11	EPR Labs, 88/234,2nd Floor, Rangarajapuram Main Rd, Kodambakkam, Chennai, Tamil Nadu 600024	Application of Python program	28	25	2022-2023
12	TVS Maarine Pvt Ltd, No 116/146, 4th Main Road, Ponnammanmedu, Chennai - 600110 (Prakash Nagar)	Application of light weight materials in ship construction.	40	30	2022-2023

The workshops, seminars and conferences involving industry experts organized by the department from 2022 are shown in the table below.

Table B.2.2.4(g) Workshops, seminars and conferences involving industry experts

S.No	Date	Program	Title of the Program	Industry Collaboration	Resource Person
------	------	---------	----------------------	------------------------	-----------------

1	4.06.2021	National seminar	Coastal Hydrodynamics and Structures	1. National Institute of Ocean Technology, Chennai. 2. CWPRS, Pune	1. Dr. Arun V Bapat, Eminent Seismologist, CWPRS, Pune. 2.Dr.S.V.S.Phani Kumar, Scientist-G, NIOT
2	24.06.2021-25.06.2021	International Conference	Shipbuilding and Offshore Structures ICSOE'21	1. National Institute of Ocean Technology, Chennai. 2. SPI Ship Constructor Software solutions, 3. UAE-INDIA-Singapore. 4. Aryatech Marine & Offshore Services (p) Ltd, Mumbai, INDIA. 5. Tminus-AI Ltd, UK	1.Dr.Purnima jaliha, Group Head-EFW, National institute of Ocean technology, Chennai, INDIA 2. Dr.S.Neelamani, Senior Research Scientist, Kuwait institute of Scientific Research, KUWAIT. 3. Mr. Manoj singh, Senior Global Expert, Siemens Digital industries software, INDIA. 3. Mr. Umesh Angadi, Director Sales and Operations, SPI Ship Constructor Software solutions, UAE-INDIA-SINGAPORE 4. Mr.Sakthi Nattesh.S & Mr.Subham Borkar.S, Naval Architect & Offshore Structural Engineer, Aryatech Marine & Offshore Services (p) Ltd, Mumbai, INDIA. 5. Mr. Bobby Ananda, Director, Tminus-AI Ltd, UK
3	10.01.2022	Workshop	Project Management using primavera	CMCS Pvt.Ltd	Mr.Bose Venkat/CMCS Training Manager
4	2.03.2022	International seminar	Innovations in design and analysis of marine structures	1. Cochin ship yard Ltd. 2. Helmsmar Design and Engineering Solutions, Cochin.	1.Mr.Sudhanshu, University of Delaware, USA. 2.Mr.Anish.S, DGM, Cochin ship yard Ltd. 3.Dr.Savin Viswanath k, Norwegian University of science and Technology (NTNU), Norway. 4.Mr.Vinod Vincent, Project Manager, Helmsmar Design and Engineering Solutions, Cochin.
5	20.04.2022-21.04.2022	Workshop	Renewable Energy Sources	Larsen & Turbo Pvt.Ltd.	Mr. SagarSharan R, AEM, Larsen & Turbo Pvt.Ltd.
6	12.08.2022	Workshop	Application of CAD in Ship design	CADD Centre – Sholinganallur	Mr.Ram Kumar,CAD Designer
7	22.09.2022	Workshop	Fishing Vessel Technology	Tamil Nadu Dr.Jayalalitha Fisheries University,Nagapattinam	Mr.Deepak,C.R.,Scientist,NIOT
8	17.10.2022	Workshop	Marine heating, Ventilation and Air Conditioning	AVEVA Information technology Pvt.Ltd.	1.Mr.Elize Chency,Training Manager.

9	28.10.2022	National conference	Aquatic Foods & Fishing Vessel Technology	Tamil Nadu Dr.Jayalalitha Fisheries University,Nagapattinam	1.Mr.Mahiban, MD, ABJ IMPEX, Chennai, India 2.Dr.N.Manimehalai, CEEE, TNJFU, Nagapattinam, India 3. Dr Hema, CFNFT,TNJFU, Madhavaram, India 4. Dr G Nagamaniammai, SRM University, Chennai, India. 5. Mr G K Chenthil Kannan, Managing Partner, Lockheed Engineering Works, Chennai, India
10	16.11.2022	Workshop	Marine & Offshore Design	Conceptia Software Technologies Private Limited	Mr.Prashanth, Training manager, Conceptia.
11	30.11.2022	National Seminar	Advances in Offshore Structures	National Institute of Ocean Technology, Chennai.	1. Dr.Subrata Kumar Bhattacharya, Professor, Department of Ocean Engg, IIT, Chennai. 2. Dr.Srinivasan Chandrasekaran, Professor, Department of Ocean Engg, IIT, Chennai. 3. Dr.Rajiv Sharma, Professor, Department of Ocean Engg, IIT, Chennai. 4. Mr.M.Palaniappan, Scientist F, National Institute of Ocean Technology, Chennai.
12	5.12.2022	Workshop	Hydrodynamic Analysis using STAR CCM+	Tamil Nadu Centre of Excellence in Advance Manufacturing.	Mr.Kanagaraj, Training manager, TANSAM.
13	5.01.2023	Workshop	Ship piping & Instrumentation	AVEVA Information technology Pvt.Ltd.	Mr.James, Training Manager
14	24.01.2023	National Seminar	Ship systems design.	AVEVA Information technology Pvt.Ltd	1.Dr Hema, CFNFT,TNJFU, Madhavaram, India 2.Dr G Nagamaniammai, SRM University, Chennai, India

Feedback of the Seminar/Guest lecture format is shown below.

Feedback of Guest Lecture/Seminar



DEPARTMENT OF NAVAL ARCHITECTURE AND OFFSHORE ENGINEERING
GUEST LECTURE/SEMINAR FEEDBACK FORM

Name of the student:

Semester:

Group:

Date of Event:

Event/Speaker Name:

S.No	Parameters	Below Average	Average	Good	Excellent	Outstanding
1	The topic					
2	The choice of topic was relevant to me					
3	The Lecturer/Speaker					
4	Self confidence					
5	Communication Skills					
6	Doubt/Queries were answered satisfactorily					
7	The Content(Topic)					
8	Refers to the latest development in the field					
9	Career oriented					
10	Innovative Learning, If any					

Signature of the Student

Impact Analysis of Industry Institute interaction and actions taken thereof

Table B.2.2.4(h) Impact Analysis of Industry Institute Interaction:

Initiative	Implementation	Impact Analysis
------------	----------------	-----------------

Industry-Institute Interaction	Arranging Guest Lectures, Workshop and Value-added training to students inside the University with different industrial Expert	Industry-Institute Interaction is the most preferred activity for mutual benefit and growth of industries as well as institutions.
		It encourages the students to undertake the final year projects in the industry with a joint supervisor from the industry.
		It increases the research and development activities in both industries as well as institutions.
		Teaching-learning processes also improved by integrating industrial training to the students which also provides an exposure of the corporate world.
		It promotes development of entrepreneurs which further leads to rapid industrialization.
		To produce proficient graduates ready for the industry, it is necessary to know the requirements of the industries through industry-institute interaction.
		Hands on experience on latest technology.

2.2.5 Initiatives related to industry internship/summer training (10)

Institute Marks : 10.00

Industrial training/tours for students**Industry Visit**

The Department organizes industry visit for students to relevant organizations/companies so that students experience the practical implementation of theoretical knowledge in real world. This gives them an insight of the industrial environment, work culture and ethics in industries. The visits also help the students to learn about people management, which is essential in any organization. Students are taken to industry visits periodically. The details of industry visit by students are shown in the tables below.

Table B.2.2.5(a) Industry Visit

S.No	Company Name	Date of Visit	No. of Students
1	Fishing Harbour, Kasimedu, Chennai	14.06.21	28
2	L&T ship building, Kattupalli, Chennai	25.06.21	28
3	National Institute of Ocean Technology, Chennai	09.08.21	32
4	Chennai Port trust, Chennai	8.09.21	40
5	Diamond Engineering Pvt.Ltd	06.10.21	40
6	Ultramarine yachts Pvt.Ltd	19.01.22	50
7	Chidambaram Shipcare Private Limited	10.02.22	32
8	ESAB India Limited	09.03.22	50
9	Fishing Harbour, Kasimedu, Chennai	22.06.22	28
10	L&T ship building, Kattupalli, Chennai	24.06.22	28
11	National Institute of Ocean Technology, Chennai	09.08.22	32
12	Chennai Port trust, Chennai	13.09.22	40
13	Diamond Engineering Pvt.Ltd	13.10.22	40
14	Ultramarine yachts Pvt.Ltd	27.01.23	35
15	Chidambaram Ship care Private Limited	14.02.23	32
16	ESAB India Limited	16.03.23	35

Feedback of the Industrial visit is shown below.

Feedback of the Industrial visit



DEPARTMENT OF NAVAL ARCHITECTURE AND OFFSHORE ENGINEERING
INDUSTRIAL VISIT FEEDBACK FORM

Name of the student:

Semester:

Group

Date of Visit:

Name & Address of Company Visited:

Tick the most appropriate option:

	Strongly Disagree				Strongly Agree	
1. The visit was technology oriented	1	2	3		4	5
2. The program was applicable for my future needs	1	2	3		4	5
3. Enhancement in Skills	1	2	3		4	5
4. The program was well placed within the allotted time	1	2	3		4	5
5. The Resource/Industry person was a good communicator	1	2	3		4	5
6. The material was presented in an organized manner	1	2	3		4	5
7. I would be interested in attending such visits in future.	1	2	3		4	5
8. Any suggestions for improvement?						

Signature of the Student

Impact Analysis of Industry visit

1. Exposure to work place realities, challenges and culture, thereby ensuring that on completion of the program, the students are job ready.
2. Students understand the importance of ethical practices at the work place.
3. It enables the students and faculty to analyze gaps in the knowledge and skill sets being imparted at the university. It is then supplemented by additional courses and training during the remaining duration of the program.
4. Select projects to find solutions to the problems faced by the industry.
5. The student is able to identify emerging job opportunities.
6. Students also develops a network of associations and relationships in the organizations which translates into industry mentor-mentee relationships.
7. Students learn to appreciate the interdisciplinary nature of work environment.
8. Students gain an insight into managerial approaches and importance of team work.

Industrial /internship /summer training of more than two weeks and post training assessment

Students undergo industry internship/summer training for a minimum duration of 15 days during third and fourth years. Faculty give the guidelines, suggestions, scope and contact details to students. They also help the students by interacting with the industry experts and provide the necessary support to the students.

Initiatives / Implementation of industry Internship/ Summer Training are as under

- The internships are arranged collaboratively by the industrial internship coordinator of the Department with the industry associates and student volunteers.
- On completion of internship, a report, completion certificate and student feedback are submitted to the industrial internship coordinator / HoD.
- A presentation of the internship report is made by the student before a panel constituted by the HoD.

The details of students who had undergone internship from academic year 2021-2022 onwards are given in the table below.

Table B.2.2.5(b) Internship details in the Academic Year 2021-2022

S.No	Roll No	Reg. No	Student Name	Industry Name	From	To	Duration (Days)
1	NA1083	ANA17036	MOHAMMED ZAMA	SEA BLUE SHIPYARD LTD	11.02.2022	28.02.2022	18
2	NA1134	ANA18001	BABITHA S	ULTRA MARINE YACHTS PVT LTD	28.06.2021	14.07.2021	18
3	NA1135	ANA18002	CHRISTINAL B	PACIFIC BLU	17.02.2022	28.02.2022	12
4	NA1136	ANA18003	DHANUSHMA	SUMMIT MARINE PVT LTD	1.07.2021	2.08.2021	30
5	NA1137	ANA18004	A K VISHNUNATH	MERCURY SHIP REPAIRS PVT LTD	4.01.2021	18.01.2021	15
7	NA1139	ANA18006	AKKASH S	ULTRA MARINE YACHTS PVT LTD	28.06.2021	14.07.2021	15
8	NA1140	ANA18007	ANDREWS JEBARAJ A	MANNSCHAFT ENGINEERING SOLUTIONS	26.07.2021	31.07.2021	5
9	NA1141	ANA18008	B ARAVIND	SUMMIT MARINE PVT LTD	1.07.2021	2.08.2021	30
10	NA1142	ANA18009	ASHWIN VINOD	HINDUSTAN SHIPYARD LTD	1.03.2022	7.03.2022	8

11	NA114 3	ANA1801 0	ASUTOSH PADHY	HINDUSTAN SHIPYARD LTD SYNERGY GROUP	19.07.20 21 14.02.20 22	31.07.20 21 31.05.20 22	12 16
12	NA114 4	ANA1801 1	ASWIN S NAIR	SUMMIT MARINE PVT LTD	1.07.202 1	2.08.202 1	30
13	NA114 5	ANA1801 2	BALAJI P	MANNSCHA FT ENGINEERI NG SOLUTIONS	8.07.202 1	19.07.20 21	12
14	NA114 6	ANA1801 3	BASKAR ANTONY PETLEES A	MANNSCHA FT ENGINEERI NG SOLUTIONS LOFTSMAN S MARINE SERVICES	8.07.202 1 11.04.20 22	19.07.20 21 11.07.20 22	12 3 months
15	NA114 7	ANA1801 4	BHUVANESWA RAN S B	LIBRAA SHIP MANAGEM ENT PVT LTD	25.03.20 22	24.04.20 22	30
16	NA114 8	ANA1801 5	GOTTAM KARTHIK REDDY	MANNSCHA FT ENGINEERI NG SOLUTIONS	8.07.202 1	19.07.20 21	12
18	NA115 0	ANA1801 7	KISHORE V	MANNSCHA FT ENGINEERI NG SOLUTIONS	8.07.202 1	19.07.20 21	12
19	NA115 1	ANA1801 8	KRISHNA PRASAD THILAKAN	AL YASH MARINE CONSULTA NTS	13.01.20 21	26.01.20 21	16
20	NA115 2	ANA1801 9	MOHAMMED THAMIMULLA H K	MANNSCHA FT ENGINEERI NG SOLUTIONS	8.07.202 1	19.07.20 21	12

21	NA115 3	ANA1802 0	MOHAMMAD ARSHAD P	HINDUSTAN SHIPYARD LTD	28.07.20 21	16.08.20 21	18
22	NA115 4	ANA1802 1	MURALI KRISHNAN G	HINDUSTAN SHIPYARD LTD	22.07.20 21	7.08.202 1	16
23	NA115 6	ANA1802 3	PRAVEENBAL AJI S	MANNSCHA FT ENGINEERI NG SOLUTIONS	8.07.202 1	19.07.20 21	12
24	NA115 7	ANA1802 4	PREMNATH O L	SUMMIT MARINE PVT LTD	1.07.202 1	2.08.202 1	30
25	NA115 9	ANA1802 6	RAGESH R	HINDUSTAN SHIPYARD LTD	22.07.20 21	7.08.202 1	16
26	NA116 0	ANA1802 7	REGIS BENNY AROCKIAM S	HINDUSTAN SHIPYARD LTD	19.07.20 21	31.07.20 21	12
27	NA116 1	ANA1802 8	RISABH BANTIYA S	HINDUSTAN SHIPYARD LTD	19.07.20 21	31.07.20 21	12
28	NA116 2	ANA1802 9	V ROHAN KUMAR	HINDUSTAN SHIPYARD LTD	22.07.20 21	7.08.202 1	16
29	NA116 3	ANA1803 0	SANKAR LINGAM V	HINDUSTAN SHIPYARD LTD	22.07.20 21	7.08.202 1	16
30	NA116 4	ANA1803 1	SARANG.S.A	SUMMIT MARINE PVT LTD	1.07.202 1	2.08.202 1	30
31	NA116 5	ANA1803 2	SATHISH P	HINDUSTAN SHIPYARD LTD	19.07.20 21	31.07.20 21	12
32	NA116 6	ANA1803 3	SHAIK SHAJAN	HINDUSTAN SHIPYARD LTD	28.07.20 21	16.08.20 21	18
33	NA116 7	ANA1803 4	SHARUK BASHA S	SUMMIT MARINE PVT LTD	1.07.202 1	2.08.202 1	30

34	NA1168	ANA18035	SHOBHIT SHARMA	HINDUSTAN SHIPYARD LTD	19.07.2021	31.07.2021	12
35	NA1169	ANA18036	SREERAG.A.C	MERCURY SHIP REPAIRS PVT LTD	4.01.2021	18.01.2021	15
36	NA1170	ANA18037	SRI SANJAI V	MANNSCHAFT ENGINEERING SOLUTIONS	8.07.2021	19.07.2021	12
37	NA1171	ANA18038	SRIHARIRAM D	SUNNY BOATS PVT LTD	25.04.2022	29.04.2022	5
38	NA1172	ANA18039	SRILESH M P	MERCURY SHIP REPAIRS PVT LTD	4.01.2021	18.01.2021	15
39	NA1173	ANA18040	VIGNESH L	HINDUSTAN SHIPYARD LTD	22.07.2021	7.08.2021	16
40	NA1174	ANA18041	VIKRAM V	HINDUSTAN SHIPYARD LTD	22.07.2021	7.08.2021	16
41	NA1175	ANA18042	VIVEK KUMAR CHAUHAN	HINDUSTAN SHIPYARD LTD	19.07.2021	31.07.2021	12
42	NA1176	ANA18043	VIVEK KUMAR SINGH	HINDUSTAN SHIPYARD LTD	19.07.2021	31.07.2021	12
43	NA1177	ANA18044	VIVEK R	SUMMIT MARINE PVT LTD MANNSCHAFT ENGINEERING SOLUTIONS	1.07.2021 15.02.2022	2.08.2021 30.05.2021	30 3 months
44	NA1178	ANA18045	YOGI P	ULTRA MARINE YACHTS PVT LTD	15.07.2021	31.07.2021	15

45	NA1180	ANA18046	HARI ARJUN VIGNESH KUMAR	MANNSCHAFT ENGINEERING SOLUTIONS	8.07.2021	19.07.2021	12
47	NA1182	ANA18048	JEGANATHAN M	SUMMIT MARINE PVT LTD	1.07.2021	2.08.2021	30
48	NA1219L	ANA18049L	HEMACHANDRAN R	MANNSCHAFT ENGINEERING SOLUTIONS	8.07.2021	19.07.2021	12
49	NA1220L	ANA18050L	RAMAIAH MUTHU.S	SEA BLUE SHIPYARD LTD	02.02.2022	15.02.2022	12

Table B.2.2.5(c) Questionnaires for industrial internship

Q1	To what extend does the training help you to learn the recent advances in technology?
Q2	How far the training does equip you with necessary technical skills to face the industry?
Q3	To what extend does the training encompass job oriented content that will help you in getting a job?
Q4	How far does the training provide the knowledge that you are expected to be required?
Q5	To what extend does the training extend an opportunity to develop your personality, communication, leadership skills and critical thinking skills?
Q6	To what degree does the training facilitate you in improving your ability to formulate, analyse and solve problems?
Q7	How far did the training inculcate necessary values to have a concern for the society?
Q8	To what extend does the training expand your understanding on the ethical and environmental issues concerning engineering in society?
Q9	How far did laboratory enhance your understanding of the concepts?
Q10	To what extend does the laboratory enable you to apply the learnt theory practically?
Q11	Acceptance by fellow workers and coordination with them, opportunity to develop your human relations.
Q12	How far did the training provide you levels of responsibility consistent with your ability and growth?

Impact analysis of industrial internship

Although there are several tangible and intangible benefits of Industrial internship, the implementation and the impact analysis are listed below.

1. Students are exposed to real time practical experience of the subjects studied in the class room and realized the practical importance of the subjects.
2. Industrial training inculcated more interest in the subjects.
3. Student's technical skills have improved.
4. Communication skills of the students have improved.
5. Students gain valuable hands-on experience.

3 COURSE OUTCOMES AND PROGRAM OUTCOMES (175)**Total Marks 175.00****Define the Program specific outcomes**

:

PSO1	Develop innovative and contemporary designs of ships and offshore structures through advanced analysis and experiments to meet the challenges of marine operations in the shipbuilding, offshore, and other maritime industries.
PSO2	Implement engineering design and construction concepts to produce solutions in the field of Naval Architecture and Offshore Engineering.
PSO3	Apply research based knowledge and research methods to analyse and interpret data using modern tools towards the solution of complex problems in the area of Naval Architecture and Offshore Engineering.

3.1 Establish the correlation between the courses and the Program Outcomes (POs) & Program Specific Outcomes (25)**Total Marks 25.00**

No. of Core Courses : 6	C2 : 2	C3 : 2	C4 : 2
-------------------------	--------	--------	--------

Note : Number of Outcomes for a Course is expected to be around 6.

Course Name :	C2 04	Course Year :	2019-2020
----------------------	--------------	----------------------	------------------

Course Name	Statements
C2 04.1	Explain the basic principles involved in fluid statics and kinematics.
C2 04.2	Discuss the ideal flow equations of motion and their practical applications.
C2 04.3	Explain viscous flow theory and the importance of boundary layer.
C2 04.4	Estimate forces on submerged bodies and application to floating structure design.
C2 04.5	Explain the linear wave theory and its application to floating body design.
C2 04.6	Identify the applications of fluid mechanics to ships.

Course Name :	C2 16	Course Year :	2019-2020
----------------------	--------------	----------------------	------------------

Course Name	Statements
C2 16.1	Explain the mechanism involved in stability of ships.
C2 16.2	Identify the effect of cargo on ship stability.
C2 16.3	Explain the small angle and large angles stability of ship.
C2 16.4	Estimate the dynamical stability of ship.
C2 16.5	Examine the longitudinal stability of ship.
C2 16.6	Examine the damage stability of ship for various conditions.

Course Name :	C3 03	Course Year :	2020-2021
----------------------	--------------	----------------------	------------------

Course Name	Statements
C3 03.1	Explain the basic concepts of fluid mechanics and its relevance with ship resistance and propulsion.
C3 03.2	Classify the ship resistance components.
C3 03.3	Estimate ship resistance using empirical methods and model test data.
C3 03.4	Explain propeller theories used in propeller design.
C3 03.5	Design propeller using KT-KQ-J charts.
C3 03.6	Develop energy efficient hull forms and select the engine.

Course Name :	C3 17	Course Year :	2020-2021
----------------------	--------------	----------------------	------------------

Course Name	Statements
C3 17.1	Explain the Beaufort scale and Seastate spectrum.
C3 17.2	Estimate the ship response in regular waves.

C3 17.3	Analyse the dynamic effects in seakeeping.
C3 17.4	Interpret the linear hydrodynamic derivatives in ship manoeuvring.
C3 17.5	Design of rudder.
C3 17.6	Predict ship motions in a random seaway and manoeuvring characteristics of a hull form.

Course Name :	C4 03	Course Year :	2021-2022
---------------	-------	---------------	-----------

Course Name	Statements
C4 03.1	Explain the static and kinematic indeterminacy of structures.
C4 03.2	Apply the virtual work and energy-based methods to structural problems.
C4 03.3	Outline flexibility and stiffness matrix methods for structural analysis.
C4 03.4	Apply Finite Element Method to structural problems.
C4 03.5	Apply Finite Element Method to ship structures.
C4 03.6	Design ship structures using Finite Element Method.

Course Name :	C4 07	Course Year :	2021-2022
---------------	-------	---------------	-----------

Course Name	Statements
C4 07.1	Explain types of high performance marine vehicles.
C4 07.2	Illustrate high speed displacement craft design.
C4 07.3	Outline design features of planing craft and hydrofoil.
C4 07.4	Classify types of air cushion vehicles.
C4 07.5	Outline the design features of special crafts.
C4 07.6	Design and develop hybrid crafts.

.....

Course Articulation Matrix

:

1 . course name : C204

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C204.1	Explain the	3 ▾	3 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾
C204.2	Discuss the	3 ▾	3 ▾	2 ▾	2 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	2 ▾
C204.3	Explain visc	2 ▾	2 ▾	2 ▾	1 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	2 ▾
C204.4	Estimate fo	3 ▾	3 ▾	3 ▾	3 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	2 ▾
C204.5	Explain the	3 ▾	3 ▾	3 ▾	3 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	2 ▾
C204.6	Identify the	3 ▾	2 ▾	2 ▾	2 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	2 ▾
Average		2.83	2.67	2.40	2.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.00

2 . course name : C216

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C216.1	Explain the	3 ▾	2 ▾	2 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾
C216.2	Identify the	3 ▾	2 ▾	2 ▾	2 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	2 ▾
C216.3	Explain the	3 ▾	2 ▾	2 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	2 ▾
C216.4	Estimate th	3 ▾	3 ▾	3 ▾	3 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	2 ▾
C216.5	Examine th	3 ▾	3 ▾	2 ▾	2 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	2 ▾
C216.6	Examine th	3 ▾	3 ▾	2 ▾	2 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	2 ▾
Average		3.00	2.40	2.20	2.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.00

3 . course name : C303

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C303.1	Explain the	2 ▾	1 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾
C303.2	Classify the	3 ▾	3 ▾	2 ▾	2 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾
C303.3	Estimate st	3 ▾	3 ▾	3 ▾	3 ▾	3 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	3 ▾
C303.4	Explain pro	3 ▾	2 ▾	2 ▾	2 ▾	2 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	3 ▾
C303.5	Design proj	3 ▾	3 ▾	3 ▾	3 ▾	3 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	3 ▾
C303.6	Develop en	3 ▾	2 ▾	2 ▾	2 ▾	2 ▾	- ▾	3 ▾	- ▾	- ▾	- ▾	- ▾	2 ▾
Average		2.83	2.33	2.40	2.40	2.50	0.00	3.00	0.00	0.00	0.00	0.00	2.75

4 . course name : C317

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C317.1	Explain the	2 ▾	2 ▾	2 ▾	2 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾

C317.2	Estimate th	3	▼	3	▼	3	▼	3	▼	3	▼	-	▼	-	▼	-	▼	-	▼	-	▼	-	▼	-	▼	3	▼
C317.3	Analyse the	3	▼	3	▼	2	▼	2	▼	2	▼	-	▼	-	▼	-	▼	-	▼	-	▼	-	▼	-	▼	3	▼
C317.4	Interpret the	3	▼	3	▼	3	▼	3	▼	3	▼	-	▼	-	▼	-	▼	-	▼	-	▼	-	▼	-	▼	3	▼
C317.5	Design of n	3	▼	3	▼	3	▼	3	▼	3	▼	-	▼	-	▼	-	▼	-	▼	-	▼	-	▼	-	▼	3	▼
C317.6	Predict ship	3	▼	3	▼	3	▼	3	▼	3	▼	-	▼	-	▼	-	▼	-	▼	-	▼	-	▼	-	▼	3	▼
Average		2.83		2.83		2.67		2.67		2.80		0.00		0.00		0.00		0.00		0.00		0.00		0.00		3.00	

5 . course name : C403

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12												
C403.1	Explain the	2	▼	2	▼	-	▼	-	▼	2	▼	-	▼	-	▼	-	▼	-	▼	-	▼	-	▼	-	▼
C403.2	Apply the v	3	▼	2	▼	2	▼	-	▼	-	▼	-	▼	2	▼	-	▼	-	▼	-	▼	-	▼	-	▼
C403.3	Outline flex	3	▼	2	▼	2	▼	-	▼	-	▼	2	▼	2	▼	3	▼	-	▼	-	▼	-	▼	-	▼
C403.4	Apply Finite	3	▼	2	▼	2	▼	-	▼	-	▼	2	▼	2	▼	3	▼	-	▼	-	▼	2	▼	2	▼
C403.5	Apply Finite	3	▼	2	▼	2	▼	2	▼	2	▼	2	▼	2	▼	2	▼	-	▼	-	▼	-	▼	3	▼
C403.6	Design ship	3	▼	3	▼	3	▼	3	▼	3	▼	3	▼	-	▼	-	▼	-	▼	-	▼	-	▼	3	▼
Average		2.83		2.17		2.20		2.50		2.50		2.20		2.25		2.40		0.00		0.00		2.00		2.67	

6 . course name : C407

Course	Statements	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C407.1	Explain typ	2 ▾	2 ▾	1 ▾	1 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾
C407.2	Illustrate hiq	2 ▾	2 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	2 ▾
C407.3	Outline des	3 ▾	3 ▾	3 ▾	3 ▾	3 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	2 ▾
C407.4	Classify typ	3 ▾	3 ▾	2 ▾	2 ▾	2 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	2 ▾
C407.5	Outline the	3 ▾	3 ▾	2 ▾	2 ▾	2 ▾	- ▾	- ▾	- ▾	- ▾	- ▾	- ▾	2 ▾
C407.6	Design and	3 ▾	3 ▾	3 ▾	3 ▾	3 ▾	- ▾	- ▾	2 ▾	- ▾	- ▾	- ▾	2 ▾
Average		2.67	2.67	2.20	2.20	2.50	0.00	0.00	2.00	0.00	0.00	0.00	2.00

1 . Course Name : C204

Course	PSO1	PSO2	PSO3
C204.1	3 ▾	3 ▾	3 ▾
C204.2	3 ▾	3 ▾	3 ▾
C204.3	3 ▾	3 ▾	3 ▾
C204.4	3 ▾	3 ▾	3 ▾
C204.5	3 ▾	3 ▾	3 ▾
C204.6	3 ▾	3 ▾	3 ▾
Average	3.00	3.00	3.00

2 . Course Name : C216

Course	PSO1	PSO2	PSO3
C216.1	3 ▾	3 ▾	3 ▾
C216.2	3 ▾	3 ▾	3 ▾
C216.3	3 ▾	3 ▾	3 ▾
C216.4	3 ▾	3 ▾	3 ▾
C216.5	3 ▾	3 ▾	3 ▾
C216.6	3 ▾	3 ▾	3 ▾
Average	3.00	3.00	3.00

3 . Course Name : C303

Course	PSO1	PSO2	PSO3
C303.1	3 ▾	3 ▾	3 ▾
C303.2	3 ▾	3 ▾	3 ▾
C303.3	3 ▾	3 ▾	3 ▾
C303.4	3 ▾	3 ▾	3 ▾
C303.5	3 ▾	3 ▾	3 ▾
C303.6	2 ▾	2 ▾	2 ▾
Average	2.83	2.83	2.83

4 . Course Name : C317

Course	PSO1	PSO2	PSO3
C317.1	3 ▾	3 ▾	3 ▾

C317.2	3	▼	3	▼	3	▼
C317.3	3	▼	3	▼	3	▼
C317.4	3	▼	3	▼	3	▼
C317.5	3	▼	3	▼	3	▼
C317.6	3	▼	3	▼	3	▼
Average	3.00		3.00		3.00	

5 . Course Name : C403

Course	PSO1	PSO2	PSO3
C403.1	3 ▼	3 ▼	3 ▼
C403.2	3 ▼	3 ▼	3 ▼
C403.3	3 ▼	3 ▼	3 ▼
C403.4	3 ▼	3 ▼	3 ▼
C403.5	3 ▼	3 ▼	3 ▼
C403.6	3 ▼	3 ▼	3 ▼
Average	3.00	3.00	3.00

6 . Course Name : C407

Course	PSO1	PSO2	PSO3
C407.1	- ▼	- ▼	2 ▼
C407.2	3 ▼	3 ▼	3 ▼
C407.3	3 ▼	3 ▼	3 ▼
C407.4	3 ▼	3 ▼	3 ▼
C407.5	3 ▼	3 ▼	3 ▼
C407.6	3 ▼	3 ▼	3 ▼
Average	3.00	3.00	2.83

 Program Articulation Matrix

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C101	PO1	PO2	PO3	PO4	PO5	2.50	2.50	PO8	2.67	2.67	PO11	2.83
C102	2.33	2.17	2.17	1.67	2.33	PO6	PO7	PO8	PO9	PO10	PO11	2.20
C103	2.67	2.67	2.67	2.00	1.67	PO6	PO7	PO8	PO9	PO10	PO11	1.75
C104	2.33	2.00	3.00	PO4	2.00	PO6	2.83	PO8	PO9	PO10	PO11	1.00
C106	PO1	PO2	PO3	PO4	PO5	2.50	2.00	2.00	3.00	3.00	PO11	3.00
C107	2.33	2.20	2.33	2.50	3.00	PO6	PO7	PO8	2.50	2.50	PO11	2.20
C108	2.33	1.67	3.00	2.67	2.50	PO6	2.80	PO8	2.50	2.50	PO11	2.00
C109	3.00	PO2	PO3	PO4	3.00	PO6	PO7	PO8	2.67	2.67	PO11	1.00
C110	2.67	2.33	2.33	2.33	2.33	PO6	PO7	PO8	PO9	PO10	PO11	2.17
C111	2.83	2.50	2.83	2.33	2.50	PO6	PO7	PO8	PO9	PO10	PO11	3.00
C112	2.83	2.83	2.50	2.67	2.67	PO6	PO7	PO8	PO9	PO10	PO11	2.67
C113	2.50	2.50	2.33	2.00	2.67	PO6	PO7	PO8	PO9	2.00	PO11	1.00
C114	PO1	PO2	PO3	PO4	PO5	2.50	3.00	2.50	1.67	PO10	PO11	2.50
C115	PO1	PO2	PO3	PO4	PO5	2.50	2.00	2.00	3.00	3.00	PO11	3.00
C116	2.00	3.00	3.00	2.00	1.00	PO6	PO7	PO8	2.33	2.33	PO11	2.50
C117	2.50	2.50	2.50	2.67	3.00	PO6	PO7	PO8	2.67	3.00	PO11	2.33
C118	2.83	2.17	2.17	PO4	PO5	PO6	PO7	PO8	PO9	3.00	PO11	3.00
C201	2.83	2.50	2.83	2.50	2.00	PO6	PO7	PO8	PO9	PO10	PO11	2.00
C202	3.00	2.83	2.67	2.67	2.80	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C203	3.00	2.17	2.00	2.00	2.50	PO6	3.00	PO8	PO9	PO10	PO11	3.00
C204	3.00	2.83	2.40	2.40	PO5	PO6	PO7	PO8	PO9	PO10	PO11	2.00
C205	3.00	2.83	2.40	2.67	2.00	PO6	PO7	2.00	PO9	PO10	PO11	3.00
C206	3.00	3.00	2.33	2.00	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C209	PO1	PO2	PO3	PO4	PO5	PO6	PO7	2.40	3.00	2.50	3.00	PO12
C210	3.00	2.00	2.33	2.83	2.00	PO6	PO7	PO8	2.83	PO10	PO11	2.00
C211	3.00	2.67	2.67	PO4	PO5	2.83	PO7	1.00	2.83	2.83	PO11	2.83
C212	3.00	2.00	2.83	2.75	3.00	PO6	PO7	PO8	2.83	PO10	PO11	3.00
C213	3.00	2.67	2.00	2.00	PO5	PO6	PO7	PO8	PO9	PO10	PO11	3.00
C214	3.00	2.00	PO3	PO4	PO5	PO6	3.00	PO8	PO9	PO10	PO11	3.00
C215	3.00	2.20	2.00	3.00	3.00	PO6	PO7	PO8	PO9	PO10	PO11	3.00
C216	3.00	2.80	2.80	2.80	PO5	PO6	3.00	PO8	PO9	PO10	PO11	2.00
C217	PO1	PO2	PO3	PO4	PO5	2.50	PO7	3.00	2.25	2.67	2.00	2.00
C221	PO1	PO2	PO3	PO4	PO5	PO6	PO7	1.00	2.00	3.00	PO11	PO12

C222	2.83	3.00	2.00	2.00	2.50	PO6	PO7	PO8	PO9	PO10	PO11	3.00
C223	2.83	2.00	2.67	2.00	2.75	PO6	PO7	PO8	2.00	PO10	PO11	2.00
C301	PO1	PO2	1.00	1.00	1.00	PO6	PO7	2.00	2.00	2.00	PO11	1.67
C302	3.00	2.83	2.83	2.00	2.83	PO6	PO7	PO8	PO9	PO10	PO11	2.67
C303	2.67	2.50	2.50	2.60	2.75	PO6	2.00	PO8	PO9	PO10	PO11	2.75
C304	2.67	2.17	2.20	2.20	1.60	PO6	PO7	1.00	PO9	PO10	PO11	PO12
C305	3.00	2.83	2.83	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C306	3.00	2.83	2.67	2.80	2.00	PO6	2.50	PO8	PO9	PO10	PO11	2.00
C307	3.00	2.67	2.80	2.67	2.67	PO6	PO7	PO8	PO9	PO10	PO11	2.50
C312	2.17	2.00	2.40	2.20	2.20	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C313	2.83	2.50	2.83	2.67	2.20	PO6	PO7	PO8	PO9	2.00	PO11	2.00
C314	3.00	3.00	2.67	2.00	2.00	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C315	3.00	3.00	PO3	PO4	2.00	2.00	PO7	3.00	3.00	3.00	PO11	PO12
C316	3.00	2.83	2.83	PO4	2.83	PO6	2.00	2.00	PO9	PO10	PO11	2.75
C317	2.83	2.67	2.50	2.67	2.83	PO6	PO7	PO8	PO9	PO10	PO11	2.80
C318	3.00	2.83	2.83	2.80	2.80	PO6	PO7	2.83	PO9	PO10	2.00	2.75
C319	3.00	2.83	2.75	PO4	2.75	PO6	PO7	PO8	PO9	PO10	PO11	2.00
C320	3.00	2.00	2.00	2.00	2.00	PO6	PO7	PO8	PO9	2.00	2.00	2.00
C321	2.83	2.67	2.67	2.00	2.50	2.00	2.33	2.50	PO9	PO10	PO11	2.00
C322	2.83	2.80	3.00	PO4	2.00	2.67	2.83	2.80	PO9	2.00	PO11	2.80
C323	2.67	2.75	2.75	2.75	2.00	2.20	2.00	2.67	PO9	PO10	PO11	2.50
C324	3.00	2.83	2.50	2.20	2.50	PO6	PO7	PO8	PO9	PO10	PO11	2.50
C325	2.00	2.00	PO3	PO4	PO5	2.00	2.00	PO8	PO9	PO10	2.00	2.00
C331	3.00	3.00	3.00	3.00	2.83	PO6	PO7	PO8	2.83	PO10	PO11	PO12
C332	3.00	2.83	2.00	2.00	2.83	PO6	PO7	PO8	2.00	2.00	PO11	2.83
C401	3.00	3.00	3.00	3.00	3.00	3.00	PO7	3.00	2.17	PO10	PO11	3.00
C402	3.00	2.83	3.00	2.50	2.80	2.17	2.00	3.00	2.50	3.00	3.00	PO12
C403	2.83	2.83	2.80	3.00	2.50	2.20	2.20	2.40	PO9	PO10	2.00	2.67
C404	3.00	2.75	2.75	2.75	2.75	2.00	2.00	2.00	2.00	2.00	2.00	3.00
C405	3.00	2.80	2.80	2.00	2.80	PO6	PO7	PO8	PO9	PO10	PO11	2.67
C406	3.00	2.83	2.83	2.83	2.00	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C407	3.00	2.00	2.67	2.67	2.80	PO6	PO7	2.00	PO9	PO10	PO11	2.00
C408	3.00	2.40	2.75	2.33	2.33	PO6	2.83	PO8	PO9	PO10	PO11	2.00
C409	3.00	2.25	2.25	2.60	2.83	2.00	PO7	2.00	PO9	2.00	PO11	2.83

C410	3.00	2.75	2.75	2.75	2.80	2.00	2.00	2.75	2.00	PO10	2.83	2.50
C411	3.00	2.67	2.67	2.50	2.83	2.00	2.00	2.00	2.00	PO10	PO11	2.67
C412	3.00	2.60	2.50	3.00	2.50	2.17	2.50	2.50	PO9	PO10	PO11	2.00
C417	3.00	3.00	2.67	2.33	2.00	PO6	PO7	PO8	2.00	2.00	PO11	2.83
C418	2.83	2.83	2.00	2.00	2.83	PO6	PO7	PO8	2.00	2.00	PO11	2.83
C419	3.00	3.00	2.00	2.50	2.67	2.67	2.67	3.00	2.17	3.00	PO11	3.00
C420	2.83	2.83	PO3	PO4	2.00	2.00	PO7	3.00	2.50	2.83	PO11	PO12
C421	3.00	3.00	2.00	2.50	2.67	2.67	2.67	3.00	2.17	3.00	2.00	3.00

Course	PSO1	PSO2	PSO3
C101	PSO1	PSO2	PSO3
C102	2.00	2.33	2.17
C103	2.17	2.17	2.17
C104	2.00	PSO2	PSO3
C106	PSO1	PSO2	PSO3
C107	2.00	2.50	2.33
C108	1.00	PSO2	PSO3
C109	PSO1	PSO2	PSO3
C110	2.00	2.00	2.00
C111	2.00	2.00	2.00
C112	2.67	2.83	2.67
C113	2.00	1.00	1.00
C114	PSO1	PSO2	PSO3
C115	PSO1	PSO2	PSO3
C116	3.00	3.00	2.17
C117	PSO1	PSO2	PSO3
C118	PSO1	2.00	PSO3
C201	2.33	2.60	2.50
C202	2.00	2.83	2.00
C203	3.00	3.00	3.00
C204	3.00	3.00	3.00
C205	3.00	3.00	3.00
C206	3.00	3.00	2.00
C209	1.50	1.67	1.00
C210	3.00	3.00	3.00

C211	PSO1	PSO2	PSO3
C212	2.00	2.33	2.00
C213	2.83	3.00	2.67
C214	3.00	2.67	2.67
C215	3.00	3.00	2.00
C216	3.00	3.00	3.00
C217	2.00	2.00	2.00
C221	PSO1	PSO2	PSO3
C222	PSO1	2.00	2.00
C223	2.83	2.83	2.83
C301	PSO1	PSO2	PSO3
C302	2.83	2.83	2.83
C303	2.83	2.83	2.83
C304	PSO1	PSO2	PSO3
C305	2.00	2.00	PSO3
C306	2.83	2.83	2.83
C307	2.83	2.83	2.83
C312	PSO1	PSO2	PSO3
C313	2.83	2.83	2.83
C314	3.00	3.00	3.00
C315	3.00	3.00	3.00
C316	2.83	2.83	2.83
C317	2.83	2.83	2.83
C318	3.00	3.00	3.00
C319	2.83	PSO2	2.83
C320	2.00	PSO2	PSO3
C321	2.00	2.83	2.83
C322	2.80	2.67	2.67
C323	2.83	2.75	2.75
C324	2.50	2.33	2.83
C325	2.83	PSO2	2.00
C331	2.83	2.00	2.00
C332	2.83	2.83	2.83
C401	3.00	3.00	3.00

C402	3.00	3.00	2.00
C403	3.00	3.00	3.00
C404	2.83	3.00	3.00
C405	3.00	3.00	3.00
C406	3.00	2.00	3.00
C407	3.00	3.00	2.83
C408	2.83	3.00	2.75
C409	3.00	3.00	3.00
C410	3.00	3.00	PSO3
C411	2.33	3.00	2.17
C412	3.00	3.00	3.00
C417	3.00	3.00	3.00
C418	3.00	3.00	3.00
C419	3.00	3.00	3.00
C420	3.00	3.00	3.00
C421	3.00	3.00	3.00

3.2 Attainment of Course Outcomes (75)

Total Marks 75.00

Assessment Process

In the Outcome Based Education, evaluation of COs are based on both direct and indirect assessment methods. Direct assessment includes evaluation of the students' knowledge gained and skills acquired based on their performance in the internal and external assessments such as Continuous Assessment Test 1 (CAT1), Continuous Assessment Test 2 (CAT2), Model Examination, End Semester Examination (ESE), assignments and any other activity-based learning. Indirect assessment is done through Course End Survey.

Assessment Tools

The direct assessment tools are used for evaluation is shown in Table B 3.3 and indirect assessment tool of evaluation is shown in Table B 3.4.

Table B 3.3: Direct Assessment Tools

Theory Course		
Assessment Process	Frequency	Method
Continuous Assessment Test (CAT)	Two per semester	CAT is a 1½ hr test conducted by the department. The question paper is set by the course faculty and approved by HoD/PAC.
Model Examination	One per semester	The Model examination is a 3 hrs exam and covers the entire syllabus. The question paper is set by the course faculty and approved by HoD/PAC.
End Semester Examination (ESE)	As per University Examination Regulations	The ESE is 3 hrs exam and covers the entire syllabus. It is conducted at the University level by the office of Controller of Examinations.
Assignments	Three or more per course	Assignments are given to assess the analytical and higher order skills of the students. They are used as a tool to assess the attainment of COs.
Seminar or activity-based learning	As required for the course	Seminar improves the communication skills of the students, their knowledge, networking skills, motivation and confidence. All these contribute to the attainment of COs.

Practical / Laboratory Courses		
Assessment Process	Frequency	Method
Performance evaluation	For every individual experiment / exercise	Continuous evaluation of individual experiment / exercise as per the lab schedule.
End Semester Examination (ESE)	As per University examination regulations	The ESE is of 3 hr duration satisfying all COs.

Mini Project / Major Project		
Assessment Process	Frequency	Method
Reviews	Two/three reviews	Evaluation by the project evaluation committee of the Department. Rubrics based assessment is carried out.
Viva-Voce	One	Evaluation by the project evaluation committee of the Department. Rubrics based assessment is carried out.

Internship / Industrial Visit		
Assessment Process	Frequency	Method

Viva-Voce	One	Rubrics based evaluation to determine the level of learning.
-----------	-----	--

Table B 3.4: Indirect Assessment Tools

Assessment Process	Frequency	Method
Course end survey	One	Reflective feedback on the level of understanding and feedback on teaching. A survey is taken which is considered in CO attainment.

COURSE OUTCOME ASSESSMENT PROCESS

CO Assessment process for Theory, Laboratory courses and Project work are shown in Table B 3.5, B 3.6 and B 3.7

Table B 3.5: CO Assessment Process for Theory Courses

Theory						
Assessment Type	Assessment Tool	Maximum Marks	COs	Marks Scaled to	Computation of CO Attainment in the Course	
Direct Attainment (DA)	CAT-I Type: Continuous assessment Test (CAT)	50	1,2	10	40%	90%
	CAT-II Type: Continuous assessment Test (CAT)	50	3,4	10		
	Model Examination Type: Continuous assessment Test (CAT)	100	1-6	10		
	Assignment	60	1-6	10		
	University Examination Type: End Semester Examinations (ESE)	100	1-6	60	60%	
Indirect Attainment (IDA)	Course End Survey				100%	10%
Overall Attainment (OA) = DA converted to 90% + IDA converted to 10%						100%

Table B 3.6 CO Assessments for Practical/Laboratory Courses

Practical / Laboratory Courses						
Assessment Type	Course Type	Assessment Tool	Maximum Marks	COs	Marks Scaled to	Computation of CO Attainment in the Course

Direct Attainment (DA)	Laboratory	Lab Experiments/Exercises, Observation, Record & Viva Type: Continuous Internal Assessment (CIA)	10 Marks per Experiment	1-6	40	40%	90%
		University Examination Type: End Semester Examinations (ESE)	100	1-6	60	60%	
Indirect Attainment (IDA)	Course End Survey					100%	10%
Overall Attainment (OA) = DA converted to 90% + IDA converted to 10%							100%

Table B 3.7: CO Assessment for Project Work

Mini Project and Major Project							
Assessment Type	Course Type	Assessment Tool	Maximum Marks	COs	Marks Scaled to	Computation of CO Attainment in the Course	
Direct Attainment (DA)	Project/ Internship	Reviews, Presentation, Viva-voce Type: Continuous Internal Assessment (CIA)	40	1-6	40	40%	90%
		University Examination Report & Viva- voce Type: End Semester Examinations (ESE)	60	1-6	60	60%	
Indirect Attainment (IA)	Course End Semester Survey					100%	10%
Overall Attainment (OA) = DA converted to 90% + IDA converted to 10%							100%

3.2.2 Record the attainment of Course Outcomes of all courses with respect to set attainment levels (65)

Institute Marks : 65.00

- Direct Attainment of COs is measured from the assessment of students in Continuous Internal Assessment (CIA), i.e CAT-I, CAT-II, Model Examination and assignment, as well as End Semester Examination (ESE).
- Indirect Attainment of COs is measured using course end survey.
- Overall Attainment is considered with 90% of Direct Attainment, and 10% of Indirect Attainment.
- The COs are mapped to specific questions, and target attainment levels are fixed for each CO for the type of course, namely, theory, practical or project.

Attainment levels vs. targets

Table B 3.8 Methodology for setting attainment for Courses

Theory / Laboratory / Project / Internship Courses		
CO Attainment	Scale	Level
>70%	3	Substantial
>60 % to 70 %	2	Moderate
50 % to 60 %	1	Low
below 50%	0	Not Attained

Based on the tools and processes described in section 3.2.1, the assessment of COs is carried out for all courses. Table below gives the attainment of COs for all the courses for Batch – 12 (2018- 2022).

Table B 3.9 CO Attainment for Batch-12 (2018-22)

NBA Course code	CO	Course Name	Direct CO Attainment	Direct CO Attainment (90%)	Indirect CO Attainment	Indirect CO Attainment (10%)	Total attainment (90% DA+10%IDA)	Overall CO Attainment level
C101	1	TECHNICAL ENGLISH-I	69.57%	62.61%	86.96%	8.70%	71.30%	3
	2		81.74%	73.57%	86.96%	8.70%	82.26%	3
	3		78.26%	70.43%	73.91%	7.39%	77.83%	3
	4		76.52%	68.87%	73.91%	7.39%	76.26%	3
	5		86.09%	77.48%	86.96%	8.70%	86.17%	3
	6		80.43%	72.39%	78.26%	7.83%	80.22%	3
C102	1	ENGINEERING PHYSICS- I	64.35%	57.91%	82.61%	8.26%	66.17%	2
	2		73.91%	66.52%	82.61%	8.26%	74.78%	3
	3		72.61%	65.35%	76.09%	7.61%	72.96%	3
	4		69.57%	62.61%	76.09%	7.61%	70.22%	3
	5		66.52%	59.87%	82.61%	8.26%	68.13%	2
	6		70.00%	63.00%	82.61%	8.26%	71.26%	3
C103	1	ENGINEERING MATHEMATICS-I	66.96%	60.26%	93.48%	9.35%	69.61%	2
	2		74.78%	67.30%	93.48%	9.35%	76.65%	3
	3		54.35%	48.91%	91.30%	9.13%	58.04%	1
	4		51.30%	46.17%	82.61%	8.26%	54.43%	1
	5		68.26%	61.43%	84.78%	8.48%	69.91%	2
	6		61.74%	55.57%	82.61%	8.26%	63.83%	2
C104	1	ENGINEERING CHEMISTRY	53.48%	48.13%	78.26%	7.83%	55.96%	1
	2		63.48%	57.13%	69.57%	6.96%	64.09%	2
	3		39.13%	35.22%	69.57%	6.96%	42.17%	0
	4		44.35%	39.91%	69.57%	6.96%	46.87%	0
	5		56.96%	51.26%	78.26%	7.83%	59.09%	1
	6		47.39%	42.65%	73.91%	7.39%	50.04%	1

C106	1	ENGLISH LANGUAGE LABORATORY-I	100.00%	90.00%	100.00%	10.00%	100.00%	3
	2		100.00%	90.00%	100.00%	10.00%	100.00%	3
	3		100.00%	90.00%	100.00%	10.00%	100.00%	3
	4		100.00%	90.00%	100.00%	10.00%	100.00%	3
	5		100.00%	90.00%	100.00%	10.00%	100.00%	3
	6		100.00%	90.00%	100.00%	10.00%	100.00%	3
C107	1	ENGINEERING PHYSICS LABORATORY	77.83%	70.04%	67.39%	6.74%	76.78%	3
	2		78.70%	70.83%	67.39%	6.74%	77.57%	3
	3		79.57%	71.61%	67.39%	6.74%	78.35%	3
	4		80.43%	72.39%	73.91%	7.39%	79.78%	3
	5		79.57%	71.61%	73.91%	7.39%	79.00%	3
	6		77.83%	70.04%	73.91%	7.39%	77.43%	3
C108	1	ENGINEERING CHEMISTRY LABORATORY	80.00%	72.00%	73.91%	7.39%	79.39%	3
	2		82.61%	74.35%	73.91%	7.39%	81.74%	3
	3		81.74%	73.57%	73.91%	7.39%	80.96%	3
	4		83.48%	75.13%	86.96%	8.70%	83.83%	3
	5		82.61%	74.35%	86.96%	8.70%	83.04%	3
	6		82.61%	74.35%	86.96%	8.70%	83.04%	3
C109	1	WORKSHOP PRACTICES	95.65%	86.09%	46.46%	4.65%	90.73%	3
	2		93.04%	83.74%	46.46%	4.65%	88.39%	3
	3		92.17%	82.96%	46.46%	4.65%	87.60%	3
	4		93.91%	84.52%	46.46%	4.65%	89.17%	3
	5		93.04%	83.74%	46.46%	4.65%	88.39%	3
	6		93.91%	84.52%	46.46%	4.65%	89.17%	3
C110	1	ENGINEERING PHYSICS- II	60.00%	54.00%	86.96%	8.70%	62.70%	2
	2		60.87%	54.78%	86.96%	8.70%	63.48%	2
	3		40.87%	36.78%	78.26%	7.83%	44.61%	0
	4		44.35%	39.91%	86.96%	8.70%	48.61%	0
	5		50.87%	45.78%	86.96%	8.70%	54.48%	1
	6		66.96%	60.26%	78.26%	7.83%	68.09%	2
C111	1	ENGINEERING MATHEMATICS - II	64.08%	57.67%	87.76%	8.78%	66.45%	2
	2		62.04%	55.84%	87.76%	8.78%	64.61%	2
	3		68.16%	61.35%	83.67%	8.37%	69.71%	2
	4		52.24%	47.02%	85.71%	8.57%	55.59%	1
	5		62.04%	55.84%	87.76%	8.78%	64.61%	2
	6		62.45%	56.20%	85.71%	8.57%	64.78%	2

C112	1	BASIC ELECTRICAL ENGINEERING	83.48%	75.13%	89.13%	8.91%	84.04%	3
	2		79.13%	71.22%	86.96%	8.70%	79.91%	3
	3		72.61%	65.35%	84.78%	8.48%	73.83%	3
	4		70.43%	63.39%	84.78%	8.48%	71.87%	3
	5		75.22%	67.70%	82.61%	8.26%	75.96%	3
	6		83.91%	75.52%	80.43%	8.04%	83.57%	3
C113	1	PROGRAMMING FOR PROBLEM SOLVING	63.91%	57.52%	89.13%	8.91%	66.43%	2
	2		61.30%	55.17%	84.78%	8.48%	63.65%	2
	3		43.48%	39.13%	76.09%	7.61%	46.74%	0
	4		42.61%	38.35%	82.61%	8.26%	46.61%	0
	5		50.00%	45.00%	89.13%	8.91%	53.91%	1
	6		63.04%	56.74%	69.57%	6.96%	63.70%	2
C114	1	ENVIRONMENTAL SCIENCE	49.57%	44.61%	86.96%	8.70%	53.30%	1
	2		44.78%	40.30%	84.78%	8.48%	48.78%	0
	3		46.09%	41.48%	78.26%	7.83%	49.30%	0
	4		40.00%	36.00%	82.61%	8.26%	44.26%	0
	5		53.04%	47.74%	84.78%	8.48%	56.22%	1
	6		60.87%	54.78%	80.43%	8.04%	62.83%	2
C115	1	ENGLISH LABORATORY-II	100.00%	90.00%	89.13%	8.91%	98.91%	3
	2		98.26%	88.43%	95.65%	9.57%	98.00%	3
	3		99.13%	89.22%	89.13%	8.91%	98.13%	3
	4		99.13%	89.22%	89.13%	8.91%	98.13%	3
	5		98.26%	88.43%	95.65%	9.57%	98.00%	3
	6		99.13%	89.22%	89.13%	8.91%	98.13%	3
C116	1	BASIC ELECTRICAL ENGINEERING LABORATORY	100.00%	90.00%	95.65%	9.57%	99.57%	3
	2		100.00%	90.00%	95.65%	9.57%	99.57%	3
	3		100.00%	90.00%	95.65%	9.57%	99.57%	3
	4		100.00%	90.00%	95.65%	9.57%	99.57%	3
	5		100.00%	90.00%	95.65%	9.57%	99.57%	3
	6		100.00%	90.00%	95.65%	9.57%	99.57%	3
C117	1	PROGRAMMING FOR PROBLEM SOLVING LABORATORY	80.43%	72.39%	91.30%	9.13%	81.52%	3
	2		78.70%	70.83%	93.48%	9.35%	80.17%	3
	3		80.43%	72.39%	91.30%	9.13%	81.52%	3
	4		78.70%	70.83%	91.30%	9.13%	79.96%	3
	5		81.30%	73.17%	91.30%	9.13%	82.30%	3
	6		78.70%	70.83%	93.48%	9.35%	80.17%	3

C118	1	ENGINEERING GRAPHICS	59.13%	53.22%	91.30%	9.13%	62.35%	2
	2		63.04%	56.74%	86.96%	8.70%	65.43%	2
	3		50.43%	45.39%	86.96%	8.70%	54.09%	1
	4		66.96%	60.26%	86.96%	8.70%	68.96%	2
	5		54.35%	48.91%	86.96%	8.70%	57.61%	1
	6		55.22%	49.70%	86.96%	8.70%	58.39%	1
C201	1	ENGINEERING MATHEMATICS – III	64.08%	57.67%	87.76%	8.78%	66.45%	2
	2		62.04%	55.84%	87.76%	8.78%	64.61%	2
	3		68.16%	61.35%	83.67%	8.37%	69.71%	2
	4		52.24%	47.02%	85.71%	8.57%	55.59%	1
	5		62.04%	55.84%	87.76%	8.78%	64.61%	2
	6		62.45%	56.20%	85.71%	8.57%	64.78%	2
C202	1	INTRODUCTION TO NAVAL ARCHITECTURE	59.18%	53.27%	85.71%	8.57%	61.84%	2
	2		63.67%	57.31%	77.55%	7.76%	65.06%	2
	3		53.88%	48.49%	85.71%	8.57%	57.06%	1
	4		64.08%	57.67%	85.71%	8.57%	66.24%	2
	5		62.04%	55.84%	77.55%	7.76%	63.59%	2
	6		58.37%	52.53%	85.71%	8.57%	61.10%	2
C203	1	MARINE MATERIALS AND WELDING TECHNOLOGY	48.57%	43.71%	81.63%	8.16%	51.88%	1
	2		49.80%	44.82%	81.63%	8.16%	52.98%	1
	3		53.06%	47.76%	81.63%	8.16%	55.92%	1
	4		55.10%	49.59%	81.63%	8.16%	57.76%	1
	5		68.16%	61.35%	81.63%	8.16%	69.51%	2
	6		48.57%	43.71%	81.63%	8.16%	51.88%	1
C204	1	MARINE HYDRODYNAMICS	54.29%	48.86%	77.55%	7.76%	56.61%	1
	2		57.96%	52.16%	77.55%	7.76%	59.92%	1
	3		46.12%	41.51%	79.59%	7.96%	49.47%	0
	4		57.96%	52.16%	79.59%	7.96%	60.12%	2
	5		48.16%	43.35%	77.55%	7.76%	51.10%	1
	6		48.98%	44.08%	79.59%	7.96%	52.04%	1
C205	1	STRENGTH OF MATERIALS	53.47%	48.12%	85.71%	8.57%	56.69%	1
	2		49.80%	44.82%	85.71%	8.57%	53.39%	1
	3		57.55%	51.80%	85.71%	8.57%	60.37%	2
	4		56.33%	50.69%	85.71%	8.57%	59.27%	1
	5		53.47%	48.12%	85.71%	8.57%	56.69%	1
	6		44.49%	40.04%	85.71%	8.57%	48.61%	0

C206	1	ENGINEERING MECHANICS	53.88%	48.49%	93.88%	9.39%	57.88%	1
	2		55.92%	50.33%	93.88%	9.39%	59.71%	1
	3		50.20%	45.18%	91.84%	9.18%	54.37%	1
	4		53.88%	48.49%	93.88%	9.39%	57.88%	1
	5		52.65%	47.39%	93.88%	9.39%	56.78%	1
	6		52.65%	47.39%	93.88%	9.39%	56.78%	1
C209	1	INTERPERSONAL COMMUNICATION	54.00%	48.60%	92.00%	9.20%	57.80%	1
	2		62.40%	56.16%	92.00%	9.20%	65.36%	2
	3		62.00%	55.80%	90.00%	9.00%	64.80%	2
	4		59.60%	53.64%	78.00%	7.80%	61.44%	2
	5		62.40%	56.16%	78.00%	7.80%	63.96%	2
	6		49.60%	44.64%	90.00%	9.00%	53.64%	1
C210	1	STRENGTH OF MATERIALS LABORATORY	53.47%	48.12%	91.84%	9.18%	57.31%	1
	2		54.29%	48.86%	91.84%	9.18%	58.04%	1
	3		53.47%	48.12%	89.80%	8.98%	57.10%	1
	4		57.55%	51.80%	91.84%	9.18%	60.98%	2
	5		53.47%	48.12%	89.80%	8.98%	57.10%	1
	6		56.73%	51.06%	91.84%	9.18%	60.24%	2
C211	1	ELECTRICAL WORKSHOP	57.55%	51.80%	91.84%	9.18%	60.98%	2
	2		57.55%	51.80%	93.88%	9.39%	61.18%	2
	3		57.55%	51.80%	91.84%	9.18%	60.98%	2
	4		57.55%	51.80%	91.84%	9.18%	60.98%	2
	5		59.18%	53.27%	89.80%	8.98%	62.24%	2
	6		58.37%	52.53%	91.84%	9.18%	61.71%	2
C212	1	SHIP DRAWING- LINES PLAN	83.27%	74.94%	85.71%	8.57%	83.51%	3
	2		84.08%	75.67%	85.71%	8.57%	84.24%	3
	3		85.71%	77.14%	83.67%	8.37%	85.51%	3
	4		82.45%	74.20%	87.76%	8.78%	82.98%	3
	5		84.90%	76.41%	83.67%	8.37%	84.78%	3
	6		85.71%	77.14%	83.67%	8.37%	85.51%	3
C213	1	THERMODYNAMICS AND MARINE MACHINERY	81.22%	73.10%	93.88%	9.39%	82.49%	3
	2		85.71%	77.14%	91.84%	9.18%	86.33%	3
	3		93.88%	84.49%	93.88%	9.39%	93.88%	3
	4		80.41%	72.37%	81.63%	8.16%	80.53%	3
	5		92.65%	83.39%	91.84%	9.18%	92.57%	3
	6		88.16%	79.35%	81.63%	8.16%	87.51%	3

C214	1	MARINE PRODUCTION TECHNOLOGY	77.96%	70.16%	91.84%	9.18%	79.35%	3
	2		77.14%	69.43%	89.80%	8.98%	78.41%	3
	3		80.00%	72.00%	91.84%	9.18%	81.18%	3
	4		76.73%	69.06%	89.80%	8.98%	78.04%	3
	5		87.35%	78.61%	89.80%	8.98%	87.59%	3
	6		82.45%	74.20%	89.80%	8.98%	83.18%	3
C215	1	ELEMENTS OF OFFSHORE ENGINEERING	58.37%	52.53%	83.67%	8.37%	60.90%	2
	2		62.04%	55.84%	79.59%	7.96%	63.80%	2
	3		71.02%	63.92%	95.92%	9.59%	73.51%	3
	4		74.29%	66.86%	79.59%	7.96%	74.82%	3
	5		79.18%	71.27%	83.67%	8.37%	79.63%	3
	6		68.57%	61.71%	83.67%	8.37%	70.08%	3
C216	1	THEORY OF SHIPS	72.24%	65.02%	93.88%	9.39%	74.41%	3
	2		79.18%	71.27%	85.71%	8.57%	79.84%	3
	3		91.43%	82.29%	81.63%	8.16%	90.45%	3
	4		89.80%	80.82%	85.71%	8.57%	89.39%	3
	5		84.90%	76.41%	93.88%	9.39%	85.80%	3
	6		84.90%	76.41%	93.88%	9.39%	85.80%	3
C217	1	ETHICS AND VALUES	71.43%	64.29%	89.80%	8.98%	73.27%	3
	2		69.80%	62.82%	81.63%	8.16%	70.98%	3
	3		61.22%	55.10%	97.96%	9.80%	64.90%	2
	4		75.51%	67.96%	81.63%	8.16%	76.12%	3
	5		71.84%	64.65%	89.80%	8.98%	73.63%	3
	6		69.39%	62.45%	89.80%	8.98%	71.43%	3
C221	1	PROFESSIONAL COMMUNICATION	100.00%	90.00%	100.00%	10.00%	100.00%	3
	2		100.00%	90.00%	100.00%	10.00%	100.00%	3
	3		100.00%	90.00%	100.00%	10.00%	100.00%	3
	4		100.00%	90.00%	100.00%	10.00%	100.00%	3
	5		100.00%	90.00%	100.00%	10.00%	100.00%	3
	6		92.65%	83.39%	100.00%	10.00%	93.39%	3
C222	1	HYDROSTATICS & STABILITY LABORATORY	92.24%	83.02%	79.59%	7.96%	90.98%	3
	2		92.24%	83.02%	77.55%	7.76%	90.78%	3
	3		92.24%	83.02%	79.59%	7.96%	90.98%	3
	4		93.06%	83.76%	79.59%	7.96%	91.71%	3
	5		92.24%	83.02%	77.55%	7.76%	90.78%	3
	6		92.24%	83.02%	79.59%	7.96%	90.98%	3

C223	1	SURFACE MODELLING AND ANALYSIS - SOFTWARE LABORATORY	94.29%	84.86%	87.76%	8.78%	93.63%	3
	2		92.65%	83.39%	87.76%	8.78%	92.16%	3
	3		92.65%	83.39%	87.76%	8.78%	92.16%	3
	4		92.65%	83.39%	87.76%	8.78%	92.16%	3
	5		92.65%	83.39%	87.76%	8.78%	92.16%	3
	6		95.10%	85.59%	87.76%	8.78%	94.37%	3
C301	1	BIOLOGY FOR ENGINEERS	65.20%	58.68%	86.00%	8.60%	67.28%	2
	2		60.40%	54.36%	84.00%	8.40%	62.76%	2
	3		71.20%	64.08%	86.00%	8.60%	72.68%	3
	4		63.60%	57.24%	84.00%	8.40%	65.64%	2
	5		66.00%	59.40%	86.00%	8.60%	68.00%	2
	6		81.20%	73.08%	86.00%	8.60%	81.68%	3
C302	1	STRENGTH OF SHIPS	79.20%	71.28%	78.00%	7.80%	79.08%	3
	2		80.00%	72.00%	84.00%	8.40%	80.40%	3
	3		78.40%	70.56%	82.00%	8.20%	78.76%	3
	4		80.80%	72.72%	78.00%	7.80%	80.52%	3
	5		80.80%	72.72%	78.00%	7.80%	80.52%	3
	6		85.60%	77.04%	78.00%	7.80%	84.84%	3
C303	1	SHIP RESISTANCE AND PROPULSION	78.40%	70.56%	80.00%	8.00%	78.56%	3
	2		85.20%	76.68%	88.00%	8.80%	85.48%	3
	3		78.80%	70.92%	86.00%	8.60%	79.52%	3
	4		81.20%	73.08%	80.00%	8.00%	81.08%	3
	5		78.00%	70.20%	80.00%	8.00%	78.20%	3
	6		84.80%	76.32%	80.00%	8.00%	84.32%	3
C304	1	OBJECT ORIENTED PROGRAMMING	86.80%	78.12%	88.00%	8.80%	86.92%	3
	2		89.20%	80.28%	88.00%	8.80%	89.08%	3
	3		86.80%	78.12%	88.00%	8.80%	86.92%	3
	4		89.20%	80.28%	88.00%	8.80%	89.08%	3
	5		88.00%	79.20%	88.00%	8.80%	88.00%	3
	6		92.00%	82.80%	88.00%	8.80%	91.60%	3
C306	1	WAVE MECHANICS	47.20%	42.48%	68.00%	6.80%	49.28%	0
	2		53.60%	48.24%	64.00%	6.40%	54.64%	1
	3		51.20%	46.08%	72.00%	7.20%	53.28%	1
	4		56.00%	50.40%	68.00%	6.80%	57.20%	1
	5		58.40%	52.56%	68.00%	6.80%	59.36%	1
	6		66.40%	59.76%	68.00%	6.80%	66.56%	2

C307	1	LIFTING SURFACES FOR MARINE APPLICATIONS	78.40%	70.56%	100.00%	10.00%	80.56%	3
	2		66.40%	59.76%	100.00%	10.00%	69.76%	2
	3		64.80%	58.32%	100.00%	10.00%	68.32%	2
	4		78.40%	70.56%	100.00%	10.00%	80.56%	3
	5		76.80%	69.12%	100.00%	10.00%	79.12%	3
	6		95.20%	85.68%	100.00%	10.00%	95.68%	3
C312	1	OBJECT ORIENTED PROGRAMMING LAB	100.00%	90.00%	98.00%	9.80%	99.80%	3
	2		99.20%	89.28%	98.00%	9.80%	99.08%	3
	3		100.00%	90.00%	98.00%	9.80%	99.80%	3
	4		100.00%	90.00%	98.00%	9.80%	99.80%	3
	5		99.20%	89.28%	98.00%	9.80%	99.08%	3
	6		100.00%	90.00%	98.00%	9.80%	99.80%	3
C313	1	SHIP STRENGTH LAB	70.80%	63.72%	96.00%	9.60%	73.32%	3
	2		71.60%	64.44%	94.00%	9.40%	73.84%	3
	3		69.20%	62.28%	96.00%	9.60%	71.88%	3
	4		67.60%	60.84%	96.00%	9.60%	70.44%	3
	5		64.40%	57.96%	94.00%	9.40%	67.36%	2
	6		66.80%	60.12%	98.00%	9.80%	69.92%	2
C314	1	STRUCTURAL MODELLING-SOFTWARE LABORATORY	90.80%	81.72%	98.00%	9.80%	91.52%	3
	2		88.40%	79.56%	98.00%	9.80%	89.36%	3
	3		87.60%	78.84%	98.00%	9.80%	88.64%	3
	4		88.40%	79.56%	98.00%	9.80%	89.36%	3
	5		87.60%	78.84%	98.00%	9.80%	88.64%	3
	6		90.80%	81.72%	98.00%	9.80%	91.52%	3
C315	1	INTERNSHIP-I	64.00%	57.60%	100.00%	10.00%	67.60%	2
	2		54.00%	48.60%	100.00%	10.00%	58.60%	1
	3		62.00%	55.80%	100.00%	10.00%	65.80%	2
	4		62.00%	55.80%	100.00%	10.00%	65.80%	2
	5		68.00%	61.20%	100.00%	10.00%	71.20%	3
	6		64.00%	57.60%	100.00%	10.00%	67.60%	2
C316	1	DESIGN OF OFFSHORE STRUCTURES	92.00%	82.80%	92.00%	9.20%	92.00%	3
	2		88.40%	79.56%	88.00%	8.80%	88.36%	3
	3		90.40%	81.36%	92.00%	9.20%	90.56%	3
	4		90.00%	81.00%	92.00%	9.20%	90.20%	3
	5		91.60%	82.44%	92.00%	9.20%	91.64%	3
	6		88.00%	79.20%	92.00%	9.20%	88.40%	3

C317	1	SHIP MOTIONS AND CONTROL	48.40%	43.56%	94.00%	9.40%	52.96%	1
	2		52.00%	46.80%	94.00%	9.40%	56.20%	1
	3		52.80%	47.52%	94.00%	9.40%	56.92%	1
	4		46.80%	42.12%	92.00%	9.20%	51.32%	1
	5		48.80%	43.92%	94.00%	9.40%	53.32%	1
	6		57.60%	51.84%	92.00%	9.20%	61.04%	2
C318	1	SHIP DESIGN	64.00%	57.60%	94.00%	9.40%	67.00%	2
	2		77.20%	69.48%	94.00%	9.40%	78.88%	3
	3		76.00%	68.40%	94.00%	9.40%	77.80%	3
	4		62.00%	55.80%	92.00%	9.20%	65.00%	2
	5		59.60%	53.64%	94.00%	9.40%	63.04%	2
	6		80.40%	72.36%	92.00%	9.20%	81.56%	3
C319	1	CAD/CAM IN SHIPBUILDING	71.85%	64.67%	96.30%	9.63%	74.30%	3
	2		84.44%	76.00%	96.30%	9.63%	85.63%	3
	3		84.44%	76.00%	96.30%	9.63%	85.63%	3
	4		73.33%	66.00%	96.30%	9.63%	75.63%	3
	5		66.67%	60.00%	96.30%	9.63%	69.63%	2
	6		87.41%	78.67%	96.30%	9.63%	88.30%	3
C321	1	FISHING VESSEL TECHNOLOGY	62.61%	56.35%	91.30%	9.13%	65.48%	2
	2		64.35%	57.91%	91.30%	9.13%	67.04%	2
	3		65.22%	58.70%	91.30%	9.13%	67.83%	2
	4		53.04%	47.74%	86.96%	8.70%	56.43%	1
	5		45.22%	40.70%	91.30%	9.13%	49.83%	0
	6		57.39%	51.65%	86.96%	8.70%	60.35%	2
C323	1	INLAND WATER TRANSPORTATION	68.80%	61.92%	96.00%	9.60%	71.52%	3
	2		78.40%	70.56%	92.00%	9.20%	79.76%	3
	3		76.80%	69.12%	96.00%	9.60%	78.72%	3
	4		70.40%	63.36%	92.00%	9.20%	72.56%	3
	5		60.80%	54.72%	96.00%	9.60%	64.32%	2
	6		81.60%	73.44%	96.00%	9.60%	83.04%	3
C325	1	SHIP SYSTEMS ENGINEERING	71.20%	64.08%	92.00%	9.20%	73.28%	3
	2		83.20%	74.88%	76.00%	7.60%	82.48%	3
	3		85.60%	77.04%	92.00%	9.20%	86.24%	3
	4		71.20%	64.08%	76.00%	7.60%	71.68%	3
	5		68.00%	61.20%	92.00%	9.20%	70.40%	3
	6		84.00%	75.60%	92.00%	9.20%	84.80%	3

C331	1	MARINE HYDRODYNAMICS LABORATORY	62.40%	56.16%	80.00%	8.00%	64.16%	2
	2		63.20%	56.88%	76.00%	7.60%	64.48%	2
	3		60.80%	54.72%	80.00%	8.00%	62.72%	2
	4		64.80%	58.32%	80.00%	8.00%	66.32%	2
	5		60.80%	54.72%	76.00%	7.60%	62.32%	2
	6		63.20%	56.88%	80.00%	8.00%	64.88%	2
C332	1	OFFSHORE STRUCTURE DESIGN – SOFTWARE LABORATORY	94.69%	85.22%	100.00%	10.00%	95.22%	3
	2		94.69%	85.22%	100.00%	10.00%	95.22%	3
	3		94.69%	85.22%	100.00%	10.00%	95.22%	3
	4		94.69%	85.22%	100.00%	10.00%	95.22%	3
	5		97.14%	87.43%	100.00%	10.00%	97.43%	3
	6		95.51%	85.96%	100.00%	10.00%	95.96%	3
C401	1	ADVANCED SHIP TECHNOLOGY	65.60%	59.04%	92.00%	9.20%	68.24%	2
	2		73.60%	66.24%	88.00%	8.80%	75.04%	3
	3		68.00%	61.20%	92.00%	9.20%	70.40%	3
	4		69.60%	62.64%	90.00%	9.00%	71.64%	3
	5		79.60%	71.64%	92.00%	9.20%	80.84%	3
	6		83.20%	74.88%	92.00%	9.20%	84.08%	3
C403	1	COMPUTER AIDED STRUCTURAL DESIGN (FEA)	57.14%	51.43%	92.86%	9.29%	60.71%	2
	2		52.86%	47.57%	92.86%	9.29%	56.86%	1
	3		57.14%	51.43%	92.86%	9.29%	60.71%	2
	4		64.29%	57.86%	92.86%	9.29%	67.14%	2
	5		67.14%	60.43%	92.86%	9.29%	69.71%	2
	6		60.00%	54.00%	92.86%	9.29%	63.29%	2
C405	1	DYNAMICS OF OFFSHORE STRUCTURES	64.44%	58.00%	91.67%	9.17%	67.17%	2
	2		62.22%	56.00%	77.78%	7.78%	63.78%	2
	3		58.89%	53.00%	91.67%	9.17%	62.17%	2
	4		62.78%	56.50%	86.11%	8.61%	65.11%	2
	5		48.33%	43.50%	91.67%	9.17%	52.67%	1
	6		68.33%	61.50%	91.67%	9.17%	70.67%	3
C407	1	HIGH PERFORMANCE MARINE VEHICLES	80.00%	72.00%	90.91%	9.09%	81.09%	3
	2		60.00%	54.00%	90.91%	9.09%	63.09%	2
	3		65.45%	58.91%	90.91%	9.09%	68.00%	2
	4		83.64%	75.27%	90.91%	9.09%	84.36%	3
	5		63.64%	57.27%	90.91%	9.09%	66.36%	2
	6		89.09%	80.18%	90.91%	9.09%	89.27%	3

C408	1	MARINE CORROSION AND CONTROL	69.73%	62.76%	94.59%	9.46%	72.22%	3
	2		64.86%	58.38%	86.49%	8.65%	67.03%	2
	3		68.65%	61.78%	94.59%	9.46%	71.24%	3
	4		69.19%	62.27%	94.59%	9.46%	71.73%	3
	5		58.92%	53.03%	94.59%	9.46%	62.49%	2
	6		81.08%	72.97%	94.59%	9.46%	82.43%	3
C409	1	THEORY AND PRACTICE IN MARINE CFD	90.77%	81.69%	76.92%	7.69%	89.38%	3
	2		60.00%	54.00%	76.92%	7.69%	61.69%	2
	3		58.46%	52.62%	76.92%	7.69%	60.31%	2
	4		55.38%	49.85%	76.92%	7.69%	57.54%	1
	5		66.15%	59.54%	76.92%	7.69%	67.23%	2
	6		81.54%	73.38%	76.92%	7.69%	81.08%	3
C410	1	SHIPYARD PRACTICES AND PROJECT MANAGEMENT	66.67%	60.00%	97.44%	9.74%	69.74%	2
	2		65.13%	58.62%	97.44%	9.74%	68.36%	2
	3		51.79%	46.62%	97.44%	9.74%	56.36%	1
	4		68.72%	61.85%	97.44%	9.74%	71.59%	3
	5		63.08%	56.77%	97.44%	9.74%	66.51%	2
	6		78.97%	71.08%	97.44%	9.74%	80.82%	3
C417	1	SHIP SYSTEM DRAWING & LAUNCHING CALCULATIONS LABORATORY	84.80%	76.32%	89.36%	8.94%	85.26%	3
	2		84.00%	75.60%	89.36%	8.94%	84.54%	3
	3		82.40%	74.16%	89.36%	8.94%	83.10%	3
	4		84.80%	76.32%	89.36%	8.94%	85.26%	3
	5		84.00%	75.60%	89.36%	8.94%	84.54%	3
	6		84.00%	75.60%	89.36%	8.94%	84.54%	3
C418	1	NUMERICAL SHIP HYDRODYNAMICS – SOFTWARE LABORATORY	71.20%	64.08%	97.83%	9.78%	73.86%	3
	2		72.80%	65.52%	97.83%	9.78%	75.30%	3
	3		74.40%	66.96%	97.83%	9.78%	76.74%	3
	4		72.80%	65.52%	97.83%	9.78%	75.30%	3
	5		55.20%	49.68%	97.83%	9.78%	59.46%	1
	6		55.20%	49.68%	97.83%	9.78%	59.46%	1
C419	1	MINI PROJECT	66.00%	59.40%	86.00%	8.60%	68.00%	2
	2		68.00%	61.20%	86.00%	8.60%	69.80%	2
	3		70.00%	63.00%	86.00%	8.60%	71.60%	3
	4		74.00%	66.60%	86.00%	8.60%	75.20%	3
	5		72.00%	64.80%	86.00%	8.60%	73.40%	3
	6		78.00%	70.20%	86.00%	8.60%	78.80%	3

C420	1	INTERNSHIP - II	84.80%	76.32%	89.36%	8.94%	85.26%	3
	2		84.00%	75.60%	89.36%	8.94%	84.54%	3
	3		82.40%	74.16%	89.36%	8.94%	83.10%	3
	4		84.80%	76.32%	89.36%	8.94%	85.26%	3
	5		84.00%	75.60%	89.36%	8.94%	84.54%	3
	6		84.00%	75.60%	89.36%	8.94%	84.54%	3
C421	1	MAJOR PROJECT / INDUSTRY INTERNSHIP PROJECT	50.00%	45.00%	86.00%	8.60%	53.60%	1
	2		52.00%	46.80%	86.00%	8.60%	55.40%	1
	3		56.00%	50.40%	86.00%	8.60%	59.00%	1
	4		58.00%	52.20%	86.00%	8.60%	60.80%	2
	5		74.00%	66.60%	86.00%	8.60%	75.20%	3
	6		76.00%	68.40%	86.00%	8.60%	77.00%	3

The Assessment process for CO attainment for a sample Course C325 / Ship System Engineering is shown in Table 3.10 and is graphically represented in Fig.3.1.

Table B 3.10 CO Assessment Process for the Course C325 / Ship System Engineering (Sample)

S.No	Process for CO attainment	CO attainment					
		CO1	CO2	CO3	CO4	CO5	CO6
1	% of Direct Attainment through Internal Exams(CAT I & II, Model Exam, Assignment I,II,III & ALM)	100.00%	100.00%	100.00%	100.00%	92.00%	84.00%
2	40% from Direct Attainment through Internal Exams	40.00%	40.00%	40.00%	40.00%	36.80%	33.60%
3	% of Direct Attainment through End Semester Exam	52.00%	72.00%	76.00%	52.00%	52.00%	84.00%
4	60% from Direct Attainment through End Semester Exam	31.20%	43.20%	45.60%	31.20%	31.20%	50.40%
5	40% + 60% of Direct Attainment	71.20%	83.20%	85.60%	71.20%	68.00%	84.00%
6	90% from Direct Attainment	64.08%	74.88%	77.04%	64.08%	61.20%	75.60%
7	% of Indirect Attainment Through Course End Survey	92.00%	76.00%	92.00%	76.00%	92.00%	92.00%
8	10% from Indirect Attainment Through Course End Survey	9.20%	7.60%	9.20%	7.60%	9.20%	9.20%
9	90% from Direct + 10% from Indirect For Co Attainment	73.28%	82.48%	86.24%	71.68%	70.40%	84.80%
Attainment Level		3	3	3	3	3	3

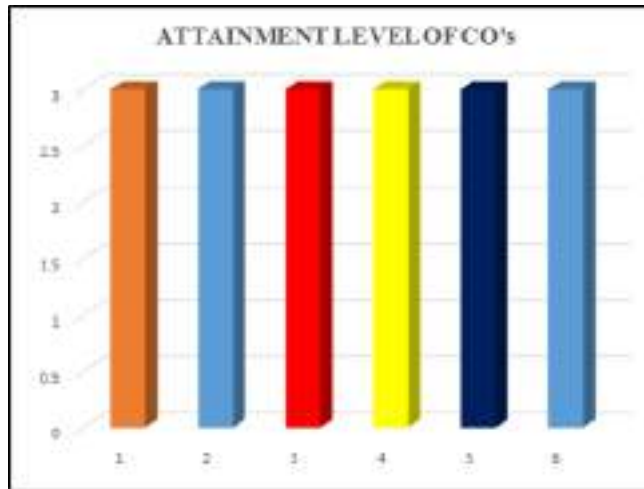


Fig. 3.1 CO Attainment level for the course C325 / Ship System Engineering

3.3 Attainment of Program Outcomes and Program Specific Outcomes (75)

Total Marks 75.00

3.3.1 Describe assessment tools and processes used for measuring the attainment of each Program Outcome and Program Specific Outcomes (10)

Institute Marks : 10.00

Direct assessment tools:

- Internal assessment examinations
- Assignments / seminar - rubrics based evaluation
- End semester examinations
- Rubrics based performance evaluation in laboratory classes
- End semester practical examination
- Project - rubrics based evaluation

Indirect assessment tools:

- Employer survey
- Program exit survey
- Alumni survey

The PO assessment process is elucidated in the Fig 3.2 and the same procedure is followed for calculating the attainment of PSOs. The direct and indirect assessment tools are tabulated in Table 3.11, which is used for the evaluation of POs and PSOs.

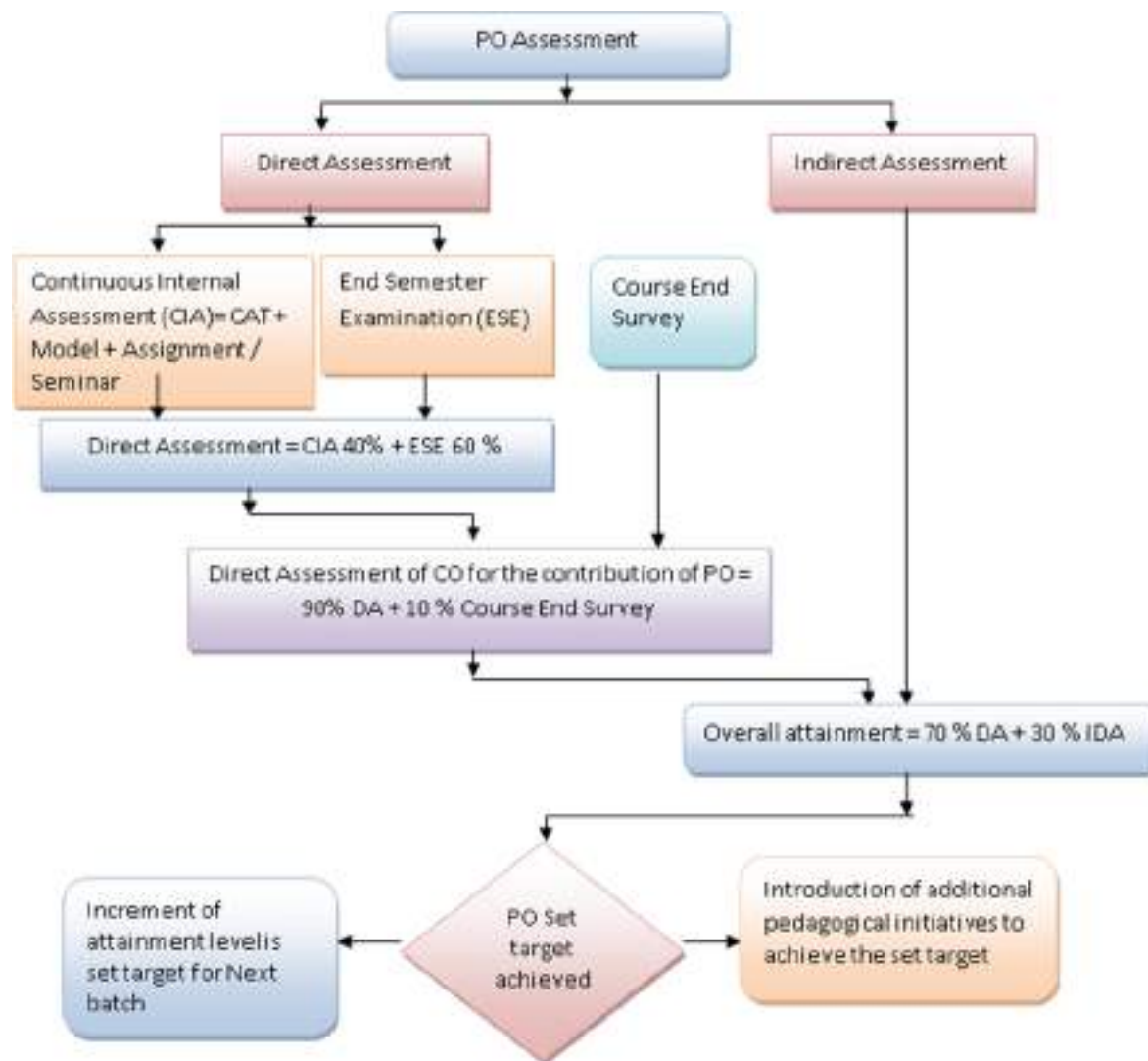


Fig 3.2 PO Assessment process

Table B 3.11. PO and PSO Attainments through Assessment Tools, Assessment Frequency & Responsibility

Assessment Type	Assessment Tool	Assessment frequency	Assessed by	Reviewed by	Weightage

Direct Attainment (DA)	CAT Exam	Twice in Semester	Department	HOD	40%	70%
	Assignment	2 - 4 in Semester	Course Faculty	Course Faculty		
	Model Examination	Once in Semester	Department	HOD		
	University Examination	Once in Semester	Office of COE	Institute, COE	60%	
	Laboratory Examination	Once in Semester	COE, Department	Institute, COE	100%	
	Internship / Seminar	Twice in Four year	Department	HOD	100%	
	Mini Project	In the third year Even Semester	Department	HOD	100%	
	Major Project	In Final-year	Department	HOD	100%	
Indirect Attainment (IDA)	Employer Survey (ES)	Once in Year	Department	HOD	5%	30%
	Program Exit Survey (PES)	At the end of the Fourth year second semester	Alumni Coordinator	HOD	15%	
	Alumni Survey (AS)	Yearly once after graduation	Alumni Coordinator	HOD	10%	
Overall Attainment (OA)	DA converted to 70% +IDA converted to 30%					100%

QUALITY/RELEVANCE OF ASSESSMENT TOOLS AND PROCESS

Direct assessment tool:

Direct assessment tools used are described in Table B 3.3. Initially the attainment of each course outcome is determined using internal assessment and External Assessment as described in section 3.2.1. The PO attainment for each course is determined from the attainment values obtained for each course outcome related to that PO and CO-PO mapping values. Similarly, the PSO attainment is also determined.

Indirect assessment tool:

Indirect assessment is done through student exit survey, Alumni survey and employer survey. Program exit survey is given a weightage of 15%, alumni survey a weightage of 10% and employer survey of 5% during the assessment process.

Student exit survey:

An exit survey was conducted for each student, who has completed the programme of study. Relevant questionnaire of the student exit survey form to evaluate the attainment of PO and PSO.

Employer Survey:

A survey is conducted from the company where the graduates are working, after one or two years of their graduation. A questionnaire prepared containing all the aspects of the program attainment and the feedback is collected from the employers of various reputed industries/ organizations graduated from the University. Based on the feedback, the POs and PSOs assessments, the curriculum and syllabus for the program are modified.

PO attainment Target Levels

The set attainment value for PO and PSO is fixed based on the performance of students in the Direct and Indirect assessments. Once the set attainment level is reached, the set attainment value is increased for the successive academic years. The PO attainment Target levels are shown below in Table B 3.12.

Table B 3.12: PO attainment Target Levels

Scale	Level	PO Attainment to target in %
3	Substantial	≥70 %
2	Moderate	>60 – 70 %
1	Low	50 - 60 %

3.3.2 Provide results of evaluation of each PO & PSO (65)

Institute Marks : 65.00

PO Attainment

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C101	PO1	PO2	PO3	PO4	PO5	1.97	1.97	PO8	2.10	2.10	PO11	2.24
C102	1.64	1.53	1.53	1.18	1.64	PO6	PO7	PO8	PO9	PO10	PO11	1.54
C103	1.75	1.75	1.75	1.31	1.10	PO6	PO7	PO8	PO9	PO10	PO11	1.15
C104	1.24	1.06	1.50	PO4	1.00	PO6	1.50	PO8	PO9	PO10	PO11	0.53
C106	PO1	PO2	PO3	PO4	PO5	2.50	2.00	2.00	3.00	3.00	PO11	3.00
C107	1.82	1.72	1.82	1.96	2.34	PO6	PO7	PO8	1.95	1.95	PO11	1.72
C108	1.92	1.37	2.49	2.19	2.05	PO6	2.29	PO8	2.05	2.05	PO11	1.59
C109	2.67	PO2	PO3	PO4	2.67	PO6	PO7	PO8	2.37	2.37	PO11	0.89
C110	1.52	1.32	1.36	1.37	1.34	PO6	PO7	PO8	PO9	PO10	PO11	1.25
C111	1.81	1.62	1.81	1.50	1.61	PO6	PO7	PO8	PO9	PO10	PO11	1.93
C112	2.22	2.21	1.97	2.08	2.09	PO6	PO7	PO8	PO9	PO10	PO11	2.10
C113	1.43	1.43	1.32	1.14	1.52	PO6	PO7	PO8	PO9	1.14	PO11	0.57
C114	PO1	PO2	PO3	PO4	PO5	1.31	1.57	1.31	0.89	PO10	PO11	1.32
C115	PO1	PO2	PO3	PO4	PO5	2.46	1.96	1.96	2.95	2.95	PO11	2.95
C116	1.99	2.99	2.99	1.99	1.00	PO6	PO7	PO8	2.32	2.32	PO11	2.49
C117	2.02	2.02	2.02	2.16	2.43	PO6	PO7	PO8	2.16	2.43	PO11	1.89
C118	1.74	1.32	1.32	PO4	PO5	PO6	PO7	PO8	PO9	1.83	PO11	1.83
C201	1.81	1.62	1.81	1.59	1.29	PO6	PO7	PO8	PO9	PO10	PO11	1.28
C202	1.87	1.77	1.67	1.67	1.76	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C203	1.70	1.22	1.13	1.18	1.18	PO6	1.56	PO8	PO9	PO10	PO11	1.72
C204	1.56	1.48	1.31	1.21	PO5	PO6	PO7	PO8	PO9	PO10	PO11	1.09
C205	1.68	1.59	1.32	1.41	0.97	PO6	PO7	0.97	PO9	PO10	PO11	1.58
C206	1.72	1.72	1.33	1.14	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C209	PO1	PO2	PO3	PO4	PO5	PO6	PO7	1.47	1.84	1.52	1.84	PO12
C210	1.75	1.75	1.56	1.56	1.17	PO6	PO7	PO8	1.65	PO10	PO11	1.17
C211	1.84	1.64	1.64	PO4	PO5	1.74	PO7	0.61	1.74	1.74	PO11	1.74
C212	2.53	1.69	2.39	2.32	2.57	PO6	PO7	PO8	2.39	PO10	PO11	2.57
C213	2.62	2.32	1.74	1.74	PO5	PO6	PO7	PO8	PO9	PO10	PO11	2.80
C214	2.44	1.66	PO3	PO4	PO5	PO6	2.41	PO8	PO9	PO10	PO11	2.63
C215	2.11	1.56	1.50	2.39	2.39	PO6	PO7	PO8	PO9	PO10	PO11	2.24
C216	2.53	2.42	2.42	2.42	PO5	PO6	2.71	PO8	PO9	PO10	PO11	1.73
C217	PO1	PO2	PO3	PO4	PO5	1.82	PO7	2.15	1.64	1.97	1.43	1.43

C221	PO1	PO2	PO3	PO4	PO5	PO6	PO7	0.99	1.87	2.97	PO11	PO12
C222	2.58	2.73	1.82	1.82	2.27	PO6	PO7	PO8	PO9	PO10	PO11	2.73
C223	2.63	1.86	2.47	1.87	2.55	PO6	PO7	PO8	1.86	PO10	PO11	1.85
C301	PO1	PO2	0.70	0.70	0.70	PO6	PO7	1.39	1.39	1.39	PO11	1.18
C302	2.42	2.28	2.28	1.61	2.28	PO6	PO7	PO8	PO9	PO10	PO11	2.18
C303	2.16	2.02	2.02	2.11	2.21	PO6	1.69	PO8	PO9	PO10	PO11	2.21
C304	2.36	1.92	1.96	1.96	1.43	PO6	PO7	0.89	PO9	PO10	PO11	PO12
C306	1.70	1.59	1.51	1.61	1.22	PO6	1.52	PO8	PO9	PO10	PO11	1.22
C307	2.37	2.08	2.17	2.11	2.23	PO6	PO7	PO8	PO9	PO10	PO11	2.14
C312	2.16	1.99	2.39	2.19	2.19	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C313	2.02	1.77	2.02	1.84	1.58	PO6	PO7	PO8	PO9	1.44	PO11	1.44
C314	2.70	2.70	2.38	1.78	1.77	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C315	1.98	1.98	PO3	PO4	1.36	1.32	PO7	2.00	2.03	1.98	PO11	PO12
C316	2.71	2.56	2.56	PO4	2.56	PO6	1.80	1.80	PO9	PO10	PO11	2.49
C317	1.57	1.47	1.38	1.47	1.56	PO6	PO7	PO8	PO9	PO10	PO11	1.55
C318	2.17	2.03	2.03	2.03	2.03	PO6	PO7	2.03	PO9	PO10	1.44	1.95
C319	2.40	2.25	2.17	PO4	2.17	PO6	PO7	PO8	PO9	PO10	PO11	1.60
C321	1.73	1.46	1.46	1.22	1.45	1.30	1.44	1.45	PO9	PO10	PO11	1.22
C323	2.00	2.03	2.03	2.03	1.49	1.66	1.51	1.94	PO9	PO10	PO11	1.92
C325	1.01	1.01	PO3	PO4	PO5	1.18	0.83	PO8	PO9	PO10	0.79	1.01
C331	1.28	1.28	1.82	1.28	1.82	1.28	1.28	1.28	1.82	1.28	1.29	PO12
C332	2.87	2.71	1.91	1.91	2.71	PO6	PO7	PO8	1.91	1.91	PO11	2.71
C401	2.25	2.29	2.33	2.47	2.25	2.33	PO7	2.05	1.64	PO10	PO11	2.47
C403	1.79	1.79	1.79	2.00	1.65	1.41	1.41	1.52	PO9	PO10	1.34	1.78
C405	1.91	1.75	1.75	1.26	1.75	PO6	PO7	PO8	PO9	PO10	PO11	1.65
C407	2.26	1.48	1.98	1.98	2.05	PO6	PO7	1.79	PO9	PO10	PO11	1.48
C408	2.14	1.71	1.95	1.74	1.74	PO6	2.00	PO8	PO9	PO10	PO11	1.42
C409	2.09	1.48	1.48	1.68	1.95	1.62	PO7	1.62	PO9	1.62	PO11	1.95
C410	2.07	1.86	1.86	1.86	1.90	1.37	1.39	1.86	1.37	PO10	1.93	1.81
C417	2.54	2.54	2.26	1.97	1.69	PO6	PO7	PO8	1.69	1.69	PO11	2.40
C418	2.00	2.00	1.40	1.40	2.00	PO6	PO7	PO8	1.40	1.40	PO11	2.00
C419	2.18	2.18	1.46	1.79	1.94	1.90	1.95	2.18	1.57	2.18	1.46	2.18
C420	2.54	2.54	2.26	1.97	1.69	PO6	PO7	PO8	1.69	1.69	PO11	2.40
C421	3.00	3.00	2.00	2.50	2.67	2.67	2.67	3.00	2.17	3.00	2.00	3.00

PO Attainment Indirect

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Employer S	2.80	2.70	2.80	2.50	2.80	2.80	2.80	2.80	2.40	2.60	2.60	2.70
Program Ex	2.84	2.76	2.72	2.64	2.74	2.90	2.88	2.60	2.58	2.54	2.60	2.76
Alumni Sur	2.79	2.84	2.68	2.58	2.79	3.00	2.84	2.53	2.84	2.53	2.53	2.74

PO Attainment Level

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
InDirect Attainment	2.81	2.77	2.73	2.57	2.78	2.90	2.84	2.64	2.61	2.56	2.58	2.73
Direct Attainment	2.06	1.88	1.84	1.75	1.82	1.76	1.78	1.66	1.91	2.00	1.50	1.84

PSO Attainment

Course	PSO1	PSO2	PSO3
C101	PSO1	PSO2	PSO3
C102	1.41	1.65	1.53
C103	1.43	1.43	1.43
C104	1.06	PSO2	PSO3
C106	PSO1	PSO2	PSO3
C107	1.56	1.95	1.82
C108	0.82	PSO2	PSO3
C109	PSO1	PSO2	PSO3
C110	1.14	1.14	1.14
C111	1.29	1.29	1.29
C112	2.10	2.23	2.09
C113	1.14	0.57	0.57
C114	PSO1	PSO2	PSO3
C115	PSO1	PSO2	PSO3
C116	2.99	2.99	2.16
C117	PSO1	PSO2	PSO3
C118	PSO1	1.22	PSO3
C201	1.50	1.63	1.59
C202	1.27	1.77	1.25
C203	1.70	1.70	1.70
C204	1.65	1.65	1.65
C205	1.68	1.68	1.68
C206	1.72	1.72	1.14

C209	0.91	1.02	0.61
C210	1.75	1.75	1.75
C211	PSO1	PSO2	PSO3
C212	1.69	1.97	1.69
C213	2.46	2.62	2.32
C214	2.44	2.17	2.17
C215	2.11	2.11	1.41
C216	2.53	2.53	2.53
C217	1.43	1.43	1.43
C221	PSO1	PSO2	PSO3
C222	PSO1	1.82	1.82
C223	2.63	2.63	2.63
C301	PSO1	PSO2	PSO3
C302	2.28	2.28	2.28
C303	2.30	2.30	2.30
C304	PSO1	PSO2	PSO3
C306	1.59	1.59	1.59
C307	2.21	2.21	2.21
C312	PSO1	PSO2	PSO3
C313	2.02	2.02	2.02
C314	2.70	2.70	2.70
C315	1.98	1.98	1.98
C316	2.56	2.56	2.56
C317	1.56	1.56	1.56
C318	2.17	2.17	2.17
C319	2.25	PSO2	2.25
C321	1.22	1.73	1.73
C323	2.11	2.03	2.03
C325	1.45	PSO2	1.01
C331	1.82	1.28	1.28
C332	2.71	2.71	2.71
C401	2.25	2.25	2.25
C403	1.89	1.89	1.89
C405	1.91	1.91	1.91
C407	2.23	2.23	2.13
C408	2.02	2.16	1.98

C409	2.09	2.09	2.09
C410	2.07	2.07	PSO3
C417	2.54	2.54	2.54
C418	2.10	2.10	2.10
C419	2.18	2.18	2.18
C420	2.54	2.54	2.54
C421	3.00	3.00	3.00

PSO Attainment Indirect

Survey	PSO1	PSO2	PSO3
Employer Survey	2.60	2.70	2.50
Program Exit Survey	2.60	2.62	2.58
Alumni Survey	2.63	2.68	2.79

PSO Attainment Level

Course	PSO1	PSO2	PSO3
Direct Attainment	1.93	1.98	1.89
InDirect Attainment	2.61	2.67	2.62

4 STUDENTS' PERFORMANCE (100)

Total Marks 74.47

Table 4.1

Item (Information to be provided cumulatively for all the shifts with explicit headings, wherever applicable)	2022-23 (CAY)	2021-22 (CAYm1)	2020-21 (CAYm2)	2019-20 (CAYm3)	2018-19 (CAYm4)	2017-18 (CAYm5)	2016-17 (CAYm6)
Sanctioned intake of the program(N)	30	30	54	60	120	65	90
Total number of students admitted in first year minus number of students migrated to other programs/ institutions plus No. of students migrated to this program (N1)	30	28	17	36	47	63	83
Number of students admitted in 2nd year in the same batch via lateral entry (N2)	0	5	14	7	3	9	7
Separate division students, If applicable (N3)	0	0	0	0	0	0	0
Total number of students admitted in the programme(N1 + N2 + N3)	30	33	31	43	50	72	90

Table 4.2

Year of entry	Total No of students admitted in the program (N1 + N2 + N3)	Number of students who have successfully graduated without backlogs in any semester/ year of study (Without Backlog means no compartment or failures in any semester/ year of study)			
		I year	II year	III year	IV year
2022-23 (CAY)	30				
2021-22 (CAYm1)	33	13			
2020-21 (CAYm2)	31	12	10		
2019-20 (CAYm3)	43	20	16	14	
2018-19 (LYG)	50	40	26	22	18
2017-18 (LYGm1)	72	26	25	24	15
2016-17 (LYGm2)	90	27	26	22	20

Table 4.3

Year of entry	Total No of students admitted in the program (N1 + N2 + N3)	Number of students who have successfully graduated in stipulated period of study) [Total of with Backlog + without Backlog]			
		I year	II year	III year	IV year
2022-23 (CAY)	30				
2021-22 (CAYm1)	33	20			
2020-21 (CAYm2)	31	15	14		
2019-20 (CAYm3)	43	28	26	19	
2018-19 (LYG)	50	47	46	45	41
2017-18 (LYGm1)	72	62	61	60	58
2016-17 (LYGm2)	90	79	76	75	72

4.1 Enrolment Ratio (20)

Total Marks 14.00

Institute Marks : 14.00

	N (From Table 4.1)	N1 (From Table 4.1)	Enrollment Ratio [(N1/N)*100]
2022-23 (CAY)	30	30	100.00
2021-22 (CAYm1)	30	28	93.33
2020-21 (CAYm2)	54	17	31.48

Average [(ER1 + ER2 + ER3) / 3] : 74.94

Assessment : 14.00

4.2 Success Rate in the stipulated period of the program (20)

Total Marks 7.95

4.2.1 Success rate without backlogs in any semester / year of study (15)

Institute Marks : 3.90

Item	Latest Year of Graduation, LYG (2018-19)	Latest Year of Graduation minus 1, LYGm1 (2017-18)	Latest Year of Graduation minus 2 LYGm2 (2016-17)
X Number of students admitted in the corresponding First year + admitted in 2nd year via lateral entry and seperated division, if applicable	50.00	72.00	90.00
Y Number of students who have graduated without backlogs in the stipulated period	18.00	15.00	20.00
Success Index [SI = Y / X]	0.36	0.21	0.22

Average SI [(SI1 + SI2 + SI3) / 3] : 0.26

Assessment [15 * Average SI] : 3.90

4.2.2 Success rate in stipulated period (5)

Institute Marks : 4.05

Item	Latest Year of Graduation, LYG (2018-19)	Latest Year of Graduation minus 1, LYGm1 (2017-18)	Latest Year of Graduation minus 2 LYGm2 (2016-17)
X Number of students admitted in the corresponding First year + admitted in 2nd year via lateral entry and seperated division, if applicable	50.00	72.00	90.00
Y Number of students who have graduated in the stipulated period	41.00	58.00	72.00
Success Index [SI = Y / X]	0.82	0.81	0.80

Average SI[(SI1 + SI2 + SI3) / 3]: 0.81

Assessment [5 * Average SI] : 4.05

Note : If 100% students clear without any backlog then also total marks scored will be 20 as both 4.2.1 & 4.2.2 will be applicable simultaneously.**4.3 Academic Performance in Second Year (10)**

Total Marks 4.22

Institute Marks : 4.22

Academic Performance	CAYm2 (2020-21)	CAYm3 (2019-20)	LYG (2018-19)
Mean of CGPA or mean percentage of all successful students(X)	6.20	6.10	5.60
Total number of successful students (Y)	14.00	26.00	46.00
Total number of students appeared in the examination (Z)	29.00	35.00	50.00
API [X * (Y/Z)]	2.99	4.53	5.15

Average API [(AP1 + AP2 + AP3)/3] : 4.22

Assessment [AverageAPI] : 4.22

4.4 Placement, Higher Studies and Entrepreneurship (30)

Total Marks 28.30

Item	LYG(2018-19)	LYGm1(2017-18)	LYGm2(2016-17)
Total No of Final Year Students(N)	45.00	60.00	75.00
No of students placed in the companies or goverment sector(X)	38.00	52.00	65.00
No of students admitted to higher studies with valid qualifying scores(GATE or equivalent State or National Level tests, GRE, GMAT etc.) (Y)	2.00	5.00	6.00
No of students turned enterpreneur in engineering/technology (Z)	1.00	1.00	0.00
Placement Index [(X+Y+Z)/N] :	0.91	0.97	0.95

Average Placement [(P1 + P2 + P3)/3] : 0.94

Assessment [30 * Average Placement] : 28.30

Program Name : Marine Engg.
Assessment Year : 2021-22 (CAYm1)

S.No	Student Name	Enrollment No	Employee Name	Appointment No
1	Babitha S	NA1134	Unique Solutions, Kochi	26-05-2022 (By email)
2	Christinal B	NA1135	Capital Ship Solutions-CSS	05-04-2022 (By email)
3	Dhanushma	NA1136	Accel Design, Kochi	11-04-2022 (By email)
4	A K Vishnunath	NA1137	Larsen and Toubro Pvt Ltd	26-04-2022 (By email)
5	Akash A	NA1138	Zener Maritime solutions Pvt Ltd	18-07-2022 (By email)
6	Andrews Jebaraj A	NA1140	Larsen and Toubro Pvt Ltd	26-04-2022 (By email)
7	Aravind B	NA1141	Vik Sandvik design India Pvt Ltd	26-04-2022 (By email)
8	Ashwin Vinod	NA1142	Accel Design, Kochi	22-02-2022 (By email)
9	Asuthosh Padhy	NA1143	Zener Maritime solutions Pvt Ltd	30-03-2022 (By email)
10	Ashwin S.Nair	NA1144	Accel Design, Kochi	18-07-2022(By email)
11	Balaji P	NA1145	Albion Marine Solutions	04-04-2022 (By email)
12	Baskar Antony Petlees A	NA1146	Optio Marine Pvt Ltd	25-05-2022(By email)
13	Gottam Karthik Reddy	NA1148	Cyber Marine Knowledge Systems Pvt Ltd	CM/22-600/001/05
14	Jegatheeswaran R	NA1149	Optio Marine Pvt Ltd	25-05-2022 (By email)
15	Kishore V	NA1150	Cyber Marine Knowledge Systems Pvt Ltd	CM/22-600/001/09
16	Kriishnaprasad thilakan	NA1151	Vik Sandvik design India Pvt Ltd	22-02-2022 (By email)
17	Mohammed Thamimullah K	NA1152	Optio Marine Pvt Ltd	25-05-2022 (By email)
18	Mohammed Arshad P	NA1153	Cyber Marine Knowledge Systems Pvt Ltd	CM/22-600/001/08
19	Murali Krishnan G	NA1154	Optio Marine Pvt Ltd	25-05-2022 (By email)
20	Praveen Balaji S	NA1156	PacMak DB Naval Charterers & Consultants Pvt Ltd	27-10-2022 (By email)
21	Yogi P	NA1178	Synergy Maritime Pvt Ltd	25-05-2022 (By email)
22	Regis Benny Arokham S	NA1160	Vedam Design & Technical Consultancy Pvt Ltd	V/HR/21-22/O-51
23	Rishab Bhantiya S	Na1161	Capital Ship Solutions-CSS	05-04-2022 (By email)
24	Rohan Kumar V	NA1162	PacMak DB Naval Charterers & Consultants Pvt Ltd	27-10-2022 (By email)
25	Sankar Lingam V	NA1163	Ammus Marine services, Karnataka	AMS/HR/12
26	Sarang S A	NA1164	Optio Marine Pvt Ltd	5-04-2022 (By email)
27	Sharukh Basha S	NA1167	Zener Maritime solutions Pvt Ltd	18-07-2022 (By email)
28	Shobit Sharma	NA1168	KDU Marine Equipment Trading & Maintenance LLC, dubai	25-07-2022 (By email)
29	Sri Sanjay V	NA1170	KDU Marine Equipment Trading & Maintenance LLC, Dubai	25-07-2022 (By email)
30	Srilesh M P	NA1172	Unique Solutions, Kochi	26-05-2022 (By email)
31	Vignnesh L	NA1173	Vedam Design & Technical Consultancy Pvt Ltd	V/HR/21-22/O-52
32	Vikram V	NA1174	Cognizant Solutions	16-02-2022 (By email)
33	Vivek Kumar chauhan	NA1175	Vedam Design & Technical Consultancy Pvt Ltd	V/HR/21-22/O-53
34	Vivek Kumar Singh	NA1176	Unique Solutions, Kochi	16-02-2022 (By email)
35	Vivek R	NA1177	PacMak DB Naval Charterers & Consultants Pvt Ltd	27-10-2022 (By email)
36	Kola Yasho Krishna Deo	NA1181	Unique Solutions, Kochi	26-05-2022 (By email)

37	Hemachandran R	NA1219L	PacMak DB Naval Charterers & Consultants Pvt Ltd	04-04-2022 (By email)
38	Ramiah Muthu S	NA1220L	Capital Ship Solutions-CSS	05-04-2022 (By email)

Assessment Year : 2020-21 (CAYm2)

S.No	Student Name	Enrollment No	Employee Name	Appointment No
1	Ardra MP	NA1048	Navgathi Marine Design and Constructions Pvt Ltd	30-6-2021 (By eMail)
2	Indumathi S	NA1049	Fietz Global Engineering and Technology Pvt Ltd	26-05-2021 (By eMail)
3	Shreyanka K	NA1052	Varuna Marine Services	03-08-2021 (By email)
4	Sivaranjani A	NA1053	3- Fish Informatics Pvt Ltd	02-08-2021 (By email)
5	Aakash B	NA1054	Bernard Schulte Ship Managment Pvt Ltd	20-7-2021 (By email)
6	Abhinav Ramesh S	NA1055	Vik -Sandwik Designs India Pvt Ltd	10-12-2021 (By email)
7	Abraham Noel	NA1056	Bernard Schulte Ship Managment Pvt Ltd	20-07-2021 (By email)
8	Achyuth Ravi Raj	NA1057	Conceptia Software Technologies Pvt Ltd	13-12-2020 (By email)
9	Afreeth S	NA1059	Larsen and Toubro Pvt Ltd	28-04-2021 (By email)
10	Aman Tripathi	NA1063	Ammus Marine Services, Karnataka	AMS/HR/08
11	Arigela Vara Prasad	NA1064	Albion Engineering India Pvt Ltd	17-03-2021(By email)
12	Athul Sunny	NA1068	Boat Craft Services Pvt Ltd	22-06-2021 (By email)
13	Balachandar B	NA1069	Varuna Marine Services	03-08-2021 (By email)
14	Bill John	NA1072	Conceptia Software Technologies Pvt Ltd	13-12-2020 (By email)
15	Chandrasekar K	NA1073	3- Fish Informatics Pvt Ltd	02-08-2021 (By email)
16	Christian Britto S	NA1074	Larsen and Toubro Pvt Ltd	28-04-2021 (By email)
17	Deepak K	NA1075	Larsen and Toubro Pvt Ltd	28-04-2021 (By email)
18	Gnanaruban S	NA1077	Larsen and Toubro Pvt Ltd	28-04-2021 (By email)
19	Mohammed Nooh	NA1081	Conceptia Software Technologies Pvt Ltd	13-12-2020 (By email)
20	Mohammed Binyamine	NA1082	Unique Solutions	20-04-2021 (By email)
21	Mohammed Hanan Haneef	NA1084	Ammus Marine Services, Karnataka	AMS/HR/06
22	Mohan Prasad A	NA1085	Vedam Designs Pvt Ltd	V/HR/20-21/O-19
23	Nishanth S A	NA1087	Unique Solutions	20-04-2021 (By email)
24	Nithish Balaji J	NA1088	Vedam Designs Pvt Ltd	V/HR/20-21/O-15
25	Prithiman V	NA1090	Vedam Designs Pvt Ltd	V/HR/20-21/O-18
26	Prithvi Raj Chauhan	NA1091	Unique Solutions	20-04-2021 (By email)
27	Ravindra Singh	NA1095	Albion Engineering India Pvt Ltd	17-03-2021 (By email)
28	Sachin Kumar	NA1097	Unique Solutions	20-04-2021 (By email)
29	Samuel M	NA1099	Vik -Sandwik Designs India Pvt Ltd	10-12-2021 (By email)
30	Sankar Raj A	NA1100	Conceptia Software Technologies Pvt Ltd	13-12-2020 (By email)
31	Saravana Prasath K	NA1101	Unique Solutions	20-04-2021 (By email)
32	Sathish B	NA1102	3- Fish Informatics Pvt Ltd	02-08-2021 (By email)
33	Shibin Shyju Thekkekkara	NA1105	Ammus Marine Services, Karnataka	AMS/HR/04
34	Sree Veeramanee N	NA1107	Vik -Sandwik Designs India Pvt Ltd	10-12-2020 (By email)
35	Ujjwal Raj	NA1111	Vik -Sandwik Designs India Pvt Ltd	10-12-2020 (By email)
36	Venu Aravindh T	NA1114	Vedam Designs Pvt Ltd	V/HR/20-21/O-34

37	Yashwanth C	NA1117	Vik -Sandwik Designs India Pvt Ltd	10-12-2020 (By email)
38	Karthik M	NA1121	Conceptia Software Technologies Pvt Ltd	13-12-2020 (By email)
39	Abishek P	NA1126L	Boat Craft Services Pvt Ltd	22-06-2021(By email)
40	Athira Unnikrishnan	NA1127L	3- Fish Informatics Pvt Ltd	02-08-2021(By email)
41	Hari Prasanth N	NA1130L	Boat Craft Services Pvt Ltd	22-06-2021(By email)
42	Sreenivas S	NA1131L	Albion Engineering India Pvt Ltd	17-03-2021 (By email)
43	Surender G	NA1132L	Vedam Designs Pvt Ltd	V/HR/20-21/O-14
44	Vikram Selvaganesh S	NA1133L	Vik -Sandwik Designs India Pvt Ltd	10-12-2020 (By email)
45	Akarsh S	NA1060	Fietz Global Engineering and Technology Pvt Ltd	26-05-2021 (By email)
46	Arunmughi M	NA1065	Navgathi Marine Design and Constructions Pvt Ltd	30-06-2021 (By email)
47	Balamurugan M	NA1070	Fietz Global Engineering and Technology Pvt Ltd	26-05-2021 (By email)
48	Mamidi Deva Kiran	NA1076	Fietz Global Engineering and Technology Pvt Ltd	26-05-2021 (By email)
49	Kaja Gowtham Sathya Kumar	NA1079	Zener Maritime solutions Pvt Ltd	05-7-2021 (By email)
50	Hari K	NA1080	Fietz Global Engineering and Technology Pvt Ltd	26-05-2021 (By email)
51	Ram Kumar M	NA1094	Ambition Marine Services Pvt Ltd	06-04-2021(By email)
52	Sabeer Mohammed S	NA1096	Act Infraport Pvt Limited	17-03-2021(By email)
53	Senthilmurugan U	NA1103	Ambition Marine Services Pvt Ltd	26-05-2021(By email)
54	Shinan Kareem	NA1106	Act Infraport Pvt Limited	17-03-2021(By email)
55	Sreeraj A S	NA1108	Act Infraport Pvt Limited	17-03-2021(By email)
56	Subham Padhy	NA1110	Ambition Marine Services Pvt Ltd	06-04-2021(By email)
57	Vijay P	NA1116	Zener Maritime solutions Pvt Ltd	05-7-2021 (By email)
58	Deepak S	NA1128L	Ambition Marine Services Pvt Ltd	06-04-2021(By email)

Assessment Year : 2019-20 (CAYm3)

S.No	Student Name	Enrollment No	Employee Name	Appointment No
1	Prithvi A R	RNA901	Act Infraport Pvt Ltd, Gujarat	04-1-2020 (By email)
2	Bindhiya. R	NA955	Act Infraport Pvt Ltd, Gujarat	04-1-2020 (By email)
3	Elsarisha Manih	NA956	Cybermarine Knowledge Systems Pvt Ltd	15-07-2020 (By email)
4	Varada Saidas	NA957	Cybermarine Knowledge Systems Pvt Ltd	15-07-2020 (By email)
5	Abhiroop K	NA959	Vedam Design & Technical Consultancy Pvt Ltd	V/HR/19-20/O-44
6	Affin Roy	NA960	Albion Engineering India Pvt Ltd	21-10-2020 (By email)
7	Akbar Babu P	NA962	3 - Fish Informatics Pvt Ltd	30-2-2020 (By email)
8	Akhil M	NA963	Larsen and Toubro Pvt Ltd	19-03-2020(By email)
9	Aman Sign Bisht	NA964	Act Infraport Pvt Ltd, Gujarat	04-1-2020 (By email)
10	Anugrah Raj P S	NA967	Albion Engineering India Pvt Ltd	21-10-2020 (By email)
11	Aravind K	NA968	Unique Solutions, Kochi	08-11-2020 (By email)
12	Aravindha Kumar. M	NA969	Vik- Sandwik Designs India Pvt Ltd	10-05-2020 (By email)
13	Ashlin. I	NA970	Act Infraport Pvt Ltd, Gujarat	04-1-2020 (By email)
14	Ashwin S	NA971	Ambition Marine Services, Chennai	23-04-2020(By email)
15	Austin Ajith	NA972	Vedam Design & Technical Consultancy Pvt Ltd	V/HR/19-20/O-46
16	Berin Raja AJ	NA973	Cybermarine Knowledge Systems Pvt Ltd	15-07-2020 (By email)
17	Putha Bhavya Natha Reddy	NA974	3 - Fish Informatics Pvt Ltd	30-2-2020 (By email)
18	Dhanasekaran A	NA975	Albion Engineering India Pvt Ltd	21-10-2020 (By email)
19	Dheepan Raj R	NA977	Vik- Sandwik Designs India Pvt Ltd	10-05-2020 (By email)
20	Dipanjan Nath	NA978	Expert United Marine Services LLC, Dubai	EUMS/EO/201981244
21	Glen Aloysius Fernandes	NA980	Larsen and Toubro Pvt Ltd	19-03-2020(By email)
22	Gopala Krishnan	NA981	Albion Engineering India Pvt Ltd	21-10-2020 (By email)
23	Gowrishankar P	NA982	Act Infraport Pvt Ltd, Gujarat	04-1-2020 (By email)
24	Harshil Fernandez	NA983	Unique Solutions, Kochi	08-11-2020 (By email)
25	Himanshu Khutiyare	NA984	Vedam Design & Technical Consultancy Pvt Ltd	V/HR/19-20/O-49
26	Jagadishan B	NA985	Cybermarine Knowledge Systems Pvt Ltd	15-07-2020 (By email)
27	Jeffry Isaacs R S	NA987	Synergy Maritime Private Limited	13-07-2020(By email)
28	Jeremy Joseph	NA988	Vik- Sandwik Designs India Pvt Ltd	10-05-2020 (By email)
29	Jerin Shibu	NA989	Varuna Marine Services	02-22-2020 (By email)
30	Jins Joy	NA990	Expert United Marine Services LLC, Dubai	EUMS/EO/201981246
31	Kavin X	NA991	Act Infraport Pvt Ltd, Gujarat	04-1-2020 (By email)
32	Kishore Kumar B	NA993	Larsen and Toubro Pvt Ltd	19-03-2020(By email)
33	Manish M	NA994	Ambition Marine Services, Chennai	23-04-2020(By email)
34	Mervin Israel G	NA995	Larsen and Toubro Pvt Ltd	19-03-2020(By email)
35	Mohammed Afrin	NA996	Cybermarine Knowledge Systems Pvt Ltd	15-07-2020 (By email)
36	Mohammed Issa Azeez	NA997	Expert United Marine Services LLC, Dubai	EUMS/EO/201981248

37	Muhammed Ajnas K	NA999	Vedam Design & Technical Consultancy Pvt Ltd	V/HR/19-20/O-41
38	Muhammed Nilsham	NA1000	3 - Fish Informatics Pvt Ltd	30-1-2020 (By email)
39	Bargi Pranav Sanjay	NA1004	Vik- Sandwik Designs India Pvt Ltd	10-05-2020(By email)
40	Preet Punit	NA1005	Vedam Design & Technical Consultancy Pvt Ltd	V/HR/19-20/O-51
41	Rajamanickam M	NA1006	Unique Solutions, Kochi	08-11-2020(By email)
42	Rakshat	NA1007	Synergy Maritime Private Limited	13-07-2020 (By email)
43	Ram Kumar R	NA1008	Albion Engineering India Pvt Ltd	21-10-2020(By email)
44	Rohit R	NA1010	Expert United Marine Services LLC, Dubai	EUMS/EO/201981249
45	Sachu Antony	NA1011	Ambition Marine Services, Chennai	23-04-2020(By email)
46	Sathiya Murugan	NA1013	Vedam Design & Technical Consultancy Pvt Ltd	V/HR/19-20/O-47
47	Shijo C	NA1014	Unique Solutions, Kochi	08-11-2020 (By email)
48	Shubham Yadav	NA1015	Synergy Maritime Private Limited	13-07-2020 (By email)
49	Sudar Nilavan S	NA1017	Vik- Sandwik Designs India Pvt Ltd	10-05-2020 (By email)
50	Akhil Shrikant Thik	NA1021	Cybermarine Knowledge Systems Pvt Ltd	15-07-2020 (By email)
51	Veera Ragavan A	NA1024	Unique Solutions, Kochi	08-11-2020(By email)
52	Veerathamilmahan. S	NA1025	Expert United Marine Services LLC, Dubai	EUMS/EO/201981251
53	Venkatesh K	NA1026	Varuna Marine Services	2-11-2020 (By email)
54	Vijay Anandh K	NA1028	Synergy Maritime Private Limited	13-07-2020 (By email)
55	Vijay K	NA1029	Vedam Design & Technical Consultancy Pvt Ltd	V/HR/19-20/O-42
56	Vinesh T	NA1031	3 - Fish Informatics Pvt Ltd	30-1-2020 (By email)
57	Vivek G	NA1032	Varuna Marine Services	2-11-2020 (By email)
58	Vivekananth R	NA1033	Albion Engineering India Pvt Ltd	21-10-2020(By email)
59	Yadav P.P	NA1035	Vedam Design & Technical Consultancy Pvt Ltd	V/HR/19-20/O-48
60	Vinil PS	NA1038	Synergy Maritime Private Limited	13-07-2020 (By email)
61	Puvya Ravi	NA1040	Albion Engineering India Pvt Ltd	21-10-2020(By email)
62	Shruti Aditi Tiwari	NA1041T	Vedam Design & Technical Consultancy Pvt Ltd	V/HR/19-20/O-45
63	Niyamathullah A	NA1044L	3 - Fish Informatics Pvt Ltd	30-1-2020 (By email)
64	Shivam Mehrotra	NA1045L	Vedam Design & Technical Consultancy Pvt Ltd	V/HR/19-20/O-43
65	Gogul Sathyan	NA1047L	Ambition Marine Services, Chennai	23-04-2020

4.5 Professional Activities (20)

Total Marks 20.00

List of Professional Societies/ Chapters

S. No	Professional Societies	Students (Nos.)	Faculty (Nos.)
1	Institution of Naval Architects India	120	5
2	Marine Technology Society (MTS)	10	2
3	International Association of Ports and Harbors (IAPH)	25	4

List of Student Members of Professional Societies

S.NO	Name of the Student	Membership Details
1.	Tumma Uma Venkata Naga Varalakshmi	MTS
2.	Mohammed Arham Sheriff	MTS
3.	Anoop George	MTS
4.	Adhipathi S	MTS
5.	Pravesh Dattaram Dhuri	MTS
6.	Chandana Saran Venkata Amarnath Swamy	MTS
7.	Judith Eromy J	MTS
8.	Maria Infant Vinobel	MTS
9.	Pratyush	MTS
10.	Sanjana J T	MTS
11.	Adarsh Kumar Thakur	INA
12.	Ajeena Steaphen	INA
13.	Amirudeen A	INA
14.	Aswin V	INA
15.	Atul Krishna Ka	INA
16.	Bhupesh S	INA
17.	Daniel Judah C	INA
18.	Deepak S	INA
19.	Dhayanihi Shanmugam	INA
20.	Fidal Sankar Ts	INA
21.	Gopinath V	INA
22.	Griffith Gonsalves	INA
23.	Jayal Joy J	INA
24.	Krithik Kumar R	INA
25.	Masik M	INA
26.	Mejin Melvin	INA
27.	Mothilal Jayanth M	INA

28.	Pranav	INA
29.	Rithik Ashok	INA
30.	Saheer Ali V	INA
31.	Sanket Umesh Patel	INA
32.	Shaurya Dutta Singh	INA
33.	Sibin Edward	INA
34.	Sreethu Ps	INA
35.	Srikaviyadharshini S	INA
36.	Sruthi S	INA
37.	Yokesk	INA
38.	Adarsh Kumar Thakur	INA
39.	Ajeena Steaphen	INA
40.	Amirudeen A	INA
41.	Aswin V	INA
42.	Atul Krishna Ka	INA
43.	Bhupesh S	INA
44.	Daniel Judah C	INA
45.	Deepak S	INA
46.	Dhayanithi Shanmugam	INA
47.	Fidal Sankar Ts	INA
48.	Gopinath V	INA
49.	Bhalekar Athar	INA
50.	Ebenezer Abrah	INA
51.	Mohamed Faizh	INA
52.	Rushil Karthick	INA
53.	Sateesh Kumar	INA
54.	Abisheik J	INA
55.	Anna Biju	INA
56.	Jeena Kristika J	INA
57.	Mabel Mariya H	INA
58.	Mohammed Asik M	INA
59.	Muthurajan S	INA
60.	Narvekar Raviraj	INA
61.	Nataraj K	INA
62.	Navneeth M	INA
63.	Prasana Kumar R	INA
64.	Rohith S	INA

65.	Sabarivasan S	INA
66.	Yedida Venkata Sesha Sai Lavanya	INA
67.	Tumma Uma Venkata Naga Varalakshmi	INA
68.	Kirthika K	INA
69.	Ramprakash V	INA
70.	Adhi Pathi S	INA
71.	Mohammed Arham Shariff	INA
72.	Akash T	INA
73.	Anoop George	INA
74.	Avinash S	INA
75.	Sathish K	INA
76.	Pravesh Dattaram Dhuri	INA
77.	Abhilash S	INA
78.	Tharik S	INA
79.	A.Diksan Anto	INA
80.	Abishek Nanda Kumar	INA
81.	Ajith P	INA
82.	Albinus Savio	INA
83.	Amal Sebastian	INA
84.	Aswin N	INA
85.	Avines S	INA
86.	Bharath Kumar S	INA
87.	Bhasil T	INA
88.	Chandana Saran Venkata Amarnath Swamy	INA
89.	Cyril Tom	INA
90.	Dhanush S	INA
91.	Edwin E	INA
92.	Emil Jude Poul	INA
93.	Fayez Mohammed Khan	INA
94.	Jayasankar Chandraseril Vijayan	INA
95.	Jubin	INA
96.	Judith Eromy J	INA
97.	Krishnaraj T	INA
98.	Kumtakar Yash Sanjay	INA
99.	Manu Khare	INA
100.	Maria Infant Vinobel	INA
101.	Murugesan M	INA

102.	N.M.Mohamed Sulaiman	INA
103.	Nilapu Karan Rao	INA
104.	Nishant	INA
105.	Pratyush	INA
106.	S.Raja Gopal	INA
107.	Sanjana J T	INA
108.	Sanjay S	INA
109.	Sarang Madhusudhanan	INA
110.	Shiva S T	INA
111.	Sneha G	INA
112.	Varun Bharat Singh	INA
113.	Yaswanth Babu	INA
114.	Gupta Sachin Kumar	INA
115.	Sainath Prakash Salgaonkar	INA
116.	Jagadeeshwaran T	INA
117.	Jagadeesvaran S	INA
118.	Vetrivel M R A	INA
119.	Abhishek S	INA

List of Engineering Events

S. No	Academic year	Name of the Professional Society	Title of the Program	Resource Person Details	Event date
2022-2023					
1	2022-2023	MTS	National Seminar on "Advances in Offshore Engineering"	1)Mr. M. Palaniappan, Scientist F, National Institute of Ocean Technology, Chennai 2) Dr. Srinivasan Chandrasekaran, Professor ,Department of Ocean Engg ,IIT ,Chennai	November 30, 2022

2	2022-2023	INA	National Seminar on “Aquatic Foods & Fishing Vessel Technology”	1.Mr.Mahiban, MD, ABJ IMPEX, Chennai, India 2.Dr.N.Manimehalai, CEEE, TNJFU, Nagapattinam, India	28 th , October 2022
3	2022-2023	MTS	Webinar on “Challenges in Dry Docking”	Mr.Vasantha Kumar Rajan (MRINA) Docking & Hasmat Manager 21 Knots Design & Consulting Pvt.Ltd	June 6, 2022
4	2022-2023	MTS	Guest Lecture on “Current Trend in Naval Architecture & Offshore Engineering”	Mr.Sriram.S Naval Architect/Structural Engineer TUFF Offshore Energy & Engineering PVT Ltd	November 19, 2022
2021-2022					
1	2021-2022	MTS	Webinar on “Design and analysis of marine structures”	Mr. Rogero Fernanado CEO, OPTIO Marine Pvt Ltd	February 6 th , 2022
2	2021-2022	MTS	Guest Lecture on “Green ship recycling”	Mr. P.Sriram Krishnan, Recycling supervisor & IHM Surveyor, Sea Sentinels Pvt Ltd, Singapore	February 25 th , 2022
3	2021-2022	INA	National Webinar on “International happiness day”	Dr.S.Asokan, M S (General Surgery), Director CEO, GEM Hospital, Chennai.	March 20, 2022
4	2021-2022	MTS	Guest Lecture on “Detailed ship design “	Mrs. Shiva Shankari & Mr. Tamizharuvi, Maniyan, Cyber Marine, Mumbai	April 24 th , 2022

5	2021-2022	MTS	Webinar on “ Innovation and Intellectual Property Right ”	Dr. Deepa Bopanna, Indian Registered Patent Attorney	June 25 th , 2021
6	2021-2022	INA	International Conference on “ Trends in Marine and Ocean Technology ”	Dr G.A. Ramadass, Director, National Institute of Ocean Technology, Chennai, INDIA	24 -25 th June 2021
7	2021-2022	MTS	National Seminar on “ Coastal Hydrodynamics and Structures ”	1. Dr. Arun V Bapat, Eminent Seismologist, CWPRS, Pune. 2.Dr.S.V.S.Phani Kumar, Scientist-G, NIOT	4 th June 2021
8	2021-2022	INA	International Conference on “ Trends in Marine Science and Technology ”	Dr. K M Sivakolundhu	4-5 th November 2021
2020-2021					
1	2020-2021	INA	Guest Lecture on “ Nondestructive testing methods ”	Mr. Mugunthan Chennai Engineers Academy	May 6, 2021
2	2020-2021	MTS	Guest Lecture on “ Ship General arrangement ”	Mrs. K. Kusuma, Manager, VIKSANDVIK Visakhapatnam	May 7 th 2021
3	2020-2021	MTS	Guest Lecture on “ Design Parameters and Structural Design of Fishing Vessel ”	Mr. M. Jagannatham Raju DGM, VIKSANDVIK Visakhapatnam	May 8 th 2021

4	2020-2021	MTS	Guest lecture on "Role and Relevance of Numerical tools in the Maritime Industry"	Dr. D. Doreen Robin Founder and Director Computational Intelligence Research Foundation	May 24 th 2021
5	2020-2021	MTS	Guest Lecture on "Ship structural inspection and dry dock supervision"	Mr. Naveen Shanmugan Assistant Technical Superintendent LDPL Middle East Shipping ,Dubai	May 29 th , 2021
6	2020-21	INA	Guest Lecture on "Project Management using Primavera"	Mr.Vivek Infinity PMC Solution Pvt Ltd	June 13 th , 2021

4.4.2 Publication of technical magazines, newsletters, etc. (5)

Institute Marks : 5.00

List of Newsletters Published

S. No	Name of the Newsletter	Editorial Members	Periodicity	Year of publishing
1	Navi-Technics	<u>Faculty Members</u> Dr. P Sivabalan Prof .MSP. Raju Mr. R. Jaya Surya <u>Student Members</u> V. Kishore R. Jagadeeswaran Sanjana J T	Biannual	Dec 2022
2	Navi-Technics	<u>Faculty Members</u> Dr. P Sivabalan Prof. MSP. Raju Mr. R. Jaya Surya <u>Student Members</u> V. Kishore Yogi P R. Ragesh	Biannual	June 2022
3	Navi-Technics	<u>Faculty Members</u> Dr. P Sivabalan Prof. MSP.Raju Mr. R.Jaya Surya <u>Student Members</u> Rishab Bantiya R. Jagadeeswaran A K Vishnunath	Biannual	Dec 2021
4	Navi-Technics	<u>Faculty Members</u> Prof. MSP.Raju Dr P Sivabalan Mr.R.Jaya Surya <u>Student Members</u> V.Kishore Dhanushma Aman Tripathi	Biannual	June 2021

5	Navi-Technics	<u>Faculty Members</u> Mr.MSP.Raju Dr.P.Sivabalan <u>Student Members</u> V.Rohan Kumar Abhinav S Ramesh P.Balaji	Biannual	Dec 2020
6	Navi-Technics	<u>Faculty Members</u> Mr.MSP.Raju Dr.P.Sivabalan <u>Student Members</u> Sivaranjani A Abraham Noel P.Balaji	Biannual	June 2020

List of Magazines Published

S. No	Name of the Newsletter	Editorial Members	Periodicity	Year of publishing
1	KNOTS	<u>Faculty Members</u> Prof. MSP Raju Dr.P.Sivabalan Dr. M. Subha, Director, Centre for Internal Publications Dr. R. Lilly	Biannual (Even Sem)	2022
2	KNOTS	<u>Faculty Members</u> Prof. MSP Raju Dr.P.Sivabalan Dr. M. Subha, Director, Centre for Internal Publications Dr. R. Lilly	Biannual (Odd Sem)	2022
3	KNOTS	<u>Faculty Members</u> <u>Prof. MSP Raju</u> Dr.P.Sivabalan Dr. M. Subha, Director, Centre for Internal Publications Dr. R. Lilly	Biannual (Even Sem)	2021

4	KNOTS	<u>Faculty Members</u> Prof. MSP Raju Dr.P.Sivabalan Dr. M. Subha, Director, Centre for Internal Publications Dr. R. Lilly	Biannual (Odd Sem)	2021
5	KNOTS	<u>Faculty Members</u> Prof. MSP Raju Dr.P.Sivabalan Dr. M. Subha, Director, Centre for Internal Publications	Biannual (Even Sem)	2020
6	KNOTS	<u>Faculty Members</u> Prof. MSP Raju Dr.P.Sivabalan Dr. M. Subha, Director, Centre for Internal Publications	Biannual (Odd Sem)	2020

List of Conference proceedings published by the Department

S. No	Name of the Conference Proceeding	Editorial Members	ISBN Number	Year of publishing
1	ICSOE 2021 (International Conference on Shipbuilding and Offshore Engineering)	Prof.MSP Raju, Dr. B.M. Shameem, Dr.P. Sivabalan, and Mr. K. Sujit Kumar	978-93-85434-91-4	24 – 25 th July 2021
2	AFT 2020 (National Conference on Research and Innovation in Aquatic foods and Fishing Vessel Technology)	Prof M S P Raju, Mrs. K Sasi Praba, and Dr. N Manimehalai	978-93-85434-78-5	6 th March 2020
3	MARINCO 2019 (Maritime International Conference)	Mr. S.V. Durga Prasad, Dr. B.M. Shameem and Mr. Akshar Patel	978-93-85434-76-1	4-5 th November 2019

4.4.3 Participation in inter-institute events by students of the program of study (10)

Institute Marks : 10.00

List of Participants in Symposium, Workshops and Engineering Events

S.No	Name of the Student	Name of the Event	Place	Month & Year	Prize
2022-23					
1	Yedida Venkata Sesha Sai Lavanya	Smart India Hackthon (SIH) 2022	Chandigarh Group of Colleges, Chandigarh	August 25-26, 2022	Participated
2	Anoop George	Smart India Hackthon (SIH) 2022	Chandigarh Group of Colleges, Chandigarh	August 25-26, 2022	Participated
3	Sanjana J T	Smart India Hackthon (SIH) 2022	Chandigarh Group of Colleges, Chandigarh	August 25-26, 2022	Participated
4	Mohammed Arham Shariff	Smart India Hackthon (SIH) 2022	Chandigarh Group of Colleges, Chandigarh	August 25-26, 2022	Participated
5	Chandana Saran Venkata Amarnath Swamy	Smart India Hackthon (SIH) 2022	Chandigarh Group of Colleges, Chandigarh	August 25-26, 2022	Participated
6	Muthurajan S	SAMRT AGROTHON 2022	Karpagam College of Engineering, Coimbatore	23 November 2022	1 st Prize
7	Sabarivasan	Archeoworld 2022	Asian School of Architecture and Design Innovations (ASADI), Kochi	27 November , 2022	Participated -
8	Maria Infant Vinobel	TEXARITA' 2K23	Bannari Amman Institute of Technology, Erode	09 th January, 2023	3 rd Prize
9	Krithika K	EMINENCE 2K23 ver. 10.0	St. Marys College Manarcadu, Kottayam	January 13, 2023	Participated
10	Judith Eromy	One Day Hands-on Training on Smart automation using IoT 2023	Sengunthar Engineering College, Tiruchengode	25 th January, 2023	Participated
11	Ajeena Stephen	ELCOWARZ '2K23	PSR Engineering College, Sivakasi	27 January 2023	Participated
12	Nilapu Karan Rao	QUINTESENCE'2023	SRM Eswari Engineering College	28 th January , 2023	Participated
13	Rajagopal	TECHNOTRONZ' 2023	PSG College of Technology, Coimbatore	Feb 1 st , 2023	Participated
14	Mohammed Sulaiman	TECHNOTRONZ' 2023	PSG College of Technology, Coimbatore	Feb 1 st , 2023	Participated
15	Abishek S	TECHNOTRONZ' 2023	PSG College of Technology, Coimbatore	Feb 1 st , 2023	Participated
16	Jagadeeshwaran T	TECHNOTRONZ' 2023	PSG College of Technology, Coimbatore	Feb 1 st , 2023	Participated
17	Viswanathan S	Global Science for Global Wellbeing	AMET University	February 2023	2 nd Prize
18	Swaved Walimbe	Global Science for Global Wellbeing	AMET University	February 2023	2 nd Prize
19	Ariba Ahmed	Global Science for Global Wellbeing	AMET University	February 2023	2 nd Prize
20	Chirag Muralidharan	Global Science for Global Wellbeing	AMET University	February 2023	2 nd Prize

21	Abishek Nanda Kumar	International Conference on Sustainable Development, Innovation and Information Exchange	Savitha Institutions	16 th - 18 th December 2022	Participated
22	Chandana Saran	Role of Engineers for Creating a Sustainable& Self Reliant, India	The Institution of Engineers (India)	16 th - 18 th December 2022	Participated
23	Dhanush S	International Conference on Electrical engineering and Multidisciplinary Research [ICOFT 2021]	Chennai Institute of Technology ,Kundrathur ,Chennai India	23 rd -24 th June 2022	Participated
24	Edwin E	International Conference on Sustainable Development,Innovation and Information Exchange	Savitha Institutions	16 th -18 th December 2022	Participated
25	Kumtakar yash	International Conference on Sustainable Development, Innovation and Information Exchange	Savitha Institutions	16 th -18 th December 2022	Participated
26	Pratyush	International Conference on Electrical engineering and Multidisciplinary Research [ICOFT 2021]	Chennai Institute of Technology ,Kundrathur ,Chennai India	23 rd - 24 th June 2022	Participated
27	Nilapu karan rao	International Conference on Sustainable Development ,Innovation and Information Exchange	Savitha Institutions	16 th - 18 th December 2022	Participated
28	Yaswanth babu K	Role of Engineers for Creating a Sustainable& Self Reliant ,India	The Institution of Engineers (India)	16 th -18 th December 2022	Participated
29	Jagadeeshvaran S	Role of Engineers for Creating a Sustainable& Self Reliant ,India	The Institution of Engineers (India)	16 th - 18 th December 2022	Participated
30	Sanjana J T	International Conference on Electrical engineering and Multidisciplinary Research [ICOFT 2021]	Chennai Institute of Technology ,Kundrathur ,Chennai India	23 rd - 24 th June 2022	Participated
31	Judith eromy J	International Conference on Electrical engineering and Multidisciplinary Research [ICOFT 2021]	Chennai Institute of Technology ,Kundrathur ,Chennai India	23 rd - 24 th June 2022	Participated
2021-22					
1.	Yedida Venkata Sesha Sai Lavanya	International Yoga Day	AMET	21 June 2021	Participated
2.	Sankar Lingam	Virtual First International Conference on Social Work, Science and Technology ICSST 2021	SDNB Vaishnav College for Women	29 th July 2021	Participated
3.	Sathish P	Advanced Research in Mechanical Sciences 2021	SRM Institute of Science and Technology Vadapalani Campus, Chennai	6 th August 2021	Participated

4.	Chandana Saran Venkata Amarnath Swamy	Study Tour To Atal Tunnel Yuvak Scheme	Kulu Manali, Himachal Pradesh	15- 20th November, 2021	Participated
5.	Yogi P	Study Tour To Atal Tunnel Yuvak Scheme	Kulu Manali, Himachal Pradesh	15- 20th November, 2021	Participated
6.	Chandana Saran Venkata Amarnath Swamy	Anveshan (National Student Research Convention)	AMET	27-28 March 2022	Participated
7.	Sanjana JT	Anveshan (National Student Research Convention)	AMET	27-28 March 2022	Participated
8.	Sanjay S	Virtual Webinar on Crypto Currency 2022	KV Institute of Management and Information Studies, Coimbatore	8 th April 2022	Participated
9.	Rishab Bantiya S	One day Workshop on Non Destructive Testing and its Techniques 2022	AVS College of Technology, Salem	12 th April 2022	Participated
10.	Regis Benny Arockiam	KRATORQ' 22	SRM Institute of Science and Technology Vadapalani Campus, Chennai	13 th April 2022	2 nd Prize
11.	Ragesh R	Webinar on How to Plan for Start-up – Legal and Ethical Steps 2022	Vel Tech Rangarajan Dr Sagunthala R and D Institute of Science and Technology, Chennai	23 rd April 2022	Participated
12.	Jegatheeswaran R	Mariners aboard 2k22	AMET University	26 th April 2022	Participated
13.	Gottam Karthik Reddy	Mariners aboard 2k22	AMET University	26 th April 2022	Participated
14.	Baskar Antony Petlees A	Mariners aboard 2k22	AMET University	26 th April 2022	3 rd Prize
15.	Bhuvaneswaran S B	Mariners aboard 2k22	AMET University	26 th April 2022	Participated
16.	Babitha A	International Virtual Conference on "New Technologies for Greener Shipping"	AMET University	May 2022	Participated
17.	Christinal B	International Virtual Conference on "New Technologies for Greener Shipping"	AMET University	May 2022	Participated
18.	Dhanushma	International Virtual Conference on "New Technologies for Greener Shipping"	AMET University	May 2022	Participated

19.	Ashwin S Nair	International Virtual Conference on "New Technologies for Greener Shipping"	AMET University	May 2022	Participated
20.	Sneha G	Technite' 2022	St. Mother Theresa Engineering College, Tuticorin	3rd June 2022	Participated
21.	Shiva S T	National Conference on Recent Trends in Engineering, Science and Management 2022	Kurinchi College of Engineering, Manapparai	9th June 2022	Participated
22.	Muthurajan S	Workshop and Summer internship on Virtual Reality Development in Python 2022	SRM Institute of Science & Technology, Chennai	18th June 2022	Participated
23.	Aby Joseph	International conference on ship building and Offshore Engineering	AMET University	June 24th - 25th ,2021	Participated
24.	Akkash S	International Conference on Future Technologies in Manufacturing, Automation, Design and Energy[ICOFT 2021]	National Institute of Technology ,Puducherry, India	16 th -18 th December 2021.	Participated
25.	Balaji P	Challenges and Oppurtunities in Renewable Energy, Smart Systems and E-Mobility[ICCORSE 2020]	Easwari Engineering College, Ramapuram, Chennai	7 th - 8 st October 2021.	Participated
26.	Nithish Balaji	International conference on ship building and Offshore Engineering	AMET University	June 24th -25th ,2021	Participated
27.	Rohan kumar V	Challenges and Oppurtunities in Renewable Energy, Smart Systems and E-Mobility[ICCORSE 2020]	Easwari Engineering College,Ramapuram,Chennai	7 th - 8 st October 2021.	Participated
28.	Sreerag A C	Challenges and Oppurtunities in Renewable Energy, Smart Systems and E-Mobility[ICCORSE 2020]	Easwari Engineering College,Ramapuram,Chennai	7 th - 8 st October 2021.	Participated
29.	Vivek kumar chauhan	Challenges and Oppurtunities in Renewable Energy, Smart Systems and E-Mobility[ICCORSE 2020]	Easwari Engineering College, Ramapuram, Chennai	7 th -8 st October 2021.	Participated
30.	Aryan Bharmani Ayush Dave Nishil Gandhi Aadil Khan	International conference on ship building and Offshore Engineering	AMET University	June 24th - 25th ,2021	Participated
31.	Vikram.V	Challenges and Oppurtunities in Renewable Energy, Smart Systems and E-Mobility[ICCORSE 2020]	Easwari Engineering College, Ramapuram, Chennai	7 th - 8 st October 2021.	Participated

32.	Sri hari ram D	International Conference on Future Technologies in Manufacturing, Automation, Design and Energy[ICOFT 2021]	National Institute of Technology ,Puducherry, India	16 th - 18 th December 2021.	Participated
33.	Vignesh.B	International conference on ship building and Offshore Engineering	AMET University	June 24 th - 25 th ,2021	Participated
34.	Vivek R	2021 IEEE Conference	IEEE Conference on Norbert Wiener in the 21st Century (21CW)	22 nd - 25 th July 2021.	Participated
35.	Asutosh padhy	2021 IEEE Conference	IEEE Conference on Norbert Wiener in the 21st Century (21CW)	22 nd - 25 th July 2021.	Participated
36.	Dhanushma	International Conference on Future Technologies in Manufacturing, Automation, Design and Energy[ICOFT 2021]	National Institute of Technology ,Puducherry, India	16 th - 18 th December 2021.	Participated
2020-21					
1.	K Shreyanka	Zenista TechFest 2020	Sri Sairam Engineering College, Chennai	October 8, 2020	3 rd Prize
2.	Ganesh T	PRAESTANTIA 2020	Sri Sairam Institute of Technology, Chennai	October 14, 2020	Participated
3.	Ardra MP	A COURSE ON CYBER SAFETY and SECURITY FOR GIRLS & WOMEN 2021	Coimbatore Institute of Technology, Coimbatore	January 2, 2021	Participated
4.	Vasimkhan B	SKILLUP: Gamers Connect 2021	VIT Bhopal	January 9, 2021	Participated
5.	Nithish Balaji J	SHOUT'2021	Jai Hind College, Mumbai	January 21, 2021	Participated
6.	Aadil khan	Third International Conference In Shipbuilding & Offshore Engineering 2021(Icsoe 2021)	AMET	June 24-25, 2021	Participated
7.	Prithiman V	Rolling Reels Film Festival 2021	NIT Tiruchirappalli	January 23, 2021	Participated
8.	Balachandar B	MESCIA 2k21	Kongu Engineering College Erode	March 27, 2021	2 nd Prize
9.	A Sivaranjani	ELIXIR 21	Karpaga Vinayaga College of Engineering and Technology, Kanchipuram	March 30, 2021	1 st Prize
10.	Aby Joseph	Third International Conference In Shipbuilding & Offshore Engineering 2021(Icsoe 2021)	AMET	June 24-25, 2021	Participated
11.	Nithish Balaji J	Third International Conference In Shipbuilding & Offshore Engineering 2021(Icsoe 2021)	AMET	June 24-25, 2021	Participated
12.	Aryan Bharmani	Third International Conference In Shipbuilding & Offshore Engineering 2021(Icsoe 2021)	AMET	June 24-25, 2021	Participated

13.	Ayush Dave	Third International Conference In Shipbuilding & Offshore Engineering 2021(Icsoe 2021)	AMET	June 24-25, 2021	Participated
14.	Nishil Gandhi	Third International Conference In Shipbuilding & Offshore Engineering 2021(Icsoe 2021)	AMET	June 24-25, 2021	Participated
15.	Aadil khan	Third International Conference In Shipbuilding & Offshore Engineering 2021(Icsoe 2021)	AMET	June 24-25, 2021	Participated
16.	B. Vignesh	Third International Conference In Shipbuilding & Offshore Engineering 2021(Icsoe 2021)	AMET	June 24-25, 2021	Participated
17.	Ardra M P	Emerging Technology on Energy System[NCETES 2020]	Prasad V. Potluri Siddhartha Institute of Technology, Kanuru, Vijayawada - 520 007	11 th September 2020.	Participated
18.	Indumathi S	Machining Behavior of Fly ash composites	The Quaide Milleth College for Men, Medavakkam, Chennai, Tamil Nadu, India	4 th - 5 th July 2020.	Participated
19.	Deebak K	Machining Behavior of Fly ash composites	The Quaide Milleth College for Men, Medavakkam, Chennai, Tamil Nadu, India	4 th - 5 th July 2020.	Participated
20.	Sharon shaju	Machining Behavior of Fly ash composites	The Quaide Milleth College for Men, Medavakkam, Chennai, Tamil Nadu, India	4 th - 5 th July 2020.	Participated
21.	Prithiv raj	Challenges and Opportunities in Renewable Energy, Smart Systems and E-Mobility[ICCORSE 2020]	Easwari Engineering College, Ramapuram, Chennai	30 th - 31 st July 2020.	Participated
22.	Samuel	Challenges and Opportunities in Renewable Energy, Smart Systems and E-Mobility[ICCORSE 2020]	Easwari Engineering College, Ramapuram, Chennai	30 th - 31 st July 2020.	Participated
23.	Venu Aravindh	International Conference on Sustainable Innovation in Science and Technology[ICSIST 2021]	G.H.Raisoni University, Sai kheda	26 th - 27 st February 2021	Participated
24.	S.vikramselvagaesh	International Conference on Sustainable Innovation in Science and Technology[ICSIST 2021]	G.H.Raisoni University, Sai kheda	26 th - 27 st February 2021	Participated
25.	Ujjwal raj	International Conference on Sustainable Innovation in Science and Technology[ICSIST 2021]	G.H.Raisoni University, Sai kheda	26 th - 27 st February 2021	Participated
26.	Balachandar B	International e-Conference on Recent Trends and Technologies in Soft Computing[ICRTSC 20]	St.Peters Institute of Higher Education and Research, Avadi, Chennai	28 th - 29 th September 2020	Participated
27.	Gowtham satya kumar	International e-Conference on Recent Trends and Technologies in Soft Computing[ICRTSC 20]	St.Peters Institute of Higher Education and Research, Avadi, Chennai	28 th - 29 th September 2020	Participated
28.	Mohamed Nooh	International Conference on Sustainable Innovation in Science and Technology[ICSIST 2021]	G.H.Raisoni University, Sai kheda	26 th - 27 st February 2021	Participated

29.	Mohammed Binyamine	International Conference on Sustainable Innovation in Science and Technology[ICSIST 2021]	G.H.Raisoni University,Sai kheda	26 th - 27 st February 2021	Participated
-----	--------------------	--	----------------------------------	--	--------------

5 FACULTY INFORMATION AND CONTRIBUTIONS (200)

Total Marks 198.00

Sr. No	Name	PAN No.	University Degree	Date of Receiving Degree	Area of Specialization	Research Paper Publications	Ph.D Guidance	Faculty receiving Ph.D during the assessment year	Current Designation	Date (Designated as Prof / Assoc. Prof.)	Initial Date of Joining	Association Type	At present working with the Institution (Yes / No)	Date of Leaving	IS HOD?
1	Dr.S.K.Bhattacharya	AADPB4666L	MS and PhD	01/07/1989	Ship & Offshore Structures	6	6	0	Professor		25/07/2022	Regular	Yes		Yes
2	Mr.M.Surya Prakasa Raju	ADUPM7258B	M.E/M.Tech	01/07/1998	Ocean Engineering	2	0	0	Assistant Professor		17/07/2017	Regular	Yes		No
3	Dr.K.Thiruvengatasamy	AOIPK0475M	MS and PhD	30/09/1999	Ocean Engineering	0	0	0	Professor		21/04/2009	Regular	Yes		No
4	Dr.A.Adinarayanan	AIAPA0445R	ME/M. Tech and PhD	28/12/2018	Production Engineering	0	0	0	Professor	25/07/2022	24/01/2019	Regular	Yes		No
5	Dr.P.Sivabalan	BWPPS4385E	ME/M. Tech and PhD	20/07/2018	Ship Motion and Control	6	4	0	Associate Professor	15/07/2019	21/01/2019	Regular	Yes		No
6	Dr.R.Lilly	AEQPL2643J	ME/M. Tech and PhD	07/08/2020	Water Resources	10	0	0	Associate Professor	23/08/2021	18/08/2020	Regular	Yes		No
7	Dr.Prasob PA	APDPA7118R	ME/M. Tech and PhD	08/06/2021	Composite Materials	0	0	1	Assistant Professor		01/07/2019	Regular	Yes		No
8	Dr.Suneela J	BBNPJ5237M	ME/M. Tech and PhD	24/09/2021	Naval Architecture	5	0	0	Assistant Professor		08/08/2022	Regular	Yes		No
9	Dr.V.Arularasi	AONPV7336P	ME/M. Tech and PhD	24/11/2022	Advanced Construction Technology	0	0	1	Assistant Professor		25/07/2022	Regular	Yes		No
10	Mr.A.Prem Anadh	BIAPP3891H	MS	20/09/2018	Offshore Structures	4	0	0	Assistant Professor		02/07/2012	Regular	Yes		No
11	Mr.Kolli Sujit Kumar	CYCPK8035K	M.E/M.Tech	08/06/2015	Naval Architecture and Ocean Engineering	0	0	0	Assistant Professor		01/07/2019	Regular	Yes		No
12	Mr.R.Jaya Surya	CAPPJ9653R	M.E/M.Tech	07/06/2019	Naval Architecture and Ocean Engineering	1	0	0	Assistant Professor		26/02/2020	Regular	Yes		No
13	Mr.S.Thalapathy Vigneshwar	FATPS5715N	M.E/M.Tech	19/01/2016	Naval Architecture and Ocean Engineering	0	0	0	Assistant Professor		17/08/2022	Regular	Yes		No
14	Dr.G.Vennila	BJVPV7104F	ME/M. Tech and PhD	12/10/2022	Information Technology	0	0	0	Assistant Professor		02/09/2009	Regular	Yes		No
15	Dr.B.M.Shameem	FPMP0005F	ME/M. Tech and PhD	24/07/2015	Ship Hydrodynamics	2	0	0	Associate Professor	15/07/2019	13/07/2016	Regular	No	30/07/2022	No
16	Mr.Kari Jagadish	CMBPK6312A	M.E/M.Tech	17/12/2018	Naval Architecture	0	0	0	Assistant Professor		01/07/2019	Regular	No	30/06/2022	No

17	Mr.Srivathsan B	CQGPS7456C	M.E/M.Tech	08/06/2015	Naval Architecture and Ocean Engineering	0	0	0	Assistant Professor		10/07/2019	Regular	No	30/06/2022	No
18	Ms.M.Geethapriya	BOJPG3900H	M.E/M.Tech	10/06/2019	Naval Architecture and Ocean Engineering	0	0	0	Assistant Professor		22/07/2019	Regular	No	30/06/2022	No
19	Dr.J.Rajaraman	ATXPR7782A	MS and PhD	03/08/2015	Ocean Engineering	0	0	0	Professor		01/10/2010	Regular	No	30/06/2022	No
20	Mr.S.Ramaswami	ADEPR3779G	M.E/M.Tech	08/08/2014	Computer Integrated Manufacturing	0	0	0	Assistant Professor		02/08/2007	Regular	Yes		No
21	Cdr Prashant Kumar (Retd)	ACTPK2572Q	M.E/M.Tech	18/08/1989	Naval Construction	0	0	0	Assistant Professor		10/09/2014	Regular	No	30/06/2022	No
22	Mr.Patel Akshar Kumar Pankajbhai	BFP3924P	MS	22/07/2016	Ship Design	0	0	0	Assistant Professor		03/07/2014	Regular	No	04/05/2021	No
23	Dr.Subir Kumar Satsangi	AKWPS9919C	ME/M. Tech and PhD	09/03/1987	Naval Architecture	0	0	0	Professor		01/07/2017	Regular	No	04/05/2020	No
24	Mr.Himanshu Uppal	ACEPU2293F	M.E/M.Tech	08/06/2015	Naval Architecture and Ocean Engineering	0	0	0	Assistant Professor		01/07/2015	Regular	No	04/05/2020	No
25	Mr.Vijith P P	AJYPV6646C	M.E/M.Tech	19/07/2013	Ocean Engineering	0	0	0	Assistant Professor		23/02/2015	Regular	No	04/05/2020	No
26	Mr.Ranjeet Kumar Verma	AOEPV8922Q	MS	22/07/2016	Structural Vibrations	0	0	0	Assistant Professor		22/10/2015	Regular	No	04/05/2020	No
27	Mr.Vinod Vincent	BKYPV6033K	M.E/M.Tech	13/06/2016	Naval Architecture and Ocean Engineering	0	0	0	Assistant Professor		09/01/2017	Regular	No	04/05/2020	No
28	Dr. A. Rajesh Kanna	ARMPR9693K	ME/M. Tech and PhD	22/06/2016	Petroleum and Offshore Engineering	7	0	0	Associate Professor	02/01/2019	17/07/2017	Regular	Yes		No
29	Dr. Ponmani Swaminathan	GPWPS0868R	ME/M. Tech and PhD	23/07/2015	Petroleum and Offshore Engineering	2	1	0	Assistant Professor		16/07/2018	Regular	Yes		No

5.1 Student-Faculty Ratio (SFR) (20)

Total Marks 20.00

UG

No. of UG Programs in the Department 1

B.E - Naval Architecture and Offshore Engineering						
Year of Study	CAY		CAYm1		CAYm2	
	(2022-23)		(2021-22)		(2020-21)	
	Sanction Intake	Actual admitted through lateral entry students	Sanction Intake	Actual admitted through lateral entry students	Sanction Intake	Actual admitted through lateral entry students
2nd Year	30	5	54	14	60	7
3rd Year	54	14	60	7	120	2
4th Year	60	7	120	2	65	9
Sub-Total	144	26	234	23	245	18
Total	170		257		263	
Grand Total	170		257		263	

PG

No. of PG Programs in the Department 1

M.E - Naval Architecture and Offshore Engineering			
Year of Study	CAY(2022-23)	CAYm1(2021-22)	CAYm2 (2020-21)
	Sanction Intake	Sanction Intake	Sanction Intake
1st Year	16	16	16
2nd Year	16	16	18
Total	32	32	34
Grand Total	32	32	34

SFR

No. of UG Programs in the Department 1

No. of PG Programs in the Department 1

Description	CAY(2022-23)	CAYm1 (2021-22)	CAYm2 (2020-21)
Total No. of Students in the Department(S)	202 Sum total of all (UG+PG) students	289 Sum total of all (UG+PG) students	297 Sum total of all (UG+PG) students
No. of Faculty in the Department(F)	17 F1	19 F2	20 F3
Student Faculty Ratio(SFR)	11.88 SFR1=S1/F1	14.85 SFR2=S2/F2	15.21 SFR3=S3/F3
Average SFR	13.98 SFR=(SFR1+SFR2+SFR3)/3		
F=Total Number of Faculty Members in the Department (excluding first year faculty)			

Note: All the faculty whether regular or contractual (except Part-Time), will be considered. The contractual faculty (doing away with the terminology of visiting/adjunct faculty, whatsoever) who have taught for 2 consecutive semesters in the corresponding academic year on full time basis shall be considered for the purpose of calculation in the Faculty Student Ratio. However, following will be ensured in case of contractual faculty:

1. Shall have the AICTE prescribed qualifications and experience.
2. Shall be appointed on full time basis and worked for consecutive two semesters during the particular academic year under consideration.
3. Should have gone through an appropriate process of selection and the records of the same shall be made available to the visiting team during NBA visit

5.1.1. Provide the information about the regular and contractual faculty as per the format mentioned below:

	Total number of regular faculty in the department	Total number of contractual faculty in the department
CAY(2022-23)	17	0
CAYm1(2021-22)	19	0
CAYm2(2020-21)	20	0

Average SFR for three assessment years : 13.98

Assessment SFR : 20

5.2 Faculty Cadre Proportion (20)

Total Marks 20.00

Institute Marks : 20.00

Year	Professors		Associate Professors		Assistant Professors	
	Required F1	Available	Required F2	Available	Required F3	Available
CAY(2022-23)	1.00	3.00	2.00	3.00	6.00	11.00
CAYm1(2021-22)	1.00	2.00	3.00	5.00	9.00	12.00
CAYm2(2020-21)	1.00	2.00	3.00	4.00	9.00	14.00
Average Numbers	1.00	2.33	2.67	4.00	8.00	12.33

Cadre Ratio Marks $[(AF1 / RF1) + [(AF2 / RF2) * 0.6] + [(AF3 / RF3) * 0.4]] * 10$: 20.00

5.3 Faculty Qualification (20)

Total Marks 20.00

	X	Y	F	$FQ = 2 \times [(10X + 4Y) / F]$
2022-23(CAY)	11	6	10.00	26.80
2021-22(CAYm1)	9	10	14.00	18.57
2020-21(CAYm2)	8	12	14.00	18.29

Average Assessment : 21.22

5.4 Faculty Retention (10)

Total Marks 8.00

Institute Marks : 8.00

Description	2021-22 (CAYm1)	2022-23 (CAY)
No of Faculty Retained	19	13
Total No of Faculty	20	20
% of Faculty Retained	95	65

Average : 80.00

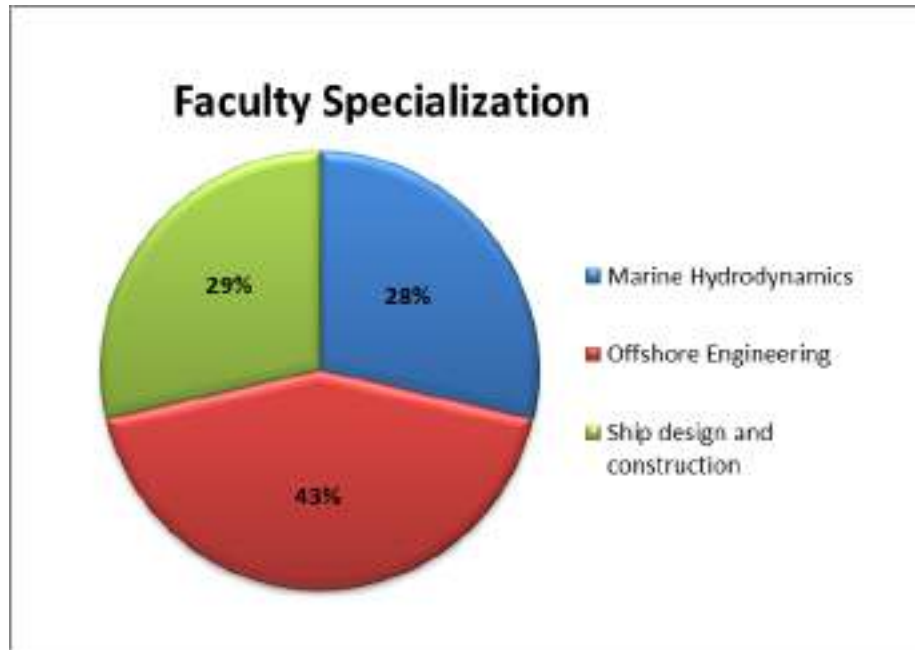
Assessment Marks : 8.00

5.5 Faculty competencies in correlation to Program Specific Criteria (10)

Total Marks 10.00

Program specific criteria research groups:

- a. Marine Hydrodynamics
- b. Ship Design and Construction
- c. Offshore Engineering

Figure: Program specific criteria research groups**Table 5.5 Faculty Competencies to Program Specific Criteria**

S.No	Name of the Faculty	Engineering (Specialization)	Areas of Research Expertise	No of Publications in refereed Journals	Number of Books & Book Chapters Published	Relevant Courses in Curriculum
1.	Dr. S. K. Bhattacharyya	Marine Hydrodynamics/ Offshore Engineering	Ship & Offshore Structures	80	1	Wave Mechanics Introduction to Naval Architecture Marine Hydrodynamics Strength of Ships
2.	Dr. K. Thiruvengadasamy	Marine Hydrodynamics/ Offshore Engineering	Wave hydrodynamics	25	1	Dynamics of Offshore Structures Advanced Offshore Engineering Wave Mechanics Offshore Structure Design - Software Laboratory Elements of Offshore Engineering Elements of Coastal Engineering

3.	Mr. M Surya Prakash Raju	Ship Design and Construction	Marine Vehicle Hydrodynamics	10	3	Ship Resistance and Propulsion High Performance Marine Vehicles Ship Design Fishing Vessel Technology Statutory Regulations and Classification Rules
4.	Dr. P. Sivabalan	Marine Hydrodynamics	Ship Motion Control	9	3	Introduction to Naval Architecture Marine Materials and Welding Technology Ship Drawing - Lines Plan Elements of Offshore Engineering Theory of Ships Hydrostatics & Stability Laboratory Ship System Drawing & Launching Calculations Laboratory Wave Mechanics Computer Aided Structural Design (FEA) Basic Principles of Marine Vehicle Design Elements of Marine Structure Marine Design
5.	Dr. Prasob PA	Offshore Engineering	Composite Materials	6	-	Marine Materials and Welding Technology Thermodynamics & Marine Machinery Marine Production Technology Strength of Ships Ship Strength Laboratory Ship Systems Engineering Strength of Ocean Structures
6.	Dr. R. Lilly	Offshore Engineering	Ground Water	16	4	Strength of Materials Engineering Mechanics Elements of Offshore Engineering Ethics and Values Inland Water Transportation Computer Aided Structural Design (FEA)
7.	Mr. A. PremAnandh	Ship Design and Construction	Naval Architecture	8	3	Quality, Health, Safety and Environmental Management Shipyard Practices and Project Management Surface Modelling and Analysis - Software Laboratory
8.	Mr. Kolli Sujit Kumar	Offshore Engineering	Naval Architecture	3	-	Advanced Ship Technology Design of Offshore Structures Hydrostatics & Stability Laboratory Ship Drawing - Lines Plan

9.	Mr. R. Jaya Surya	Ship Design and Construction	Offshore Structures, CFD	-	-	Lifting Surfaces for Marine Applications Surface Modelling and Analysis - Software Laboratory Structural Modelling - Software Laboratory Design of Offshore Structures
10.	Mr. Thalapati Vigneshwar	Ship Design and Construction	Naval Architecture	-	-	Marine Materials and Welding Technology Ship System Drawing & Launching Calculations Laboratory Introduction to Underwater Technology
11.	Mrs. D. Venila	IT	IT	15	-	Data Science Object Oriented Programming Object Oriented Programming Artificial Intelligence
12.	Dr. A. Adinarayanan	Materials	Mechanical	9	-	Strength of Materials Engineering Mechanics Strength of Materials Lab CAD/CAM in ship building Marine CAD & CAE Lab
13.	Dr. Suneela J	Marine Hydrodynamics	Ship hydrodynamics	6	1	Ship Drawing - Lines Plan Ship Motion and Control Numerical Ship Hydrodynamics – Software Laboratory Theory and Practice in Marine CFD Guidance and Control of Marine Vehicles Applied Hydrodynamics
14.	Dr. V. Arularasi	Offshore Engineering	Advanced Construction Techniques	7	2	Elements of Offshore Engineering Strength of Materials Strength of Materials Laboratory Ocean Energy
15.	Dr. Ponmani Swaminathan	Offshore Engineering	Offshore Drilling Engineering	15	1	Elements of Offshore Engineering
16.	Dr. A. Rajesh Kanna	Offshore Engineering	Offshore Engineering- Oil recovery	36	1	Mechanics of Materials, Engineering Fluid Mechanics
17.	Mr. S. Ramaswami	Ship Design and Construction	Manufacturing	3	-	Introduction to Naval Architecture Thermodynamics & Marine Machinery Marine Production Technology CAD/CAM in ship building
18.	Mr. Kari Jagadish	Ship Design and Construction	Naval Architecture	-	-	Lifting Surfaces for Marine Applications Inland Water Transportation High Performance Marine Vehicles

19.	Mr. Srivathsan B	Marine Hydrodynamics	Naval Architecture	3	-	Design of Offshore Structures Elements of Offshore Engineering Surface Modelling and Analysis - Software Laboratory
20.	Ms. M. Geethapriya	Offshore Engineering	Naval Architecture	-	-	Lifting Surfaces for Marine Applications Shipyard Practices and Project Management Marine Pollution Regulations
21.	Dr. J. Rajaraman	Offshore Engineering	Wave hydrodynamics, Ocean Engineering	10	-	Design of Offshore Structures Wave Mechanics Fluid Structure Interaction Elements of Coastal Engineering
22.	Cdr. Prashant Kumar (Retd)	Marine Hydrodynamics	Naval Architecture	3	-	Statutory Regulations and Classification Rules Ship Systems Engineering Marine Corrosion and Control
23.	Dr. B. M. Shameem	Marine Hydrodynamics	Ship Hydrodynamics	8	2	Theory and Practice in Marine CFD Guidance and Control of Marine Vehicles Numerical Ship Hydrodynamics – Software Laboratory Marine Hydrodynamics Laboratory Ship Motion and Control Marine Hydrodynamics Applied Hydrodynamics Computational Hydrodynamics Lab

Course Development

The following faculty members have involved in the course content development.

Table 5.5.1 Course Development

S. No	Name of the faculty	Name of the Course
1	Dr. S. K. Bhattacharya	Strength of Ships
2	Dr K.Thiruvengatasamy	Wave Mechanics
3	Mr M Surya Prakash Raju	Ship Resistance and Propulsion
4	Dr.P.Sivabalan	Introduction to naval architecture
5	Dr. A. Rajesh Kanna	Engineering fluid mechanics
6	Dr Prasob PA	Lifting Surfaces for marine Application
7	Dr.R.Lilly	Theory and Practice in Marine CFD
8	Mr A.PremAnandh	Shipyard Practices and Project Management
9	MrKolliSujit Kumar	Advanced Ship Technology
10	Mr. R Jaya Surya	Dynamics of Offshore Structures
11	Mr.Thalapathy Vigneshwar	Strength of Ships
12	Dr.G.Venila	Data Science

13	Dr.A. Adinarayanan	Marine Materials and Welding Technology
14	Dr Suneela J	Guidance and Control of Marine Vehicles
15	Dr.Arularasi	Mechanics of materials
16	Mr S Ramaswami	Computer Aided Structural Design
17	Dr. Ponmani Swaminathan	Elements of Offshore Engineering

5.6 Innovations by the Faculty in Teaching and Learning (10)**Total Marks 10.00**

The instructional material developed by the faculty are shown in the table below.

Table 5.6 Instructional Materials Developed by Faculty

S.No	Name of the Faculty	Course Name	Topic	Youtube link of the video	Views
1.	Mr. A. Prem Anandh	Marine Materials and Welding Technology	Welding in Shipbuilding	https://youtu.be/AlmCcYhe5-c?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj (https://youtu.be/AlmCcYhe5-c?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj)	1000
2.	Mr. P. Akshar Patel	Strength of Ships	Strength of ships	https://youtu.be/ruVzFTE9FPk?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj (https://youtu.be/ruVzFTE9FPk?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj)	632
3.	Mr.S.Tamilarasan	Surface Modelling and Analysis - Software Lab	Software lab maxsurf	https://youtu.be/7z8NYJAJh4M?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj&t=2 (https://youtu.be/7z8NYJAJh4M?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj&t=2)	187
4.	Dr.B.M.Shameem	Marine Hydrodynamics	Basic concepts in hydrodynamics	https://youtu.be/jTjyQ1ymvtI?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj (https://youtu.be/jTjyQ1ymvtI?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj)	473
5.	Mr.S.Ramaswami	Theory of Ships	Ship Stability	https://youtu.be/FmAM-QT--Yc?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj (https://youtu.be/FmAM-QT--Yc?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj)	139
6.	Mr.MSP Raju	Ship Resistance and Propulsion	Ship Frictional Resistance	https://youtu.be/vgVXiql1uk0?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj (https://youtu.be/vgVXiql1uk0?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj)	1100
7.	Cdr Prashant Kumar (Retd)	Ship Systems Engineering	Cargo Handling System	https://youtu.be/YB_Aobv2D80?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj (https://youtu.be/YB_Aobv2D80?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj)	324
8.	Mr.Himanshu Uppal	Strength of Ships	Loads on Ship And marine structures	https://youtu.be/XSVlyM-674w?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj (https://youtu.be/XSVlyM-674w?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj)	311
9.	Mr.Kolli sujit Kumar	Theory of ships	Transverse stability of ships	https://youtu.be/rSdRCTQRcQg?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj (https://youtu.be/rSdRCTQRcQg?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj)	81
10.	Mr. Prasob PA	Marine Materials and Welding Technology	Dynamic behaviour of composites	https://youtu.be/V6KfLbiLgV0?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj (https://youtu.be/V6KfLbiLgV0?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj)	45
11.	Dr.P Sivabalan	Ship construction	Ship construction	https://youtu.be/h8aHqKVrPow?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj (https://youtu.be/h8aHqKVrPow?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj)	175
12.	Dr. Ponmani Swaminathan	Offshore Engineering	BASICS OF DRILLING ENGINEERING	https://youtu.be/TSNBnRV6oRY	550
13.	Mr.KJagadish	Wave mechanics	wave mechanics	https://youtu.be/TO3SMH-LV6w?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj (https://youtu.be/TO3SMH-LV6w?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj)	39
14.	Mr. B.Srivathsan	Theory of Ships	Hull form and geometry	https://youtu.be/v2UWCZkf3X8?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj (https://youtu.be/v2UWCZkf3X8?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj)	189
15.	Mr.Gopi krishna	Introduction to naval architecture	Introduction to naval architecture	https://youtu.be/pwvKXBehPRk?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj (https://youtu.be/pwvKXBehPRk?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj)	175
16.	Ms.Geetha priya	Lifting Surfaces for Marine Applications	Lifting Surfaces for Marine Applications	https://youtu.be/CTU8LJDArhU?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj (https://youtu.be/CTU8LJDArhU?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj)	73
17.	Mr.P.Akshar Patel	Theory of Ships	floating body and its stability	https://youtu.be/nO2ePV5_irA?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj (https://youtu.be/nO2ePV5_irA?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj)	54
18.	Mr.Himanshu Uppal	Marine Production Technology	welding parameters	https://youtu.be/DitSGInO42Y?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj (https://youtu.be/DitSGInO42Y?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj)	148
19.	Mr.Saravanakumar	Ship construction	ship construction double bottom structure	https://youtu.be/7US2av8foik?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj (https://youtu.be/7US2av8foik?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj)	1300
20.	Mr.Vinod Vincent	Ship construction	structural application	https://youtu.be/G3Rt3CsaUc0?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj (https://youtu.be/G3Rt3CsaUc0?list=PLE_LzT-IIxNRmQLWie_1Q4_Thmg370Lj)	66

The use of innovative methods in educational institutions has the potential not only to improve the quality of education, but also to empower people and strengthen governance of an institution. In this regard, apart from traditional teaching methods, various modern teaching methods, techniques and tools are used by the faculty of the Department in the teaching-learning process.

Modern teaching-learning techniques consist of

- Flipped class activities – animation, group discussion, etc.
- Project based learning, industrial visits, field visits.
- NPTEL courses and MOOCS.
- Lectures by industrial experts.
- E-books, e-journals, e-lecture notes.
- Platform for interaction of students with industrial experts through special lectures, workshops, seminars, conferences, industrial visits and internship.
- Value added courses and skill development programs.
- Video lectures by faculty.
- Student clubs that engages students in designing and developing innovative ideas and activities.

Modern teaching learning tools like,

- LCD / LED projectors.
- ICT Enabled class rooms.
- Smart board, PPT, Internet enabled computer systems in digital library.
- Wi-Fi enabled laptops are usually employed in classrooms and other student learning environments.
- Online teaching of Faculty to students by using software like Google Meet, Cisco-Webex, Microsoft Teams, Mycamu & Google class rooms.

5.7 Faculty as participants in Faculty development/training activities/STTPs (15)

Total Marks 15.00

Name of the faculty	Max 5 Per Faculty		
	2021-22(CAYm1)	2020-21(CAYm2)	2019-20(CAYm3)
Mr.MSP. Raju	5.00	3.00	3.00
Dr.P.Sivabalan	5.00	5.00	5.00
Dr.R.Lilly	5.00	5.00	0.00
Mr.A. Prem Anandh	5.00	5.00	5.00
Mr. Kolli Sujit Kumar	5.00	5.00	5.00
Dr. P.A.Prasob	5.00	5.00	5.00
Dr. B. M. Shameem	5.00	5.00	5.00
Dr. A. Rajesh Kanna	5.00	5.00	5.00
Mr. Kari Jagadish	5.00	5.00	5.00
Ms. M. Geethapriya	5.00	5.00	5.00
Mr. Akshar Patel	0.00	5.00	5.00
Mr. S Ramaswami	5.00	5.00	5.00
Mr.R Jayasurya	5.00	5.00	0.00
Srivathsan. B	5.00	5.00	5.00
Mr. Himanshu Uppal	0.00	0.00	5.00
Cdr Prashant Kumar (Retd)	0.00	0.00	5.00
Sum	65.00	68.00	68.00
RF = Number of Faculty required to comply with 20:1 Student Faculty Ratios as per 5.1	10.00	14.00	14.00
Assessment [$3 \times (\text{Sum} / 0.5\text{RF})$]	39.00	29.14	29.14

Average assessment over 3 years: 15.00

5.8 Research and Development (75)

Total Marks 75.00

5.8.1 Academic Research

Table 5.8.1 Number of papers published by faculty

Academic Year	2022-2023	2021-2022	2020-2021
Number of papers published in journals	6	9	13
Number of papers published in conferences	6	11	19
Number of Books & Book Chapters Published	22	2	-
Total	34	22	32

Table 5.8.1.1 List of papers published by faculty in refereed journals (2022 – 2023)

S.No	Title of the paper	Name of the author/authors	Name of the journal	Year of publication	ISSN number	Indexing
1.	Benchmarking of VIV numerical analysis with prototype response for fatigue assessment of inverse catenary coldwater pipelines	S. K. Bhattacharyya	SN Applied Sciences	2023	2523-3971	Scopus
2.	Numerical analysis of natural fibre composite for marine structures	Mr.A.Prem Anandh	Journal of Survey in Fisheries Sciences	2023	2368-7487	Scopus
3.	Experimental Investigation Of Geopolymer Flexible Pavement with Waste Plastics Aggregates	Dr.Lilly. R,Aravind Raj P.S,Divahar	Nature Environment and Pollution Technology	2022	2395.3454	Scopus
4.	Efficiency of ferritin Bio-Nano material in reducing the pollutants level of water in the underground corridors of metro rail using GIS	Dr.Lilly R,Prabhakaran.S	Scientific reports	2022	2045-2322	Scopus
5.	CFD based prediction on hydrodynamic effects of Interceptor and flap combination on planing hull	Dr.Suneela J	Ocean Engineering (https://www.sciencedirect.com/journal/ocean-engineering)	2022	0029-8018	Scopus
6.	Investigation of chia based copper oxide nanofluid for water based drilling fluid: An experimental approach	Dr.Ponmani S	Journal of Natural Gas Science and Engineering (https://www.sciencedirect.com/journal/journal-of-natural-gas-science-and-engineering)	2022	1875-5100	SCI

Table 5.8.1.2 List of papers published by faculty in conferences (2022 – 2023)

S.No	Name of the Author/Authors	Title of the Paper	Title of the Conference Proceedings	Title of the Conference	National/ International	Year of Publication	ISBN No
1	Dr. P.Sivabalan	Importance of hydrodynamic design of a drillship with rectangular moonpool	Proceedings of International conference on Innovation and advances in Inter discipline	International conference on Innovation and advances in Inter discipline	International	2022	978-93-91535-29-2
2	Mr.A.Prem Anandh	Numerical Analysis of Natural Fiber Composite for the marine structure	Proceedings of International Multidisciplinary Conference on Recent Innovation in Science, Engineering, Management and Humanities (RISEMH-2022)	International Multidisciplinary Conference on Recent Innovation in Science, Engineering, Management and Humanities (RISEMH-2022)	International	2022	978-93-91535-34-6
3	Mr.JayaSurya	Estimation of Water level in three different phases of underground tunnel construction	Proceedings of International Conference on computer science, Engineering & Information Technology (ICSITY 2022)	Proceedings of International Conference on computer science, Engineering & Information Technology (ICSITY 2022)	International	2022	978-625-8284-23-2
4	Dr.Lilly	Estimation of Water level in three different phases of underground tunnel construction	Proceedings of International Conference on computer science, Engineering & Information Technology (ICSITY 2022)	Proceedings of International Conference on computer science, Engineering & Information Technology (ICSITY 2022)	International	2022	978-625-8284-23-2
5	Dr. A. Rajesh Kanna	Enhanced oil recovery using Pseudomonas putida MTCC 2422 for heavy oil reservoir for improved Production	Proceedings of International Web Conference on Emerging Trends in Arts, Science, Engineering & Technology, (IWCETASET 2022)	International Web Conference on Emerging Trends in Arts, Science, Engineering & Technology, (IWCETASET 2022)	International	2022	978-81-945642-3-2
6	Dr. A. Rajesh Kanna	Exploration and estimating recovery rate of oil shale by in-situ combustion	Proceedings of International Web Conference on Emerging Trends in Arts, Science, Engineering & Technology, (IWCETASET 2022)	International Web Conference on Emerging Trends in Arts, Science, Engineering & Technology, (IWCETASET 2022)	International	2022	978-81-945642-3-2

Table 5.8.1.3 List of Papers Published by Faculty in Refereed Journals (2021 – 2022)

S.NO	Title of the paper	Name of the author/authors	Name of the journal	Year of publication	ISSN number	Indexing
1.	Use of Carbon Nano Tube(CNT) as a wrapping Material to the building Structures to control Salt Damage'	Lilly. R , Prabhakaran .S	Stochastic Modeling and Applications	2022	0972-3641	UGC
2.	Pollution Of pH In Ground Water due to Under Ground Tunnel below the Ground Surface Using GIS Analysis	Lilly. R, Aravind Raj P.S, Divahar.R	Indian Journal of Environmental Protection	2022	0253-7141	Scopus
3.	Bio-medical Waste Remediation by Environmental Safe Gelatin Coated Blod Sample Paper and its effective utilization	Jayakumar.P, Padmanabhan.S, Lilly.R, Suthendran.K	Nature Environment and Pollution Technology	2022	2395-3454	Scopus

4.	Estimating Recovery Rate of Oil Shale by In-Situ Combustion	Rajesh Kanna, Kumaraguru	Journal of Xidian University	2022	1001-2400	Scopus
5.	Pressure distribution on planing hull bottom with stern interceptor	Suneela J	Applied Ocean Research (https://www.sciencedirect.com/journal/applied-ocean-research)	2021	0141-1187	Scopus
6.	"A Numerical Analysis on Use of Eco-friendly Composites for Shipbuilding"	A.Prem anandh	Journal of Huazhong University of Science and Technology	2021	1671-4512	Scopus
7.	The hydrodynamic analysis of multiple hydrofoils translating in tandem in presence of a free surface	Suneela J	The Journal of Engineering for the Maritime Environment	2021	1475-0902	Scopus
8.	Hydrodynamic performance of planing craft with interceptor-flap hybrid combination	Suneela J	Journal of Ocean Engineering and Marine Energy (https://www.researchgate.net/journal/Journal-of-Ocean-Engineering-and-Marine-Energy-2198-6452) (Springer Nature)	2021	2198-6444	Scopus
9.	Experimental investigation of aloe-vera-based CuO nanofluid as a novel additive in improving the rheological and filtration properties of water-based drilling fluid	Ponmani S	SPE Drilling and Completion	2021	1064-6671	Sci index

Table 5.8.1.4 List of Papers Published by Faculty in Conferences (2021 – 2022)

S.No	Name of the Author/Authors	Title of the Paper	Title of the Conference Proceedings	Title of the Conference	National/ International	Year of Publication	ISBN No/ ISSN No
1.	Lilly.R, Prabhakaran.S	Estimation of Ground Water Storage before and after the Construction of Underground Metro Rail Corridor, Chennai	Proceedings of International Conference on Recent Advances in Sustainable Construction Technologies and Environment	ICRASCE-2022	International	2022	978-93-8921-221-1
2.	Lilly.R, Prabhakaran.S	Comparison of minimum and maximum Values of Water Levels above mean sea level in three different phases of underground tunnel construction	Proceedings of International Conference on Emerging Trends in Engineering and Technology	ICETET 2021	International	2022	978-93-90214-02-0
3.	Lilly. R , Prabhakaran .S	Use of Carbon Nano Tube(CNT) as a wrapping Material to the building Structures to control Salt Damage	Proceedings of 10th International Conference on Contemporary Engineering and Technology (ICCET 2022)	ICCET 2022	International	2022	978-93-81288-22-1
4.	Dr. A. Rajesh Kanna	Crude Oil Spillage: Causes and Prevention Techniques	Proceedings of National Level Conference on Advanced Technologies in WasteWater Treatment	Conference on Advanced Technologies in WasteWater Treatment	National	2022	978-93-5620-748-6
5.	Dr. A. Rajesh Kanna	Production of Biosurfactant Using Pseudomonas Putida MTCC 1273 For Improved Oil Recovery	International Conference on Chemical, Bio and Environmental Engineering	International Conference on Chemical, Bio and Environmental Engineering	International	2022	

6.	Dr. A. Rajesh Kanna	Utilization of Bacteria <i>Pseudomonas putida</i> MTCC 1273 for Possible Hydrocarbon Recovery	National Conference on Recent Advances in Chemical and Environmental Sciences (RACES – 2022)	National Conference on Recent Advances in Chemical and Environmental Sciences (RACES – 2022)	National	2022	
7.	Dr. A. Rajesh Kanna	Experimental Investigation on Performance of a Nanoparticle Induced Drilling Mud for Effective Oil Recovery	International Conference on Sustainable Materials and Technologies for Bio and Energy Applications (SMTBEA-2022)	International Conference on Sustainable Materials and Technologies for Bio and Energy Applications (SMTBEA-2022)	International	2022	
8.	Dr. A. Rajesh Kanna	Potential Micro-nano Structures to Improve the Efficiency of Hydrocarbon Extraction	International Conference on Sustainable Materials and Technologies for Bio and Energy Applications (SMTBEA-2022)	International Conference on Sustainable Materials and Technologies for Bio and Energy Applications (SMTBEA-2022)	International	2022	
9.	Lilly.R, Prabhakaran.S	GIS in Urban Planning-A Review	Proceedings of 7 th International Conference on Research trends in Engineering and Technology	ICRTET 2021	International	2021	978-93-92105-20-3
10.	Dr.SK Bhattacharya	Numerical study of aerodynamic noise and spatial coherence lengths for circular cylinders" Advances in Noise and Vibration Control Technology	49th International Congress and Exposition on Noise Control Engineering (INTERNOISE 2020)	Advances in Noise and Vibration Control Technology	International	2021-2022	978-1-7138-2223-3
11.	Dr.SK Bhattacharya	Estimation of aero/hydrodynamic noise using large eddy simulations" Oceans 2020	IEEE Oceanographic Society	IEEE Oceanographic Society	International	2021-2022	978-91-946521-3-2

Table 5.8.1.5 List of Papers Published by Faculty in Refereed Journals (2020 – 2021)

S.No	Title of the paper	Name of the author/authors	Name of the Journal	Year of publication	ISSN number	Indexing
1.	Numerical and experimental studies of heave damping and added mass of spar with heave plates using forced oscillation	Dr.SK Bhattacharya	Applied Ocean Research	2021	0141-1187	Scopus
2.	CFD approach to heave damping of spar with heave plates with experimental validation Numerical prediction of roll damping of ships with and without bilge keel	Dr.SK Bhattacharya	Applied Ocean Research	2021	0141-1187	Scopus
3.	Numerical investigation of the influence of wave parameters on maneuvering hydrodynamic derivatives in regular head waves	Suneela J	Brodogradnja	2021	0007-215X	Scopus
4.	Numerical and experimental studies of heave damping and added mass of spar with heave plates using forced oscillation	Bhattacharyya, S. K	Applied Ocean Research	2021	0141-1187	Scopus
5.	Happiness	Cdr Prashant Kumar (Retd)	International Journal of Psychosocial Rehabilitation	2020	1475-7192	Scopus

6.	CFD Analysis on Propeller Performance with Propeller Boss Cap Fin	Swapnil D Sankpal., B M Shameem	International Journal of Recent Technology and Engineering	2020	2277-3878	Scopus
7.	Experimental Research on Water Column Oscillations in Moonpool of a Drillship in Downtime Period	MSP Raju., Sivabalan Ponnappan	International Journal of Recent Technology and Engineering	2020	2277-3878	Scopus
8.	Numerical Validation of Frictional Coefficient for a Flat Plate with Various Anti-Fouling Coatings	K Abhiroop., B M Shameem	International Journal of Recent Technology and Engineering	2020	2277-3878	Scopus
9.	Effect of Moonpool on Drillship Motions	P.Sivabalan and MSP Raju	Journal of Offshore Structure and Technology	2020	2349-8986	Scopus
10.	Drillship Response on Different loading condition	P.Sivabalan, A.Prem Anandh and N.Sivakumar	Journal of TEST Engineering and Management	2020	0193-4120	Scopus
11.	Numerical investigation on the hydrodynamic performance of high-speed planing hull with transom interceptor	Suneela J	ships and Offshore Structures	2020	1744-5302	Scopus

Table 5.8.1.6 List of Papers Published by Faculty in Conferences (2020 – 2021)

S.No	Name of the Author/Authors	Title of the Paper	Title of the Conference Proceedings	Title of the Conference	National/ International	Date & Year	ISBN No
1.	P.Sivabalan	Practical Tidal Turbine Design Consideration And Tidal Energy Generation	Proceedings of Third International Conference in Shipbuilding & Offshore Engineering 2021	ICSOE 2021	International	24-25, June 2021	978-93-85434-91-4
2.	A.Prem Anandh	'Numerical Analysis of NFC deck plate of a RO-RO ship'	Proceedings of National conference on advanced functional composite materials and applications	conference on advanced functional composite materials and applications	National	28-29 th June 2021	978-93-85434-37-2
3.	Sivabalan Ponnappan, and Yogi Patcheappane	Research on Motion Response of The Fishing Boat In Different Heading of Waves	Proceedings of National Conference on Research & Innovation in Aquatic Foods & Fishing Vessel Technology	National Conference on Research & Innovation in Aquatic Foods & Fishing Vessel Technology	National	6 March 2020	978-93-85434-78-5
4.	P.Sivabalan, A.Prem Anandh and N.Sivakumar	Drillship Response on Different Loading Condition	Proceedings of International Conference on Computing, Communication and Control	I4C - 2020	International	28-29 February 2020.	978-93-89107-79-1
5.	A.Prem Anandh, Dr. P.Sivabalan	Numerical Analysis of Natural Fiber Hybrid Composite for the Construction of Ship Structure	Proceedings of International Conference on Advancements in Materials Science and Technology	iCAM 2020	International	23-25 November 2020	978-93-83409-61-7

6.	M S P Raju., A. Prem Anandh	Sustainability in Ship Recycling Process	Proceedings of International Conference on Sustainable Computing in Science, Technology and Management	International Conference on Sustainable Computing in Science, Technology and Management	International	2020	978-93-7261-24-2
7.	B. Srivathsan	Retail Sector and Agriculture e-commerce Business in India: An Overview	Proceedings of 4th International Web Conference on Emerging Trends in Arts, Science, Engineering and Technology, Organized by DK International Foundation (IWCETASET-2020)	IWCETASET-2020	International	2020	978-81-945642-3-2
8.	M Geetha Priya	Retail Sector and Agriculture e-commerce Business in India: An Overview	Proceedings of 4th International Web Conference on Emerging Trends in Arts, Science, Engineering and Technology, Organized by DK International Foundation (IWCETASET-2020)	IWCETASET-2020	International	2020	978-81-945642-3-2
9.	Kolli Sujit Kumar	Retail Sector and Agriculture e-commerce Business in India: An Overview	Proceedings of 4th International Web Conference on Emerging Trends in Arts, Science, Engineering and Technology, Organized by DK International Foundation (IWCETASET-2020)	IWCETASET-2020	International	2020	978-81-945642-3-2
10.	A.Prem Anandh	Economic Trend of Indian Shipbuilding and recycling Industry	Proceedings of 4th International Web Conference on Emerging Trends in Arts, Science, Engineering and Technology, Organized by DK International Foundation (IWCETASET-2020)	IWCETASET-2020	International	2020	978-81-945642-3-2
11.	Dr.Sivabalan.P	Economic Trend of Indian Shipbuilding and recycling Industry	Proceedings of 4th International Web Conference on Emerging Trends in Arts, Science, Engineering and Technology, Organized by DK International Foundation (IWCETASET-2020)	IWCETASET-2020	International	2020	978-81-945642-3-2
12.	Yarabala Gopi Krishna	Case study on siltation affects the fishing trade in india	Proceedings of 4th International Web Conference on Emerging Trends in Arts, Science, Engineering and Technology, Organized by DK International Foundation (IWCETASET-2020)	IWCETASET-2020	International	2020	978-81-945642-3-2
13.	Kolli sujit kumar	Comparitive study of six sigma, tqm and iso manerial based approaches	Proceedings of 4th International Web Conference on Emerging Trends in Arts, Science, Engineering and Technology, Organized by DK International Foundation (IWCETASET-2020)	IWCETASET-2020	International	2020	978-81-945642-3-2
14.	M Geethapriya	A comparative study between conventional commerce and e-commerce	Proceedings of 4th International Web Conference on Emerging Trends in Arts, Science, Engineering and Technology, Organized by DK International Foundation (IWCETASET-2020)	IWCETASET-2020	International	2020	978-81-945642-3-2

15.	B Srivathsan	A comparative study between conventional commerce and e-commerce	Proceedings of 4th International Web Conference on Emerging Trends in Arts, Science, Engineering and Technology, Organized by DK International Foundation (IWCETASET-2020)	IWCETASET-2020	International	2020	978-81-945642-3-2
16.	M.S.P. Raju	Applications of Concepts of E-mobility in fishing vessels	Proceedings of 4th International Web Conference on Emerging Trends in Arts, Science, Engineering and Technology, Organized by DK International Foundation (IWCETASET-2020)	IWCETASET-2020	International	2020	978-81-945642-3-2
17.	B.Vignesh	Applications of Concepts of E-mobility in fishing vessels	Proceedings of 4th International Web Conference on Emerging Trends in Arts, Science, Engineering and Technology, Organized by DK International Foundation (IWCETASET-2020)	IWCETASET-2020	International	2020	978-81-945642-3-2
18.	Jagadish Kari	An investigation on supramax vessel efficiency by considering hull parameters	Proceedings of 4th International Web Conference on Emerging Trends in Arts, Science, Engineering and Technology, Organized by DK International Foundation (IWCETASET-2020)	IWCETASET-2020	International	2020	978-81-945642-3-2
19.	Jagadish Kari	Optimization of ship safety carrying toxic wastes	Proceedings of 4th International Web Conference on Emerging Trends in Arts, Science, Engineering and Technology, Organized by DK International Foundation (IWCETASET-2020)	IWCETASET-2020	International	2020	978-81-945642-3-2

Table 5.8.1.9 Citation details of faculty from Google Scholar

S.No	Name of the Faculty	Citation	i10	h-index
1.	Dr.S. K. Bhattacharyya	1373	38	21
2.	Dr K.Thiruvenkatasamy	217	5	7
3.	Mr M Surya Prakash Raju	2	0	1
4.	Dr.P.Sivabalan	17	0	2
5.	Dr.R.Lilly	192	6	6
6.	MrA.PremAnandh	02	0	1
7.	Dr. V Arularasi	42	1	4
8.	Dr.Suneela J	29	1	4
9.	Dr. Ponmani S	559	4	7
10.	Dr. A.Rajesh Kanna	159	5	7

Table 5.8.1 B Total number of books published by faculty

S.No	Title of the Book	Name of the Authors	Academic Year
1.	Maritime Technology 4.0 (ISBN: 978-93-94958-57-9)	Mr.MSP Raju	2022-2023
2.	Recent Trends in Offshore Engineering	Mr.MSP Raju	2022-2023
3.	ICSOE 2021 (International Conference on Shipbuilding and Offshore Engineering) (ISBN : 978-93-85434-91-4)	Mr.MSP Raju, Dr. B.M. Shameem, Dr.P. Sivabalan, and Mr. K. Sujit Kumar	2021-2022
4.	Managing Self (ISBN :979-8-6180-8408-6)	Mr.MSP Raju	2019-2020
5.	MARINCO 2019 (Maritime International Conference) (ISBN : 978-93-85434-76-1)	Mr. S.V. Durga Prasad, Dr. B.M. Shameem and Mr. Akshar Patel Mr.MSP Raju	2019-2020
6.	AFT 2020 (National Conference on Research and Innovation in Aquatic foods and Fishing Vessel Technology) (ISBN : 978-93-85434-78-5)	Mr.MSP Raju, Mrs. K Sasi Praba, and Dr. N Manimehalai	2019-2020

Table 5.8.1.1 B Total number of book chapters published by faculty

S.No	Title of the Book Chapter	Name of the Book	Name of the Publisher	Name of the Authors	Academic Year
1.	Introduction to Maritime Technology 4.0: Concepts and Overview	Maritime Technology 4.0 (ISBN: 978-93-94958-57-9)	Archers And Elevators Publishing House	MSP Raju	2022-23
2.	Big Data for Big Ships	Maritime Technology 4.0 (ISBN: 978-93-94958-57-9)	Archers And Elevators Publishing House	P. Sivabalan	2022-23
3.	Digitization and it's trends in maritime technology	Maritime Technology 4.0 (ISBN: 978-93-94958-57-9)	Archers And Elevators Publishing House	Suneela. J	2022-23
4.	Green Materials in Shipbuilding	Maritime Technology 4.0 (ISBN: 978-93-94958-57-9)	Archers And Elevators Publishing House	Arularasi .V	2022-23
5.	Introduction to voyage planning	Maritime Technology 4.0 (ISBN: 978-93-94958-57-9)	Archers And Elevators Publishing House	Lilly R	2022-23
6.	Revolutionizing Shipbuilding: The Role of Digital Technologies	Maritime Technology 4.0 (ISBN: 978-93-94958-57-9)	Archers And Elevators Publishing House	A Prem Anandh	2022-23

7.	Data science for ship design	Maritime Technology 4.0 (ISBN: 978-93-94958-57-9)	Archers And Elevators Publishing House	Thalapathy Vigneshwar	2022-23
8.	Application of microbial remediation in the treatment of offshore oil pollution	Maritime Technology 4.0 (ISBN: 978-93-94958-57-9)	Archers And Elevators Publishing House	Rajesh Kanna	2022-23
9.	Loads for offshore structure	Maritime Technology 4.0 (ISBN: 978-93-94958-57-9)	Archers And Elevators Publishing House	Ponmani S	2022-23
10.	Offshore Renewable Energy: Recent developments and Challenges	Recent Trends in Offshore Engineering	Archers And Elevators Publishing House	MSP Raju	2022-23
11.	Submerged arc welding (SAW)-An overview	Recent Trends in Offshore Engineering	Archers And Elevators Publishing House	P. Sivabalan	2022-23
12.	Materials and corrosion trends in offshore structures-	Recent Trends in Offshore Engineering	Archers And Elevators Publishing House	Arularasi.V	2022-23
13.	Mooring and riser lines design and it's development	Recent Trends in Offshore Engineering	Archers And Elevators Publishing House	Suneela. J	2022-23
14.	The Future of Offshore Engineering with Digital Twins	Recent Trends in Offshore Engineering	Archers And Elevators Publishing House	A Prem Anandh	2022-23
15.	Data science application for underwater vehicle design-	Recent Trends in Offshore Engineering	Archers And Elevators Publishing House	Thalapathy Vigneshwar	2022-23
16.	The importance of RoV's in the offshore oil & gas industry	Recent Trends in Offshore Engineering	Archers And Elevators Publishing House	Rajesh Kanna	2022-23
17.	Well completion method for offshore oil & gas operation-	Recent Trends in Offshore Engineering	Archers And Elevators Publishing House	Ponmani S	2022-23
18.	Engineering challenges for floating offshore turbine	Recent Trends in Offshore Engineering	Archers And Elevators Publishing House	Jayasuya R	2022-23
19.	Impact of Multidisciplinary research in Maritime sector Voume-14	Title: Application of block chain marine Insurance	Redshine Publisher (ISBN: 978-91-987981-11)	P Sivabalan	2022-2023
20.	Impact of Multidisciplinary research in Maritime sector Voume-13	Title: Impact of Multidisciplinary research in Maritime sector	Redshine Publisher (ISBN : 978-91-987981-04)	A.Prem Anandh	2022-2023
21.	Field and Laboratory Investigations Carried out in the Underground Metro Rail Corridors	Emerging Trends in Engineering and Technology	Integrated publications (ISBN: 978-93-93502-73-5)	Lilly. R , Prabhakaran .S	2022-2023
22.	Effect of Climate change in communities, Habitats and Eco-systems'	Nature Resource Management and Environment Security	Integrated publications (ISBN: 978-93-93502-73-5)	Lilly. R , Prabhakaran .S	2022-2023

23.	Review on the different stages of voyage planning	Emerging Technologies and Innovative Research in Science ,Engineering and Management	AMET (ISBN: 978-93-90956-62-3)	Lilly. R , Prabhakaran .S	2021-2022
-----	---	--	--------------------------------	---------------------------	-----------

Table 5.8.1 C List of faculty pursuing Ph.D

S. No	Name	Ph.D., Scholar /PDF – Post Doc	Registration Number	Date of Registration	Research Area	University	Status
1	Mr.R.Jayasurya	Ph.D-PartTime	2022294101	21/06/2022	Wind turbine performance analysis	AMET	1 st DC
2	Mr. A. Prem Anandh	Ph.D-PartTime	2018194306	12/07/2018	Application of natural fiber composites in marine structures.	AMET	Course work completed
3	Mr.MSP.Raju	Ph.D-PartTime	2018294401	03/07/2018	Optimization design for fishing vessels and rational utilization of their production capacity	AMET	Course work completed

Table 5.8.1 D List of faculty guiding Ph.D

S.No	Name of the Faculty	Name of the Scholar	Research Area	University
1	Dr.P.Sivabalan	Mr.MSP.Raju	Optimization design for fishing vessels and rational utilization of their production capacity	AMET
		Mr. A. Prem Anandh	Application of natural fiber composites in marine structures.	AMET
		Mr.R.Jayasurya	Wind turbine performance analysis	AMET
		Mr.GowthamGopal	Ship performance optimization	AMET
2	Dr.SK. Bhattacharya	Roni Francis	Underwater acoustics	IIT-M
		Mahesh Rao	Hydrodynamic damping of offshore structures	IIT-M
		Joemon Jacob	Hydrodynamic noise	IIT-M
		R. Saravanan	Flow induced vibration of pipelines	IIT-M
		Arun Kumar	Thermal forming of plates	IIT-M
		Mohsin Irkal	Experimental and Numerical Study of Roll Damping of Ships	IIT-M
3	Dr. Ponmani Swaminathan	Hammed Hussain	Drilling Engineering	AMET

Table 5.8.1 E List of faculty awarded Ph.D

S. No	Name	Registration Number	Thesis Title	University	Year of Award
-------	------	---------------------	--------------	------------	---------------

1	V. Arularasi	18291191106	Rheological characteristics of ready mix concrete and its impact energy consumption measured in RMC plant	Anna University/College of Engineering, Guindy, Chennai	24th November 2022
2	Prasob Ashoka	14PHD1102	Impact behavior & characterization of jute epoxy hybrid composites under subzero temperature conditions	VIT, Chennai	8 th June 2021

5.8.2 Sponsored Research (20)

Institute Marks : 20.00

2021-22 (CAYm1)

Project Title	Duration	Funding Agency	Amount(in Rupees)
Assessment and Optimizati	12 months	Enervi Eneviro service prive	419000.00
Development and Evaluatio	12 months	Enervi Eneviro service prive	416000.00
			Total Amount(X): 835000.00

2020-21 (CAYm2)

Project Title	Duration	Funding Agency	Amount(in Rupees)
Design and mooring analys	6	JJ marine Service pvt Ltd	65000.00
A study on Ballast free ship	6	JJ marine Service pvt Ltd	56000.00
			Total Amount(Y): 121000.00

2019-20 (CAYm3)

Project Title	Duration	Funding Agency	Amount(in Rupees)
Numerical modeling on hyd	36 months	Ministry of Earth Sciences	2179000.00
An Eco friendly Technology	36 months	DST- SERB	2112000.00
			Total Amount(Z): 4291000.00

Cumulative Amount(X + Y + Z) = 5247000.00

5.8.3 Development activities (15)

Institute Marks : 15.00

5.8.3 Product Development:

The Faculty of the Department involve in product development through AMET CEII(Centre for Entrepreneurship, Innovation and Incubation). Faculty of the department by act as coordinators and facilitate developing products. Faculty who develop novel ideas and projects also patent the same through IPR Cell of the university

Table 5.8.3 List of Seed Money Projects

S.No	Name of the Faculty	Title of the Project	Amount (INR)	Duration (Months)	
1	Dr.Suneela J	Numerical study on planing hull with interceptor in waves	50000	6	
2	Mr.Jayasurya.R	Thrust augmentation of Fletner rotor using dynamic flaps on pressure side.	50000	6	
3	Dr.R.Lilly	Computation of storage of the aquifer due to the construction of underground Metrorail corridor using Visual Mod Flow	55,000	4	
4	Mr.S.Thalapathy Vigneshwar	Viscous Resistance reduction through ultrasonic wave induced Bubbles.	85,000	4	
5	Dr.P.Sivabalan	Numerical and Experimental Analysis of Steel Plates with Rectangular Openings Subjected to Axial Stress	1,00,000	6	
6	Dr.V.Arularasi	Optimization of Energy Consumption in manufacture of Ready Mix Concrete by admixtures and using the concrete for the development of rural areas	1,21,300	6	
7	Mr. A.Prem Anandh	Investigation of Natural Fiber Composites as Sustainable Alternatives in Shipbuilding	60,000	6	
8	Mr.MSP Raju	Design of Fishing Vessel 4.0	15,000	6	
9	Dr.P.Sivabalan	Modelling of Moonpool with Baffle plates for drillship to minimize the water motion and evaluation of its optimum	10,000	6	
10	Cdr Prashant Kumar (Retd)	Optimization of container ship by undertaking stability analysis of container ships using Maxsurf suite of software	40,000	6	
11	Dr.B.M.Shameem	Prediction of frictional resistance of antifouling coating and biofouling conditions for a tanker vessel using CFD	10,000	6	
12	Dr. K. Thiruvenkataswamy	A generic approach on short term morphological changes of shoreline due to construction of groins using Delft3D modeling in Estuarine mouths	1,34,000	6	
13	Dr.J.Rajaraman	Effect of Brine Intrusion on Soil Bearing Capacity in Sandy Coastal Areas	1,00,000	6	
14	Mr.Patel Akshar	A stereotype approach towards an optimization of a barge hull in National waterways of India	50,000	6	
15	Dr. K . Thiruvenkatasamy	Hydrodynamic Analysis of Scour around coastal Structures	45,000	12	
16	Dr. A. Rajesh Kanna	Bioremediation of Oil Sludge using Crude Biosurfactant	40,000	6	

5.8.3.1D Faculty Patent Details**Total Number of Patents: 4**

1. Dr. Ponmani Swaminathan ,Water Based Drilling Fluid Additive and a Preparation Method Thereof, Patent application number- 201841026429 A; 2022- Granted
2. Dr.P.Sivabalan, Parachute Deployment System for Ship, Application Number: 202241050784, 2022.
3. Dr. Ponmani Swaminathan and Dr. A. Rajesh Kanna, Composition for Water Based Mud, Patent application number- 2021411029037A; 09/07/2021
4. Dr. A. Rajesh Kanna and Dr. Ponmani Swaminathan, Microbial Enhanced Oil Recovery Induced by Purified Bio surfactant, Patent application number- 2019414004475A,

5.8.3.2 Research laboratories

The Department has software facilities, which are used for the purpose of research. Below is the list of Major software laboratories used for research purpose.

Table : 5.8.3.2 Major Equipment in laboratories used for research purpose

S.No	Name of the Laboratory	Equipment

1	Research Lab	1. Bentley-MAXSURF(30 2. users) 3. Ship Constructor (30 users) 4. Bentley-SACS (30 users) 5. Bentley-MOSES (30 users) 6. STAR-CCM+ (15 users) 7. Scilab (Open source) 8. FEAST SMT (Open source)	Hardware configuration: Computer No's: 10 Processor: Intel I5 Ram: 16 GB Ram Graphics: 8 GB Intel® UHD graphics 630 Hard disk: 1 TB
2	Marine Hydrodynamics Laboratory	Wave Flume (Wave Generation Software)	Hardware configuration: Computer Nos: 1 Processor: Intel I5 Ram: 16 GB Ram Graphics: 8 GB Intel® UHD graphics 630 Hard disk: 1 TB

5.8.3.3 Instructional materials

Instructional materials like course materials, video lectures, lab manuals and working models are developed for different courses by the respective faculty which are used by the students. Lab manuals are prepared by the Faculty, which is being approved by the HoD for all laboratories. The details of instructional materials developed by faculty are as follows,

Table : 5.8.3.3 List of Lab Manuals prepared by Faculty

S.No	Name of the Laboratory Manual	Name of the Faculty
1	Hydrostatics & Stability Laboratory Manual	Dr.P.Sivabalan
2	Ship Strength Laboratory Manual	Dr.P.A.Prasob
3	Structural Modelling - Software Laboratory Manual	Mr.R.Jaya Surya
4	Marine Hydrodynamics Laboratory Manual	Dr.B.M.Shameem
5	Offshore Structure Design - Software Laboratory Manual	Dr.R.Lilly
6	Ship System Drawing & Launching Calculations Laboratory Manual	Dr.P.Sivabalan
7	Numerical Ship Hydrodynamics – Software Laboratory Manual	Dr.J.Suneela

5.8.3.4 Working models/charts/monograms

Working Models are available in practical courses which could provide a better understanding of course. Working models of different shapes are also used in Engineering Graphics Laboratory.

The details of working models are as follows,

Table : 5.8.3.4 Details of Working Models

S.No	Name of the Working Model	Name of the Laboratory
1.	Working Models of shapes	Engineering Graphics
2.	Wave Tank	Marine Hydrodynamics Laboratory
3.	Bulkhead cross section	Ship design & Drawing
4.	Bulk carrier model	Ship design & Drawing

5.	Jacket Platform	Offshore Structure Design
6.	TLP- Tension Leg platform	Offshore Structure Design
7.	SPAR Platform	Offshore Structure Design
8.	Cross section of cargo hold	Ship design & Drawing
9.	Inclining Experiment- Model	Ship design & Drawing
10.	Folding cargo hatch cover	Ship design & Drawing

5.8.4 Consultancy (from Industry) (20)

Institute Marks : 20.00

2021-22 (CAYm1)

Project Title	Duration	Funding Agency	Amount(in Rupees)
Optimization of	12	Jayashree Con	115000.00
Determination of	12	Jayashree Con	112000.00
Strength asses	12	JJ marine Serv	117000.00
A study on frict	12	JJ marine Serv	114000.00
Performance o	12	LL marine serv	121000.00
Hull form optim	12	LL marine serv	119000.00
			Total Amount(X): 698000.00

2020-21 (CAYm2)

Project Title	Duration	Funding Agency	Amount(in Rupees)
Design and mc	6	JJ marine Serv	65000.00
A study on Ball	6	JJ marine Serv	56000.00
			Total Amount(Y): 121000.00

2019-20 (CAYm3)

Project Title	Duration	Funding Agency	Amount(in Rupees)
CFD Analysis for pollution c	6	Hindustan Ship	106000.00
The CFD Analysis of fast pe	6	Hindustan Ship	100000.00
Design and Detailing work-I	6	Lockheed Engi	25000.00
An Analysis on natural com	6	Sarojini Resear	150000.00
Hydrodynamic Analysis of S	6	Suvedha Cons	45000.00
			Total Amount(Z): 426000.00

Cumulative Amount(X + Y + Z) = 1245000.00

5.9 Faculty Performance Appraisal and Development System (FPADS) (10)

Total Marks 10.00

Key Parameters for Faculty Appraisal in terms of API (Academic Performance Indicator) Score

Three important parameters are evaluated for appraisal of Faculty in terms of API Score

Parameter – I

Teaching, Learning and Evaluation Related Activities Which Includes the Following,

- Lectures, Seminars, Tutorials, Practical, Contact Hours Undertaken as Percentage of Lectures Allotted
- Lectures or Other Teaching Duties in Excess of The UGC Norms
- Preparation and Imparting of Knowledge / Instruction as Per Curriculum; Syllabus Enrichment by Providing Additional Resources to Students
- Use of Participatory and Innovative Teaching-Learning Methodologies; Updating of Subject Content, Course Improvement Etc.
- Examination Duties (Invigilation; Question Paper Setting, Evaluation of Answer Scripts) As Per Allotment

Parameter – II

Co-Curricular, Extension and Professional Development Related Activities,

- Student Related Co-Curricular, Extension and Field Based Activities (Such as Extension Work Through Eco Club and Other Channels, Cultural Activities, Subject Related Events, Advisement and Counseling)
- Contribution to Corporate Life and Management of the Department and Institution Through Participation in Academic and Administrative Committees and Responsibilities
- Professional Development Activities

Parameter -III

Research and Academic Contributions

- Refereed Journals
- Non-Refereed but Recognized and Reputable Journals and Periodicals, Having ISBN/ISSN Numbers
- Conference Proceedings as Full Papers, Etc
- Other Research Publications (Books, Chapters in Books, Other Than Refereed Journal Articles)
- Text or Reference Books by Other Local Publishers with ISBN/ISSN Numbers
- Completed/Ongoing Sponsored Projects
- Projects Outcome/Output: Patent/Technology Transfer/ Product/Process
- Research Guidance
- Participation in Training Courses/Conference/Seminar/Workshop
- Training Courses / Conference /Seminar/Workshop Organized
- Award/Honours/Recognition



S.No	Parameter	Expected Requirement /Year	Response (From date of Joining at AMET to June 2021)	Response From July 2021 to Till date	Total
1.	Average Pass percentage in all the subjects handled				
2	Average Student satisfaction feedback (%)				
3	Publication in Journals Indexed by UGC (Group 1 & Group 2 CARE list)/Scopus/Web of Science				

4	Papers Presented in National/International Conference (with ISBN Number)				
5	Book(s)/Chapter(s) published With ISBN Number				
6	Research Projects/Consultancies/Seed money Project				
7	Video Lectures given for University LMS				
8	Training Courses, Teaching-Learning Evaluation Programme, Faculty Development Programme Attended				
9	Seminar/Workshop/Conference Attended outside the University				
10	Guest Lectures Arranged				
11	Academic/Professional/Industrial Membership				
12	National/International Innovation/Research Award				
13	National/International Academic Award Government Bodies only				
14	Patents filed/published/granted				
15	E-Contents created for Private /Govt. MOOC Platforms				
16	Ph.D., scholars guided				
17	No. of Conferences/Seminars/Workshops organized				
18	No. of Extension/Social Service events organized				

Declaration

I hereby declare that the above information are correct and are supported by documents. In case needed, I shall produce necessary documents.

Signature of the Faculty

Verified and Forwarded by

Date of Submission:

(Seal and Signature of HoD)

Table : 5.9.1. Incentive received by the faculty

	Name of the faculty	Category	Amount	AY
1.	Mr.MSP Raju	FDP	2,200	2022-23
2.	Dr.Lilly	Journal Paper publication	25,000	2022-23
3.	Dr.Lilly	Book Chapter Publication	5,000	2022-23
4.	Dr.Ponmani S	Patent	53,600	2022-23
5.	Dr.Ponmani S	Paper	25,000	2022-23
6.	Dr.P Sivabalan	Book Chapter Publication	5,000	2022-23

8.	Mr. .A.Prem Anandh	Book Chapter Publication	5,000	2022-23
11.	Mr.A.Prem Anandh	Award	5,000	2020-21
12.	Mr.A.Prem Anandh	Paper publication	10,000	2019-20
13.	Mr.MSP Raju	Conference	15,000	2019-20

5.10 Visiting/Adjunct/Emeritus Faculty etc. (10)

Total Marks 10.00

Table 5.10 A Courses Handled by Visiting Faculty (2020-2021)

S.No	Name of the Visiting Faculty	Designation	No of hours handled	Course Handled	Semester
1	Dr. R. Natarajan	Visiting Faculty	45	Introduction to Naval Architecture	III
2	Dr. R. Natarajan	Visiting Faculty	45	Deign of offshore structures	VI

Table 5.10 B Courses Handled by Visiting Faculty (2021-2022)

S.No	Name of the Visiting Faculty	Designation	No of hours handled	Course Handled	Semester
1	Dr. R. Natarajan	Visiting Faculty	45	Dynamics of offshore structures	VII
2	Dr. R. Natarajan	Visiting Faculty	45	Marine Pollution Regulation	IV

Table 5.10 C Courses Handled by Visiting Faculty (2022-2023)

S.No	Name of the Visiting Faculty	Designation	No of hours handled	Course Handled	Semester
1	Dr. R. Natarajan	Visiting Faculty	45	Dynamics of offshore structures	VII

Table 5.10 D Visiting Faculty Details

S.No	Name of the Faculty Member	Qualification		Associati on with the Institutio n	Designation	Department	Specialization	Academic Research			Currently Associated (Y/N) Date of Leaving (In case Currently Associated is ("No"))
		Degree (highest degree)	University					Research Paper Publicati ons	Ph.D. Guidan ce	Faculty Receivin g Ph.D. during the Assessm ent Years	
1	Dr. R. Natarajan	Ph.D	IIT-Madras	Y	Visiting Faculty	NA&OE	Naval Architecture & Offshore Engineering	2	-	-	Y

6 FACILITIES AND TECHNICAL SUPPORT (80)**Total Marks 80.00****6.1 Adequate and well equipped laboratories, and technical manpower (40)**

Total Marks 40.00

Institute Marks : 40.00

Sr. No	Name of the Laboratory	Number of students per set up(Batch Size)	Name of the Important Equipment	Weekly utilization status(all the courses for which the lab is utilized)	Technical Manpower Support		
					Name of the Technical staff	Designation	Qualification
1	Ship Design La	1	Hardware conf	20 hrs.	Mr.S.Sivaram	Lab Instructor	B.E(NAOE)
2	Ship Drawing L	1	Drawing Aids a	8 hrs.	Mr.Praveenkun	Lab Instructor	B.E(MARINE)
3	Marine Simulat	1	Hardware conf	12 hrs.	Mr.J.Jayagopal	Lab Instructor	B.E(NAOE)
4	Marine Hydrod	3	Hardware conf	20 hrs.	Mr.J.Jayagopal	Lab Instructor	B.E(NAOE)
5	Ship Systems I	1	Drawing Aids a	2 hrs.	Mr.Ronaldo An	Lab Instructor	B.E(MARINE)
6	Numerical Mari	1	Hardware conf	18 hrs.	Mr.S.Vignesh	Lab Instructor	B.E(NAOE)
7	Project Lab	1	Hardware con	Open	Mr. J.Rajkumar	Lab Instructor	B.E(MARINE)
8	Research Lab	1	Hardware conf	Open	Mr.Siju Francis	Lab Instructor	B.E(MARINE)

6.2 Laboratories maintenance and overall ambiance (10)

Total Marks 10.00

Daily maintenance	• Work tables and equipment are cleaned and the working condition of the equipment/system is checked by the lab technician daily. • The labs and equipment are kept clean by the housekeeping department of the college. • Wave flume is monitored for water leakage.
Weekly maintenance	• Floor mopping of Labs is done twice in a week by the housekeeping department. • Electrical repair work is done by the electrician. • The batteries of the UPS are cleaned to remove dust and prevent corrosion.
Monthly maintenance	• The required consumables are purchased as per requirement. • All equipment is serviced once a month.
Yearly maintenance	• Fire extinguishers are checked. • Painting of wave flume to avoid corrosion. • Wave maker is overhauled. • The calibration of equipment is done every year.
Preventive maintenance	• The working conditions of the equipment are periodically checked • The students are given instructions on handling the equipment before doing the experiments. • UPS backup is provided for all system-based laboratories. • Anti-virus software is updated regularly.
Breakdown maintenance	• Service registers are maintained in each laboratory. • Breakdown maintenance of the personal computer is done by IT support.

Overall Ambience
• All laboratories are well furnished and spacious with good ventilation and lighting. • List of experiments as per the curriculum is maintained in each laboratory. • Soft and hard copies of laboratory manuals are available in the laboratory. • Regular checkup of equipment is carried out at the beginning of every semester. • Laboratory equipment is regularly inspected in academic and administrative audits. • Breakdown and maintenance registers are maintained in the laboratories. • All labs are provided with a UPS in case of power failure. • All laboratories are provided with fire extinguishers. • All laboratories have whiteboard/ smart board/ projector and Wi-Fi connection. • ISO system practices are used for auditing laboratories regularly.

6.3 Safety measures in laboratories (10)

Total Marks 10.00

Sr. No	Laboratory Name	Safety Measures
1	Ship Design Laboratory/ Marine Simulation Laboratory/ Numerical Ship Hydrodynamics – Software Laboratory/ Research Lab/ Project Lab.	1. Laboratory is equipped with a first aid box and fire extinguisher. Lab layout (emergency exit plan), fire-fighting equipment, and safety measures are displayed. 2. Students are instructed to report any broken plugs or exposed electrical wires to your lecturer/laboratory technician immediately. 3. Students are advised to avoid stepping on electrical wires or any other computer cables. 4. Students are advised not to insert metal objects such as clips, pins, and needles into the computer casings. They may cause a fire. 5. Students are instructed not to touch, connect or disconnect any plug or cable without informing the lecturer/laboratory technician. Only qualified, certified, and authorized persons may connect and disconnect any cables, devices, computers, or anything in the computer lab. Others is not allowed to connect or disconnect machines or components 6. It is advised to use vacuum cleaner while cleaning computer laboratory. 7. No one is allowed to delete information from the computer unless otherwise it is given as an instruction. 8. Students are advised not to install unauthorized software or troubleshoot computer systems. Any problem or issues should be reported. 9. Unless otherwise required, external storage devices are not allowed in the laboratory.
2	Ship Drawing Laboratory/ Ship System Design Laboratory	1. Laboratory is equipped with a first aid box and fire extinguisher. Lab layout (emergency exit plan), fire-fighting equipment, and safety measures are displayed. 2. Do not eat or drink in the laboratory. 3. Students are instructed to report any fault in lighting and ventilation. 4. Students are advised to handle sharp drawing tools with utmost care.
3	Marine Hydrodynamics lab.	1. Laboratory is equipped with a first aid box and fire extinguisher. Laboratory layout (emergency exit plan), fire-fighting equipment, and safety measures are displayed. 2. Students are instructed to report the fires or accidents to your lecturer/laboratory technician immediately. 3. Students are asked to report any broken plugs or exposed electrical wires to your lecturer/laboratory technician immediately. 4. Students are advised to avoid stepping on electrical wires. 5. Students are advised to avoid any act of contaminating the wave flume water. 6. Students are instructed to wear shoes to avoid electric shocks. Wear shoes to avoid electric shocks.

6.4 Project laboratory (20)

Total Marks 20.00

1. Project/Research laboratories are utilized by UG students, PG students, research scholars, and faculty members for their projects and research activities.
2. Discussions and implementations of innovative ideas about mini-projects and major projects are carried out in the project/research laboratory.
3. Project and research labs are utilized exclusively for the student project and research work with the hardware and software facilities listed below.

a. Hardware configuration:

Computer No's: 20

Processor: Intel i5

RAM: 16 GB

Graphics: 8 GB Intel® UHD graphics 630

Hard disk: 1 TB

b) List of Software Available

- a. Bentley-MAXSURF (30 users)
- b. Ship Constructor (30 users)
- c. Bentley-SACS (30 users)
- d. Bentley-MOSES (30 users)
- e. STAR-CCM+ (15 users)
- f. Scilab (Open source)
- g. Python (Open source)
- h. FEAST^{SMT} (Open source)

Utilization:

1. Utilization for Project labs start from 9.00 A.M-4.00 P.M during working hours and from 4.00 P.M -5.00 P.M after the working hours.
2. Upon prior request and permission from HOD by the students can access the facility during non- working days also.
3. Laboratories are also operated beyond the college hours for the convenience of the students.
4. Students can utilize the project labs for final year projects and mini projects.

Research Facilities:

The Department of Naval Architecture and Offshore Engineering has setup research environment to utilize the knowledge resources to carryout research activities. The detailed infrastructural facilities are listed in table below.

S.No	Infrastructural facilities	Configurations	No. of Equipment
1	Personal Computers	Intel Core i5 processors	10
2	Bentley-MAXSURF	Academic -Ultimate	30
3	Bentley-MOSES	Academic -Ultimate	30
4	Bentley-SACS	Academic -Ultimate	30
5	Ship Constructor	Academic	30
6	STAR-CCM+	Academic	15
7	Wave flume with Wave maker		1

7 CONTINUOUS IMPROVEMENT (75)

Total Marks 75.00

7.1 Actions taken based on the results of evaluation of each of the COs, POs & PSOs (30)

Total Marks 30.00

POs Attainment Levels and Actions for Improvement- (2021-22)

POs	Target Level	Attainment Level	Observations
PO 1 : Engineering Knowledge			
PO 1	2.1	2.29	Target is achieved, still some courses have scope for further improvements. These courses are: C104-Engineering Chemistry: 1.24; The level is not attained because the students' performance was poor in Non-Conventional Energy Sources, Electro chemical cell and series, Carbon nano tubes and fullerenes, Reverse Osmosis and Electro Dialysis. C325-Ship System Engineering: 1.01; The level is not attained because the students' performance was poor in Conventional propulsion system and Sewage Treatment Plant.
C104-Engineering Chemistry: Action1: Extra tutorial classes will be conducted for slow learners to clarify their doubts. Action 2: Students will be given more assignments for better understanding. C325-Ship System Engineering: 1.01: Action1: Extra tutorial classes will be conducted for slow learners to clarify their doubts. Action 2: Guest lectures by industry experts will be arranged for the students to improve their learning.			
PO 2 : Problem Analysis			
PO 2	2.1	2.15	Target is achieved, still some courses have scope for further improvements. These courses are: C104-Engineering Chemistry: 1.06; The level is not attained because the students' performance was poor in Non-Conventional Energy Sources, Electro chemical cell and series, Carbon nano tubes and fullerenes, Reverse Osmosis and Electro Dialysis. C203-Marine Materials and Welding Technology: 1.22; The level is not attained because the students' performance was poor in Non Destructive Test, Residual Stress and Distortion mechanism.
C104-Engineering Chemistry: Action 1: Extra tutorial classes will be conducted for slow learners to clarify their doubts. Action 2: Students will be given more assignments for better understanding. C203-Marine Materials and welding technology: Action 1: Concepts will be explained with the help of demo/practical sessions. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts. PO3: Design/development of solutions: Design solutions for complex Engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.			
PO 3 : Design/development of Solutions			
PO 3	2.1	2.11	Target is achieved, still some courses have scope for further improvements. These courses are: C205-Strength of Materials: 1.32; The level is not attained because the students' performance was poor in Bulkhead analysis, Bending stress calculation and Working stress design. C206-Engineering Mechanics: 1.33; The level is not attained because the students' performance was poor in Varignon's Theorem, support and their reactions, Static, Dynamic Friction and Relative motion. C301- Biology for Engineers: 0.7; The level is not attained because students' performance was poor in Mendelian Genetics, Genetic code, Principles of energy transactions and Bio Sensors.
C205- Strength of Materials: Extra tutorial classes will be conducted for slow learners to clarify their doubts. Action 2: Students will be given more assignments for better understanding. C206-Engineering Mechanics: Action 1: Extra tutorial classes will be conducted for slow learners to clarify their doubts. Action 2: Assignments will be given to the students for more practice before the examination. C301- Biology for Engineers: Action 1: Concepts will be explained with the help of demo/practical sessions. Action 2: Assignments will be given to the students for more practice before the examination.			
PO 4 : Conduct Investigations of Complex Problems			
PO 4	2.1	2.02	Target is almost achieved and hence some courses have scope for further improvements. These courses are: C103-Engineering Mathematics-I: 1.31; The level is not attained because the students' performance was poor in diagonalization of matrices, and Orthogonal transformation, Limit continuity and partial derivatives and Integration of standard functions. C206-Engineering Mechanics: 1.14; The level is not attained because the students' performance was poor in Varignon's Theorem, support and their reactions, Static, Dynamic Friction and Relative motion. C301-Biology for Engineers: 0.7; The level is not attained because students' performance was poor in Mendelian Genetics, Genetic code, Principles of energy transactions and Bio Sensors.
C103-Engineering Mathematics-I: Action1: Assignments will be given to the students for more practice before the examination. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts. C206-Engineering Mechanics: Action 1: Students will be given more assignments on problems for better understanding. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts. C301-Biology for Engineers: Action 1: Concepts will be explained with the help of demo/practical sessions. Action 2: Assignments will be given to the students for more practice before the examination.			
PO 5 : Modern Tool Usage			

PO 5	2.1	2.10	Target is achieved, still some courses have scope for further improvements. These courses are: C201- Engineering Mathematics – III: 1.29; The level is not attained because the students' performance was poor in expansion of functions in Fourier series, – Lagrange's Linear First order equation and Nonlinear equations of First order. C205-Strength of Materials: 0.97; The level is not attained because the students' performance was poor in Bulkhead analysis, Bending stress calculation and Working stress design. C306-Wave Mechanics: 1.22; The level is not attained because students' performance was poor in Stokes wave theory, Cnoidal wave theory, Solitary wave theory, Wave forces: – Morison equation and Diffraction theory.
C201- Engineering Mathematics – III: 1.28; Action 1: Students will be given more assignments on problems for better understanding. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts. C205-Strength of Materials: Action 1: Students will be given more assignments on Strength of Materials problems for better understanding. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts. C306- Wave Mechanics: Action 1: Guest lecture and Industrial visit will be arranged for giving more practical exposure. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts.			

PO 6 : The Engineer and Society

PO 6	2.1	2.11	Target is achieved, still some courses have scope for further improvements. These courses are: C325-Ship Systems Engineering: 1.18; The level is not attained because the students' performance was poor in Conventional propulsion system and Sewage Treatment Plant. C410-Shipyard Practices and Project Management: 1.37; The level is not attained because the students' performance was poor in Shipyard Capacity estimation, Role of process and procedure in Shipyard Capacity evaluation, CPM & PERT, Scheduling and Resource planning and Primavera.
C325-Ship Systems Engineering: Action 1: Students will be given more assignments on for better understanding. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts. C410 - Shipyard Practices and Project Management: Action 1: Students will be given more assignments on problems for better understanding. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts.			

PO 7 : Environment and Sustainability

PO 7	2.1	2.10	Target is achieved, still some courses have scope for further improvements. These courses are: C104-Engineering Chemistry: 1.5; The level is not attained because the students' performance was poor in Non-Conventional Energy Sources, Electro chemical cell and series, Carbon nano tubes and fullerenes, Reverse Osmosis and Electro Dialysis C321-Fishing Vessel Technology: 1.3; The level is not attained because the students' performance was poor in analysis of fishing nets and Resistance & Propulsion calculations. C325-Ship Systems Engineering: 1.18; The level is not attained because the students' performance was poor in Conventional propulsion system and Sewage Treatment Plant.
C104-Engineering Chemistry: Action 1: Students will be given more assignments on problems for better understanding. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts. C321-Fishing Vessel Technology: Action1: Guest lecture and Industrial visit will be arranged for giving more practical exposure. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts. C325-Ship Systems Engineering: Action 1: Students will be given more assignments on for better understanding. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts.			

PO 8 : Ethics

PO 8	2.1	1.94	Target is almost achieved and hence some courses have scope for further improvements. These courses are: C205-Strength of Materials: 0.97; The level is not attained because the students' performance was poor in Bulkhead analysis, Bending stress calculation and Working stress design. C301-Biology for Engineers: 1.39; The level is not attained because the students' performance was poor in Mendelian Genetics, Genetic code, Principles of energy transactions and Bio Sensors.
C205 - Strength Of Materials: Action 1: Assignments will be given to the students on problems and equations to get more practice before the examination. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts. C301- Biology for Engineers: Action 1: Students will be given more assignments on problems for better understanding. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts.			

PO 9 : Individual and Team Work

PO 9	2.1	2.13	Target is achieved, still some courses have scope for further improvements. These courses are: C210-Strength of Materials Laboratory: 1.65; The level is not attained because the students' performance was poor in Rockwell's Hardness and Single or double shear test on M.S. bar to finding the resistance of material to shear load. C217-Ethics and Values: 1.64; The level is not attained because the students were unable to read vast theory. C410-Shipyard Practices and Project Management: 1.37; The level is not attained because the students' performance was poor in Shipyard Capacity estimation, Role of process and procedure in Shipyard Capacity evaluation, CPM & PERT, Scheduling and Resource planning and Primavera.
C210-Strength of Materials Laboratory: Action 1: Concepts will be explained with the help of demo sessions. Action 2: Extra practical classes will be conducted for slow learners to clarify their doubts. C217- Ethics And Values: Action 1: Action 1: Students will be given more assignments on problems for better understanding. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts. C410-Shipyard Practices and Project Management: Action 1: Students will be given more assignments on problems for better understanding. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts.			

PO 10 : Communication

PO 10	2.1	2.16	Target is achieved, still some courses have scope for further improvements. These courses are: C301-Biology for Engineers: 1.39; The level is not attained because the students' performance was poor in Mendelian Genetics, Genetic code, Principles of energy transactions and Bio Sensors. C313 - Ship Strength Lab:1.44; The level is not attained because the students' performance was poor in Scantling calculations and Drawing of Mid ship Section and Section Modulus Calculations of the ship as per Classification Rules. C331-Marine Hydrodynamics Laboratory: 1.28; The level is not attained because the students' performance was poor in determination of CG of the ship model, inclining experiment, and estimating the radius of gyration of a ship model.
C301- Biology For Engineers: Action 1 Action 1: Students will be given more assignments on problems for better understanding. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts. C313- Ship Strength Lab: Action 1: Extra practical classes will be conducted to the students for more practice. Action 2: Concepts will be explained with the help of demo sessions. C331-Marine Hydrodynamics Laboratory: Action 1: Students will be given more assignments for better understanding. Action 2: Extra tutorial classes will conducted for slow learners to clarify their doubts.			

PO 11 : Project Management and Finance

PO 11	2.1	1.82	Target is not achieved and hence some courses have scope for further improvements. These courses are: C325 - Ship Systems Engineering: 0.79; The level is not attained because the students' performance was poor in Conventional propulsion system and Sewage Treatment Plant. C217 - Ethics and Values: 1.64; The level is not attained because the students were unable to read vast theory.
C325 - Ship Systems Engineering: Action 1: Group discussion will be conducted to the students for more practice. Action 2: Individual paper presentation will be given to the students to improve the communication skill. C217- Ethics And Values: Action 1: Action 1: Students will be given more assignments on problems for better understanding. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts.			

PO 12 : Life-long Learning

PO 12	2.1	2.11	Target is achieved, still some courses have scope for further improvements. These courses are: C321 - Fishing Vessel Technology: 1.22; The level is not attained because the students' performance was poor in analysis of fishing nets and Resistance & Propulsion calculations. C325 - Ship Systems Engineering: 1.01; The level is not attained because the students' performance was poor in Conventional propulsion system and Sewage Treatment Plant.
C321- Fishing Vessel Technology: Action 1: Concepts will be explained with the help of demo/practical sessions. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts. C325-Ship Systems Engineering: Action 1: Extra tutorial classes will be conducted for slow learners to clarify their doubts. Action 2: Concepts will be explained with the help of demo/practical sessions.			

PSOs Attainment Levels and Actions for Improvement- (2021-22)

PSOs	Target Level	Attainment Level	Observations
PSO 1 : Develop innovative and contemporary designs of ships and offshore structures through advanced analysis and experiments to meet the challenges of marine operations in the shipbuilding, offshore, and other maritime industries.			
PSO 1	2.1	2.13	Target is achieved, still some courses have scope for further improvements. These courses are: C321 - Fishing Vessel Technology: 1.22; The level is not attained because the students' performance was poor in analysis of fishing nets and Resistance & Propulsion calculations. C306- Wave Mechanics: 1.59; The level is not attained because the students' performance was poor in Stokes wave theory, Cnoidal wave theory, Solitary wave theory, Wave forces: – Morison equation and Diffraction theory.
C321-Fishing Vessel Technology: Action 1: Value added courses and skill based trainings will be provided to the students with the help of industrial experts to improve their problem solving skills. Action 2: Industrial visits and Internships will be organized to get core industrial exposures and Ship visit will be arranged for the students to gain the knowledge of practical applications. C306- Wave Mechanics: Action 1: Guest lecture and Industrial visit will be arranged for giving more practical exposure. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts.			
PSO 2 : Implement engineering design and construction concepts to produce solutions in the field of Naval Architecture and Offshore Engineering.			
PSO 2	2.1	2.19	Target is achieved, still some courses have scope for further improvements. These courses are: C306- Wave Mechanics: 1.59; The level is not attained because the students' performance was poor in Stokes wave theory, Cnoidal wave theory, Solitary wave theory, Wave forces: – Morison equation and Diffraction theory. C317-Ship Motion Control: 1.56; The level is not attained because the students' performance was poor in Determination of hydrodynamic coefficients using Strip theory, Experimental determination of hydrodynamic derivatives and Estimation of maneuverability in ship design.
C306- Wave Mechanics: Action 1: Guest lecture and Industrial visit will be arranged for giving more practical exposure. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts. C317-Ship Motion Control: Action 1: Extra tutorial classes will be conducted for slow learners to clarify their doubts. Action 2: Concepts will be explained with the help of demo/practical sessions.			
PSO 3 : Apply research based knowledge and research methods to analyse and interpret data using modern tools towards the solution of complex problems in the area of Naval Architecture and Offshore Engineering.			
PSO 3	2.1	2.12	Target is achieved, still some courses have scope for further improvements. These courses are: C206-Engineering Mechanics: 1.14; The level is not attained because the students' performance was poor in Varignon's Theorem, support and their reactions, Static, Dynamic Friction and Relative motion. C306-Wave Mechanics: 1.59; The level is not attained because the students' performance was poor in Stokes wave theory, Cnoidal wave theory, Solitary wave theory, Wave forces: – Morison equation and Diffraction theory.
C206-Engineering Mechanics: Action 1: Concepts will be explained with the help of demo/practical sessions. Action 2: Students will be encouraged to organize and participate in technical events to know about latest technology. C306- Wave Mechanics: Action 1: Guest lecture and Industrial visit will be arranged for giving more practical exposure. Action 2: Extra tutorial classes will be conducted for slow learners to clarify their doubts.			

Academic Auditing

Periodical academic audits are conducted based on Teaching-Learning Process. The audits are then evaluated by the Department level and Institutional level committees. Generally, the academic audit comprising of internal and external audits.

Objectives of Academic Auditing

- To ensure that Teaching-Learning process is carried out effectively in the department because of which, there is a measurable improvement in the POs and COs attainment.
- To maintain the quality of teaching learning process, the evaluation approaches (with specific focus on question papers) is carried out in the department based on Bloom's taxonomy.
- Sample based screening of evaluation of CAT answer scripts thereby ensures the maintenance of quality in the teaching-learning process.

The auditing team verifies the following components during the visit of audit.

- Teaching-Learning process
- Quality of teaching
- Effective utilization of laboratory
- Quality of internal/ external assessment
- Quality of students project
- Co-curricular activities:
 - Seminar
 - Guest lecture
 - Conference
 - Workshop
 - Faculty development programme
 - Value added course
 - Internship
 - In-plant training
 - Industrial visit
- Research activities: publications, patent, proposal
- Review of faculty members
- Feedback from the students

The institution has an Internal Quality Assurance Cell (IQAC) to monitor and maintain the standard of academicians of the University. The cell is chaired by the Vice-Chancellor and the activities of the cell are being planned, executed and coordinated by a Senior Professor with the assistance of representative members of faculty from each department. The details of IQAC is provided on institute website www.ametuniv.ac.in with various actions taken.

Performance evaluation of Internal Quality Assurance Cell (IQAC)

- Improvement in quality of teaching and research by regular inputs to all concerned based on feedback from students.
- Providing inputs for best practices in administration for efficient resource utilization and better services to students and staff.
- Providing inputs for Academic and Administrative Audit (AAA) and analysis of results for improvement in areas.

The flow chart for academic audit is given in the figure below.

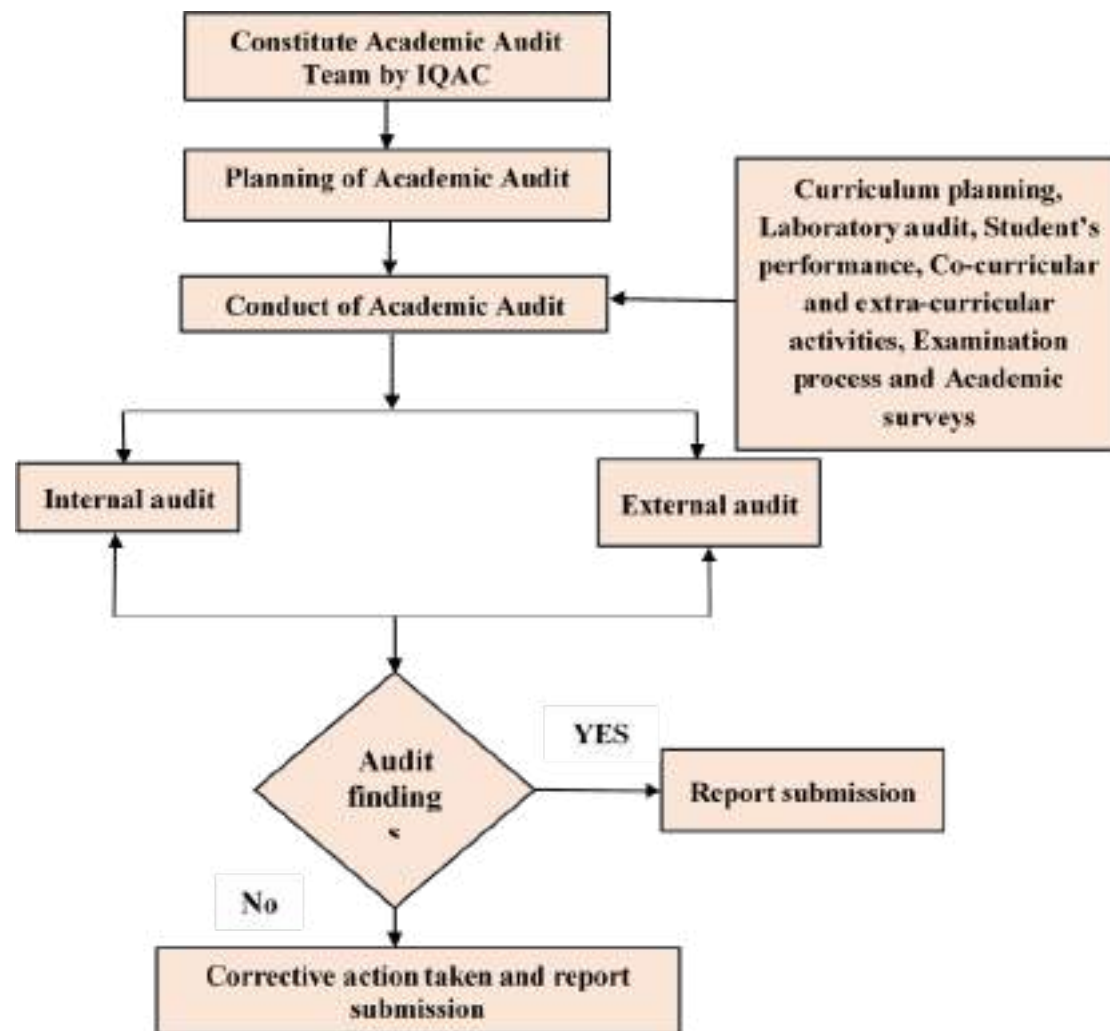


Figure 7.1 Flow chart for academic audit

The following Table shows the Academic and Administrative Audit conducted in the Department for the three academic years.

Table B.7.2: Internal and External Experts audited AAA activities of Department of NA&OE

Academic Year	Auditee	Auditor
2019 - 2020	Mr.MSP.Raju Head of the Department (in charge), Department of Naval Architecture and Offshore Engineering, AMET (Deemed to be University).	Prof.R. Natrajan(Retd) Professor, Department of Ocean Engineering IIT Madras, Chennai.
2020 - 2021	Mr.MSP.Raju Head of the Department (in charge), Department of Naval Architecture and Offshore Engineering, AMET (Deemed to be University).	Dr .Mathiazhagan A, Professor, Cochin University of Science and Technology, Cochin.
2021 - 2022	Mr.MSP.Raju Head of the Department (incharge), Department of Naval Architecture and Offshore Engineering, AMET (Deemed to be University).	Dr.Rajiv Sharma, Professor, Department of Ocean Engineering, IIT Madras, Chennai.

Outcome of the audit shall be placed before the Internal Quality Assurance Cell (IQAC) and the Governing Bodies of the institution. Plan of action will be prepared to implement the suggestions accepted by IQAC and Governing Bodies.

The following guidelines for practicing Academic Audit for continuous monitoring of all activities is given in the below Table 7.2.

Table B 7.2 (a) List of Audit elements

S. No	Academic Activities	Associated practices	Responsible person to carry out audit	Frequency of Audit
1	Curriculum planning and execution	- Adherence to academic Calendar - Deviations if any and corrective measures	Head of the Department	Once in a semester

2	Course Files	• Lesson Plan and Execution • Lecture materials • List of assignments / Tutorial sessions/Seminar Topics/Active Learning Methods • Attendance Register • Question Papers • Sample assignments / tutorials • Samples of answer scripts for internal Examination • Guest lectures conducted • CO-PO Attainment Document • Overall result analysis for Internal and External Examinations • Identification of slow and advance learners. • Activities for slow and advance learners.	Head of the Department Course File In charge	Once in a Semester
3	Teaching learning pedagogy	• Flipped Class activity • Industrial Visit • Model demonstration • Simulation • Project based learning for laboratory course	Head of the Department	Once in a Semester
4	Laboratory Audit	• Laboratory manuals • Functioning Status of Equipment • Working Status of the Software • Student's laboratory Records	Head of the Department Lab-In-Charges, Senior Faculty	Once in a Semester
5	Outcome based practices	• Question papers as per COs with Blooms Taxonomy Levels. • Timely conduct of Class Committee meeting. • Timely conduct of PAC meeting.	Head of the Department Review Committee/PAC	Once in a Semester
6	Students performance	• Performance analysis of students in Continuous Assessment Tests • Performance analysis and Attendance Reports of each student	Head of the Department Mentors	Twice in a Semester
7	Attendance monitoring	• Communication to parents regarding shortage of attendance. • Actions for improving the attendance of students.	Mentors	Once in a month
8	Co-curricular and extra-curricular activities	• Events organized by the Department. • Students participation in (a) Conferences (b) Workshops (c) Seminars. • Industrial visits.	Head of the Department	Once in a Semester
9	Student Counselling	• Mentoring the students. • Counselling sessions	Mentors Student counsellor of the University	Twice in a semester
10	Academic surveys	• Program Exit Survey • Alumni survey • Employer survey	Head of the Department	Once in a Year
11	Feedback system	• Collection of feedbacks from the students. • Analysis of the feedback and corrective actions.	Head of the Department	Once in a Semester

12	Good practices	Program Assessment Committee (PAC) for identifying the gaps between the curriculum and recent trends in the industry and recommends the BoS to make appropriate changes in the curriculum through the Head of the Department	Head of the Department PAC	Once in a Semester
13	Industry linkage	Continuous interaction with various industries for Collaborative programs and Value added courses.	Head of the Department	Once in a Year

Table B 7.2(b) List of Audit Reports and Corrective actions

S. No	Academic Activities	Associated practices	Audit Reports	Corrective Actions
1	Curriculum planning and execution	Adherence to academic calendar and Course File preparation. All these files are verified and approved by the Head of the Department.	Course files and lesson plans are prepared before the commencement of the new semester by the respective subject faculty. Quality of assignment, tutorials and quiz programmes are checked by PAC.	Necessary action is taken and re-auditing will be done within a weak time for the faculty members who are not able to comply with standards.
2	Laboratory audit	Lab manuals and Evaluation scheme	Laboratory manuals and rubrics based evaluation scheme are prepared prior to the commencement of the semester by respective faculty. All these files are verified PAC and approved by the Head of the Department.	
		Student lab records	Sample records are checked by Audit members	
		Status of Equipment and Software	Laboratory audit is done once in a semester by Audit Team Members and check whether the regular maintenance is done by Lab in-charges and lab technicians periodically and verify any shortfall in equipment/ up gradation of software and also verify the working status of the equipment/software.	It is suggested by Audit members to ensure the faculty to maintain the quality in the lab records.
3	Student performance	Mark statements are collected from the faculty members and the results are analysed.	Audit team checks the result analysis CO-PO attainment and checks the corrective measures taken.	If any deviations found, that will be brought to the attention of Head of the Department and PAC members to monitor effectively
		Identify the students learning capacity based on previous semester results/assessment test/class performance.	Audit team check the list of slow learners and advance learners and also checks whether the selection criterion is followed properly and also verify whether the remedial classes are conducted properly	
		Improve the performance of slow learners and encourage Advanced learners.	Audit team checks the performance of slow learners and advanced learners in the End Semester Examination	
5	Co-curricular and Extra-curricular activities	The data of students who participated in seminars, workshops, symposiums inside and outside the campus, is recorded.	Audit team checks data provided	Suggested for more participation in the seminars, workshops, conferences and workshops and also suggested to involve students to take part in organising workshops and international conference
		The industrial visits are planned in order to provide exposure on real time industry working environment.		Suggested for more industrial visits

7	Examination Process	After the internal examinations, answer sheets are evaluated and marks are submitted to the Head of the Department by the respective subject faculty. Results are analysed.	Audit team checks the continuous improvement in the performance of the students by comparing the previous academic year results	If found any deviation in the improvement, the particular faculty will be counselled by the Head of the Department and the performance of the particular faculty will be monitored.
		University results are reviewed by Head of the Department.		
8	Academic Surveys	Student exit survey, Alumni survey and Employer survey forms are collected and aggregated by the Head of the Department for PO attainment.	The audit team verify the collected survey and analysis for attainment of PO.	Corrective measures are taken for the improvement of PO attainment analysis for forthcoming batch.

7.3 Improvement in Placement, Higher Studies and Entrepreneurship (10)

Total Marks 10.00

Training and Placement

Intensive placement training is offered only to those who have opted for campus recruitment. The campus recruitment comes under the purview of the placement director. The placement director is assisted by a faculty member from the department who act as placement coordinator. Placement coordinators plans and coordinates the activities related to placement training and other activities related to career guidance. The intensive training for campus recruitment covers the following aspects.

- General and math aptitude tests
- Communication skills enhancement (mandatory in I & II years)
- Technical aptitude tests during the III year
- Soft skills training including group discussions and mock interviews

The student statistics for employment, higher studies and entrepreneurship are given in the table below.

Table 7.3(a) The student statistics for employment, higher studies and entrepreneur

Item	CAY m1 (2021-22)	CAY m2 (2020-21)	CAY m3 (2019-20)
Total no. of final year students (N)	50	72	86
No. of students placed in companies or government Sector (x)	40	58	65
No. of students admitted to higher studies with valid qualifying scores (GATE or equivalent State or National Level Tests, GRE, GMAT etc.) (y)	2	5	6
No. of students turned entrepreneur in engineering/technology (z)	1	1	0
$x + y + z =$	43	64	71
Placement Index : $(x + y + z)/N$	0.86	0.89	0.83
Average placement= $(P1 + P2 + P3)/3$	0.85		
Assessment = $30 \times$ average placement	25.74		

Table 7.3 (b) Percentage of students got placement in Core Company

Item	CAY m1 (2021-22)	CAY m2 (2020-21)	CAY m3 (2019-20)
Total no. of final year students (N)	50	72	86
No. of students placed in companies	40	58	65
% of students placed in core companies	100	100	100

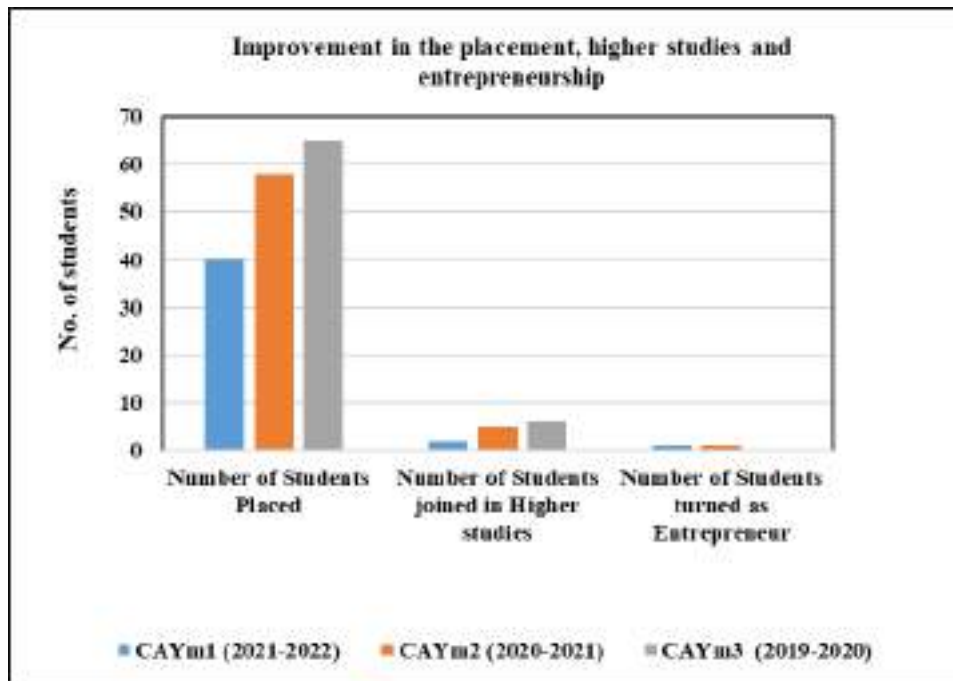


Figure 7.3(a) Improvement in Placement, Higher Studies and Entrepreneurship

7.4 Improvement in the quality of students admitted to the program (20)

Total Marks 20.00

Item		2022-23	2021-22	2020-21
National Level Entrance Examination	No of students admitted	0	0	0
	Opening Score/Rank	0	0	0
	Closing Score/Rank	0	0	0
State/ University/ Level Entrance Examination/ Others AMET CET	No of students admitted	44	28	20
	Opening Score/Rank	83	87.4	92
	Closing Score/Rank	60	50	43
Name of the Entrance Examination for Lateral Entry or lateral entry details	No of students admitted	0	4	8
	Opening Score/Rank	0	86.2	82.9
	Closing Score/Rank	0	78.6	70
Average CBSE/Any other board result of admitted students(Physics, Chemistry&Maths)		71.5	75.5	71.97

8 FIRST YEAR ACADEMICS (50)

Total Marks 47.38

8.1 First Year Student-Faculty Ratio (FYSFR) (5)

Total Marks 5.00

Please provide First year faculty information considering load

Name of the faculty member	PAN No.	Qualification	Date of Receiving Highest Degree	Area of Specialization	Designation	Date of joining	Teaching load (%)			Currently Associated (Yes / No)	Nature Of Association (Regular / Contract)	Date Of leaving(In case Currently Associated is 'No')
							CAY	CAYm1	CAYm2			
Dr.S.Meher Taj	GTHPS8759E	M.Sc. and PhD	20/12/2020	Mathematics	Assistant Professor	09/08/2010	100	100	100	Yes	Regular	
Dr.M.Suresh K	BBCPS8597Q	M.Sc. and PhD	24/07/2021	Physics	Assistant Professor	01/08/2008	100	100	100	No	Regular	30/01/2023
Dr.K.Manigand	BXNPM9080M	M.Sc. and PhD	08/10/2021	English	Assistant Professor	01/08/2011	100	100	100	Yes	Regular	
Dr.Anita Warrie	ABYPW0832H	M.Sc. and PhD	10/10/2011	Physics	Assistant Professor	15/10/2014	100	100	100	Yes	Regular	
Dr.K.Anandan	BSUPA4846D	M.Sc. and PhD	17/04/2015	Physics	Assistant Professor	14/09/2015	100	100	100	Yes	Regular	
Dr.L. Senthilna	CHZPS6906B	M.Sc. and PhD	26/02/2016	Environmental Science	Assistant Professor	19/10/2015	100	100	100	Yes	Regular	
Dr.I.Paulraj Jay	BTRPP2139Q	M.Sc. and PhD	09/06/2017	Mathematics	Associate Professor	04/07/2016	100	100	100	Yes	Regular	
Dr.K. Rajesh	CNGPR3372F	M.Sc. and PhD	11/04/2015	Physics	Assistant Professor	04/07/2016	100	100	100	Yes	Regular	
Dr.P. Balagane	AFIPB8703L	M.Sc. and PhD	22/07/2017	Mathematics	Associate Professor	12/07/2017	100	100	100	Yes	Regular	
Dr. N.R.Ramku	AKJPR5949J	M.A and Ph.D	25/07/2012	Physical Education	Professor	17/08/2016	100	100	100	Yes	Regular	
Dr. A.Manoj Ku	CYOPM1400Q	M.A and Ph.D	25/06/2012	Physical Education	Assistant Professor	10/08/2012	0	0	100	No	Regular	29/05/2021
Dr.K.Gayathri	BQSPG5031Q	M.Sc. and PhD	14/03/2015	Physics	Assistant Professor	12/07/2017	100	100	100	Yes	Regular	
Dr. T.Dhivya	DYNPD9911F	M.Sc. and PhD	22/08/2022	Mathematics	Assistant Professor	17/07/2017	100	100	100	Yes	Regular	
Dr.T.R. Heera	AIHPH6816Q	M.Sc. and PhD	04/02/2010	Physical Chemistry	Assistant Professor	26/07/2017	100	100	100	Yes	Regular	
Dr.C.Vairavan	AHJPV8538J	M.A and Ph.D	19/04/2017	English	Assistant Professor	03/10/2017	100	100	100	Yes	Regular	
Dr.D.Jaisankar	AJIPJ7831K	M.A and Ph.D	17/10/2018	English	Associate Professor	06/08/2018	100	100	100	Yes	Regular	
Dr.K.K.Siva Ku	ALEPK0825G	M.Sc. and PhD	04/10/2013	Physical Chemistry	Professor	01/07/2019	100	100	100	Yes	Regular	
Dr.S.Rafi Aharr	AURPA4279J	M.Sc. and PhD	03/10/2017	Physics	Associate Professor	01/07/2019	100	100	100	Yes	Regular	
Dr.A.Mohamed	AWYPM1495R	M.Sc. and PhD	12/06/2017	Physics	Associate Professor	01/07/2019	100	100	100	Yes	Regular	
Dr.M.Subha	BGZPS7475R	M.A and Ph.D	17/10/2018	English	Assistant Professor	23/10/2018	100	100	100	Yes	Regular	

Dr.N.Maheswa	BWJPM7256A	M.Sc. and PhD	05/04/2019	Physics	Assistant Professor	24/07/2019	100	100	100	Yes	Regular	
Mrs.V.Ramabh	FAQPR4767C	M.E/M.Tech	15/06/2014	Environmental Science	Assistant Professor	04/01/2020	100	100	100	Yes	Regular	
Mr. V. Kurinji	CZBPK2115E	M.E/M.Tech	14/11/2006	Mechanical Engineering	Assistant Professor	02/01/2018	0	0	100	No	Regular	31/12/2020
Dr.Jayakumar	ARQPJ2405H	ME/M. Tech and PhD	10/01/2020	Mechanical Engineering	Assistant Professor	16/07/2018	100	100	0	Yes	Regular	
Mr.B.Prabhush	AOHPP4187C	M.E/M.Tech	17/04/2010	Computer Science and Engineering	Assistant Professor	16/08/2018	0	0	100	No	Regular	25/07/2021
Ms.P. Indhuma	AFEPI0418R	M.Phil	02/02/2017	English	Assistant Professor	22/10/2018	0	0	100	No	Regular	31/12/2020
Dr. A. Ananthi	AHZPA2377A	ME/M. Tech and PhD	16/03/2016	Electrical Engineering	Professor	01/02/2023	100	0	0	Yes	Regular	
Dr.K.Komathy	AGPPK9366C	ME/M. Tech and PhD	02/12/2008	Computer Science Engineering	Professor	15/05/2017	100	100	100	No	Regular	02/02/2023
Mr.K.Dinesh K	AKMPD5383K	M.E/M.Tech	30/04/2013	Computer Science and Engineering	Assistant Professor	16/07/2018	100	100	100	No	Regular	30/08/2022
Dr.R.Suresh	EETPS1447N	M.A and Ph.D	20/06/2015	Physical Education	Assistant Professor	09/03/2022	100	100	0	Yes	Regular	
Ms. S. Sowmiy	IRHPS1180M	M.E/M.Tech	09/05/2018	Mechanical Engineering	Assistant Professor	26/02/2020	0	100	100	No	Regular	30/12/2021
Dr. Selvakuma	DPPPS7068P	M.Sc. and PhD	25/05/2013	Environmental Science	Assistant Professor	21/12/2020	0	100	100	No	Regular	28/02/2022
Dr.S.Arjunan	AEGPA0377L	M.Sc. and PhD	02/09/2008	Physics	Professor	10/07/2019	0	100	100	No	Regular	30/08/2022
Dr.S.Kalpana	AXFPK2107E	M.Sc. and PhD	24/05/2019	Physics	Associate Professor	27/08/2019	0	100	100	No	Regular	05/08/2022
Dr.M.Sudha	DZEPS1084N	M.Sc. and PhD	18/12/2020	Mathematics	Assistant Professor	02/12/2021	100	100	0	No	Regular	17/12/2022
Dr.C.M.Ramak	AOVPR7013P	M.Sc. and PhD	04/03/1994	Environmental Science	Professor	13/05/2022	100	0	0	Yes	Regular	
Dr.C.Chitra	CJYPC4507M	M.Sc. and PhD	09/10/2015	Environmental Science	Assistant Professor	06/10/2022	100	0	0	Yes	Regular	
Dr. K. Bharathi	APJPB2323R	ME/M. Tech and PhD	25/02/2022	Electrical Engineering	Associate Professor	30/01/2023	100	0	0	Yes	Regular	
Dr. W. Sylveste	BPVPS8466R	M.Sc. and PhD	07/03/2014	Environmental Science	Assistant Professor	15/11/2017	0	0	100	No	Regular	25/02/2021
Dr.G. Mayakka	CNZPM0314H	M.Sc. and PhD	17/08/2013	Environmental Science	Assistant Professor	10/03/2021	0	100	100	No	Regular	28/02/2022
Mr.S.Ramaraju	ALFPR7893C	M.Phil	01/08/2002	English	Assistant Professor	12/07/2022	100	0	0	Yes	Regular	
Mr.Balamuruga	AYQPB1947F	M.Phil	02/07/2012	English	Assistant Professor	16/07/2018	100	100	100	Yes	Regular	

Dr.Anitha Rexa	AQGPA6291D	M.Sc. and PhD	30/08/2014	Physics	Assistant Professor	24/09/2022	100	0	0	Yes	Regular	
Mrs. R.K. Padr	BZYPP2672P	M.E/M.Tech	07/07/2008	Electrical Engineering	Assistant Professor	06/08/2018	100	100	100	Yes	Regular	
Mr.P.Ramesh	AWXPR8154P	M.E/M.Tech	28/04/2014	Mechanical Engineering	Assistant Professor	01/07/2019	0	0	100	No	Regular	10/01/2023
Dr.C.Dhanalak	BGNPD9345R	M.A and Ph.D	18/03/2022	Physical Education	Assistant Professor	12/10/2022	100	0	0	Yes	Regular	
Ms.Shanmuga	EGDPS4302E	M.E/M.Tech	01/06/2016	Environmental Science	Assistant Professor	05/02/2021	100	100	0	Yes	Regular	
Dr.G.Jenitha	BSVPG7440N	M.Sc. and PhD	24/01/2022	Mathematics	Assistant Professor	04/09/2014	100	100	100	Yes	Regular	
Dr. N.Sivakum	GOOPS5559Q	M.Sc. and PhD	27/07/2012	Mathematics	Assistant Professor	28/10/2014	100	100	100	Yes	Regular	
Dr.Altaff Karee	AAEPA4302R	M.Sc. and PhD	10/05/1990	Environmental Science	Professor	13/09/2017	100	100	100	Yes	Regular	
Dr.L.Rajendran	ABUPR6656R	M.Sc. and PhD	28/01/2012	Mathematics	Professor	21/05/2018	100	100	100	Yes	Regular	
Ms.U. Pushpal	AWCPP7334G	M.Phil	05/03/2011	English	Assistant Professor	01/11/2018	100	100	100	Yes	Regular	
Dr. S. Krishna	KFCPK0006M	M.A and Ph.D	18/09/2018	English	Assistant Professor	24/02/2020	100	100	100	Yes	Regular	
Mr. R. Karthick	BMOPK2366E	M.E/M.Tech	12/04/2012	Electrical Engineering	Assistant Professor	24/02/2020	0	0	100	Yes	Regular	
Mr. J. S. Ashwi	AZXPA0303Q	M.E/M.Tech	01/06/2014	Electrical Engineering	Assistant Professor	24/02/2020	100	100	100	Yes	Regular	

Year	Number Of Students(approved intake strength) N	Number of Faculty members(considering fractional load) F	FYSFR (N/F)	*Assessment=(5*20)/FYSFR(Limited to Max.5)
2020-21(CAYm2)	606	44	14	5
2021-22(CAYm1)	558	41	14	5
2022-23(CAY)	618	43	14	5
Average	594	42	14	5

AverageFYSFR: 0.00

Assessment [(5 * 15) / AverageFYSFR]: 5.00

8.2 Qualification of Faculty Teaching First Year Common Courses (5)

Total Marks 5.00

Year	x (Number Of Regular Faculty with Ph.D)	y (Number Of Regular Faculty with Post graduate Qualification)	RF (Number Of Faculty Members required as per SFR of 20:1)	Assessment Of Faculty Qualification [(5x + 3y) / RF]
2020-21	24	10	30	5.00
2021-22	26	7	27	5.00
2022-23	27	7	30	5.00

Average Assessment: 5.00

8.3 First Year Academic Performance (10)

Total Marks 7.38

Institute Marks : 7.38

Academic Performance	CAYm1(2021-22)	CAYm2(2020-21)	CAYm3 (2019-20)
Mean of CGPA or mean percentage of all successful students(X)	6.77	7.61	7.76
Total Number of successful students(Y)	27.00	17.00	33.00
Total Number of students appeared in the examination(Z)	27.00	17.00	33.00
API [X*(Y/Z)]	6.77	7.61	7.76

Average API[(AP1+AP2+AP3)/3] : 7.38

Assessment = Average API : 7.38

8.4 Attainment of Course Outcomes of first year courses (10)

Total Marks 10.00

8.4.1 Describe the assessment processes used to gather the data upon which the evaluation of Course Outcomes of first year is done (5)

Institute Marks : 5.00

CO Assessment Processes**A] Assessment process**

In the Outcome Based Education, evaluation of Course Outcome is done through internal and external assessment processes such as CAT1, CAT2, Model Examination, End Semester Examination, Assignments and Seminar or any Activity Learning Methods. Direct assessment includes evaluation of the student's knowledge and skills based on their performance in the internal and external assessments and indirect assessment is done through Course End Survey.

B] Assessment Tools: -

The Direct assessment tools which are used for evaluation is shown in Table B 8.4.1 (a) and the Indirect assessment tool of evaluation is shown in Table B 8.4.1 (b).

Table: 8.4.1 (a) Direct Assessment Tools

Assessment Process	Frequency	Process
DIRECT ASSESSMENT TOOLS		
THEORY		
Continuous Assessment Test (CAT)	Two per semester	The internal assessments are carried out in a centralized manner in the department level and are so designed to facilitate the assessment of the attainments of the outcome.
Model Exam	One per semester	The Model examination is of three hour duration and covers the entire syllabus of the course and satisfies all course outcomes for the particular course
End Semester Examination	As per the End Semester Exam schedule	The end-semester examination is of three hour duration and covers the entire syllabus of the course and satisfies all course outcomes for the particular course.
Assignments	Two/three Assignments per course	To develop the critical and analytical skills of the students, assignments are given and is also taken as a criterion to assess the attainment of Cos
Seminar/ALM	One per Course	In view of improving communication skills, gaining expert knowledge, networking with others and renewing motivation and confidence, seminars are presented by students and is also taken as a criterion to assess the attainment of COs
LABORATORY COURSES		
Performance Evaluation	Continuous Monitoring	Continuous evaluation of laboratory courses as per rubrics
End Semester Exam	As per the End Semester Examination schedule	The end-semester examination is of three hours duration and covers all the experiments of the course and satisfies all course outcomes.
Communicative and Soft Skills Laboratory/English Laboratory	Throughout the semester	Assessment process includes interactions with questions structured properly and analyzing student's communication and writing abilities
PT/Parade/Games	Continuous Monitoring	Student assessment is done in alignment with University Standards and Rubrics.

Table B: 8.4.1 (b) Indirect Assessment Tool

S.No	Assessment Tool	Frequency	Method Description
------	-----------------	-----------	--------------------

1	Course End Survey	End of semester	To enhance the Teaching Learning Process, Students are provided with an opportunity to reflect and provide feedback based on their understanding level.
---	-------------------	-----------------	---

CO Assessment Process for Theory Courses:**Table B 8.4.1. (c) CO Assessment Process for Theory Courses**

Assessment		CO1	CO2	CO3	CO4	CO5	CO6
	IA (CAT-I, CAT-II, Model exam, Assignments and Seminar/ALM)	40%	40%	40%	40%	40%	40%
	ESE	60%	60%	60%	60%	60%	60%
	Total Direct assessment (DA)	100%	100%	100%	100%	100%	100%
DA (90%) = DA x 0.9		90%	90%	90%	90%	90%	90%
INDA = CES x 0.1		10%	10%	10%	10%	10%	10%
CO attainment % = DA + INDA		100%	100%	100%	100%	100%	100%
IA : Internal Assessment			ESE : End Semester Examination				
CAT I : Continuous Assessment Test -I			CAT II : Continuous Assessment Test –II				
DA : Direct Assessment			INDA : Indirect Assessment				
CES : Course End Survey							
ALM : Activity Learning Method							

CO Assessment Process for Laboratory Courses:**Table B 8.4.1 (d) CO Assessment for Laboratory Courses**

Assessment		CO1	CO2	CO3	CO4	CO5	CO6
	IA	40%	40%	40%	40%	40%	40%
	(Observation+ Performance Evaluation+ Record)						
	ESE	60%	60%	60%	60%	60%	60%
	Total Direct Assessment (DA)	100%	100%	100%	100%	100%	100%
DA (90%) = TDA x 0.9		90%	90%	90%	90%	90%	90%
INDA = CES x 0.1		10%	10%	10%	10%	10%	10%
CO attainment % = DA + INDA		100%	100%	100%	100%	100%	100%
IA : Internal Assessment ESE: : End Semester Examination							
CES : Course End Survey DA : Direct Assessment							
INDA : Indirect Assessment CES : Course End Survey							

8.4.2 Record the attainment of Course Outcomes of all first year courses (5)

Institute Marks : 5.00

Criteria for Attainment Level of Course Outcomes:**Table B: 8.4.2 (a) The Criteria for Attainment Levels of Course Outcomes**

ATTAINMENT LEVEL FOR COURSE OUTCOMES	
Theory Courses & Laboratory Courses	
Overall Attainment Level (Direct assessment + Indirect Assessment)	Level
50% - 60% students scoring more than the subject threshold	1
>60% - 70% students scoring more than the subject threshold	2
> 70% of students scoring more than the subject threshold	3

Table B 8.4.2 (b) gives the Course outcome statements for a sample course 'Engineering Mathematics - I (C103)' and Table B 8.4.2 (c) gives the CO articulation matrix for a sample course 'Engineering Mathematics - I (C103)'.

Table B: 8.4.2 (b) Course Outcome Statements for the Course**Engineering Mathematics - I (C103)**

CO No.	CO Statement	BTL
CO1	Apply the theorems of matrices to solve linear equations.	K3
CO2	Solve problems using vector calculus.	K3
CO3	Apply the theorems and formulae for solving problems in differential calculus.	K3
CO4	Classify the functions of several variables.	K2
CO5	Apply integral calculus on engineering problems	K3
CO6	Demonstrate the concepts of Calculus matrices for engineering applications	K2

Table B: 8.4.2 (c) CO Articulation matrix for the Course**Engineering Mathematics - I (C103)**

CO No.	PROGRAM OUTCOMES											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2	1	-	-	-	-	-	-	2
CO2	3	3	2	2	2	-	-	-	-	-	-	2
CO3	3	3	2	2	1	-	-	-	-	-	-	2
CO4	3	3	3	2	2	-	-	-	-	-	-	2
CO5	3	3	2	2	2	-	-	-	-	-	-	2

CO6	3	3	3	2	2	-	-	-	-	-	-	2
Avg	3.00	3.00	2.50	2.00	1.67	0	0	0	0	0	0	2.00

Indirect Assessment Process for Course Outcomes:

Course end survey is an integral part of assessment process. Surveys are conducted for all courses as a step towards quality measure. The survey includes a questionnaire set for each course outcome. This system gives a measure of the program outcomes attained indirectly which forms an essential element to improvise on course goals. The strategy by which indirect assessment is carried out is shown in Table 8.4.2 (b) and the Overall Assessment Process in Table 8.4.2 (c) for a sample course 'Engineering Mathematics - I (C103)'. The average of the course end survey in the overall attainment calculation of course outcomes is 10%.

Table B: 8.4.2 (b): Indirect Assessment Process for Course Outcomes**(Engineering Mathematics I - C103)**

Indirect Course Outcomes	CO1	CO2	CO3	CO4	CO5	CO6
Number of Students Answered - 1 (Low)	2	4	3	1	3	2
Number of Students Answered - 2 (Medium)	11	10	11	10	10	13
Number of Students Answered - 3 (High)	14	13	13	16	14	12
Total Number of Students Participated	27	27	27	27	27	27
Total Number of Students Answered 2 & 3	25	23	24	26	24	25
Indirect CO Attainment Percentage	93%	85%	89%	96%	89%	93%

Table B: 8.4.2 (c): The Overall CO Attainment Process for the Course**(Engineering Mathematics I - C103)**

S.No	Process for CO attainment	CO attainment					
		CO1	CO2	CO3	CO4	CO5	CO6
1	% Of Direct Attainment Through Internal Exams(CAT I & II, Model exam, Assignment I,II & ALM)	92.59%	59.26%	92.59%	55.56%	55.56%	51.85%
2	40% From Direct Attainment Through Internal Exams	37.04%	23.70%	37.04%	22.22%	22.22%	20.74%
3	% Of Direct Attainment Through End Semester Exam	88.89%	62.96%	59.26%	77.78%	81.48%	77.78%
4	60% From Direct Attainment Through End Semester Exam	53.33%	37.78%	35.56%	46.67%	48.89%	46.67%
5	40% + 60% Of Direct Attainment	90.37%	61.48%	72.59%	68.89%	71.11%	67.41%
6	90% From Direct Attainment	81.33%	55.33%	65.33%	62.00%	64.00%	60.67%
7	% Of Indirect Attainment Through Course End Survey	92.59%	85.19%	88.89%	96.30%	88.89%	92.59%
8	10% From Indirect Attainment Through Course End Survey	9.26%	8.52%	8.89%	9.63%	8.89%	9.26%
9	90% From Direct + 10% From Indirect For Co Attainment	90.59%	63.85%	74.22%	71.63%	72.89%	69.93%
10	Attainment Level	3	2	3	3	3	2

CO Attainment for First Year Courses (Academic Year: 2021-2022).

Based on the tools and processes described in section 8.4.1, the assessment of COs is carried out for all the first-year courses of B.E-Naval Architecture and Offshore Engineering. Table 8.4.2(d) gives the attainment of course outcomes of first year courses of B.E-Naval Architecture and Offshore Engineering for the academic year 2021-22.

Table B: 8.4.2. (d): The CO Attainment of First Year Courses of B.E-Naval Architecture and Offshore Engineering

FIRST YEAR	Theory Courses (% of students crossing threshold)									
Course Outcome s	CIO1	C102	CIO3	CIO4	C109	C110	C111	C112	C113	C114
CO1	77.56%	90.59%	90.59%	92.30%	70.59%	71.56%	85.93%	61.63%	71.26%	91.93%
CO2	78.59%	80.96%	63.85%	91.93%	70.96%	70.52%	77.93%	70.89%	68.89%	79.93%
CO3	80.52%	85.19%	74.22%	87.56%	61.85%	72.96%	81.56%	65.85%	77.93%	79.56%
CO4	81.56%	81.56%	71.63%	81.26%	61.93%	68.59%	84.30%	61.19%	71.56%	82.96%
CO5	81.63%	77.63%	72.89%	70.96%	70.89%	62.22%	84.89%	72.59%	73.63%	78.22%
CO6	81.26%	83.93%	69.93%	84.52%	78.89%	71.93%	82.52%	63.85%	70.52%	84.30%
Overall CO attainment	80.19%	83.31%	73.85%	84.75%	69.19%	69.63%	82.85%	66.00%	72.30%	82.81%
ATT LEVEL	3	3	3	3	2	2	3	2	3	3
FIRST YEAR	Laboratory Courses (% of students crossing threshold)									
Course Outcome s	C105	C106	C107	C108	C115	C116	C117	C118		
CO1	75.26%	73.93%	99.26%	92.89%	86.30%	85.56%	90.89%	66.89%		
CO2	73.56%	76.95%	99.63%	87.93%	81.19%	82.52%	91.26%	65.63%		
CO3	74.30%	73.19%	98.89%	88.52%	63.26%	85.93%	91.26%	65.63%		
CO4	85.56%	77.19%	98.52%	94.22%	82.30%	82.89%	90.89%	69.26%		
CO5	84.59%	77.93%	99.63%	94.59%	99.26%	85.93%	90.30%	68.89%		
CO6	82.52%	76.22%	99.26%	96.30%	98.52%	82.89%	87.26%	61.26%		
Overall CO attainment	79.30%	75.90%	99.20%	92.41%	85.14%	84.28%	90.31%	66.26%		
ATT LEVEL	3	3	3	3	3	3	3	2		

8.5.1 Indicate results of evaluation of each relevant PO and/or PSO if applicable (10)

Institute Marks : 10.00

POs Attainment:

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C101	0	0	0	0	0	2.4	2.4	0	2.41	2.41	0	2.40
C102	2.49	2.51	2.46	2.48	2.49	0	0	0	0	0	0	2.49
C103	2.22	2.22	2.24	2.22	2.16	0	0	0	0	0	0	2.22
C104	2.54	2.53	2.56	2.54	2.56	0	0	0	0	0	0	2.56
C105	0	0	0	0	0	2.37	2.38	2.38	2.38	2.38	0	2.38
C106	2.28	2.27	2.27	2.27	2.28	0	0	0	2.28	2.27	0	2.29
C107	2.98	2.98	2.98	2.97	2.97	0	2.97	0	2.98	2.98	0	2.98
C108	2.77	2.77	2.78	2.78	2.78	0	0	0	2.78	2.78	0	2.77
C109	2.09	2.08	2.10	2.12	2.10	0	0	0	0	0	0	2.10
C110	2.08	2.09	2.08	2.08	2.08	0	0	0	0	0	0	2.09
C111	2.49	2.48	2.49	2.48	2.49	0	0	0	0	0	0	2.48
C112	1.98	1.98	1.98	1.98	0	0	0	0	0	0	0	0
C113	2.17	2.19	2.16	2.13	2.15	0	0	0	0	0	0	0
C114	0	0	0	0	0	2.49	2.48	2.46	2.50	0	0	2.48
C115	0	0	0	0	0	2.54	2.55	2.55	2.55	2.55	0	2.55
C116	2.53	2.53	2.53	2.53	2.53	0	0	0	2.53	2.53	0	2.53
C117	2.70	2.70	2.70	2.70	2.70	0	0	0	0	0	0	0
C118	1.99	1.99	1.99	0	0	0	0	0	0	1.99	0	1.99

PO Attainment Level**PSOs Attainment:**

Course	PSO1	PSO2	PSO3
C101	0	0	0
C102	0	2.49	2.49
C103	2.22	0	0
C104	0	2.76	0
C105	0	0	0
C106	0	2.26	0
C107	2.98	2.98	2.98
C108	0	0	0
C109	2.12	2.12	0
C110	2.13	2.13	0
C111	0	0	0
C112	1.98	1.98	0
C113	0	0	0
C114	2.62	2.62	0
C115	0	0	0
C116	0	0	0
C117	0	0	0
C118	1.99	1.99	1.99

PSO Attainment Level

Course	PO1	PO2	PO3
Direct Attainment	2.29	2.37	2.49
PSO Attainment	2.29	2.37	2.49

8.5.2 Actions taken based on the results of evaluation of relevant POs and PSOs (10)

Institute Marks : 10.00

POs Attainment Levels and Actions for Improvement- (2021-22)

POs	Target Level	Attainment Level	Observations
PO 1 : Engineering Knowledge			
PO 1	2.4	2.38	C103-2.22 : Engineering Mathematics - I has contributed moderately to the attainment as the students were unable to understand the basic mathematical formulation in problem solving. C106-2.28: Engineering Physics Laboratory has contributed moderately to the attainment as the students find difficulty in applying basic concept learnt in theory in hands on experiment. C112-1.98: Engineering Mechanics attainment level is moderately attained as the students finds difficulty in gaining an inductive understanding of the underlying principles in subject.
Action 1: Tutorial classes will be conducted for the needy students to make them understand the basic application of mathematical formulas in problem solving Action 2: As much of engineering has its roots in physics, Bridge courses to be conducted for the students to enhance their knowledge towards understanding the basics and application of physics. Action 3: Students will be supported in the learning of Engineering Mechanics with theoretical material, general key techniques, and a sufficient number of solved sample problems.			
PO 2 : Problem Analysis			
PO 2	2.4	2.38	C109- 2.08: Engineering Physics II has contributed moderately to the attainment as the student lack in applying the learnt concept during problem analysis. C110- 2.09: Engineering Mathematics II has contributed moderately as students found difficulty in solving complex mathematical problems.
Action 1: Additional classes to be conducted for the students to enrich their knowledge towards the problem-solving strategies. More visual based materials with animations were given to improve the learning level of the students. Action 2: Through ALM techniques, students must be trained to focus on the application of physics in problem analysis. Real life applications and practical examples will support them in understanding the concept better.			
PO 3 : Design/development of Solutions			
PO 3	2.4	2.38	C112- 1.98: Engineering Mechanics has contributed moderately as students felt hard in modelling and analysis of Engineering mechanics problems with an emphasis on real world engineering applications and problem solving. C113- 2.16: Python for Problem Solving Attainment level is moderately attained as the students lack industrial knowledge towards the application of python in development of solutions. C118- 1.99: Engineering Graphics and Design has contributed moderately due to lack of understanding of students with basic graphic language through which they can visualize three dimensional objects in two-dimensional representation.
Action 1: Video lectures to be shown for learning and applying the principles required to solve engineering mechanics problems. Action 2: ALM techniques are to be followed to realize effective learning practices Python programming language. Action 3: Video lectures to be arranged for the better visualization and more hands on practice to be given to understand and perceive a graphic representation.			
PO 4 : Conduct Investigations of Complex Problems			
PO 4	2.4	2.41	This attribute has reached the set attainment level but still attainment level of few courses need to be improved. C110- 2.08: Engineering Mathematics II has contributed moderately to the attainment as the student's understanding in solving complex mathematical problems and arriving at the solution was average. C112- 1.98: Engineering Mechanics has contributed moderately as students felt hard in modelling and analysis of Engineering mechanics problems with an emphasis on real world engineering applications and problem solving.
Action 1: Practice sessions provided for the needy students to acquire the problem solving skill support them in understanding complex problems. Action 2: Enriching the students through online videos and demos with application of mechanics to solve problems involving common engineering elements			
PO 5 : Modern Tool Usage			
PO 5	2.4	2.44	This attribute has reached the set attainment level and will be further strengthened in higher semesters
PO 6 : The Engineer and Society			
PO 6	2.4	2.45	This attribute has reached the set attainment level and will be further strengthened in higher semesters
PO 7 : Environment and Sustainability			

PO 7	2.4	2.56	This attribute has reached the set attainment level and will be further strengthened in higher semesters
PO 8 : Ethics			
PO 8	2.4	2.46	This attribute has reached the set attainment level and will be further strengthened in higher semesters
PO 9 : Individual and Team Work			
PO 9	2.4	2.55	This attribute has reached the set attainment level and will be further strengthened in higher semesters
PO 10 : Communication			
PO 10	2.4	2.49	This attribute has reached the set attainment level and will be further strengthened in higher semesters
PO 11 : Project Management and Finance			
PO 11	2.4	0	This attribute will be strengthened in higher semesters.
PO 12 : Life-long Learning			
PO 12	2.4	2.42	This attribute will be strengthened by emphasizing on learning goals, improving personal and professional skills and keeping the student's well educated with the latest technological advancements.

PSOs Attainment Levels and Actions for Improvement- (2021-22)

PSOs	Target Level	Attainment Level	Observations
PSO 1 : Develop innovative and contemporary designs of ships and offshore structures through advanced analysis and experiments to meet the challenges of marine operations in the shipbuilding, offshore, and other maritime industries.			
PSO 1	2.4	2.29	C103 – 2.22: Engineering Mathematics I has contributed moderately to the attainment as the students were unable to understand the basic mathematical formulation in problem solving. C112 – 1.98: Engineering Mechanics has contributed moderately to the attainment as the students were facing challenges with multi-disciplinary subjects.
Action 1: Conducted bridge courses for the students to get strong foundation on basics. Action 2: Arrange special lectures and seminars for the students to motivate and to improve the understanding and application of subject knowledge in maritime sector.			
PSO 2 : Implement engineering design and construction concepts to produce solutions in the field of Naval Architecture and Offshore Engineering.			
PSO 2	2.4	2.37	C110-2.13: Engineering Mathematics II has contributed moderately as students lack hands-on experience in mathematical modelling and problem solving for development of solutions. C118– 1.99: Engineering Graphics and Design has contributed moderately due to lack of understanding of students with basic graphic language through which they can visualize three dimensional objects in two-dimensional representation.
Action 1: Tutorial Classes are conducted and more assignment problems are given for practice. Action 2: Video lectures to be arranged for the better visualization and more hands-on practice to be given to understand and perceive a graphic representation.			
PSO 3 : Apply research based knowledge and research methods to analyse and interpret data using modern tools towards the solution of complex problems in the area of Naval Architecture and Offshore Engineering.			
PSO 3	2.4	2.49	This attribute has reached the set attainment level which will be further strengthened in higher semesters.
-			

9 STUDENT SUPPORT SYSTEMS (50)

Total Marks 50.00

9.1 Mentoring system to help at individual level (5)

Total Marks 5.00

An effective mentoring relationship is characterized by mutual respect, trust, understanding, and empathy. Our mentors are able to share life experiences and wisdom, as well as technical expertise with the students. They are good listeners, good observers, and good problem-solvers. They make an effort to know, accept, and respect the goals and interests of a student. In the end, they establish an environment in which the students accomplishment is limited only by the extent of his or her talent.

In AMET, we follow the best possible mentoring system (i.e.) Total Development Mentoring System for the benefit of the students. The student mentor ratio is 15:1 to 20:1 which greatly helps the mentor to focus not only on the issues raised by the students but enables them to communicate the desired and constructive outcomes of the students. The mentors meet the students once in a week and discuss a range of topics from academics to personal life and other beneficiary topics.

Table 9.1. (a) Frequency of Meetings

Mentoring System	Yes
Type of Mentoring	All Round Development
Number of Faculty Mentors	170
Number of Students per Mentor	15 to 20
Frequency of Meeting	Weekly

Details of Mentoring System:

The purpose of the mentoring system is to monitor the student with regard to their academic and professional well-being. Every student is assigned to a faculty member who continues to be the mentor for that student till the completion of his/her course. Mentors identify the shortcomings with respect to punctuality, attendance, academic performance etc.

Mentors also identify the core competencies of the students and guide them to make better professionals. Students are counselled before and after conducting internal assessment examinations. Marks scored, attendance percentage with respect to every subject is analyzed and remarks are listed in the mentor file. Problems if any are suitably addressed. Students are monitored to incorporate the suggestions for further improvements. Parent Teacher meetings are conducted as and when required. An independent login is provided to every student in ERP for sharing the pertinent information like attendance, academic performance, feedback etc. All Faculty encourage the students to participate in co-curricular, extra-curricular and other professional activities, which motivates them and stimulate their growth to become all-round young professionals.

All Round Development:

- Pre motivation for all students taking part in sports activities, cultural activities, symposiums, conferences, mini projects, internships and seminars, will be given by class in charges, mentors and professional body coordinators.
- Students are also motivated by extracurricular activities and co-curricular activities.
- Mentoring is also provided for the students who participate in various activities organized by other institutions.
- Counselling for extra curricular and co-curricular activities will help the student to become an all-rounder and OD is also provided for the same.

V. Professional Guidance

- Professional counselor is available in the campus on all the working days during the college working hours.
- The counselor counsels the aggressive students and students with health issues in a private confidential environment.
- The Institution provides Professional counselling and psychological assistance for all the students and to ensure a healthy environment and an emotional well-being.

V. Career Advancement

- Pre motivation for all the students taking part in the personality development program, English reasoning and technical training will be given by the placement training coordinator of the concerned Department and the IELTS training officer.
- Pre motivation for all students taking part in campus drive and placements will be given by the placement officer.
- Placement cell coordinator plays an important role in this regard, they frequently visit each and every class to motivate and counsel students in the career aspect.
- Career based counseling is also provided on the various companies visiting the campus and kind of skill sets and preparation is required for each company.
- Mentors and alumni counsel students on the opportunities for higher studies and the preparations required for the same.

V. Course Work Specific

- The student's personal details along with their academic performance and progress is updated in the 'Green card' maintained by the mentor.
- The counselor whose primary activity is to monitor the academic progress of the students under their care and to take corrective measures for any aberration in their progress in academic pursuits.
- The students can acquire the uploaded course material from LMS/ERP uploaded by the faculty.
- Faculty members while giving the evaluated answer sheets to the students for checking, discuss about the general weakness and improvements to be done in the course. Later on weak students are individually called upon to discuss about the ways to improve their performance by identifying areas of weakness.
- These students are counseled by the faculty during the Library/counseling hour and improvement at the end of the semester is also recorded in Mentoring efficacy.
- Any discrepancy found in counseling is taken to the HOD.

V. Laboratory Specific

- Students those who are not attending Lab sessions due to Leave, O.D are monitored by the mentors. They are motivated to complete their experiments in the evening within a week.
- These students are asked to meet the faculty in person during the extra hours such as Library/Counseling /Break and will be given the importance of Lab Experiments.

- Faculty members encourage students to give seminar on the course topics to improve their technical communication and presentation.

Objectives of Mentoring Program:

- To interact with students about their need, problems, difficulties and to address them effectively.
- To mentor students to participate in various technical and cultural events for their overall development.
- The mentor regularly observes the overall growth of the student and provides counselling whenever required.
- To guide the students to ease the transition from school to work.
- To gain access to the professional community.
- To help students develop skills and dispositions such as resilience, perseverance and determination.
- To ensure that they can solve problems and work through challenges as well as take ownership of their learning and decisions
- To gain access to the professional community.

V. Responsibilities of Mentor

- Professional and Personal Guidance
- Internships, Industrial Trainings & Industry Readiness
- Higher studies exposure
- Guiding in project works
- Participation in co-curricular and extra-curricular activities. Special attention on students with learning difficulties.
- Facilitate career exploration and professional growth
- Developing self-confidence and peer relationships

V. Issues to be discussed with the students by mentor

- Personal study timetable
- Attendance in Theory and Practical classes
- Economic status
- Area of Interest
- Hostel / Food issues
- Confidence level
- Personal, Family, social and academic issues.
- Vision and goals of Career plans

Frequency of Meeting:

The mentors meet the students every week on Wednesday from 3.20 - 4.10 pm and discuss a range of topics from academics to personal life and other beneficiary topics. The students can meet the mentors on every working day after 4.00pm.

Effective mentoring system is followed in our university to create a healthy relationship between faculty members and students.

Mentor ward meetings are conducted regularly to ensure good academic performance as well as to address personal grievances of the students. Mentors maintain a separate WhatsApp group with their wards and their parents to keep them updated about their academic performance, daily tasks, upcoming technical events organized by premier Institutions, quote of the day etc.

Green Card:

AMET has Green card system to monitor the student's academic progress. This green card contains the academic performance of an individual which includes marks obtained in internal assessment for all the eight semesters and also about the end semester examinations. It also contains the personal information of the student so that the parents will be contacted as and when required. Also, during every parent teacher's meeting this green card will be shown to the parents and discuss with them about their ward's performance in studies. A Sample Green Card is given below in Fig.9.1(a) .

AMET
ACADEMY OF MARITIME EDUCATION AND TRAINING
(Approved by the Ministry of Education, Government of India)

1. Candidate Name: ALISHA KATHI

2. Current Address: 11, Maheshwari Road, Indraprastha

3. Phone: 98765 43210

4. Email: alisha.kathi@gmail.com

5. Date of Birth: 01/10/2000

6. Candidate's Signature: [Signature]

7. Date: 01/10/2022

8. Institution's Signature: [Signature]

9. Date: 01/10/2022

10. Candidate's Address: 11, Maheshwari Road, Indraprastha

11. Candidate's Phone: 98765 43210

12. Candidate's Email: alisha.kathi@gmail.com

13. Candidate's Signature: [Signature]

14. Date: 01/10/2022

15. Institution's Signature: [Signature]

16. Date: 01/10/2022

17. Candidate's Address: 11, Maheshwari Road, Indraprastha

18. Candidate's Phone: 98765 43210

19. Candidate's Email: alisha.kathi@gmail.com

20. Candidate's Signature: [Signature]

21. Date: 01/10/2022

22. Institution's Signature: [Signature]

23. Date: 01/10/2022

24. Candidate's Address: 11, Maheshwari Road, Indraprastha

25. Candidate's Phone: 98765 43210

26. Candidate's Email: alisha.kathi@gmail.com

27. Candidate's Signature: [Signature]

28. Date: 01/10/2022

29. Institution's Signature: [Signature]

30. Date: 01/10/2022

Fig.9.1. (a) Sample Green Card

AMET
ACADEMY OF MARITIME EDUCATION AND TRAINING
(Approved by the Ministry of Education, Government of India)

Department of Marine Engineering

Student's Counselling Report

Name of the Master	Year / Group	Designation	Semester	Name of the Student	Counselling date	Remarks	Signature
1							
2							
3							
4							

Fig.9.1. (b) Sample Format of Student Counseling Report

Professional Counselor:

Apart from effective counselling sessions by the faculty members, Our University has a professional Counsellor Ms. Mathew Henah.B., MA, (Psychology), to provide personal and professional counselling for our students. Having more than 20 years of professional experience, associating with the student's community in different roles, she helps students to identify goals and potential solutions to problems which cause emotional turmoil seek to

improve communication and coping skills strengthen self-esteem and promote behavioural change and optimal mental health.

Efficacy of Mentoring System:

Establishment of the above stated mentoring system helps us in the following ways,

1. Enhanced the teaching- learning process to be more student centric
2. Created a positive work environment helped the students learn to take better control of his or her career
3. Provided impartial advice and encouragement to students
4. Developed a supportive relationship between students and staff
5. Assisted with problem solving and Improved self-confidence in students
6. Obtained gradual improvement in the attendance percentage of students
7. Enables to provide individual and personal care to the students with the help of mentors
8. Information gathering and dissemination is easy.



Mentoring Efficacy

Academic Year/ sem:

Class:

Name of the Mentor:

Date:

Sr.No	Name of the Student with Register Number	Counseling I (Purpose of Counseling)	Counseling II (Purpose of Counseling along with Follow up Counseling)	Counseling III (Purpose of Counseling along with Follow up Counseling II)	Overall Improvement Through Counseling
1					
2					
3					
4					
5					

Counseling Criteria for Enhancing Student's Participation:

- | | |
|--|---|
| 1. Punctuality/less room attendance | 3. Motivation towards Higher studies/Entrepreneurship |
| 2. Performance in oral presentation exam | 6. Motivation towards PCA & CCR |
| 3. Dress code, Cleanliness & Discipline | 7. Motivation towards Internship/ In-plant Training |
| 4. Performance in Internal Assessment | 8. Motivation towards Innovation / Projects |

Signature of the Mentor

*** Note: This has to be followed along with counseling track

Fig 9.1. (c). Efficacy of Mentoring System

9.2 Feedback analysis and reward /corrective measures taken, if any (10)

Total Marks 10.00

The feedback is collected from the students through online at the end of every semester. Different kinds of feedbacks are received from the stakeholders. They are feedback on curriculum, teaching learning process, faculty and facilities.

Feedback collected for all courses : YES
 Specify the feedback collection process : Through software
 Average Percentage of students who participate : 100%.

Feedback Process:

Methodology being followed for feedback collection, analysis and corrective action taken

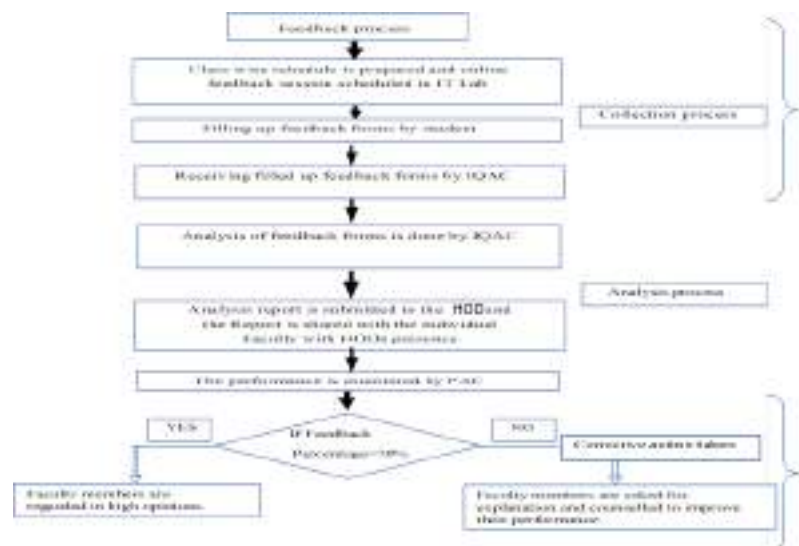


Fig: 9.2. (a) Feedback process methodology

Feedback collection process

Feedback is collected through an Online. Software with the parameters to be considered, the name of the Faculty handling the subjects for each class is prepared. A class wise schedule is prepared and the same is circulated to all the students in all the Departments. The students of each class go to the laboratories assigned for them and give confidential feedback for each faculty handling classes for them based on a standard questionnaire.

Percentage of students who participated: 100%.

Feedback assessment process

The feedback collected from students is first analyzed by the Internal Quality Assurance cell headed by the IQAC Coordinator.

- The contents of the feedback are shared with each faculty member individually by the Vice Chancellor with the presence of the concerned Head of the Department.
- Performance of each individual faculty is monitored by the Program Assessment Committee and Quality Improvement Committee.

Feedback analysis process

The software is designed that the rating for each parameter is cumulatively calculated for a total of 5 points. Based on the average points accrued for all the questions the faculty performance level is assessed. The performance of the teacher is graded as below:

More than 4.5 points : Excellent
 From 4 to 4.4 : Good
 From 3.5 to 4 : Average
 Below 3.5 : The Teacher is subjected for any corrective measures as

decided by Head of the Department.

If the faculty receives good performance feedback from the students, he is appreciated by the Vice Chancellor and HOD in the Department meetings and also rewarded with monetary benefits. If the Faculty receives average or below average performance, he/she gets counseling by the authorities and tend to improve their performances.

Metrics:

- Teaching Methodology
- Communication Skill
- Technical Skill
- Punctuality
- Class control
- Student Evaluation process

Inferred from the metrics:

- Faculty Performance
- Attitude of Faculty
- Course Delivery
- Interactive learning of students
- Interpersonal Skills
- Student satisfaction

Basis of reward:

- Performance is recorded in the Faculty appraisal.
- Performance rating of Faculty through student feedback system is one of the factors in evaluating the annual performance of the Faculty. Based on the annual performance of the Faculty the annual increments are released.

Corrective actions taken:

- The feedback on the teaching practices and methods are assessed with due attention and the reports are sent to the respective HODs. Based on the report, the HOD counsels the faculty members accordingly. The quality of teaching shall be observed periodically by the Head of the Department and the Registrar. Further guidance are given to the faculty members in person.
- The motivation and encouragement of the Head of the Department enables the Faculty to perform better. Explanation from the Faculty will be demanded for the inappropriate result and subsequent action will be processed. Counselling will be given to the concerned faculty by the HOD.

• **Effectiveness**

The improvement of the Faculty performance with respect to parameters is reflected in the subsequent feedback.

Record of corrective action taken

The following necessary corrective actions are initiated for Faculty members having Performance Index less than the university standards:

1. Necessary advice by the Head of the department.
2. Deputing faculty to the Faculty Development Program (FDP).
3. Advise the Faculty through counsellors / Improve the domain knowledge by Subject Experts.

Interactive Feedback:

Student's Council meetings are conducted as and when required by the Vice Chancellor and the Registrar along with the HODs, to share their feedback among the Faculty handling classes and facilities. Student representatives from each year will attend the meeting to give feedback and suggestions regarding academic and co-curricular activities directly. This feedback is then shared with the concerned faculty members for remedial actions.

Department level feedback collection:

Students are invited to express their view on subjects and other grievances in the department level class committee meetings by PAC Committee and the findings are communicated to the HOD for further actions. Apart from this, feedback is collected from all the students for all the subjects as and when required to improve the teaching learning methods and to address grievances of students. A Sample Feedback as shown in Fig.9.2.2

Metrics for Course evaluation:

- The course objectives, content and methods of evaluation are clearly explained at the beginning of the semester.
- Class activities include a variety of methods to hold students, maintain interest in the course and to understand the course materials.
- The assignments and activity based learning help the students understand the course materials.
- The course content and notes/handouts appeared appropriate and well structured, neatly prepared and readily understood.
- The library resources are adequate.
- Timetable, class grouping and classroom information are clearly stated sufficiently in advance.
- Percentage level of understanding that achieved for all the units in a course

Metrics based on which faculty evaluation is made:

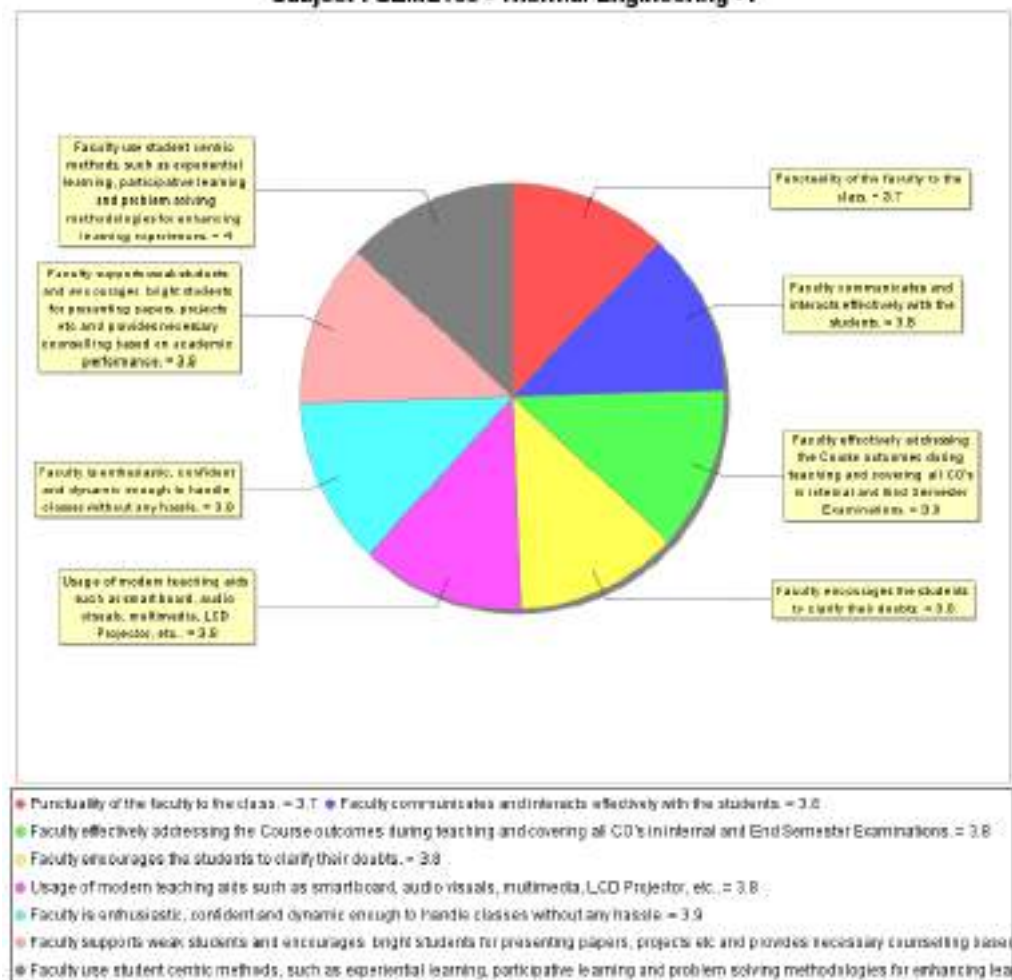
- Methodology of teaching
- Interest in the well-being and progress of the student and courses.
- Firm in evaluating the students
- Open mind discussions with the students
- Good presentation and highly Interactive teaching
- Relationship and interaction in and out of the class
- Concern for students attendance
- Timely assignments, quizzes and assessments
- Encouraging students to improve their independent level of exploration
- Clarity in explanation
- Encouraging students to ask questions and giving them appropriate answers
- Competency of faculty
- Punctuality for the classes
- Concern for students discipline and their appearance
- Understanding the concepts

Indices used for measuring quality of teaching and learning:

- [5] Strongly Agree
- [4] Agree
- [3] Neutral
- [2] Disagree
- [1] Strongly Disagree

Inferred from the metrics:

- Faculty Performance
- Attitude of Faculty
- Course Delivery
- Interactive learning of students
- Interpersonal Skills
- Student satisfaction

Subject : UEME106 - Thermal Engineering - I**Fig: 9.2. (b) Feedback Analysis of UEME106-Thermal Engineering-I**



Course: U.E. Thermal Engineering (SEM-III)
Semester: 3rd
Page Number: 4

Faculty - Feedback Analysis -Table

Sl. No.	Faculty Name	Subject Name	Semester	Feedback - Feedback/Comments										Avg.
				Availability of the Faculty in the Classroom	Faculty's communication with the students	Faculty's interaction with the students	Faculty's understanding of the students' needs	Faculty's response to the students' queries	Faculty's response to the students' suggestions	Faculty's response to the students' complaints	Faculty's response to the students' feedback	Faculty's response to the students' feedback	Faculty's response to the students' feedback	
1	Dr. Ravi Kumar	Engineering Mathematics - I	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
2	Dr. Ravi Kumar	Engineering Mathematics - II	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
3	Dr. Ravi Kumar	Engineering Mathematics - III	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
4	Dr. Ravi Kumar	Engineering Mathematics - IV	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
5	Dr. Ravi Kumar	Engineering Mathematics - V	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
6	Dr. Ravi Kumar	Engineering Mathematics - VI	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
7	Dr. Ravi Kumar	Engineering Mathematics - VII	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
8	Dr. Ravi Kumar	Engineering Mathematics - VIII	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
9	Dr. Ravi Kumar	Engineering Mathematics - IX	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
10	Dr. Ravi Kumar	Engineering Mathematics - X	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
11	Dr. Ravi Kumar	Engineering Mathematics - XI	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
12	Dr. Ravi Kumar	Engineering Mathematics - XII	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
13	Dr. Ravi Kumar	Engineering Mathematics - XIII	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
14	Dr. Ravi Kumar	Engineering Mathematics - XIV	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
15	Dr. Ravi Kumar	Engineering Mathematics - XV	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
16	Dr. Ravi Kumar	Engineering Mathematics - XVI	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
17	Dr. Ravi Kumar	Engineering Mathematics - XVII	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
18	Dr. Ravi Kumar	Engineering Mathematics - XVIII	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
19	Dr. Ravi Kumar	Engineering Mathematics - XIX	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
20	Dr. Ravi Kumar	Engineering Mathematics - XX	SEM-III	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0

Fig: 9.2. (c) Feedback Analysis

Corrective Actions taken based on Feedback:

The teachers who secure a grade less than 3.5 to 4 are counseled by the HOD on how to improve the teaching-learning process. Both study the metrics and the method for effective teaching is charted out for improvement.

In the above feedback analysis the faculty handling **UEME106-Thermal Engineering** has scored an overall average of 3.8 but he has a shortcoming in some of the aspects. So that the following faculty was counseled by the Head of the Department and gave him the following suggestions.

• Punctuality to the class (3.7)	Advised the faculty to come 5-minutes before the commencement of class and it was monitored by the class advisor.
• Student Interaction(3.8)	Advised the faculty to conduct the class more interactive.
• Clarification of Doubts(3.8)	Asked the faculty to clarify doubts then and there.

• Subject Knowledge(3.8)	The faculty is encouraged to attend FDP in this subject by subject expert.
• Attitude of faculty(3.8)	Advised the faculty to improve personel bonding with students.

The counseling has a strong positive influence on teacher and in his work. His teaching become more interactive and he comes to class on the time.

Class Committee Meeting:

Each class shall have a Class Committee which is constituted by a Chairperson, Class Counsellors, Faculty members teaching the courses for the classes, and student representatives. Three subsequent meetings are held in a semester at suitable intervals. The Chairperson is required to conduct the meetings and the concern HOD shall address the issues raised by the students and the same shall be brought to the notice of the higher authorities if needed. A format for Minutes of Meeting is given below in the Fig.9.2. (c) .

AMET
ASSOCIATES OF MARINE ENGINEERING TECHNOLOGISTS
(Approved by the Council of Technical Education, Government of India)
(Recognized by the Council of Technical Education, Government of India)

Minutes of Class committee meeting - E.E Marine Engineering

Date: _____

Feedback from the students regarding academics are collected orally in the meeting based on the following criteria 1. Quality of teaching 2. Syllabus coverage 3. Short learner's classes 4. Student interaction 5. Text Books/ Reference materials provided etc.

Sl. No.	Name of the Course	Name of the Faculty	Quality of teaching	Text Books/ Reference materials	Student Name	Student Signature

Academic Issues:

-
-
-
-

Non Academic Issues:

-
-
-
-

HOD Marine

Fig 9.2. (d) Sample Class Committee Meeting for the Marine Department

The University provides a conducive learning environment to the students with effective support on co-curricular and extracurricular activities to elicit the apparent and latent talents. The students are assisted throughout their course of study in the campus through the Director of Student Affairs, Proctor, Student Union Leader, activity Cells, activity Centres and Committees. Every student in the University campus is given an equal opportunity to participate in various events of their choice. The University takes special care of the SC/ST, minority students. The students are motivated to participate in National / International Conferences/Workshops with financial assistance. Every academic department has a mentoring system to counsel and support students through Green Card System, Student's Resource Centre, Career Development Centre and Remedial Coaching for slow learners. The students are offered various facilities in the University campus such as Student's Scholarship, Student's Insurance, ATM facility inside the campus, Swimming Pool facility, Cafeteria facility, Yoga Centre, Uninterrupted Power Supply Facility for comfortable learning, Internet facility, WIFI facility and well-equipped library facility. The University offers effective coaching to students in shaping their career through soft skills training and career development programs. The University maintains strict discipline in the campus because of the diversity of the student community. Having established as a Deemed to be University under de-novo category, AMET has developed its own unique strategy for students support and progression.



INTERNAL QUALITY ASSURANCE CELL (IQAC) AMET
STUDENT FEEDBACK ON CURRICULUM, TEACHING-LEARNING PROCESS AND RESOURCES
Academic year: _____

Name of Student (optional):	Department:
Programme Name:	Year & Semester:

Note: Please give your valuable feedback on curriculum to improve quality of the programmes. Select your rating on the scale of 0 to 4 (where 0 is strongly disagree and 4 is strongly agree) for each of the following parameters.

S.NO.	PARAMETER	Feedback				
		Strongly Agree	Agree	Not sure	Disagree	Strongly disagree
1.	Punctuality to the class					
2.	Subject Knowledge of the Faculty					
3.	Clarity and Availability					
4.	Writing Legibility					
5.	Understanding level of the subjects					
6.	Faculty-student interaction					
7.	Syllabus coverage					
8.	Technical assistance to students by providing Question Bank and study materials					
9.	Doubt Clarification					
10.	Class interaction with students promoted					
11.	Students issues are considered and addressed					
12.	Prescribed text books and reference books are adequate and appropriate					
13.	Value added courses are well structured and effective					
14.	The courses provided by the university supports career guidance and attend competitive examination					
15.	Online course platform like NPTEL, SWAYAM etc. useful for knowledge enrichment					
16.	AMET library is enriched with adequate facilities					
17.	Information and communication technology tools are effectively used in teaching learning process					
18.	Student support services (sports, recreation, accommodation, transport etc.) are adequate					
19.	Opportunities available for doing extension activity and community service					
20.	Overall, AMET has very good learning environment.					
Remarks if any:						

Signature of the Student (Optional)

Date: _____

Fig: 9.3(a) Sample Student Feedback Teaching Learning Process and Resources

Feedback collection process:

Feedback is collected from the students on the facilities available in the University. Based on the suggestions, actions will be taken to improve the facility and the same is presented to the Management review meeting.

Table 9.3. (a): Details of feedback collection process

Items	Description
Feedback collected on all facilities provided by the University	YES
Feedback collection process	Computerized

Feedback received	IQAC
Frequency of feedback collection	Once in an academic year
Metrics used for calculation	5-Excellent, 4-Very good, 3-Good 2-Satisfactory , 1-Below average
Purpose of comments	For improving the quality of facilities.

Feedback analysis process:

Feedback is collected from the Final year students on the facilities available in the University such as class room, infrastructure, library, labs, canteen, playground, internet facility, medical facility etc. The feedback is analysed by IQAC and the necessary corrective measures are implemented after discussions with the authorities.

Table. 9.3. (b) Analysis of Feedback on Facilities

Particulars	Attributes	Average
Assessment of Learning Environment		
Learning Environment	Standard of class room facilities	3.73
	Lighting and Ventilation of Classrooms.	
	Quality of teaching aids	
	Toilet Facilities	
	Availability of drinking water	
Assessment of Infrastructure Facilities		
Library	Adequacy of books	4.09
	Library arrangements (Reading room facilities, adequacy of journals, books stacking & identification)	
Recreational Facilities	Adequacy	3.62
	Encouragement for sports	
	Facilities for Extra-Curricular Activities.	
Swimming pool	Cleanliness	3.65
	Arrangements (Availability of adequate life jackets, etc.)	
Medical Facilities	Regular hours	3.39
	After hours	
Particulars	Attributes	Average
Canteen	Quality of food/ Quality of drinking water	3.72
	Adhering to Menu/ Adhering to scheduled time	
	Ventilation/Dining Seat arrangements	
	Hygiene/ Cleanliness of staff	
	Wash area	

Hostel Facilities	Quality of cabins	3.84
	Cabin ventilation	
	Toilet/ Bathroom/ Wash basins – Cleanliness	
	Availability of water	
	Availability of drinking water	
	Availability & Internet facilities	
	Facilities for Sports & Entertainments at Hostel	
Transport Facilities	Timeliness	3.74
	Safety	
	Condition of the Vehicles	

Action Taken for Medical Facility:

- Ø **Dr. Vijayalakshmi**, a full time Doctor is appointed in the University. She is available on all working days from morning 8:00 A.M to evening 6:00 P.M.
- Ø From evening 6:00 P.M to morning 8:00 A.M Duty Doctor is visiting our campus and is supported by Chettinad Hospitals at any emergency situation.
- Ø Well trained and a qualified Nurse is available 24X7.
- Ø Ambulance with first aid facility is available 24X7. Complete care is provided to the students with fully equipped campus clinic.

Feedback received from the final year students to gather their suggestions on the various facilities provided to them in their four-year degree program. Recommendations and Suggestions are suitably incorporated for the development of next batch students. A sample feedback analysis chart is shown below in Fig. 9.3.2.

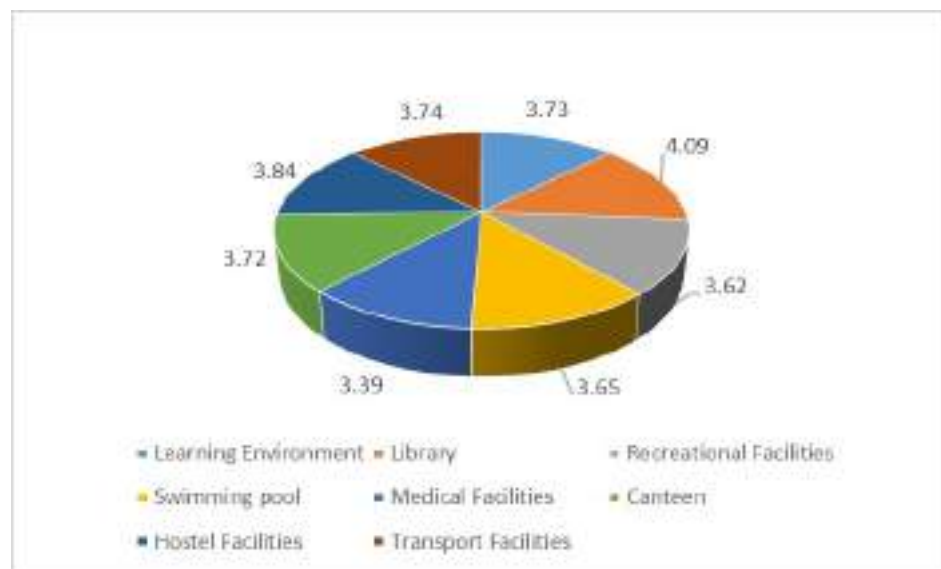


Fig.9.3(b). Feedback analysis

Feedback analysis and Corrective measures taken on facilities:

Feedback obtained from the students, alumni, industrial experts, parents are analysed and the outcome is used for enriching the system.

- Value added courses and contents beyond the syllabus are conducted on the basis of feedback from stakeholders for the betterment of the students.
- The transport coordinator takes care of the facilities related to transport.
- The Hostel facilities are improved by Proctor and Hostel Warden with the help of concerned authorities based on the feedback.
- Canteen facilities are improved and maintained with high hygiene and quality of food.

- Medical Facilities are improved and maintained with high quality of medicines.

9.4 Self-Learning (5)

Total Marks 5.00

Scope of Self Learning:

- Web based teaching and learning (teaching a course online through My CAMU)
- Learning through MOOCs, NPTEL, edX, Coursera, Webinars, YouTube etc
- Library and Digital Library
- CDs, Videos and Digital books
- Learning activities around collaborative projects (Project Based Learning)
- Learning around case descriptions (Case Study)
- Tutorials/Assignments/ Seminars/ Activity based learning
- Club activities
- Adapting to industrial needs through in plant training and industrial visits. Acquiring managerial skills and information through symposiums conducted annually.
- Participation in activities through Professional bodies, EDC, Innovation and Incubation cell.

Web based learning	The university has created Internet facility with 1000 Mbps (BSNL), 100Mbps (TIC/Fibre) leased line and 1000 computer terminals facility to promote and motivate students to self-learning. Various sources of information, media and materials such as texts, images, Video integrate as self- learning content for the students.
Library/Digital Library	Availability of course material on intranet, digital library facility, availability of video lectures in CD form as well as captured videos through Lecture Capture System. LCD projectors for presentations have been facilitated by the University. University library equips students with learning skills and latest subject knowledge. Availability of NPTEL Videos and books E-Shodh Sindhu
Professional Bodies	Joining a professional association like IEEE, IET, SAE, MTS etc. gives all the valuable information and resources to the students for the career enhancement and self-learning. Activities conducted under these professional chapters give good learning experiences
Club activities	Various club activities are organized to enhance team work and inter-personal skills of the students
Tutorials/Assignments/activity based learning	It enables students to go through the topics in a more elaborate manner in order to explore the academic topic and enhances higher order thinking.

Resources for Self-Learning:

Every week an hour in the time table is provided as Library hour and students are requested to be in the library under the supervision of a faculty member for self-learning. Besides regular tutorial classes to enrich curriculum, additional topics/courses are designed as content beyond syllabus to motivate the students towards new engineering and technology fields. Special laboratories under centre of excellence, web based and multimedia learning are the other facilities established for students to enhance them for sophisticated self-learning. The below flowchart Fig.9.4.1will explain about the various modes, availability of using the resources for self-learning along with its efficacy



Fig. 9.4. (a) Resources of Self learning

Central Library:

The Central Library has been functioning in a separate building and the total area of the library is 2936.38 Sq.m.

- Total No. of Books : 103535
- Total No. of Titles : 19444
- Total No. of Print Journal : 110
- Total No. of E-Journal : 40934
- Total No. of E-Books : 206784
- Total No. of database : 4
- Total No. of Back Volumes : 2461
- NPTEL Video and web courses : 36802
- SWAYAM PRABHA32 DTH Channels of High Quality Educational Programs
- Well protected with fire alarms and CCTV security systems.
- Separate Audio-Visual Section
- Wi-Fi enabled with library building
- Equipped with reading halls in two floors with reference section
- Organization of Book Review / Book Talk Sessions
- Organization of Book Exhibitions/Displays of new books.
- Proper system of obtaining, analysis and action pertaining to feedback from the Users of library services.

Users can submit their suggestions/grievances through library web Portal which are forwarded to the Library Committee for necessary action. Users can also submit their suggestions/ grievances through general interaction, making an application, and through e-mail.

E-Journals:

- IEEE : 205 e-Journals + Back file to 1998
- J-GATE Engineering : 35550 e-journals
- EBSCO : 1079 e-journals
- DELNET E-JOURNALS : 4100 e-journals

USE OF PLASMA TV

As an awareness the events of AMET are displayed through plasma TV. Power point slides denoting the various resources available in the library are displayed to enlight the user community.

Web OPAC (Online Public Access Catalogue)

In-house resource list can be accessed through website of AMET. Central Library has online access facility to search the availability of the required resources for the user community. The search option is unique and can be made through Author-wise, Title-wise, Subject-wise indexes using keywords and can be accessed from on and off the campus. The advanced search is provided with a more specific search for a title with Boolean operators.

ACCESS CARDS

AMET provides Identity card with Barcode Technology, for all students and staff members, which can be utilized for registering GATE Entry as well as for borrowing the library resources

Facilities / Value Added Services:

In order to meet the contemporary needs of the library users, the Central Library offers the following facilities and value-added services through Information and Communication Technologies (ICT).

Information and Communication Technology (ICT)

Preamble:

The objective of the Network and system administration is to Plan and implement the ICT (Information and Communication Technology) in AMET. The following infrastructure is implemented in order to enable the ICT.

Objective:

1. Desktops to be provided to the department staffs
2. High speed networking of the systems
3. Internet facilities for the staffs
4. Mail IDs for the staffs
5. Data centre architecture implementation
6. Security of the data
7. Firewall configuration and implementation
8. Wireless Internet access for the cadets (24*7)

9. Wireless access for top management
10. Content filtering for the internet
11. Providing teaching aids
12. Video conferencing facility
13. NPTEL
14. SMART CLASS ROOMS

Progress Planned:

- Planning to setup LAB and provide branded desktops to Staffs
- Planning to provide data centre to deploy servers and switches
- Planning to provide high speed internet line
- Provide dedicated internet with Wi-Fi for hostels
- Provide High speed connectivity with OFC for different blocks
- Implement domain controller with DHCP, DNS and an additional domain controller

Progress executed:

- Planning to setup LAB and provide branded desktops to Staffs-completed
- Planning to provide data centre to deploy servers and switches -completed
- Planning to provide high speed internet line-BSNL and TATA high speed provided
- Provide dedicated internet with Wi-Fi for hostels –Dedicated line for hostels
- Provide High speed connectivity with OFC for different blocks- Fiber connectivity provided
- Implement domain controller with DHCP, DNS and additional domain Controller-Implemented

Network and system administration activities

The main function of the Network and system administration department is to ensure that the following facilities are provided to staffs and students.

- The staffs are given desktops with Internet facility
- Email IDs are provided to teaching and non-teaching staffs on need basis.
- Maintenance of data center with Switches and Servers.
- The Entire campus is networked through OFC cabling with redundancy
- The departments are provided with sufficient Laptops and Projectors for teaching purpose
- Helpdesk management
- On need basis the teaching halls are provided with Smart class rooms
- SMS are sent to staffs and parents
- Backup space provided to departments in a server and required credentials are given to the staffs to ensure there is no single point failures
- Internet facility for the staffs with redundancy
- The Entire hostel is provided with Ruckus WIFI access points with dedicated leased line. The maintenance of the same with intelligent wireless access controller. Ensure that 24*7 Wireless Internet access is provided to all the students
- Installation and maintenance of Video conferencing equipment
- Installation and administration of Domain controller, Additional Domain controller, Configuration of Dynamic Host Configuration Server, Domain Name server
- Preventive maintenance of the systems on monthly basis
- Palo Alto Firewall administration and content filtering, load balancing between TATA and BSNL Leased line
- Installation and management of Surveillance cameras
- Maintenance of hardware and software assets
- Thin Clients in the digital Library for Internet access

Library Web Portal:

Library Portal is an interface to the Library resources and services to access e-question bank. Open source e-journals and magazines, open source e-books, e-newspapers etc., are also made available. Link: <https://ametlibrary.wordpress.com>

Digital Knowledge Centre:

Library has a separate section for digital library, which consists of 50 computer systems with an internet speed of 1 GBPS of National Knowledge Network and 100 Mbps of Tata Communications. Access of Internet in Digital Knowledge Centre enhances the facility to explore the electronic databases such as DELNET E-Journals and E-books subscribed by the Central Library.

Learning beyond Syllabus:

Content beyond syllabus is taught by each and every Faculty and is made available in the AMET LMS portal. Video conferencing lectures are organised by various Departments on content beyond the syllabus to get updated with the latest technology. The materials for content beyond syllabus, video lectures prepared by the Faculty are made available in You Tube Channels and the AMET website. The viva presentation made by Research Scholars, PG students is also available in the AMET LMS portal/My CAMU platform.

Credit allocation/ Self Study courses:

As per the AMET CBCS regulations 2018, students are permitted to earn maximum of six credits through online courses/self-study courses with the approval of Dean Academic and the Head of the Department. Online courses completed with certificates issued by recognized agencies will be considered for the credit allocation as given ,

4 weeks course = 1 credit

8 weeks course = 2credit

12 weeks course = 3 credit

Webinars/ Online Workshops:

AMET has the practice of organising many training programs for the students and the Faculty. Various Departments have organized many webinars on Intellectual Property Rights Pedagogical Initiatives and Outcome based Education etc during the lock down period.

MOOC/NPTEL/ Coursera Certification course:

Each Faculty member and student is informed to register for at least one online course in platforms such as Coursera, Edx and NPTEL, etc. per year. This enables them to enrich their subject knowledge with current trends, and also to equip themselves with multi-disciplinary domain knowledge.

National Program on Technology Enhanced Learning (NPTEL), funded by Ministry of Human Resource Development (MHRD), Government of India provides E-learning through online video courses in Engineering, Science and Humanities and Management streams. The mission of NPTEL is to enhance the quality of engineering education in the country by providing free online courseware. The collection consists of 31780 video courses.

Table.9.4(a) Student participated in SWAYAM

S.No .	SWAYAM DETAILS	2019-20 (ODD)	2019-20 (EVEN)	2020-21 (ODD)	2020-21 (EVEN)	2021-22 (ODD)	2021-22 (EVEN)	2022-23 (ODD)	2022-23 (EVEN)
1	Students Registered in SWAYAM courses	149	250	-	1	-	4	-	440
2	Students appeared in examination for SWAYAM courses	6	7	-	1	-	4	-	440
3	No of students passed in SWAYAM courses	3	7	-	1	-	4	-	Yet to attend

Seminar by Students on Innovative Topics:

Students give extramural lecture once in a week on any innovative topic of their own interest in front of all the students to demonstrate their communication and presentation skills. Students who presents extramural lecture are appreciated with certificates of participation by the Departments.

9.5 Career Guidance, Training, Placement (10)

Total Marks 10.00

Career Guidance and Training:

The University has a separate cell namely "Centre for Career Development" to guide the students to choose their right career. Career guidance provides a substantial understanding about the career that the student is considering or needs help in choosing the right career. This is useful because the students might not have an overall knowledge of careers and the kind of expectation that is put by employers on the incumbents who are opting for that career. To understand their own strengths and weaknesses and know what careers they would be best suited for, and how to pursue them, our University has a Centre for Career Guidance cell for providing Career guidance and skill development trainings. The Centre organizes various training programs and activities for the students of all the Departments.

Table.9.5.(a) Activities conducted by the Centre for Career Development

No. of placement programme conducted	Academic year- 2019-20	Academic year- 2020-21	Academic year- 2021-22
	10	7	9

Table.9.5(b).Activities conducted by the Centre for Career Development (samples)

S.No	Name of the capacity development and skills enhancement schemes	Year	Number of students enrolled	Name of the agencies/consultants involved with contact details (if any)
1	Communication Training-I	2021-2022	900	CCPD
2	Communication Training-II	2021-2022	900	CCPD
3	English - Personality Development - I	2021-22	218	CCPD
4	Personality Development Program –II	2021-22	218	CCPD
5	English - Personality Development - III	2021-22	221	CCPD
6	Personality Development Program – IV	2021-22	221	CCPD
7	Scale Up - Personality Development - I	2021-22	258	CCPD
8	Scale up – Personality Development Program –II	2021-22	258	CCPD
9	Scale up – Personality Development Program –III	2021-22	233	CCPD

Placement Training:

The Placement Cell is headed by the Placement Director. The Placement Department nominates a placement coordinator for every department of the University. The placement coordinators assist the placement cell by inviting the companies to the University and also prepare the students for the recruitment process in an efficient manner. The following placement services are provided to the students:

- Personality Development Program
- Communication skill development and soft skills Training
- Technical Training
- Value Added Training Programme
- Group Discussion
- Mock Interview

Pre-Placement and Placement Trainings are offered to the students by the Placement Cell to face interviews confidently and secure the job. Here students are guided in the following activities:

- Resume scrutiny
- Public speaking
- Written test
- Group discussion
- Personal interview
- Learning corporate culture
- Team Building etc.

The pre-placement training is given to the students on a regular basis throughout the academic year and also placement training is given before the arrival of a company to the University campus.

Table 9.5 (c) Details of Campus Placement

Name of the Program	Name of the Department	Total no of Students	No of students placed	No of Companies Visited AMET	No. of Students turned Entrepreneurs	No. of Students admitted to Higher studies
Academic Year 2019-2020						
BE (Marine Engg)	Marine Engineering	200	188	31	-	
BE (NA&OE)	Naval Architecture &Offshore Engineering	86	65	13	-	6
BE(Petroleum Engg)	Petroleum Engineering	68	46	12	-	
BE (Mechanical Engg)	Mechanical Engineering	82	34	11	-	32
BE (EEE Marine)	Electrical and Electronics Engineering	29	15	4	-	3
MBA(Shipping and Logistics Management)	Master of Business Administration	62	41	25	-	12
Academic Year 2020-2021						
BE (Marine Engg)	Marine Engineering	248	227	33	-	-
BE(NA&OE)	Naval Architecture &Offshore Engineering	72	58	15	-	5
BE(Petroleum Engg)	Petroleum Engineering	50	28	7	-	-
BE (Mechanical Engg)	Mechanical Engineering	69	44	9	-	-
BE(EEE Marine)	Electrical and Electronics Engineering	47	5	3	-	37
MBA(Shipping and Logistics Management)	Master of Business Administration	62	51	23	-	2
Academic Year 2021-2022						
BE (Marine Engg)	Marine Engineering	233	210	28	-	-
BE(NA&OE)	Naval Architecture &Offshore Engineering	50	40	15	1	2
BE(Petroleum Engg)	Petroleum Engineering	21	13	4	-	-
BE (Mechanical Engg)	Mechanical Engineering	70	38	9	-	-
BE(EEE Marine)	Electrical and Electronics Engineering	21	9	4	-	-
MBA(Shipping and Logistics Management)	Master of Business Administration	62	52	21	-	-

Centre for Higher Studies:

Centre for Higher Studies is constituted exclusively to cater to the needs of the students. Students are encouraged to write competitive exams. All the Departments offer coaching classes for Competitive Examinations such as GATE, IELTS, GMAT, GRE, TOEFL etc. The IELTS Centre has been established in AMET. This centre provides the orientation and coaching for examinations like IELTS and TOEFL.

AMET Centre for IELTS:

The AMET Centre for IELTS aims at enrichment of the language and providing in depth knowledge in communicative English and the fervour to appreciate the language in its originality. The course is designed for the students who come from different cultures and speak different languages. English for International Communication course helps the students with the necessary skills, knowledge and understanding of the language which enables them to use and apply English in everyday life effectively.

The students from various streams and disciplines attended and found the significance of IELTS to be of utmost importance. The students of all branches of Engineering, Nautical Science, HND (NS) liked International English Language Testing System to a great extent.

The students knew the need and importance of English language and understood well the benefits thereof. These students were very much motivated, decided to take up the IELTS test. They understood the need of assessment of their standard of English language with respect to listening, speaking, reading and writing skills.

Students were trained by Mr. Johnson, Mrs. Maria and Mr.T. P. Rahmathullah. The students found the training much more beneficial and they were ready to undergo good English communication training through IELTS.

The students understood that his course was initiated with an objective to enable them to be expert in all the four aspects of communication namely LSRW, Listening Speaking, Reading and Writing.

Precisely speaking the students who were trained in International Schools, Army Schools, and International Boards found the training quite easy and gave importance for deep knowledge of grammar.

Needless to say that the Students did not have an English environment at all they expressed their need for deep knowledge of English grammar. They were given assurance that grammar will also be taught with utmost importance.

Centre for Institute Industry Interaction (CIII):

The institution makes all the efforts to connect the industry with the institute by organizing visits, guest lectures, inviting the industry personnel as chief guests, arranging faculty visits to the industry, tie-up with bodies such as CIII, ICTACT, etc., and signing of MOUs with industries. The institution invariably strains to connect the academic earn and Industrial needs through several activities and ensures the industrial readiness of students. The industry interaction is headed by the experienced.

Value Added Programmes:

The students are trained with recent technologies in various courses through Value Added Programs. Value added and audit courses by the internal faculty members and Value-added certificate programs from collaborative organizations in relevant fields are provided by the University for making the student industry ready candidates.

Table.9.5 (d). Value added conducted for students detail

No. of Value Added programme conducted	Academic year- 2019-20	Academic year- 2020-21	Academic year- 2021-22
	27	27	31

Table. 9.5. (e) Shows the Value added courses conducted in AMET(sample)

S.no	Name of the value added courses (with 30 or more contact hours) offered	Course Code (if any)	Duration of course	No. of students benefited
1	BASIC SAFETY TRAINING (BST)	VCME001	30 HOURS	518
2	SECURITY TRAINING FOR SEA FARERES WITH DESIGNATED SECURITY DUTIES (STSDSD)	VCME008	30 HOURS	518
3	ECDIS Simulation	VCNS016	30 HOURS	355
4	Steering simulation	VCNS017	30 HOURS	308
5	RADAR Simulation	VCNS018	30 HOURS	12
6	Electrical machines and control	VCEE003	30 HOURS	22
7	Maintenance of electrical apparatus	VCEE015	30 HOURS	22
8	Simulation of electrical circuit using matlab	VCEE014	30 HOURS	9
9	Plc, scada and vfd for automation	VCEE013	30 HOURS	22
10	Introduction to Internet of things	VCMC015	30 HOURS	41
11	Industrial Automation	VCMC016	30 HOURS	70
12	FEA-ANSYS	VCMC017	30 HOURS	70
13	IIOT and its protocol	VCMC018	30 HOURS	8
14	IOT protocol and Architecture	VCMC019	30 HOURS	41
15	PLC and SCADA in Automation	VCMC020	30 HOURS	8
16	ANALYSIS and Simulation using ANSYS	VCMC021	30 HOURS	41

17	Modelling and Simulation in Upstream Oil and Gas	VCPE016	30 HOURS	24
18	Export and Import Documentation	VCBS015	30 HOURS	62
19	Customs Procedure	VCBS016	30 HOURS	49
20	Food Safety Training and Certification(FOSTAC)-Level 2	VCFT003	30 HOURS	8
21	Design and development of basic robots	VCEE011	30 HOURS	9
22	Application of FRP in Marine Industry	VCNA013		50
23	Advanced welding Technology in Shipbuilding	VCNA012	30 HOURS	50
24	GIS Software and Applications	VCMN010	30 HOURS	30
25	Modern Mine Survey by using Lidar and Drone Technology	VCMN011	30 HOURS	27
26	GEOVIA SURPAC (Mining Software)	VCMN009	30 HOURS	24
27	Fire Prevention and Fire Fighting (FPFF)	VCME021	30 HOURS	247
28	Personal Safety and Social Responsibilities (PSSR)	VCME022	30 HOURS	247
29	Fundamentals of CAD Modelling	VCME023	30 HOURS	216
30	CAD Modelling using AutoCAD	VCME024	30 HOURS	216
31	ISO 9001:2015	VCME025	30 HOURS	216

Memorandum of Understanding (MOU):

With a view to impart training and inculcate technical knowledge through hands on training / real time situations to the students, 140 Memoranda of Understanding have been signed with industries. The faculty members and students regularly undergo training in these companies. A MOU has been signed with Asia Pacific University of Technology Innovation, Malaysia for interacting with overseas experts.

In-plant Training

In-plant training offers a brief period of time in an industry to gain the knowledge of the work culture in industries. It is beneficial for students to gain knowledge when they work in the company after graduation. They will be familiar with the industry and help them to work better in the surroundings. The alumni coordinator constantly interacts with alumni those who are working in the industries and they also support for their junior's internship.

Industrial Visit

The department organizes industrial visits for students once in a semester to relevant organizations/companies to enable the students to experience the practical implementation of theoretical knowledge in the real world. This gives them an insight of exposure to the industrial environment and the work culture ethics in Industries.

9.6 Entrepreneurship Cell

Total Marks 5.00

The Entrepreneurship Development Cell (EDC) continuously guide and support students to participate in various activities and motivates them to become entrepreneurs. The EDC of AMET Deemed to be University was enrolled under the Academic Innovation and Entrepreneurship Development Programme (AIEDP) of Entrepreneurship Development and Innovation Institute – Tamil Nadu (EDII-TN) with effect from 26/10/2018. AMET Deemed to be University is attached as a SPOC institution with Centre for Entrepreneurship Development - CED ANNA University. Based on this, EDC students attend programmes organised by CED ANNA University. EDC organizes Entrepreneurship Awareness Camps sponsored by National Science & Technology Entrepreneurship Development Board (NSTEDB) and Department of Science & Technology (DST-NIMAT), National Entrepreneurship development centre (NEDC), National Institute of Entrepreneurship and Small Business Development (NIESBUD).

Funding from MSME – Development Centre under DC – MSME:

As the innovation ecosystem was development over a period with various activities among the students and faculty, the University is approved as the Host Institution to carry out Business Incubation (HI/ BI) from DC MSME – Micro, Small, Medium, Enterprises, New Delhi, and Government of India and is entitled to undertake innovative and entrepreneurial development activities. Under this scheme AMET is approved to undertake projects and nurture them to become start-ups by incubating them in our campus. This scheme provides Rs.3.5 crore as financial assistance, which includes up to 15 Lakh for nurturing 10 ideas (Rs.15lac x 10 ideas = 1.5 crores) to HI, Rs. 1.00 crore. As Capital Support to HI for Plant and Machines, Rs. 1.00 crore as SEED capital support to deserving idea, awareness and workshop programme.

The Incubation centre is approved as Design Centres under Incubation Scheme of DC MSME with effect from April 2020 and the centre have already tapped the opportunity, by submitting various proposals to organise awareness programmes, workshops, seminars and student design projects to a tune of Rs.3.11 Crore. This is approved under the MSME Credit Link Capital Subsidy – Technology Up Gradation Scheme (CLCS-TUS). The AMET Innovation and Incubation centre facility is equipped to accommodate incubates and have accommodated 8 external incubates over a period since 2016. Under MSME scheme 15 project ideas and Business proposal from students/ faculty are submitted and are awaiting approval in addition to 4 projects submitted by external members.

Institution Innovation Council (IIC):

In the academic year 2019-20 we have 24 faculty members and 135 student members enrolled in IIC and have so far received 2 STAR recognition for the quarter 1 and quarter 2 performance indexes with a maximum cumulative score of 37.45/50. Students of AMET participated in SMART INDIA HACKATHON (SIH -2019) organised by MHRD innovation cell (MIC). 44 team (5-software, 39-hardware) participated by submitting innovative ideas and 4 teams were selected for round 2 video presentation of the working model. 15 innovative problem-solving ideas were submitted by the students for SIH 2020. The University organised an internal Hackathon as per the norms and submitted 3- software ideas and 2-hardware ideas after internal scrutiny. In the academic year 2021-22 AMET secured a 3 star grade in IIC.

Table. 9.6. (a) Student's participation in Smart India Hackathon

S. No	Name of the Student	Prize/ Participation	Title	Event place
1	Yedida Venkata Sesha Sai Lavanya	Participation	Smart India Hackthon (SIH) 2022	Chandigarh Group of Colleges, Chandigarh
2	Anoop George	Participation	Smart India Hackthon (SIH) 2022	Chandigarh Group of Colleges, Chandigarh
3	Mohammed Arham Shariff	Participation	Smart India Hackthon (SIH) 2022	Chandigarh Group of Colleges, Chandigarh
4	Sanjana J T	Participation	Smart India Hackthon (SIH) 2022	Chandigarh Group of Colleges, Chandigarh
5	Chandana Saran Venkata Amarnath Swamy	Participation	Smart India Hackthon (SIH) 2022	Chandigarh Group of Colleges, Chandigarh

Table. 9.6. (b) Details of Programs organised on innovation/entrepreneurship

No. of innovation /entrepreneurship programme conducted	Academic year- 2019-20	Academic year- 2020-21	Academic year- 2021-22
	10	18	19

Table.9.6.(c)Details of Programs organised on innovation/entrepreneurship(samples)

Academic Year	Name of the workshop/ seminar	Date
2021-22	Idea Ignition Day 2022	24.05.2022
2021-22	Workshop on Prototype/Process Design and Development- Prototyping	13.05.2022
2021-22	Internal Hackathon 2022	22.03.2022
2021-22	Webinar series on Motivational Story of an Successful Entrepreneur – "Turn your Struggles into Stepping Stones"	02.03.2022
2021-22	Webinar series on Motivational Story of an Successful Entrepreneur	16.02.2022
2021-22	International Day of Women and Girls in Science	11.02.2022
2021-22	Expert Talk on "Process of Innovation Development & Technology Readiness Level (TRL)" & "Commercialization of Lab Technologies & Tech-Transfer.	28.01.2022
2021-22	Webinar on "Achieving Problem – Solution Fit & Product – Market Fit"	25.01.2022

Table. 9.6. (d) List of Companies incubated in AMET (samples)

S.No	Company	Name of Founder / SPOC	Designation
1	Dr.Aaliya's EasyLife Healthcare Pvt. Ltd	Dr.Aaliya Firdose	Founder /Director
2	Tems Tech Solutions	Mr.P Bhuvaneshwaran	Manager
3	Healthathon Tech Pvt Ltd	Mr. Salil Sahu	Founder & CEO
4	Wellness Mandala Pvt. Ltd.	Mr. Rakesh Sarin	Founder & Chairman
5	Polowings Healthtech Pvt. Ltd.	Mr. Mannat S	Founder & CEO
6	Renegizr Industries Pvt. Ltd.	Mr. Himanshu Gupta	Founder & CEO

7 Wenexcorp Pvt. Ltd. Mr. Saurav Chatterjee Founder & CEO

Table. 9.6. (e) List of student start- up registered under UDHYOG ADHAR (samples)

S.No	Company	Name of Founder / SPOC	UDYAM
1	Anoria Syscoms	Dr D Sivakumar	UDYAM-TN-08-0053819
2	Sgnar Consumer Products Pvt. Ltd	Dr.S.Ranganathan	UDYAM-TN-08-0054317
3	Rocks Industries	Mr.S Vishnu	UDYAM-TN-24-0060566
4	Tatolic Energy Systems	Mr.Zuben Kalyan	UDYAM-JH-06-0028250
5	Trekcodes	Mr.R S Lakshmi Balaji	UDYAM-TN-24-0062527
6	Sagaram Herbal Enterprises	Ms.Shanthi Pandurangan	UDYAM-TN-02-0023597
7	Green Prasada Food Pvt Ltd	Ms.Ponmani Swaminathan	UDYAM-TN-02-0189233
8	Go Green Uni Cycle	Dr. Duraimutharasan	UDYAM-TN-02-0190976

9.7 Co-curricular and Extra-curricular Activities

Total Marks 10.00

Co-Curricular Activities:

AMET motivates the students to take part in various co-curricular activities. Students are actively participating in the co-curricular activities such as symposia, conferences, project expo, technical events and competitions conducted by various institutions. AMET encourages the students to take part in co-curricular activities along with their regular academic commitments to keep them exposed to recent developments in the field of interest and to share their experiences among peer groups. The students are encouraged to associate themselves with the technical societies like ISTE, IEEE, SAE, MTS, etc., to present their work in conferences and to publish in reputed National and International Journals. Participation in workshops and design contests is encouraged to sharpen their knowledge. Each department has separate Technical Association through which students organize National and International level technical events under the supervision of faculty members every year.

Air-conditioned auditorium and seminar hall for organizing the guest lecture, conference, symposia, Workshops etc are available in the campus.

Table. 9.7. (a) Participation of Students in Workshop/Seminar/Symposium/Other training Program

No. of students participated Workshop/Seminar/Symposium	Academic year- 2019-20	Academic year- 2020-21	Academic year- 2021-22
	25	30	40

Table. 9.7. (b) Participation of Students in Workshop/Seminar/Symposium/Other training Program (samples)

S.NO	Name Of Students	NAME OF EVENTS	MONTH	YEAR
1	Murari Kumar	Symposium	May	2022
2	Jai Vishnu	Internship program on boilers	June	2021
3	P Jai Vishnu	Webinar	October	2021
4	Shivam Kumar Singh	Feasibility of Deep Learning	October	2021
5	P Jai Vishnu	One Day National Webinar	June	2021
6	Shivam Kumar Singh	Data Privacy and Protection	June	2021

Table.9.7.(c).Participation of students in National/ International Conferences

No. of students participated National /International Conference	Academic year- 2019-20	Academic year- 2020-21	Academic year- 2021-22
	18	22	240

Table. 9.7. (d) Participation of students in National/ International Conferences (samples)

S. No	Name of the Student	Prize/ Participation	Title of paper	Name of conference
1	Afil Mohammed Ismail	Participation	Electromagnetic Automatic Mooring System	International Virtual Conference on "New Technologies for Greener Shipping"
2	Aditya Gautam	Participation	Fabrication Of Saltwater Purifier For Marine Applications	International Virtual Conference on "New Technologies for Greener Shipping"
3	Ankit Jha	Participation	Development Of A Public Address System For Maritime Applications	International Virtual Conference on "New Technologies for Greener Shipping"

4	Chetna Saini	Participation	Controlled Pitch Propelle	International Virtual Conference on "New Technologies for Greener Shipping"
5	Arunachallam S G	Participation	Impressed Current Cathodic Protection	International Virtual Conference on "New Technologies for Greener Shipping"

Extra-Curricular Activities:

Sports and Games being an integral part of AMETs total educational programme, the following Indoor and Outdoor facilities are developed and maintained for the students.

Table 9.7. (e) Sports Facilities in AMET

S.No	Details	Location	Area in Sq. Mtrs
Outdoor			
1.	Multi-Purpose 200 Mtrs Mud Track / Cricket Field / Football Field	Ship In Campus	4225
2.	Hand Ball Court	Ship In Campus	800
3.	Kabaddi Court	Opposite to D Block	360
4.	Volley Ball Court	Opposite to D Block	600
5.	Basket Ball Court	Parade Ground	405
6.	Beach Volleyball	Ship In Campus	200
7.	Futsal Court	Ship In Campus	675
8.	Long Jump Pit	Ship In Campus	175
9.	Cricket Nets	Ship In Campus	520
10.	Swimming Pool with Gallery	Opposite to C Block	875
Indoor			
11.	Table Tennis	C Block	48
12.	Badminton Court	Yamuna Hostel	294
13.	Chess	C Block	100
14.	Carrom	C Block	100
15.	Snooker	C Block	42
Gymnasium			
16.	Boys Gym I & II	C Block & Ganga Hostel	780
17.	Girls Gym	Girls Hostel	24

Our students actively participate in the zonal, district, state and national level sports competitions and bring laurels to the University. Certificates and championships are being honored for the achievers. Travel allowances are provided to the students to participate in the Sports Meet held at various places. Sports day is conducted every year. An exclusive department headed by Physical Education Director takes care of the sports activities. Students are permitted to participate in various tournaments after getting permission from the Registrar, Vice Chancellor and the respective Head of the Department. Physical Director accompanies the students for inter/intra/zonal/state level tournaments. The participants are provided with travelling assistance, sports kit, T-shirts, etc.

Table. 9.7. (f) Availability of Sports Facilities

S.No	Game	Particulars
1.	Badminton	3 Courts with cement floors

2.	Basketball	1 Basketball court with standard marking
3.	Beach Volleyball	1 Standard Court
4.	Carrom	10 Carrom Boards
5.	Chess	15 Chess Boards
6.	Cricket	2 Cricket Nets with cement concrete
7.	Football / 200 Mtrs Athletics Track / Cricket	Multipurpose facilities
8.	Handball Court	2 Standard Courts
9.	Futsal Court	2 Nos.
10.	Gym	3 Gyms, Two Gym for Boys and 1 for Girls, all the gyms having 11 Multi Session Machines with all weight training equipments
11.	Kabaddi	2 Standard Courts and 1 mat court
12.	Snooker	1 Snooker Board
13.	Swimming Pool	One Standard Swimming Pool is available with proper safety measures (19*16 m) length (5 feet to 12 feet depth)
14.	Table tennis	6 Table tennis boards
15.	Volleyball	3 Volleyball courts protected with fencing.

The following tables show the number of awards/medals won by the students for the outstanding performance in sports activities at national/international level during the last three years.

Table.9.7. (g) Student Participation in Sports (samples)

S.No	Name of Student	Name of Event	Month	Year	Prize
1	Hemanth	Handball Fest	April	2022	Winner
2	Madasamy	Kabaddi League	May	2022	Runner Up
3	Madasamy	Cricket	May	2022	Runner Up
4	Madaswamy	Volley Ball	April	2022	Runner Up
5	Madasamy	Volley Ball	October	2021	Winner
6	Siva Subramanian	Kabaddi League	May	2022	Runner Up
7	Ignatious Antony	Volley Ball	April	2022	Winner
8	Zanetti	Football League	May	2022	Runner Up
9	Vasanth	Handball Fest	April	2022	Winner
10	Emmanuel	Handball Fest	April	2022	Winner
11	Deepak	Kabaddi League	May	2022	Runner Up
12	Gowtham Kumar	Handball Fest	April	2022	Winner
13	Jetti Sushmanth	Cricket	April	2022	Winner

Sports Day:

AMET celebrates Sport Day 'AMETUNISPO' every year. The function started with the Inaugural Address by Mrs. Sangeetha Albin, Director - Administration. The Chief Guest of the day was Mr. C.P. Senthil Kumar, Managing Director, Brizo Global Logistics Pvt. Ltd., who delivers the Chief Guest address, and the Guest of Honor was Ms. Tamilarasi Srinivasan, Paralympics State Champion. The presence of the Chief Guest and the Guest of Honors steals the student's attention. Mrs. Sangeetha Albin, Director Administration, give away the prizes for the winners and runners for the Intramural competitions. The program ends with the National anthem.

Tournaments:

The following table lists the various tournaments participated by our AMET students

Table.9.7. (h) Student's Participation in various tournaments

S.No	Context	Level
1.	AIU- South Zone Inter University TABLE TENNIS (Women) Tournament Organized By SRM University Tamil Nadu 06.01.2022	South Zone
2.	AIU- South Zone Inter University CHESS (Men) Tournament Organized By SRM University Tamil Nadu 03.04.2022	South Zone
3.	AIU- South Zone Inter University VOLLEYBALL (MEN) Tournament Organized By SRM University Tamil Nadu 25.12.2021	South Zone
4.	AIU- South Zone Inter University BASKETBALL(MEN) Tournament Organized By Christ University Karnataka 09.02.2022	South Zone
5.	AIU- South Zone Inter University FOOTBALL(MEN) Tournament Organized By MG University Kerala 09.02.2022	South Zone
6.	AIU- South Zone Inter University POWER LIFTING (MEN) Tournament Organized By JRN University, Udaipur Rajasthan 21.04.2022	All India
7.	AIU- South Zone Inter University TABLE TENNIS (WOMEN) Tournament Organized by AMET University Tamil Nadu 06.01.2022	South Zone
8.	AIU- South Zone Inter University CRICKET (MEN) Tournament Organized by Hindustan University Tamil Nadu 07.06.2022	South Zone
9.	State Level TNPESU Rolling Trophy VOLLEYBALL (MEN) Tournament Organized by Dept Physical Education TN Sports University Tamil Nadu 07.04.2022	State
10.	State Level TNPESU Rolling Trophy FOOTBALL(MEN) Tournament Organized by Dept Physical Education TN Sports University Tamil Nadu 07.04.2022	State
11.	State Level TNPESU Rolling Trophy HANDBALL(MEN) Tournament Organized by Dept Physical Education TN Sports University Tamil Nadu 07.04.2022	State
12.	State Level TNPESU Rolling Trophy CRICKET(MEN) Tournament Organized by Dept Physical Education TN Sports University Tamil Nadu 08.04.2022	State
13.	State Level TNPESU Rolling Trophy KABADDI (MEN) Tournament Organized by Dept Physical Education TN Sports University Tamil Nadu 08.04.2022	State
14.	State Level TNPESU Rolling Trophy BASKETBALL (MEN) Tournament Organized by Dept Physical Education TN Sports University Tamil Nadu 08.04.2022	State

Cultural Activities:

The cultural talents of students are identified and encouraged to participate in the cultural programs conducted by our university and other institutions. Traditional festivals like Pongal, Onam, and Saraswathi Pooja are celebrated in our campus. The students work as a team to make all needful arrangements for the celebrations.

The University conducts special drives / campaigns through

- Pongal Celebration
- Onam Celebration
- Holi Celebration
- Raksha-Bandan Celebration
- Christmas Celebration
- Pooja Festival
- Eid Celebration
- Guru Nanak Gurpurab
- Maritime Day

Teacher's day

The Teacher's day was celebrated on the 5th of September 2022. The Management appreciates the faculty members and honours the good performers. The students extend their gratitude to the faculty members.

Children's Day:

AMET Students Council organized Photography Competition in the Childrens Day Celebration function, on 14 Nov 2022.

NCC Activities:

AMET is committed to social and national responsibilities; bearing this in mind, we have raised the NCC Signal Coy for boys and army wing for girls from 2004-05 onwards. We believe that an individual will come to possess an adoring personality only by involving himself / herself in activities like NCC, NSS, etc. AMET is equipped with several infrastructure facilities like short-range firing and offers obstacle courses for NCC training program. The NCC program provides ample opportunities to the students/cadets by conducting firing camps, drill practices, personality development and voluntary service program. The mission and vision of our NCC wing is to create awareness about its importance and role in the nation building among the student community.

Table 9.7(i)Participation of Students in NCC (Samples)

Sno	Date	Activity Name	organizer
1	02/08/2019	Awareness programme on organ donation and donor registration	NCC collaboration with TRANSTEN (Government multi-specialty hospital Omandur, Chennai)
2	02/08/2019	Rally on "Importance of Organ Donation", Kanathur	NCC collaboration with TRANSTEN (Government multi-specialty hospital Omandur, Chennai)
3	03/12/2019	WORLD AIDS DAY- Ending the HIV/AIDS Epidemic: Community by Community"	NCC and Marine Biotech(AMET)
4	13/01/2020	An Awareness Campaign on Road safety& Traffic Rules	NCC and Semmenchery J10
5	29/01/2020	An awareness Programme on "Women safety with Kavalan app"	NCC & NEWS 7-ANBU PALAM (NGO)
6	22/02/2020	Free medical checkup camp for differently able people. Venue: Meston Rehabilitation Research residential special school muttukadu Chennai	NCC & NEWS 7-ANBU PALAM (NGO)
7	24/02/2020	Awareness Rally on Corona virus , Kanathur	NCC & NEWS 7- ANBU PALAM (NGO)

National Service Scheme (NSS):

The University envisages among its students the values of democratic living and upholds the need for selfless service to the society by exactly following the motto of NSS namely "Not Me But You".

Table.9.7. (j) NSS programme (samples)

Sl. No.	Name of the Activity	Organizing Agency	Year	No of Students Participated
1.	Mega Blood Donation club	National Service Scheme	2022	300
2.	Free Type Writing course	National Service Scheme	2022	100
3.	Volunteer provision for marathon race	National Service Scheme	2019	100
4.	Kanathur pond cleaning camp	National Service Scheme	2019	300
5.	Voting awareness campaign	National Service Scheme	2019	250

Club Activities:**ECO Club**

The University's Eco Club is an initiative taken by staff and students to create awareness about environmental issues. The Eco Club conducts various programs every year to address the challenges faced by the environment. The objectives of Eco Club are as follows:

- To sensitize the students about the environmental issues
- To create clean and green awareness among students through various pioneering methods
- To mobilize students towards scientific enquiry into environmental crisis and solutions
- To involve them in efforts to conserve the environment
- To motivate students to imbibe the knowledge and life style for minimizing waste generation.

Table 9.5(k) Participation of AMET Students in eco club Program (samples)

S.No	Name of the Programe	Venue of the programme	Conducted by	Date	No. of participants
1	Donate National flag	Kanathur, Reddy kuppam	Eco club and Marine Bio-technology	14.08.2022	20
2	Tree plantation on Independence day	AMET Campus (play ground)	Eco Club, AMET	15.08.2022	30
3	International Costal clean-up day – Beach cleaning	Juhu beach, Uthandi	Eco Club, AMET and Maersk	17.09.2022	140
4	Tree plantation programme to celebrate the special occasion of 72 nd Birthday of Honourable Prime Minister	AMET Campus	Eco Club, AMET	20.09.2022	25

LEO Club

The University has formed Leo Club to promote service activities for the community which will develop the individual qualities of leadership, experience and opportunity. Leo Club focuses on the spotlight on children projects with a view to enrich the lives of children through:

- Blood donation camp
- Collecting food and clothes for local street children
- Repairing playgrounds
- Implementing after school tutoring programs
- Visiting children admitted in hospitals
- Raising funds for immunization programs

Through these service activities, Leo club brings hope and laughter to the needy children, especially on December 5, the International Leo Day.

Table.9.7.(I). Leo club Programme (sample)

Sl. No.	Name of the Activity	Organizing Agency	Year	No of Students Participated
1	MEGA BLOOD DONATION CAMP	AMET LEO CLUB	2022	350

Rotaract Club

Rotaract club brings together people of ages from 18-30 to exchange ideas with leaders in the community, develop leadership and professional skills, and have fun through service. With the support of our University Rotaract members work side by side to take action through various services.

- World Water Day Awareness

- Healthy Lifestyle Awareness
- Orientation Programme
- Rotaract Club Orientation Programme
- Tree Sapling Project – One Student One Tree
- Vigilance Awareness Programme – Integrity – A Way Of Life
- Vigilance Awareness Programme – Integrity – A Way Of Life In Kovalam School

Table.9.7.(m). Rotaract club programme (samples)

Sl. No.	Name of the Activity	Organizing Agency	Year	No of Students Participated
1	Vote Awareness Campaign	AMET Rotaract Club	2019	450
2	First Aid Training	AMET Rotaract Club	2019	200
3	Organ Donation	AMET Rotaract Club	2019	120

Peace Club

The University has a joint initiative with Universal Peace Federation of Japan. The Peace Club functions with the aim of spreading peace and family welfare among people. It organizes events to bring harmony and hail humanity among the citizens.

Table.9.7.(n). Peace club programme (samples)

Sl. No.	Name of the Activity	Organizing Agency	Year	No of Students Participated
1	WORLD SUICIDE PREVENTION DAY	AMET Peace Club	2022	40
2	INTERNATIONAL PEACE DAY	AMET Peace Club	2022	50
3	UNITY DAY	AMET Peace Club	2022	40

Name of the Club	LEO CLUB	PEACE CLUB	ECO CLUB	ROTRACT CLUB	NSS/NCC/YRC	YOGA
No. of Activities	1	3	4	3	20	8

Extension Activities:

Number of extension and outreach Programmes conducted in collaboration with industries, Non. Government Organisations, NSS/NCC/Red cross/YRC Club etc., during the last three years for the benefit of the Society.

Table. 9.7.(n). Extension Activity details (samples)

S.No	Name of the activity	Organising unit/ agency/ collaborating agency	Year of the activity	Number of students participated in such activities
1	World Ocean day Celebration" Beach Clean Activity	NCC- 1 TN Naval Unit	2022	47
2	Beach Cleaning Activity- Puneet Sagar Abhiyan	NCC- 1 TN Naval Unit	2022	55
3	Free medical checkup camp for differently able people	NEWST Anbupalam	2020	51

4	An awareness program on wonlell safety with Kavalan App	NEWST Anbupalam	2020	40
5	Sapling plantation programme	KEN Foundation Society	2020	30
6	Mega Swatchta Pakwada "Awareness on ground water pollution"	Amet Eco - Club and NCC Unit	2020	95
7	Rally on Rain Water Harvesting, Kanathur	NCC & Eco club	2020	150
8	Awareness Programme on organ donation and donor registration	NCC Collaboration with Transten	2020	163
9	Rally on Importance of Organ donation	NCC Collaboration with Transten	2020	200

AMET Student's Band:

The Band is divided into three sections namely the wood wing, the brass wing and the percussion. The band performs in the Morning parades, Colour ceremonies, Sunset, VIP visits, Passing-out parades, National day celebrations and Inaugural functions.

Swimming Pool Facility:

- The University has a standardized swimming pool to practice swimming as and when required.
- The primary purpose of the swimming pool is to provide training in swimming for the cadets / students and also for the sports activities.
- The University maintains and operates the swimming pool in a safe and secured manner.
- The students are provided with life jacket during training.
- Experienced swimming coaches and life guards monitor the students keenly during the swimming classes.

Centre for Yoga and Meditation (CYM):

- The University has a Yoga Centre to keep the health and soul fit for the students.
- The University has created exclusive premises for the Yoga Centre.
- The Yoga and Meditation programs are conducted in the Yoga Centre between 3pm to 5pm.
- The University has appointed an exclusive yoga master and also invites outside yoga experts to train the students.

Yoga and meditation are the two important aspects for a perfect body and mind. The Centre for Yoga and Meditation at AMET is providing training for the students and staff for yoga and meditation by certified trainers. It offers a good ambience, technical support and easy accessibility for all. Focus areas: Yoga for Health, Yoga for Stress Management and Yoga for Personality Development.

10 GOVERNANCE, INSTITUTIONAL SUPPORT AND FINANCIAL RESOURCES (120)
Total Marks 120.00**10.1 Organization, Governance and Transparency (55)****Total Marks 55.00**

10.1.1 State the Vision and Mission of the Institute (5)

Institute Marks : 5.00

Vision

To sustain identity as a World Class Leader in Maritime Education and empower learners with wholesome knowledge through progressive innovation in training, research and development which will render students a unique learning experience and a transformation impact on the Global Society.

Mission

AMET will strive continuously to

- Impart value-based higher education and technical knowledge with uncompromising strides of an outstanding quality.
- Emerge as a Centre of Excellence inculcating skill development in recent technologies in accordance with industrial trends.
- Create World class research capabilities on par with the finest in the world and broaden student's horizons beyond classroom education.
- Nurture talent and entrepreneurship to enable all round personality development in students.
- Empower students across socio economic strata.
- Make a positive difference to society through technical education.

10.1.2 Availability of the Institutional Strategic Plan and its Effective Implementation and Monitoring (25)

Institute Marks : 25.00

AMET has a very ambitious Vision 2025 plan wherein it aims to be a one stop solution for all marine related activities happening around the World and has clearly charted out an action plan to gauge its growth towards its 2025 milestone. The Strategy document process has created a participatory process for involving stakeholders in the visioning and planning process of the University. Strategic goals were finalized keeping in view vision and mission of statements, quality policy, core values, and SWOC (Strengths, Weaknesses, Opportunities and Challenges) analysis, and current environmental, social, technological and competitive factors.

The Board of Management, Planning and Monitoring Board, Board of Research, Finance Committee and Academic Council, which have representatives from different organisations and Industries shall ensure effective planning and implementation of academic and non-academic programmes.

Availability of the Institutional Strategic Plan

AMET is progressively marching towards getting established as an **International Maritime Knowledge Hub** by expanding its all-academic, research and outreach activities in the chosen fields. The following priorities outlined in the Vision 2025 plan would help the University to achieve excellence in all facets of higher education. The AMET Vision 2025 plan is publicized through various means including hosting the same in the University website (Link: https://www.ametuniv.ac.in/mandatory_disclosure.html (https://www.ametuniv.ac.in/mandatory_disclosure.html))

- To receive and sustain the highest ranks and recognitions in all applicable accreditation and ranking frameworks such as National Assessment and Accreditation Council (NAAC), National Board of Accreditation (NBA), Common Inspection Programme (CIP) of the Directorate General of Shipping (DGS), Performance Indicators in Maritime Education and Training (PIMET) Ranking by the International Association of Maritime Universities (IAMU) and National Institutional Ranking Framework (NIRF) of Ministry of Human Resource Development (MHRD), Government of India, etc.
- To attract highly qualified and experienced personnel for Faculty positions, for imparting quality education and skills and motivating them to do research with excellent financial compensation in a congenial working environment.
- To enhance laboratory facilities with state-of-the-art equipment and advanced technological tools for teaching and research purposes.
- To reach out to industries, research laboratories and academic institutions around the world for collaboration, consultancy and joint research programmes for identifying and finding solutions to technical and technological problems.
- To concentrate on the environmental concerns plaguing the Earth due to industrial activities and deliver mitigation efforts.
- To reach out to the community through various outreach programmes for awareness, involvement and participation for the betterment of the neighboring community in particular and the entire Nation in general.

As per the AMET Vision 2025, the following are the strategic plans with identified action plans to be implemented.

Table 10.1. Strategic plan and action plan of the University towards AMET Vision 2025

Sl. No.	Strategic Plan	Action Plan
Goal 1: Provide a quality maritime and associated professional education and acquire the stature as one of the top ranking maritime Universities		
1.1	Continuous improvement in Teaching - Learning process	• Adopting Outcome Based Assessment in OBE as per the new reforms suggested by the AICTE. • Revision of curriculum on need basis and to incorporate the recent advancements such as IOT (Internet of things), Artificial intelligence (AI) and Machine learning (ML). • Encouraging all faculty to attend FDP/Conferences/Workshops/MOOCs. • Enhancing the teaching and learning experience by the use of ICT, Industry experts, mentoring system, and adopting interactive pedagogy with Active Learning Methods.

1.2	Improvement in students skills and industry readiness	• Inculcating lifelong learning skills by introducing self-learning components in the form of seminars, MOOCs in all the core departments. • Enhancing the Digital Resources in the campus, both online and offline • Incorporating project-based learning for 20% of the core courses in the curriculum • Imparting skill enhancement with competency certification to all the students • Enhancing the industry exposure to the students as part of curricular needs. • Soft Skill Training, Language Coaching, coaching for competitive examinations, career guidance, placement, training and assistance, value added courses, industry certified courses are provided to the students.
• 1.3	Centre of Excellence in	• Establishing MoUs with Industry to set up Centre of Excellence in core Departments.

	core programs	• Improving existing laboratories / establishing state of art laboratories in various departments as Centre of excellence in core programs with related industry support. • Facilitating the lab availability beyond working hours.
--	---------------	--

1.4	Enhancement of Research (Academic, Sponsored and Consultancy Research; Innovation of students) accomplishments	- Formulating procedures to improve Internationalisation for developing good research accomplishments - Academic Research shall be promoted by providing University Research Fellowship for Full time scholars - Faculty members shall be provided with fee waiver and work load reduction to promote them to do Ph. D - Seed Money shall be provided to the Faculty to promote their research activities - Incentives shall be given to Faculty with funded projects and those who publish research articles. - Promoting consultancy services in each Department by encouraging Faculty and also by appointing competent faculty with proven records earlier - Encouraging the young faculty for collaborative research with IITs and NITs and reputed foreign universities for publishing papers, international patents - Encouraging students to participate in National and International conferences to present and publish papers, insisting research scholars and Post Graduate students to publish in SCI/SCOPUS indexed journals. - Innovative ideas of students shall be encouraged to be translated into commercial products for the benefit of society by funding the student's projects through MSME and other funding agencies. - Encouraging all the students to participate, in the talent events such as Hackathons to encourage their innovations - The Institution Innovation Council shall promote the innovative ideas of students to develop in to business ideas and support their start-up ventures through Incubation Centre
1.5	Overall improvement	Policies, forms and procedures shall be defined and transparently known to all stakeholders

	through good governance practices	- Enhancing the performance of Governance through adoption of modern technologies - Promotion of participative management - A well-defined decentralized governing system is in practice for the improved transparency and speedy decision making. - Well structured, active grievance redressal mechanism is in place for students, Faculty and staff.
--	-----------------------------------	--

Key Indicators for Attainment of Goal 1

- Adopted Outcome Based Assessment in all the Engineering and Management Programmes which leads to 100% implementation by the year 2025.
- Incorporating revisions in Curriculum based on the recent advancements and needs of the employer.
- Regular consistency in all the Engineering and Management programmes which leads to 90% admissions and 100% placement in all the Programmes (Our Premier Placement Partners as on 2022 -Maersk Shipping Solutions, Flototech, Sembcorb Marine Limited, Albion Engineering India Pvt Ltd, Sea team Management India Pvt Ltd, Mediterranean Shipping Company, Vik Sandvik Ship Design and Off Shore Engineering India Pvt Ltd).
- Enhanced digital resources for Education and Research (LMS, ERP, Digital Library, E Learning portals, Digital Repositories, Software etc.) Leading towards 20% improvement in good quality and innovative projects every year.
- 100% ICT enabled class rooms.
- Fully Wi-Fi enabled Campus (hostel and classrooms).
- Enhanced Research Accomplishments leads to 50% improvement in R&D in the year 2025 (As on 2022, total Publications-3380; SCOPUS Publications:1679; h-index-53; Citations- 10276; Books-348; Patents-64; MoUs-140; Faculty with PhD-85; Faculty doing PhD-115; PhDs produced-103; Completed Projects: 7)
- Enhancing the revenue to the University by 20% through Research, Consultancy etc. (Ongoing Government Funded-18 projects; Consultancy, Seed Money and Industry sponsored projects-298; Mega Projects-4; Total worth of Grants: Rs.18.40 Crores)
- Improvement in Faculty retention ratio.

Goal 2: Enhance Interaction with Society and Industry

2.1	Enhancing	• Selecting project topics based on societal needs like energy
-----	-----------	--

	Interactions with the Society and Community	saving, green energy vehicles, remote sensing security systems, rain water harvesting, smart irrigation system, drone-modeling, retrofitting in suitable appliances and equipment. • Engaging into Village Adoption Programme to promote Government Schemes and National Missions • Expanding the interactions with Society and Community by engaging extension, social outreach and community service events • Offering short term programmes for the benefit of the community
--	---	--

2.2	Enhancing Interactions with the Industry	- Establishing MoUs and formal collaborations with Industry in Academic, Research and Extension Activities - Consulting Industry Experts in various activities of the University, including design and development of curriculum, course delivery, value added courses, guest lectures, Workshops, Seminars and joint conferences etc. - Mandatory Internships, Industrial Visits and Inplant training shall enhance industry interactions. - Addressing the needs of Industry through research proposals, thereby getting industry sponsored and consultancy projects. - Institutionalized enhanced interactions with industry through various forums such as - Industry Institution Collaboration Cell - Entrepreneurship Development Cell - AMET Incubation Centre - Business Incubator - Institution Innovation Cell - AMET Alumni Association - Career Development Centre
Key indicators for Attainment of Goal 2: - Upliftment of adopted villages by incorporating sustainable technological solutions which leads to good improvement in employment opportunities. - More than 100 MoUs with various industries are active both at Institutional and Departmental level leading to 10% of the students turned to be Entrepreneurs. (As of 2022,		

No. of Business entities incubated: 23; No. of Start Ups Promoted: 11; No. of products available with Proof of Concept: 15; No. of innovative products developed with prototypes: 15; AMET has funded 71 student innovations with overall expenditure of Rs.46 lakhs during the past three years). - Encouraging the students on innovation and start-up leads to improvement in the Ranking of ARIIA (Atal Ranking of Institutions on Innovation Achievements). - Enhancing the consultancy and industry sponsored projects based on the demands of Industries/organization. (As on 2022, No. of Industry Sponsored Projects-130; No. of Consultancy Projects-133; No. of Faculty Minor Projects-Seed Money Projects-194).
Goal 3: Generating Revenue to the University through Research, Consultancy and Collaborative initiatives

3.1	Identifying and improvising the various avenues to generate revenue to the University through Research, Consultancy and Collaborative initiatives	- Promotion of Faculty to propose major and minor projects for grants/funding from Government and Private agencies for Research, Development and Consultancy in core areas - Consultancy projects for cargo ships, cruise ship propulsion system conversion, retrofitting, etc. With talented pool of experienced faculty in Marine, Electrical, Mechanical Engineering and MBA departments - Corporate training: setting up corporate training or career enhancement training with local companies provides additional funds and increases our reach to non-traditional students.
-----	---	--

Key Indicators for Attainment of Goal 3:

- Improvement in good quality funded projects through appropriate research and retrofitting solutions from various Government funding Agencies (As on 2022, Sagarmala Project-Skill Development in Maritime Sector with a grant of Rs 814.34 Lakhs; MSME Project- Business Incubator Establishment with a grant of Rs 350.00 Lakhs; ISRO project-Capacity Building in Marine Sciences with a grant of Rs 28.42 Lakhs and DST Projects-Digital Restoration of Poompuhar with a grant of Rs 81.51 Lakhs, No. of Ongoing Government Projects-25, A total amount of Rs.18.40 Crores sanctioned through R&D).

Goal 4: Enhance the interaction with Alumni

4.1	Networking with alumni	Intensifying the activities of the AMET Alumni Association-AAA (Reg No: 135/2015; 28-09-205; Chengelpet Registrar Office, Tamil Nadu; with more than 12220 registered alumni registered; https://www.ametuniv.ac.in/alumni/Registration_Form_2016.as (https://www.ametuniv.ac.in/alumni/Registration_Form_2016.aspx) px).
-----	------------------------	---

		- Conducting University Level Alumni Association meeting in various Cities all over the World and Department Level Alumni Chapter Meetings in University Campus. - Alumni Representations in decision making bodies such as Board of Studies, IQAC etc. - Establishment of Alumni Sponsored Indoor Stadium in Yamuna Hostel. - Extending monetary support for the scheme 'Earn While you learn' beneficial to students of AMET. - Providing fund for establishing Media Lab.
--	--	--

Key indicators for Attainment of Goal 4:

- Alumni Association functioning proactively for the holistic development of the University (AAA sponsors every Department for Alumni Chapter Meeting, Alumni Guest Lecture; Besides, the AAA has contributed Rs 61 lakhs towards the establishment of a new hostel in the campus).
- Regular Alumni meetings leading to good Industrial exposure and better support to training and placement.
- Enhancement in the frequency of alumni lectures through online/offline meetings.

Goal 5: Development and maintenance of state of the art infrastructure

5.1	Maintenance of the existing Infrastructure, identification of areas for improvement	• Establishment of Cells/Centers for the effective implementation, monitoring and promotion of ICT Infrastructure in the University. • Establishing a good AI and Robotics Laboratory • Strengthening of Indoor and outdoor sports and recreation facilities. • Providing facilities for Yoga and other Life Skills in the campus. • Enhancing the features of the automated AMET Talking Library
5.2	Improvising the	• Encouraging the faculty and the students to maintain green campus through periodic awareness programmes.

	Sustainable Development	• Promoting the motto 'Clean and Green Campus" through various activities involving the Faculty and the Students both in the Campus and Hostel. • Tree Plantation and Greenery Development activities. • Maintenance of Green House with medicinal plants. • Improvising the sustainable development practices such as Rainwater harvesting, wastewater recycling and solid waste management including biogas plant and composting pit etc. • Recycle of paper wastes, safe disposal of e-wastes. • Enhancing the renewable energy capacity by promoting the solar panels, windmills etc. • Conducting Energy Audits, Environmental Audits and Green Audits.
--	--------------------------------	--

Key indicators for Attainment of Goal 5:

- 100% ICT enabled class rooms.
- Effective implementation of University policy for allocation of funds in the promotion of green energy and waste management.
- University implementing sustainable development practices. (A project on rainwater harvesting and wastewater management for water conservation is taken up).
- Every year Energy Audits and Environmental Audits are conducted and the suggested outcomes are implemented.

Effective Implementation and Monitoring of Institutional Strategic Plan

The University has Strategic planning and Steering Committee for the effective implementation of AMET Vision 2025, a strategic plan of the University for achieving greater heights and sustaining the legacy.

Table 10.2. Strategic planning and Steering Committee

S.No.	Name of the Member	Designation	Position
1	Col.Dr.G.Thiruvassagam	Vice Chancellor	Chairman
2	Dr. V.Ramachandran	Former Vice Chancellor, Anna University of Technology, Trichy	External Member
3	Dr.S. Baskar	Dean (R&D), Thiagarajar College of Engineering, Madurai.	External Member
4	Prof. Dr. S. Sathikh	Former Vice Chancellor, University of Madras, Chennai	External Member
5	Dr.R.Srinivasan	Member Secretary, Tamilnadu State Council for Science and Technology, Chennai	External Member
6	Dr. P. Devadas Manoharan	Former Vice Chancellor, Anna University of Technology, Trichy	External Member
7	Dr.M.Jayaprakashvel	Registrar i/c	Secretary

Table 10.3. Institutional Strategic planning and monitoring Committee

S.No.	Name of the Member	Designation	Position
1	Col.Dr.G.Thiruvassagam	Vice Chancellor	Chairman
2	Dr. M. Jayaprakashvel	Registrar i/c	Secretary
3	Dr. R. Rajavel	HOD - Marine Engineering	Internal Member

4	Capt.K.Karthik	Principal-Director General of Shipping Programme.	Internal Member
5	Dr.T. Sasilatha	Dean-Academics	Internal Member
6	Dr.N.R.Ramkumar	Dean-Student Welfare	Internal Member

The function of Strategic Planning Committee is

1. To provide strategic advice on global trends in marine and allied industries to help the University to design and revise its strategic plan.
2. To provide strategic plan and monitor the development of the University with the objective of implementing major goals.
3. To Monitoring and implementing of various developmental programmes of the University, including academics, administration, infrastructure and quality culture.

Frequency of Meetings, Attendance and Membership details of Strategic Planning Committee

Regular meetings of the Strategic plan and Steering Committee with stipulated attendance are held and details are presented as below.

Table 10.4 Frequency of Meetings, Attendance and Membership details of Strategic Planning Committee

S.No	Name of the Committee	Frequency of Meetings	Attendance Requirement/Minimum quorum
1	Strategic planning and Steering Committee.	1 meeting in a year	Two Third of all Members
2	Institutional Strategic planning and monitoring Committee.	2 meetings in a year	Two Third of all Members

Governing body, administrative setup and functions of various bodies

AMET Deemed to be University dedicated itself to becoming a World Class University by all virtue. The Management, administrative and executive authorities and committees like Board of Management, Academic Council, and Vice Chancellorship etc. are formed and practiced as per the requirements of the Statutory and Regulatory bodies such as the Ministry of Human Resource Development (Government of India), All India Council for Technical Education (AICTE), University Grants Commission (UGC), and Directorate General of Shipping (DGS) etc. The University is committed to follow the participatory governance to implement and keep up new and existing systems, respectively.

Following are the major Governing Bodies of the University

1. Board of Management
2. Planning and Monitoring Board
3. Academic Council
4. Board of Studies
5. Finance Committee
6. Management Review Committee

Board of Management

The Board of Management is the highest governing body of the University, which is headed by the Vice Chancellor. The Board of Management is consisting of eminent persons capable of contributing to and upholding the University ideals and traditions. All the policy matters, developmental activities, regulatory activities are reviewed by the Board of Management. Hence, The Board of Management is the apex body for the effective implementation and monitoring of institutional strategic plan of the University.

The Board of Management draws its powers and functions from the Memorandum of Agreement (MoA) which is prepared and updated in consonance with the UGC Deemed to be University Regulations (2019) and its amendments issued from time to time. The Board of Management, delegates its powers to other committees/officers, chiefly to Vice Chancellor, who is reporting to the Board of Management about the decisions taken thereupon.

Table 10.5: Members of the Board of Management

S.No.	Member of the Board of Management	Designation
1	Col. Dr. G. Thiruvassagam Vice-Chancellor	Chairperson
2	Prof. Tulsi Ram, Jawaharlal Nehru University, New Delhi	UGC Nominee
3	Capt. G Ramaswamy, CEO of OSM Fleet Management India Pvt LTD, Chennai	Nominee of the Trust - Member
4	Dr. V. Ramachandran, Former Vice-Chancellor, Anna University	Eminent Academician - Member
5	Dr. R. Srinivasan, Member Secretary, Tamil Nadu State Council for Science and Technology	Eminent Academician - Member
6	Dr. R. Venkatesan, Scientist-G and Programme Director, National Institute of Ocean Technology (NIOT)	Eminent Academician - Member
7	Dr. D. Rajasekar, Dean-Research, AMET	Member
8	Dr. K. Altaff, Chairperson, School of Biotechnology, AMET	Member

9	Dr. S. Priya, Professor - Marine Electrical and Electronics Engineering	Member
10	Capt. P. Rajendran, Associate Professor, Nautical Science	Member
11	Dr. M. Jayaprakashvel Registrar i/c	Secretary

The Board of Management meets usually twice in a semester. The Board functions as both advisory and decision making bodies. It provides strategic plans for the University, implements the same through various Committees, Programmes and Procedures. Moreover, the Board also monitors and reviews the strategic plan at periodic intervals to ensure its effective implementation.

Powers and Functions of the Board of Management

- The Board of Management is the principal organ of Management and the apex executive body of the University, with powers to make rules and regulations
- The Board of Management is the final decision making body of the University with respect to every matter, including the academic, administrative, personnel, financial and developmental matters.
- The Board of Management usually does not infringe upon the powers of the respective authorities provided under the Regulations; and where any authority has been given advisory/recommendatory powers, the Board of Management seek advice/recommendations from such authority, before deciding on any matter before it.
- The Board of Management may delegate its powers to the Vice-Chancellor or any other Officer/Faculty or to a Committee of Officers/Faculty Members of the University. However, such powers are subject to the condition that the action taken by them shall be reported at the next meeting of the Board of Management.

Planning and Monitoring Board

The Planning & Monitoring Board is the principal Planning Body of the University and is responsible for the monitoring of the developmental programmes of the University. The Vice-Chancellor and the Registrar are the Chairman and Secretary, respectively, of the Planning and Monitoring Board, which includes both internal members of the University and Experts of eminence from outside the Institution.

The Planning and Monitoring Board has the right to advise the Board of Management and the Academic Council on any matter which it considers necessary for the fulfillment of the objectives and strategic plan of the University. The Planning and Monitoring Board ensures the equitable distribution and optimal utilization of physical resources. The recommendations of the Planning and Monitoring Board will be placed before the Board of Management for consideration and approval.

Table 10.6. Composition of Planning and Monitoring Board

S.No.	Name of the Member	Designation	Position
1	Col.Dr.G.Thiruvassagam	Vice Chancellor	Chairman
2	Dr.Sushma Yadav	Member UGC, Former Vice-Chancellor, BPSMV, Haryana	External Member
3	Prof.S.P.Thiyagarajan	Chancellor Avinashilingam Institute of Home Science	External Member
4	Dr.Chitra Krishnan	Former Professor and Head, Dept. of French, University of Madras	External Member
5	Capt.K.Karthik	Principal- DGS Courses	Internal Member
6	Dr.T.Sasilatha	Dean-Academics	Internal Member
7	Dr.D.Arivazhagan	Co-ordinator- IQAC	Internal Member
8	Dr.J.Rengamani	Professor- AMET Business School	Internal Member

9	Dr.D.Velmurugan	Director-Research Advisory Council	Internal Member
10	Dr.Anita R Warriar	HoD I/c- Physics	Internal Member
11	Dr.R.Vettriselvan	Asst. Professor- AMET Business School	Internal Member
12	Dr. M. Jayaprakashvel	Registrar i/c	Secretary Ex-officio

Powers and Functions of the Planning and Monitoring Board

- The Planning and Monitoring Board is the principal Planning Body of the University and is responsible for the monitoring of the developmental programmes of the University.
- The Planning & Monitoring Board is having the right to advise the Board of Management and the Academic Council on any matter which it considers necessary for the fulfillment of the objectives of the developmental plans of the University.
- The recommendations of the Planning & Monitoring Board will be placed before the Board of Management for consideration and approval

Academic Council

Powers and Functions of the Academic Council:

The Academic Council (AC) consider matters of academic interest either on its own initiative or at the instance of the Board of Management or those proposed by the Departments / Faculty Members and to take appropriate action thereon.

- The Council usually exercises general supervision over all academic work of the University
- The AC gives direction regarding methods of instruction, evaluation, and improvements in academic standards
- The Academic Council is the apex body for the matters concerned to research, degree programmes, examinations, providing equivalence to courses/degrees, appoint examiners etc.
- The Academic Council also passes recommendations to the Board of Management on aspects such as measures for improvement of standards of teaching, research and training, institution of Fellowships/awards/medals, starting of new programmes/Departments/Centers etc.
- The Academic Council will frame rules covering the academic functioning of the University, admissions, examinations, award of fellowships and studentships, free-ships, concessions, attendance, discipline, residence etc.
- The Academic Council has the powers to appoint sub-committees to advise on such specific matters as may be referred to it by the Board of Management and consider the recommendations of the sub-committees and to take appropriate action.
- The Council may also take periodic review of the activities of the Departments / Centres and to take appropriate action with a view to maintaining and improving standards of education.

Table 10.7. Members of the Academic Council

S.No.	Name of the Member	Designation
I	Col. Dr. G. Thiruvassagam (Chairman)	Vice Chancellor
The Registrar, who shall be the secretary of the Academic Council		
II	Dr M Jayaprakashvel (Secretary)	Registrar i/c
III Dean(S) of Faculties		
1	Capt. K. Karthik	Principal-DGS Courses
2	Prof. Bhoopathy Bhaskaran	Dean- Marine Engineering
3	Prof.Dr.R.Karthikeyan	Dean-Employment Training and Progression
4	Dr. T. Sasilatha	Dean-Academics
5	Prof.Dr.K.Altaff	Chairperson - School of Biotechnology

6	Dr.N.R.Ramkumar	Dean-Student Welfare
IV Heads of the Departments		
1	Dr.S.K.Bhattacharya	Dept. of Naval Architecture and Offshore Engineering
2.	Dr.R.Rajavel	Department of Marine Engineering
3	Dr.D.Rajasekar	AMET Business School
4	Dr. C.M.Ramakritinan	Dept. of Marine Biotechnology
5	Dr.T.Nagalakshmi	Dept. of Petroleum Engineering
6	Dr.S.Priya	Dept. of Electrical and Electronics Engineering
7	Dr.K.S.Siva Subramanian	Dept. of Mining Engineering
8	Mr.R.Theertham	Dept. of HND-NS
9	Dr.John Wyson	Dept. of Food Processing Technology
10	Capt.Samson Joseph	Pre-Sea Modular Courses
11	Dr.N.Duraimutharasan	Dept. of Computer Science & Engineering
12	Dr.T.R.Heera	Dept. of Chemistry
13	Dr.P.Balaganesan	Dept. of Mathematics
14	Dr.Anitha R Warriar	Dept. of Physics
15	Dr.R.Ganesan	Dept. of English
16	Dr. K. Sekar	Librarian
V Professors other than the Heads of the Departments (By Rotation of Seniority)		
1	Dr.J.Rengamani	AMET Business School
2	Dr. D.Arivazhagan	AMET Business School
3	Dr.A.Shameem	AMET Business School
4	Dr.C.Manoharan	AMET Business School
5	Dr.G.Themozhi	Dept. of Electrical and Electronics Engineering
6	Dr.A.Suresh	Dept. of Marine Engineering
7	Dr.D.Madhesh	Dept. of Mechanical Engineering

8	Dr.K.Thiruvengkatasamy	Dept. of Naval Architecture and Offshore Engineering
VI Two Associate Professors from the Departments other than the Heads of the Departments by Rotation of Seniority		
1	Cdr.V.Ramakrishnan	Dept. of Nautical Science
2	Dr.D.Jaisankar	Dept. of English
VII Two Assistant Professors from the Departments by Rotation of Seniority		
1	Mrs.R.Divyaranjani	AMET Business School
2	Dr.P.Sivabalan	Dept. of Naval Architecture and Offshore Engineering
VIII Eminent educationist from other field related to the activities of the Institution / Deemed to be University who are not in the service of the Institution / Deemed to be University, nominated by the Vice - Chancellor		
1	Dr.K.Sivasami	Associate Professor & Head - SMET at Indian Maritime University, Chennai
2	Dr.Kalyani Desikan	Dean-Academic Research, VIT Chennai Campus
3	Dr.S.Swamynathan	Professor of Information Technology Anna University
4	Dr.S.Rajadurai	President, CEO and Head of R&D Sharda Motor Industries Ltd.
5	Dr.Chitra Krishnan	Former Professor and Head, Department of French, University of Madras
IX Three persons who are not members of the Teaching staff, co-opted by the Academic Council for their specialised knowledge		
1	Capt. Saurabh Mahesh	Head Crew Sourcing & Global Cadet Administration, Maersk Line, Singapore
2	Capt. Rohan Sabnis	Manager (Fleet Personnel & Training), Goodwood Ship Management Pvt. Ltd.
3	Capt. Sartaj Gill	Chief Executive Officer, V.Ships
4	Mr.J.ShyamSundar	Director & Chief Operating Officer Flyjac Logistics Pvt. Ltd.
Student Representatives		
1	Cadet. Bhuvaneswari	Reg.No.ASL19034, Yr:2019-21

2	Cadet. Keerthivasan	Reg.No.AME20051, 3rd year
3	Mr.Zuben Kalyan	Reg.No.AEE20015, 3rd year
4	Ms. Athira Viswanath	Reg.No.ASL21011, 1st year

Board of Studies

Composition of the Board of Studies

- Head of the Department – Chairperson
- All Professors of the Department-Members
- Two Associate Professors of the Department by rotation based on seniority
- Two Assistant Professors of the Department by rotation based on seniority
- One or Two Students of the Department
- External Experts/External Members

a. Industry Representative

b. Academic Expert

c. Alumni

Table 10.8 Membership of Board of Studies (Department of NA&OE)

S.No	Name	Membership category
1	Prof.MSP Raju	Chairman, HOD, Department of Naval Architecture & Offshore Engineering.
2	Prof. Bhoopathy Baskaran	Dean - Marine Engineering
3	Dr.T.Sasilatha	Dean - Academics
4	Dr.D.Arivazhagan	Director - Ranking & Accreditations
5	Dr.R.Karthikeyan	HoD- Dept. of Mechanical Engineering.
6	Dr.Durai Mutharasan	HoD- Dept. of Computer Science & Engineering
7	Dr.P.Balaganesan	HoD- Department of Mathematics
8	Dr.K.K.Sivakumar	HoD- Department of Chemistry
9	Dr.C.M.Ramakritinan	HoD- Dept. of Marine Biotechnology
10	<i>Dr.R.Ganesan</i>	HoD- Department of English
11	<i>Prof.N.Srinivasan</i>	Director- Placements
12	<i>Dr.K.Thiruvengatasamy</i>	Professor
13	Dr.J.Suneela	Assistant Professor
14	Dr.P.Sivabalan	Associate Professor
15	Dr.Lilly R	Associate Professor
16	Dr.Prasob PA	Assistant Professor
17	Dr.Rajiv Sharma, Professor, Department of Ocean Engineering, IIT, Chennai.	External expert – Academics
18	Commander. Kamal Kanagat, General Manager, Larsen and Toubro, Ship Building, Chennai	External expert – Industry
19	Mr.Vikram Selvaganesh, Design Engineer, Larsen &Toubro, Ship Building, Chennai.	Alumni

20	Mr.M.Gughan, Survey Coordinator, Synergy Marine Group, Ship Management and Marine Services, Chennai	Alumni
21	Mr.Maria Infant Vinobel	B.E. Final year student
22	Ms.Sanjana	B.E. Final year student
23	Dr.Rajiv Sharma, Professor, Department of Ocean Engineering, IIT, Chennai.	External expert – Academics
Special Invitees		
24	Dr.Anitha R Warriar	Associate Professor, Department of Physics

Powers and Functions of the Board of Studies

Board of Studies is one of the most important bodies of the University, which reviews the design, development and revision of curriculum and syllabus of all the programmes. There are separate Boards of Studies in each Department of the University. The Board of Studies usually consists of both internal and external members with a representation of all stakeholders.

Finance Committee

Finance Committee is a mandatory Committee as per the regulatory bodies of the University. The Committee plans, monitors and reviews the financial resources of the University. The Committee considers the annual accounts and financial estimates of the University and submits them to the Board of the Management for its approval. The Committee also recommends the annual budget and revised estimates to the Board of Management. Finance Committee has the powers to fix limits of the total recurring expenditure and the total non-recurring expenditure of the year based on the income and resources of the University. The Finance Committee meets at least twice a year to examine the accounts and to scrutinize proposals for expenditure.

Table 10.9. Members of the Finance Committee

S.No.	Name of the Member	Position
1	Col.Dr.G. Thiruvassagam Vice Chancellor	Chairperson
2	Mr. R. Thiyagarajan, B.Sc, FCA	Nominee of the Trust
3	Mr.K.Balamananandan, FCA	Nominees of the Board of Management
4	Dr. J. Rengamani, Member	Member of the Board of Management
5	Mr. K. Narayan Singh Joint Secretary, UGC	UGC Nominee
6	Mrs. K. Jayabharathi Finance Officer	Secretary

Powers and Functions of the Finance Committee

- The Finance Committee reviews and reports the annual accounts and financial estimates of the University and submits them to the Board of the Management for its approval
- The Annual Budget is prepared and submitted by each Department and functional areas. The budget proposals are reviewed in the Finance Committee and based on the merit and requirement it recommends sanction and releases of funds by the Board of Management
- The Finance Committee fixes limits of the total recurring expenditure and the total non-recurring expenditure of the year based on the income and resources of the University

Management Review Committee

AMET is certified with ISO 9001-2015 by the World renowned classification society, Det Norske Veritas (Norway) and Germanischer Lloyd (Germany)-DNV GL. As per the ISO 9001-2015, AMET has a Quality Management System (QMS) which has detailed forms and procedure for quality sustenance. As per the QMS, the Management Review Committee consisting of Internal Members alone, monitors the quality related academic, administrative system and support service activities. The system has strong Internal Audit Mechanism by which minimum two internal audits are conducted every year by qualified and trained internal auditors, while the reports are presented in the Management Review Committee Meeting. In Management Review Committee Meeting, the outcome and action taken reports of all the internal audits, academic practices and administrative processes are presented and reviewed.

Functions of the Management Review (MR) Meeting

Through the Management Review Meeting, regular meetings are conducted as per a formal agenda consisting of:

- a. The status of actions from previous management reviews
- b. Changes in external and internal issues that are relevant to the quality management system
- c. Information on the performance and effectiveness of the quality management system, including trends in:
 - i. Audit results - first party, second party and third-party audits
 - ii. The extent to which quality objectives have been met
 - iii. Process performance - result analysis
 - iv. Non-conformities and corrective actions other than the audit points
 - v. Monitoring and measurement results (course monitoring review)
 - vi. Stakeholders feedback, complaints and concerns from students and relevant interested parties
 - vii. The performance of external service providers
- d. The adequacy of resources
- e. The effectiveness of actions taken to address risks and opportunities
- f. Opportunities for improvement - recommendations from participants
- g. Any other issues related to QMS

The members shall come prepared for the review meetings with necessary inputs from their areas of responsibility. The decision taken and time bound action plans drawn at the review meetings are recorded and circulated by Management Coordinator to all members for necessary action and feedback. These decisions include resolutions made at MR Meeting for improvement of effectiveness of QMS, satisfaction of interested parties and sanction of new resources to realize the above objectives. Management Coordinator maintains records of management reviews and related correspondence.

Table 10.10. Members of the Management Review Committee

Sl. No.	Name of the Member	Designation	Position
1	Col.Dr.G.Thiruvasagam	Chairperson	Vice Chancellor
2	Dr. M. Jayaprakashvel	Secretary	Registrar i/c
3	Dr.D.Arivazhagan	Convener	Management Coordinator
4	Members	Ex Officio Heads of all Academic, Administrative and Support Service Departments	Members

Frequency of Meetings, Attendance and Membership details of Governing Bodies

Regular meetings of the governing bodies as per the UGC Regulations with stipulated attendance are held and details are presented as below.

Table 10.11 Frequency of Meetings, Attendance and Membership details of Governing Bodies

Name of the Governing Body	Frequency of Meetings	Attendance Requirement/ Minimum quorum
Board of Management	4 meetings in a year	Two Third of all Members
Finance Committee	2 meetings in a year	Two Third of all Members
Planning and Monitoring Board	2 meetings in a year	Two Third of all Members
Academic Council	3 meetings in a year	Two Third of all Members

Board of Studies	1 meeting in a year	Two Third of all Members
Management Review Committee	2 meetings in a year	Two Third of all Members

Sample meeting notice, attendance, minutes of the meeting and action taken reports of the Governing/Administrative bodies

Few sample meeting notice, attendance, minutes of the meeting and action taken reports of the Governing/Administrative bodies are given as Annexure as per the following table.

Table 10.12 List of annexure for the Governing Bodies and Administrative Bodies.

Sl.N o.	Name of the Governing / Administrat ive Body	Annexu re No.	Annexure Details (Selected Sample)	Annexure URL
1	Board of Manageme nt	10.1	1.Meeting Notice - Enclosed 2.Meeting attendance - Enclosed 3.Minutes of the Meeting- Enclosed	Annexure for the (https://drive.google.com/file/d/1Vjj8lguG9r6dtQjQaRKpA1yoKI_o_vf8L/view?usp=sharing) <u>Board of Management</u>
2	Planning and Monitoring Board	10.2	1. Meeting Notice – Enclosed 2. Meeting attendance - Enclosed 3.Minutes of the Meeting - Enclosed	Annexure for the Planning and Monitoring Board (https://drive.google.com/file/d/1CoXcnEMxgOn7zCwwQBR32G-Fcw7LYZC_/view?usp=sharing)
3	Board of Studies	10.3	Sample from one Department Board of Studies - Enclosed	Annexure for the Board of Studies (https://drive.google.com/file/d/1xRwISisFhIUDSZodpCDbgRPNH56uvAIE/view?usp=sharing)
4	Finance Committee	10.4	1. Meeting Notice - Enclosed 2. Meeting attendance – Enclosed 3. Minutes of the Meeting - Enclosed	Annexure for the (https://drive.google.com/file/d/1gQjmrKLXr9acwqn3zUx2hvFuPDZUukKM/view?usp=sharing) <u>Finance Committee</u>

Organisational Structure and Administrative Set Up

Being a Deemed to be University, AMET has a well established organizational structure by following the guidelines of the University Grants Commission. Salient features of the Organizational Structure/Administrative set up of the University are as follows:

1. Board of Management is the apex body for the University; The Vice Chancellor is the Chairperson of the Board of Management; The Board of Management shall delegate its powers to Vice Chancellor
2. The Board of Management consists of both Internal Members and eminent external experts as Members
3. The Vice Chancellor takes the recommendations of the various governing bodies and committees for taking decisions and there by strengthen the Participatory Management
4. The Board of Management and the Vice Chancellor has the powers to form various Committees and Boards, including sub committees for the effective functioning of the University through which the decentralization of powers is practiced in the University.
5. The Pro Vice Chancellor, the Registrar, Controller of Examinations and Finance Officer are the officers of the University who reports to the Board of Management through the Vice Chancellor.
6. Apart from these governing bodies and officers of the University, the Institution is having several mandatory committees/cells for the effective functioning of the University. A few such selected Committees/Cells are as follows
 - a. Committee Against Sexual Harassment
 - b. Anti-Ragging Committee
 - c. Grievance Redressal Committee
7. The University is also having many advisory Committee for the proposal, implementation and monitoring of academic and administrative improvement activities. A few of them are listed below
 - a. Internal Quality Assurance Cell (IQAC)
 - b. Research Advisory Committee
 - c. Student Council
 - d. Staff Welfare Committee
 - e. Library Committee
 - f. Women Empowerment Cell

The overall University organizational structure and administrative set up is summarized in Figure 10.1

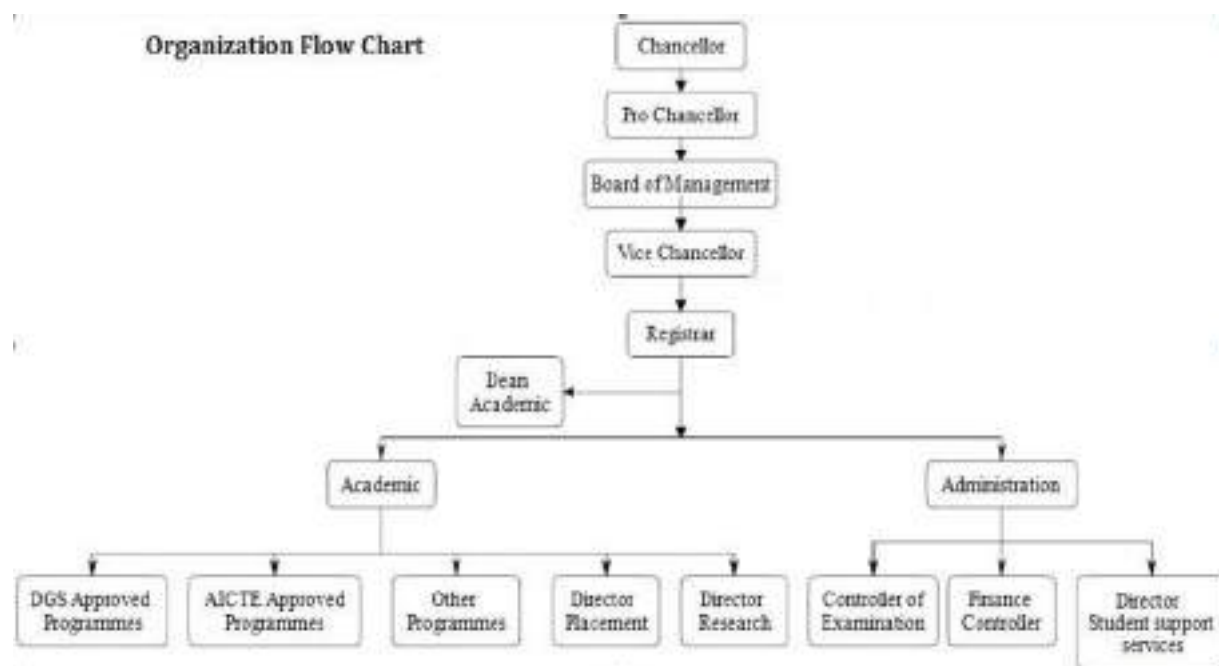


Figure 10.1 Organization Flow chart

Other Support Services/Administrative Committees

Table 10.13. Frequency of Meetings, Attendance, Membership details and Support Services/Administrative Committees

Name of the Support Service/Administrative body	Frequency of Meetings	Attendance Requirement /Minimum quorum	Membership Details
Internal Quality Assurance Cell (IQAC) of AMET	Minimum two meetings per year	Two Third of all Members	Given in Table 10.14
Anti-Ragging Committee- Key functions are to maintain AMET a ragging-free campus by creating awareness and attending ragging related complaints	As per the mandate of the UGC; As and when required; Minimum two meetings per year	Two Third of all Members	Given in Table 10.15 (Internal and External members)
Women Empowerment Committee-Creates awareness about women empowerment in campus by organizing events	Twice in a Year; However, Committee Conducts regular awareness events	Two Third of all Members	Given in Table 10.17
Grievance Redressal Committee –frames the policy and procedures for the Grievance Redressal; Monitors the same	As and when required; Minimum two meetings per year	Two Third of all Members	Given in Table 10.18
Research Advisory Council-frames the policies and procedures for the operation academic, funded and consultancy research in the University	As and when required; Minimum two meetings per year	Two Third of all Members	Given in Table 10.19
Library Committee - providing necessary inputs for the purchase of library resources	Minimum two meetings per year	Two Third of all Members	Given in Table 10.20

Internal Quality Assurance Cell (IQAC) of AMET

Quality enhancement is a continuous process; the IQAC shall become a part of the institution's system and work towards realization of the goals of quality enhancement and sustenance. The prime task of the IQAC is to develop a system for conscious, consistent and catalytic improvement in the overall performance of institutions.

Broad functions of the AMET IQAC are as follows:

- Ensuring timely, efficient and progressive performance of academic and administrative tasks
- Enhancing the quality of academic and research programmes
- Making the University environment and academic programmes accessible for all sections of the society
- Enhance the Institutional Social Responsibilities
- Advice upon the modern methods of teaching and learning
- Creation of credible feedback mechanisms
- Ensuring secure and transparent evaluation system
- Advice upon the enhancement of physical infrastructure

- Making the Green Campus by inculcating the environmental consciousness
- Ensuring the availability and maintenance of support structure and services
- Enhance the collaborations with Industry, Academia and other spheres of the society
- Development of quality benchmarks for various academic and administrative activities of the Institution
- Arrangement for feedback response from students, parents and other stakeholders on quality-related institutional processes
- Dissemination of information on various quality parameters of higher education
- Organization of inter and intra institutional workshops, seminars on quality related themes
- Documentation of the various programmes/activities leading to quality improvement
- Acting as a nodal agency of the Institution for coordinating quality-related activities, including adoption and dissemination of best practices;

Table 10.14: Members of the AMET IQAC

S.No.	Name of the Member	Designation	Position
1	Col.Dr.G.Thiruvagam	Vice Chancellor	Chairman
2	Mr.S.Karikalan	Senior Vice President-AMET,	Member (Management)
3	Dr.M.Jayaprakashvel	Registrar I/C	Member
4	Dr V Ramachandran	Director-Distance Online Education	Member
5	Dr.R.M.Chandrasekaran	Former Vice Chancellor, University of Madras, Chennai	Member
6	Dr.Abudhahir.A	Director (IQAC) B.S. Abdur Rahman Crescent Institute of Science & Technology Vandalur, Chennai	Member
7	Dr.Sanjib Kumar Patnaik	Professor, Director- Centre for Academic Courses, Dept. of EEE, Anna University, Chennai	Member
8	Dr.S.Balakrishnan	IQAC Coordinator Professor Dept. of Earth Sciences, Pondicherry University	Member
9	Capt. K. Karthik	Principal-DGS Courses	Member
10	Prof.Dr.T.Sasilatha	Dean- Academics	Member
11	Dr.S.Prabhakaran	Professor- Marine Engineering	Member
12	Mr R Sundar	Assistant Professor-Marine Engineering	Member

13	Chairperson of the Student Council AMET	Student	(Student – Member)
14	Mr B Dinesh	Crew Operator- Mearsk, Chennai	(Employer – Member)
15	Mr Arumugam	President-Fishermen Association, Kanathur	(Local Society – Member)
16	Mr. K. Muralitharan	Manager-NRR Finance, Trichy	(Parent-Member)
17	Dr R Karthik	Regional Business Manage-Provet Pharma Private Limited, Chennai	(Alumni - Member)
18	Dr D.Arivazhagan	Professor-AMET Business School	Member - Coordinator

Anti-Ragging Committee

Broad functions of the Anti-Ragging Committee are as follows:

1. To ensure compliance with the regulations as well as the provisions of any law currently in force concerning ragging and directions of the Honble Supreme Court.
2. To monitor and oversee the performance of the Anti-Ragging Squad in the prevention of ragging in the University.
3. To publicize to all students and prevalent directives and the actions that can be taken against those indulging in ragging;
4. To consider the complaints received from the students and conduct enquiry and submit a report to the Anti- Ragging Committee along with punishment recommended for the offenders;
5. Oversee the procedure of obtaining an undertaking from the students in accordance with the provisions;
6. To take all necessary measures for prevention of ragging inside the Campus/ Hostels

Table 10.15: Members of the Anti-Ragging Committee

Chairman	
Col.Dr.G.Thiruvassagam	Vice-Chancellor & Chairperson
Members	
Dr. M. Jayaprakashvel	Registrar I/C (9840529274)
Capt.K.Karthik	Principal- DGS Courses (9444084497)
Dr. N.R. Ramkumar	Dean-Students Welfare (9381080111)
Mr.Mahesh Henry	Advocate (9444319995)
Inspector of Police	Kanathur (94981 00175)
Ms. Sandhya Rani Ramadass	Lady Member of NGO and Psychologist(9840982160)
Mr. Dilli Ganesh	Local Media
Mr. Mathew Varghese	Proctor
Dr.K.Gayathri	Assistant Professor-Department of Physics (97899 33245)

Dr.R.Vetriselvan	Assistant Professor & Faculty Warden - Ganga Hostel (9788161399)
Mr.Prem Anandh	Asst. NCC Officer & Faculty Warden - Yamuna Hostel (9003360606)
Mr.Muthaiyalraj	Assistant Security Officer (98407 02091)
Dr.Meher Taj	Assistant Professor & Faculty Warden - Narmada Hostel
Cdt.Adhitya Sathish Kumar B.Sc Nautical Science, 3rd year	Chairman, Student Council
Ms.Anna Joy MBA 1st year	Secretary, Student Council
Mr.Jagadeesh B.E Marine Engineering, 4 th year	Deputy Parade Commander

Table 10.16: Members of the Anti-Ragging Squad committee

Name of the Committee Member	Profession	Mobile Number	e-mail address
Dr.M.Jayaprakeshvel	Registrar	98405 29274	registrar@ametuniv.ac.in (mailto:registrar@ametuniv.ac.in)
Dr.N.R.Ramkumar	Dean- Students Welfare	93810 80111	proctor@ametuniv.ac.in (mailto:proctor@ametuniv.ac.in)
Capt .K.Karthik	Principal- DGS Courses	94440 84497	karthik.k@ametuiv.ac.in (mailto:karthik.k@ametuiv.ac.in)
Dr.T.Sasilatha	Dean- Academics	94447 52994	deaneem@ametuniv.ac.in (mailto:deaneem@ametuniv.ac.in)
Dr.D.Rajasekar	HoD- Management Studies	98846 91744	rajasekar.d@ametuniv.ac.in (mailto:rajasekar.d@ametuniv.ac.in)
Prof.MSP. Raju	HoD-NA&OE	99896 25271	msp.raju@ametuniv.ac.in (mailto:msp.raju@ametuniv.ac.in)
Mr.Muthaiyalraj	Assistant Security Officer	98407 02091	-
Mrs. Pansy Chitty	Assistant Warden	98403 93972	-

Dr.K.Gayathri	Assistant Professor, Department of Physics	97899 33245	gayathri.k@ametuniv.ac.in (mailto:gayathri.k@ametuniv.ac.in)
Mr.P.C.Selvan	Deputy Registrar, Administration	9840196007	paul.r@ametuniv.ac.in (mailto:paul.r@ametuniv.ac.in)

Women Empowerment Committee

AMET, which is unique in offering maritime courses where girl students also get trained for shipping and maritime jobs. The WEC was established with the aim of supporting and providing resources for women (Students, Staff, Research Scholars and Supporting Staff) to make them empowered, economically independent and socially active. WEC organizes various workshops, seminars, extension activities centred towards empowering women of AMET and in the society as well.

Table 10.17 Members of the Women Empowerment Committee

S.No.	Name	Category / Roll in the Committee	Designation
1	Col.Dr.G.Thiruvagam	Chairman	Vice Chancellor
2	Dr. Sangeetha Albin	Member	Joint Registrar
3	Ms.Shanthi H.J	Member	Assistant Professor, Department of Computer Science & Engineering
4	Dr. Anitha Warriar	Member	Associate Professor, Department of Physics
5	Ms.Diviyaranjani R	Member	Assistant Professor, AMET Business School
Students office bearers:			
6	Ms.Anna Joy	Secretary	I Year MBA
7	Ms. Ruba	Member	III Year BBA
8	Ms.Sangamithra	Member	I Year B.Sc (Computer Science)
9	Ms. Sruthi	Member	II Year B.E (NA & OE)

Table 10.18. Members of the Grievance Redressal Committee.

S.No.	Name	Designation
1	Col.Dr.G.Thiruvagam	Vice-Chancellor & Chairperson
2	Dr.M.Jayaprakashvel	Registrar
3	Dr.T.Sasilatha	Dean -Academics
4	Mr.P.C.Selvan	Deputy Registrar-Administration

5	Ms.G.Bhavani	Student (II year BBA Roll No: ABA18014)
6	Ms. R. Pramiti Roy	Student III year (B.E (EEE) Roll No: AEE20017)

Online Grievance Redressal System

University has also made the provision to receive and attend various grievances of students online through University website portal ametuniv.edugrievance.com (<http://ametuniv.edugrievance.com/>). Students can register and login using their email ID /mobile number and post their grievance are being addressed by the Grievance Redressal Committee officials concerned.

Table 10.19 Members of the Research Advisory Council

S.No.	Member	Designation
1	Col.Dr.G.Thiruvasagam	Vice Chancellor, AMET (Chairperson).
2	Dr. R. Dr.William Selvamurthy	President, AMITY University, Former Chief Controller – R&D, DRDO.
3	Dr.T.Balasubramanian	Former Vice Chancellor, CARE Deemed to be University (Chettinad) Chennai.
4	Dr.R.Venkatesan	Scientist G National Institute of Ocean Technology (Rtd), Chennai.
5	Mr.Vinay Kumar	Scientist F, Department of Scientific and Industrial Research (DSIR), Government of India, New Delhi
6	Dr.S.Nallayarasu	Professor and Head, Department of Ocean Engineering, IIT Madras
7	Dr.Vasudevan	Scientist G, National Physical Oceanographic Laboratory (DRDO), Kochi
8	Dr.K.Kadirvelu	Scientist F, DRDO-BU Centre for Life Sciences Coimbatore
9	Prof.Sudhir Varadharajan	School of Interdisciplinary Design and Innovation, ITDM Kanchipuram
10	Capt.K.Karthik	Principal- DGS Courses, AMET
11	Prof.Bhoopathy Bhaskaran	Dean- Marine Engineering, AMET
12	Dr. T .Sasilatha	Dean- Academics and International Relations, AMET
13	Dr.D.Rajasekar	Dean- Academic Research, AMET
14	Dr.Anita R Warriar	Associate Professor and Head i/c, Department of Physics, AMET
15	Dr.C M.Ramakritinan	Professor and Head, Department of Marine Biotechnology, AMET
16	Dr.D.Velumurugan	Director-Research and Development Cell (RDC)- Secretary and Convener
17	Dr.M.Jayaprakashvel	Registrar, AMET (Ex-Officio Member)

Library Committee

The Prof VBS Rajan Library is the Central Library of the University in a separate building with a total area of 30,000 Sq.ft. The Library has a large collection of books with updated editions in different titles and volumes. The Library has a Digital Knowledge Centre in which research and publication databases, software etc. are available and updated every year. The Library is fully automated with RFID and Barcoding system. The Library Committee of the University is providing necessary inputs for the purchase of library resources.

Table 10.20 Members of the Library Committee

S.No.	Name of the Members	Designation
1	Dr. M. Jayaprakashvel, Registrar i/c	Chairman (Ex Officio)
2	Capt. Ramakrishnan Venkat Associate Professor Department of Nautical Science	Member
3	Dr. S. Prabhakaran, Professor Department of Marine Engineering	Member
4	Dr. V. Leela Vinodhan, Associate Professor Department of Mechanical Engineering	Member
5	Dr. S.Krishna, Assistant Professor Department of English	Member
6	Dr.Prakash I.N, Librarian, Alliance University, Bengaluru, Karnataka-India	Member
7	Dr.PPanner selvam, Librarian Central library, B.S Abdul Rahman Crescent Institute of Science and Technology Vandalur, Chennai	Member
8	Mr. R. Sundar, Assistant Professor Department of Marine Engineering	Member
9	Dr.K. Sekar, Librarian	Member Secretary (Ex Officio)

Service Rules for the Key Leadership and Officials of the University The Chancellor

The Chancellor presides over the convocations of the University but shall not be the Chief Executive Officer. The Chancellor is appointed by the AMET Trust and the duration and tenure of appointment are given in the MoA. Where power is conferred upon the Chancellor to nominate persons to authorities, he/she shall, to the extent necessary, nominate persons to represent the various interests for the furtherance of the objectives of the University.

The Pro Chancellor

The Pro Chancellor is also appointed by the AMET Trust. The role of Pro Chancellor is limited to carrying out the tasks assigned to the Chancellor, when the latter is not available for carrying out the same.

The Vice-Chancellor

The Vice-Chancellor is a whole time salaried Officer of the University and has been appointed by the Chancellor from a panel of three names suggested by the Search-cum-Selection Committee. The Vice-Chancellor is the Principal Executive Officer of the University and will exercise general supervision and control over the affairs, and will be mainly responsible for implementation of the decisions of all its Authorities.

The Vice-Chancellor is the Ex-Officio Chairperson of the Board of Management, the Academic Council, the Finance Committee, the Planning & Monitoring Board, and Selection Committees. The Vice-Chancellor is having the powers to convene or cause to be convened meeting of the various authorities of the University. As per the Regulations and Rules of the University, the Vice-Chancellor is empowered to exercise any power conferred upon any authority of the University and to take action and then report to the authority concerned on the action taken by him / her on such matter. However, if the relevant authority is of the opinion that such action ought not to have been taken, it may refer the matter to the Chancellor whose decision thereon will be final.

If any person in the service of the University is aggrieved by the action taken by the Vice – Chancellor, he / she shall have the right to appeal against such decision to the Board of Management within 30 days where the Board of Management can confirm, modify or reverse the action taken by the Vice-Chancellor.

The Vice-Chancellor can exercise all other powers delegated by the Board of Management. The Vice-Chancellor is also having the power to re-delegate some of his/her powers to any of his/her sub-ordinate Officers with the concurrence and approval of the Board of Management. It is the duty of the Vice-Chancellor to ensure that Regulations and Rules of the University are duly observed and implemented. All powers relating to the proper maintenance and discipline of the University are vested in the Vice-Chancellor.

The Pro Vice-Chancellor

The Pro Vice-Chancellor is appointed by the Board of Management on the recommendations of the Vice-Chancellor. The Pro Vice-Chancellor will hold Office co-terminus with the Office of the Vice-Chancellor. The Pro Vice-Chancellor is having the powers and duties as prescribed by the Rules of the University.

The Registrar

The Registrar is a whole time salaried Officer of the University and is appointed by the Board of Management on the recommendations of the Selection Committee.

The Registrar is the ex-officio Secretary of the Board of Management, the Academic Council and the Planning and Monitoring Board. However, The Registrar is not a member of any of these authorities.

The Registrar is directly responsible and working under the orders of the Vice-Chancellor. The following are the duties of the Registrar: -

1. To be the custodian of the records and the funds and such other property of the University as the Board of Management may commit to his / her charge
2. To conduct the official correspondence on behalf of the authorities of the University
3. To issue notices convening meetings of the authorities of the University and all Committees and sub-Committees appointed by any of these authorities
4. To maintain the minutes of the meetings of all the authorities of the University and of all the Committees and sub-Committees appointed by any of these authorities
5. To make arrangements for the examinations conducted by the University
6. To represent the University in suits or proceedings by or against the University, sign powers of attorney and perform leadings or depute his / her representatives for this purpose
7. To enter into an agreement, sign documents and authenticate records on behalf of the University
8. To make arrangements to safeguard and maintain the buildings, gardens, office, canteen, cars and other vehicles, laboratories, libraries, reading rooms, equipment and other properties of the University
9. To perform such other duties as may be specified in the Rules of the University, or as may be assigned by the Board of Management or the Vice-Chancellor from time to time

The Finance Officer/Financial Controller

The Finance Officer is a whole time salaried officer of the University appointed by the Board of Management. The Finance Officer will work under the direction of the Vice-Chancellor and shall be responsible to the Board of Management through the Vice-Chancellor. He/she is responsible for the preparation of annual budget, estimates and statements of account for submission to the Finance Committee and the Board of Management. He/she is responsible for the management of funds and investments of the University, subject to the control of Board of Management.

The Controller of Examinations

The Controller of Examinations (COE) is appointed by the Board of Management. The COE ensures that all the specific directions of the Board of Management, Academic Council and Vice- Chancellor in respect of examination and evaluation are complied with. The Controller of Examinations is the permanent invitee to the Academic Council.

Chairpersons, Deans, Directors and Coordinators

In each of the schools of the University, a Chairperson is appointed by the Vice Chancellor who is responsible for the overall guidance of the Departments in the school. The Vice Chancellor appoints Deans and Directors for specialized functions. The Deans of different schools or specialized functions will work under the orders of the Vice Chancellors. Faculty members of the University are given additional responsibility as the Directors of various Centers and Coordinators of various Cells. They carry out the specialized functions of the Cells/Centers and report to the Vice Chancellor.

The Head of the Department

There will be a Head of the Department for each of the Departments in the University who shall be appointed by the Vice-Chancellor from amongst the Professors of the Department. Provided that if there is no Professor in the Department or there is only one Professor in the Department whose term as Head of the Department is ending, the Vice-Chancellor may appoint an Associate Professor as Head of the Department. The powers and functions of the Head of Department are prescribed by the Rules of the University.

Miscellaneous matters pertaining to authorities of the University

If any question arises, as to whether any person has been duly nominated or appointed as, or is entitled to be a member of any authority or any committee of the University, the matter shall be referred to the Chancellor, whose decision thereon shall be final and binding.

Any member, other than an ex-officio member of any authority, may resign by a letter addressed to the Registrar and the resignation shall take effect as soon as it is accepted by the Chancellor or the Chairperson of the Board of Management, as the case may be.

Sudden vacancies among the members of any authority or any Committee of the University shall be filled by the respective authority, as soon as may be, and within a period of six months.

A person shall be disqualified for being chosen as, and for being a member of any of the authorities of the University;

- a. If he/she is of unsound mind
- b. If he/she is an un-discharged insolvent
- c. If he/she has been convicted by a court of law for an offense involving moral turpitude
- d. If he/she has not been appointed as per the provisions of these Regulations.

Service rules, Procedures, Recruitment and promotional policies General Services Rules

1.Hours of Work

The working hours for the teaching faculty are between 9.00 A.M and 4.30 P.M for the non-teaching and administrative staff, the hours are from 9.00 A.M to 5.00 P.M. An Employee is expected to put around 48 hours of work every week.

2.Attendance and Punctuality

Regular attendance is essential to the efficient functioning of the institution and is a necessary condition of employment. Employees are expected to report to work as scheduled and on time.

3.Identification Card

Identification cards are issued to all staff members and they are expected to wear them while they are in the campus.

4.Code of conduct

a. Dress code / Foot wear

The Institute observes a formal dress code. All Male and Female employees should use discretion in wearing attire that is appropriate for the work in the institute.

b. Foot wear

Staff members are requested to wear slip-on shoes (Cut shoes or Pump shoes) or regular slippers with normal heels.

c. Prohibited Activities

The institute prohibits the consumption of alcohol and drugs, and gambling inside the premises. In addition, any damage caused to the institute's property or cases of dishonesty and harassment or indulging in violent behaviour with students, visitors or other staff, etc. will lead to disciplinary action, which may also lead to suspension or termination for failure to comply with institute's policy.

d. Use of Telephone, Internet and Computer

The institute understands that when employees work during the week it is occasionally necessary to conduct personal business during office hours. However, employees should limit their personal use of the telephone, computer and internet during office hours.

e. Library

The library will be kept opened from 9 a.m. to 9 p.m. on all working days and from 9 a.m. to 5 p.m. on Sundays. Staff and students can make use of this facility. The Library can also be accessed from anywhere using INFED supported remote access facility.

Policies and Procedures

AMET is keen on developing and improvising various policies and proposals for the smooth functioning of the University. The Policies define the scope, aims and objectives, procedures, authorities concerned etc. These policies and procedures are uploaded in the University website (Link: <https://www.ametuniv.ac.in/policies.html> (<https://www.ametuniv.ac.in/policies.html>))

Table 10.21 Policies and Procedures of the University

Name of the Policy	Year of Implementation and publication
Admission Policy	2013
AMET Academic Integrity and Plagiarism Policy	2017
Fee Fixation Policy	2013
Governance Policy	2013
Infrastructure Policy	2013
Policy for Preventing Alcohol and Drug Abuses	2013
Policy for the Prevention, Prohibition and Punishment of Sexual Harassment of Women	2013
Policy on Curbing the Menace of Ragging	2013
Policy on Curriculum Design and Development	2013
Policy on Ethics in Research	2013

Quality Policy of AMET Deemed to be University	2013
Code of Conduct Handbook	2007
Vision 2025	2019
Recruitment and Promotional Policy	2007

Recruitment and Promotional Policies

AMET adheres the guidelines and regulations of the Regulatory and Statutory bodies such as University Grants Commission (UGC), All India Council of Technical Education (AICTE) and Directorate General of Shipping (DGS). The qualifications, emoluments and promotional avenues are also fixed based on the guidelines of UGC/AICTE/DGS.

The Recruitment and Selection process is by and large as per the guidelines of the University Grants Commission. The generalized procedure of recruitment and promotion are summarized below.

1. The Departments and Functional Areas identify the requirement of recruitment well in advance and bring it to the knowledge of Board of Management through Vice Chancellor
2. The Vice Chancellor/Board of Management reviews the recruitment requirement and process for notification of vacancy.
3. For all the teaching and key administrative positions, appropriate notification of vacancy is made by following necessary publicity measures
4. The applications are scrutinized by the University either with the help of Internal/External experts or both based on the need. The eligible applications are scrutinized for further selection process
5. Appropriate Selection Committee are formed by the Vice Chancellor for each recruitment process. The Vice Chancellor is the default Chairperson of the Selection Committee which contain both internal and external subject experts
6. The appointments are made based on the Personal Interview by the Selection Committee. Necessary wait lists are also prepared
7. Newly appointed Faculty/Staff are required to produce necessary educational qualification certificates and other documents for verification
8. The Promotion is based on Performance Based Appraisal System (PBAS) with due consideration to Academic Performance Indications (API) in case of teaching faculty and appraisal forms in case of non-teaching staff.
9. Promotional avenues based on Career Advancement Scheme are also available for the exemplary performers.
10. At appropriate sessions, University notifies the Faculty for Career Advancement Scheme in which applications are scrutinized by Scrutinizing Committee and eligible applicants are interviewed by the Selection Committee (with External Experts as Members). Based on the recommendations of the Selection Committee Board of Management/Vice Chancellor promotes the Faculty to the eligibility level.
11. University reserves the rights to cancel/revoke the appointment and promotions or terminate the probationary period of any employee for valid reasons that may deem fit to the Management.

Probationary period:

A new employee will be on probation for a period of One year from the date of joining. Employee's performance during the probationary period will be at the discretion of the management and thoroughly assessed, and only on satisfactory completion of initial or extended period of probation, new employee will be confirmed in the regular service of the University.

Resignation

Employees who wish to leave the services of AMET, he/she will be relieved only at the end of a semester, provided they have to tender three months' notice or surrender three month's salary. However, it is left to the discretion of the Management, to relieve earlier. Likewise, if employee's service is not satisfactory / required by the Management, it has the discretion to terminate the services at any point of time, with a formal notice.

Policy on Employee Welfare

1. University has established a full time medical centre for the healthcare of Employee and students
2. The Management is keen on extending the benefits of "Employees Provident Fund" to the employees of AMET. Eligible regular employees (Teaching, Non-Teaching and Supporting) are registered under the EPF scheme.
3. As per the E.S.I. scheme, Staff have to contribute an amount of 12% of their total earnings every month. The Employer (AMET Management) in turn, shall contribute a sum of 12% of the total earnings of the employee Gratuity.
4. All Employees completing 2 years of service in our Institution are eligible to cover under the Group insurance scheme with applicable benefits (Accidental hospitalization for Rs. 1,00,000/- per staff; Accidental death claim for Rs. 1,00,000/- per staff; Hospitalization expenses up to Rs. 25000)
5. Teaching Faculty are eligible for getting Financial Assistance to attend events for their professional developments ((National level – Max. limit: Rs. 3,000; International – Max. limit: Rs. 10,000)
6. University extends financial assistance to Faculty for research publications (50% Publication cost in "Journals" from Scopus Indexed, SCI Indexed) (Max. limit – Rs. 10,000)
7. University provides full fee waiver for Faculty who do part time PhD at AMET.
8. University provide incentives for paper publications, book publications, revenue generation through funded projects and consultancy.
9. University extends financial supports for Faculty who organize curricular, co-curricular and extracurricular events both in campus and outside the campus
10. University financially supports faculty for the development of educational materials such as E contents (video lectures, short term courses, MOOCs etc.)
11. All Teaching and Non-Teaching are eligible for applicable leave benefits as given in Table. 10.22

Table 10.22 Leave benefits to the Employee of AMET

Leave	Numbers of Days	Eligibility
Casual Leave	12 days with pay	All staff

Medical Leave	15 days per year with pay	All the staff who completed probationary period
Maternity Leave	3 months with full pay (for first 2 children only)	All the staff who completed probationary period
Paternity Leave	15 days with full pay	All the staff who completed probationary period
Vacation Leave	Summer and Winter vacations as notified by the University	All Teaching Faculty (Non-teaching staff can avail vacation leave on rotational basis with prior approval from the Authorities)
On Other Duty Leaves	As approved by the Authorities considering case by case and merit of the request	All the staff who completed probationary period

10.1.4 Decentralization in working and grievance redressal mechanism (5)

Institute Marks : 5.00

Decentralisation in working

1. AMET follows decentralization in the working system by delegating financial powers to the respective Heads of the Departments in case of Academics and to the Functional Heads in case of Administration. They are to make out the Academic and departmental expenses that are incurred by the Department at the Head of Department level for expenses up to a given value that is manageable within the funds allocated to the Department and close the accounts every month with the approval from the higher authorities.
2. Though Vice Chancellor is the principal authority of the University who is empowered to take decisions and implement the same as given by the Board of Management, Vice Chancellor usually constitutes various committee/cells/centre to provide recommendations, execute the decisions of the Vice Chancellor etc. This provision allows the participatory governance in the University
3. By virtue, all recommendations and resolutions pertaining to academic and administrative decisions are taken in a collective, collaborative and consultative manner through formal bodies such as Academic Council, Planning and Monitoring Board, Finance Committee, Board of Studies etc. and are approved by the Vice Chancellor/Board of Management based on merits
4. Heads of Academic Departments are given financial autonomy to operate separate Bank account for Department expenses to the limit allowed by the University.
5. Heads of Academic Departments are empowered to convene Department meetings, Board of Studies and other OBE Committees.
6. Students are empowered to be a part of decision making bodies such as Academic Council, Board of Studies besides representatives in other committees such as Cultural Club, Sports Club, Student Council, Mess Committee etc.
7. There have been several committees, which takes up decentralization of administration and academic activities. Some of the Committees are:
 - a. Research is being administered by Board of Research, Research Monitoring Committee, Department Research Committee
 - b. Library activities are administered by the Library Committee
 - c. Mess and canteen activities by Mess Committee
 - d. Disciplinary committee
 - e. Committee Against Sexual Harassment
 - f. Anti-Ragging Committee
 - g. Student Council

Table 10.23: List of the faculty members (selected) who have been delegated powers for taking administrative decisions

Sl. No.	Name of the Faculty	Functional Area	Role
1	Capt.K.Karthik	Principal-DGS Programs	Governing academic and administrative aspects of the DGS approved programmes
2	Dr T Sasilatha	Principal-Engineering & Technology Programmes	Governing academic and administrative aspects of the AICTE approved programmes
3	Prof K Seyadu	Chief Executive Officer	Operational aspects of infrastructure, student affairs and external affairs related to legal and regulatory aspects
4	Dr.D.Velumurugan	Director Research	Administration of PhD Programme, funded research, consultancy projects, IPR etc.
5	Dr.R.Karthikeyan	Director Placement	Organizing Placement Training and On campus placement interviews
6	Dr. D.Arivazhagan	IQAC Coordinator	Forms, procedures and documentation of quality initiatives in the University
7	Dr.D.S.Balaji	Director, ICT Enabled Education	Maintenance and enhancement Of ICT Infrastructure, Training of Faculty in ICT

8	Mr. Mathew Varghese	Proctor	Hostel Administration
9	Mr P Ramakrishnan	Advisor (Security and Vigilance)	Campus security and vigilance
10	Dr.R.Karthikeyan	Director Innovation and Incubation Centre	AMET Innovation, Start-up and Entrepreneurship Development activities
11	Dr. Deepa Rajesh	Director HRDC	Training and faculty empowerment
12	Mr R Sundar	Coordinator, Educational Materials Resource Development Centre	Facilitation of Faculty to prepare e contents
13	Mr J Balasubramanian	System Manager	Maintenance and enhancement of IT facilities and networking in campus
14	Mr Kannan	Transport Manager	Management of Transport Needs and transportation
15	Mr Ramanan	Public Relations Officer	Maintenance of Public Relations, Event Management and Hospitality

Grievance Redressal Mechanism

AMET has formed Grievance Redressal Committee, Committee against Sexual Harassment and Anti-Ragging Committee as per UGC regulations to obtain, analyse, resolve and recommend solutions when there are grievances of any sort and also to impose punishment on the offenders. The above committees are working as per UGC guidelines.

AMET has also made provision for online grievance representing mechanism through website.

1. **Harassment/Discrimination Complaint:** <http://www.ametuniv.ac.in/harassment-discrimination-complaint.html> (<http://www.ametuniv.ac.in/harassment-discrimination-complaint.html>)

2. Information for Students about Grievance Redressal: <http://www.ametuniv.ac.in/Students-grievance-redressal-mechanism.html> (<http://www.ametuniv.ac.in/Students-grievance-redressal-mechanism.html>)

Besides these Committees, AMET addresses the student grievances posted in the UGC Online Grievances Redressal Portal. AMET popularizes these online grievance portals by displaying boards and notifying through circulars among the stakeholders. The web link for UGC online grievance portals is posted on AMET website.

Apart from these, AMET has the following unique mechanisms with reference to grievances from stakeholders

1. Green Box for General Grievances
2. Black Box for ragging related Grievances

Director for Student Affairs is a specialized nodal officer for channeling all kinds of student grievances and also acts as a facilitator for students to register their requests and grievances.

Students Grievance Redressal Committee

This committee has been constituted as per the UGC notification published in the Gazette on 23rd March 2013.

I. Management

- The Vice Chancellor
- The Registrar
- The Director, Student Affairs

II.Heads of the Core Departments

III.Controlling Officers

- The Controller of Examination
- The Proctor
- The Director, Physical Education
- The Librarian
- The Public Relations Officer

Grievance Redressal Committee

The Grievance Redressal Committee has been formed to analyse, examine and conduct inquiries if needed on the grievances and recommend the decisions to the Authorities for appropriate redressal measures. The details and function of this committee is given in Table 10.19.

Online Grievance Redressal System

University has also made the provision to receive and attend various grievances of students online through University website portal ametuniv.edugrievance.com (<http://ametuniv.edugrievance.com/>). Students can register and login using their email ID /mobile number and post their grievance which will be addressed by the concerned Grievance Redressal Committee officials.

Registration Process for Grievance Redressal

- Students can register their grievance through the website ametuniv.edugrievance.com (<http://ametuniv.edugrievance.com/>)
- Click Students Icon.
- Click “New User Registration”.
- Enter your details such as Name, Course, Batch, Mobile No., and Email and click Register.
- Once your registration is approved, you can post a grievance.

To Post a Grievance in Online Grievance Redressal Portal

- Use Email ID or Mobile Number to Login.
- Click Post a Grievance to register your grievance.
- Select the Grievance Type - If the grievance is related to a department, select your department name. If not, select the appropriate grievance type listed.
- Enter the grievance subject with brief description. If there is any proof or documents, it can be attached.
- Click Create to submit your grievance.

To view the Status of Grievance Redressal in online portal

- Students can view the status of grievance in My Grievances section.
- To view the details of the action taken on the grievance, click the View button.
- If the student is not satisfied with the reply, they can reopen the grievance and submit it with their queries again.
- To reopen the grievance, click Reopen button and type your grievance and post it.

Other Grievance Redressal Mechanisms

Besides these generalized grievance redressal mechanisms, the University is having the following Committees for the specialized grievances.

1.Anti-Ragging Committee

2.Anti-Ragging squad committee

3.Committee Against Sexual Harassment

4.Student Grievance Cell

The summary of activities, membership details of the above committee are given in the following Table

Table 10.24 summary of activities, membership details of the Grievance Redressal Committees at AMET

Sl.No	Name of the Committee	Composition	Major functions
1	Grievance Redressal Committee	Three Senior Professors of AMET as internal Members; Details given in Table 10.18	Analyse, examine and conduct inquiries if needed on the grievances and recommend the decisions to the Authorities for appropriate redressal measures
2	Anti-Ragging Committee	Given in Table 10.15 (Internal and External members)	Key functions are to maintain AMET a ragging-free campus by creating awareness and attending ragging related complaints
3	Anti-Ragging squad committee	Given in Table 10.16 (Internal Members)	Key functions are to organize periodic and surprise monitoring of the campus to curb ragging
4	Committee Against Sexual Harassment	Internal and External Members; Details are given in Table 10.25	Receive grievances on sexual harassment from both Faculty and students Conduct inquiry and recommend action/punishment etc. Conduct events to create awareness against sexual harassment.
5	Student Grievance Cell	Internal Members; Details are given in Table 10.26	In the event of difficulties based on COVID 19 pandemic, and also guidelines from the UGC, the University has formed a Student Grievance Cell, for Academic and Examination related grievances from students. However, the committee will continue its function after the pandemic too. Students are given special e mail ID to lodge their grievances on examinations and other academic issues (egrievances@ametuniv.ac.in)

Table 10.25. Members of the Committee against Sexual Harassment

S.No	Name	Designation
1	Dr. V. Sangeetha Albin	Joint Registrar

2	Dr. A. Shameem	Professor, AMET Business School
3	Ms. Sandhya Rani Ramadass	Lady Member of NGO and Psychologist
4	Dr.N.R.Ram Kumar	Dean – Students Welfare
5	Capt.S.Kumarakuruparan	Proctor
6	Dr. K. Gayathri	Associate Professor, Dept. of Physics
7	Mrs. H.J.Shanthi	Coordinator, Women Empowerment Cell
8	Dr. Meher Taj	Assistant Professor& Faculty Warden -Narmada Hostel
9	Ms.Pramiti Roy	BE-ECE-III year
10	Ms.Janani Shetty	B.Sc Nautical Science - II year

Table 10.26. Members of the Students Grievance Cell

S.No	Name	Designation
1	Dr. K. Durai Pandian	Dean-Students Welfare
2	Mr.R. Sundar	Coordinator-Students Welfare
3	Mr.S. Ramesh Kumar	Section Officer-Students Welfare

10.1.5 Delegation of financial powers (5)

Institute Marks : 5.00

DELEGATION OF FINANCIAL POWERS

Since AMET is following the participatory governance and decentralization of administrative powers, various authorities of the University are provided with Financial Powers. Academic Departments are provided with separate bank accounts through which they can spend amounts not exceeding Rs. 50000.

AMET has permitted the Principal Investigators of the funded projects to have a separate bank account for each of the Government funded projects. They are empowered to get advance payments from the project accounts. They can make their own purchase for amounts not exceeding Rs. 50000 whereas beyond that they need to take permission of the University Authorities. They need to provide statement of expenditure to prepare the Utilization Certificate for the purpose of reporting funding agencies.

The Faculty Members provided with seed money are empowered to utilize the full project amount at their own discretion and then to give statement of expenditure to the University

Table 10.27. Details of selected Officers/Faculty of AMET who have been provided with financial powers

Sl. No.	Position	Financial Powers
1	Registrar	Up to Rs. 2 lakh
2	Principal	Up to Rs. 1 lakh
3	Deans	Up to Rs. 50000
4	Heads of Departments	Up to Rs. 50000
5	Public Relations Officer	Up to Rs. 50000
6	Director Research	◦ Approve the Honorarium for PhD Related Meetings ◦ Empowered to receive payments on behalf of University through formal manner
7	Controller of Examination	◦ Empowered to Receive Penalties and Fine amounts related to examination processes ◦ Approve honorarium for Examiners and Evaluators
8	Proctor	Up to Rs. 50000
9	Duty Officers	Up to Rs. 10000

10.1.6 Transparency and availability of correct/unambiguous information in public domain (5)

Institute Marks : 5.00

Transparency in University functioning

AMET maintains transparency in all its operations and working. At the beginning of every semester, AMET brings out calendar of events that contains information of semester activities and the same is sent to all Departments. Information on policies, rules, processes and its dissemination of this information is made available to Stakeholders on the AMET website. The required information about the Institution is made available on the AMET website www.ametuniv.ac.in (<http://www.ametuniv.ac.in/>).

Annual Reports

In every Annual Convocation, Vice Chancellor presents the Annual Report of the University comprising all the activities pertaining to entire functioning of the University. The Annual Reports are printed and distributed to all the stakeholders.

AMET Website

A website by the URL www.ametuniv.ac.in (<http://www.ametuniv.ac.in/>) is available from which the latest information and activities of the University can be accessed from anywhere anytime. University has uploaded all mandatory documents such as course approval, ranking and accreditation documents, and approval by statutory bodies, honours and awards, compliance reports to statutory body review reports and rules and regulations of the University.

The norms, procedures, circulars & all other updated relevant information are available on the Management Information Systems. Apart from this, Mission and Vision is disseminated to all the stakeholders of the programmes through faculty meetings, workshops, student induction programmes, and parent meetings.

Transparent fee

The prospectus of AMET contains transparent information about the fees and charges payable. The same is also made available in AMET website which is updated periodically.

AMET follows its own declared policy (Admission Policy and Fee Fixation Policy) in collecting any fees or charges.

AMET collects only the fee that is transparently notified in the website and prospectus may be in full or in installments. Besides, AMET does not charge/ collect any fee or other charges which are not publically stated /notified in the Prospectus.

Transparency in Appointments and Admissions

University maintains complete transparency in Employee appointments where details of all vacancy notifications are hosted in the website. During admission, the details of eligible applicants, merit/rank list etc. are publicly displayed in the notice board and University website

Transparency in Financial Transactions

AMET conducts both internal and external financial audits in every financial year. The audited account statements certified by the qualified Chartered Accountants are hosted publicly at AMET Website. The Annual Report of the University contains information about statement of income and expenditure.

Transparency in Accomplishments

University maintains all publications, funded projects, research facilities, details of research scholars and facilities for research in the University website. Besides, University maintains all necessary documents and reports submitted to regulatory, statutory and ranking bodies. The documents include Self Study Report and Annual Quality Assurance Report (AQAR) for NAAC, NIRF Ranking documents, DSIR certification, NBA related documents etc.

10.2 Budget Allocation, Utilization, and Public Accounting at Institute level (15)

Total Marks 15.00

Total Income at Institute level: For CFY,CFYm1,CFYm2 & CFYm3

CFY : (Current Financial Year),

CFYm1 : (Current Financial Year minus 1),

CFYm2 : (Current Financial Year minus 2) and

CFYm3 : (Current Financial Year minus 3)

Table 1 - CFY 2022-2023

Total Income 9075				Actual expenditure(till...): 8871			Total No. Of Students 3238
Fee	Govt.	Grants	Other sources(specify)	Recurring including salaries	Non Recurring	Special Projects/Anyother, specify	Expenditure per student
8645	0	150	280	7399	825	647	2.74

Table 2 - CFYm1 2021-2022

Total Income 8415.25				Actual expenditure(till...): 9254.90			Total No. Of Students 3398
Fee	Govt.	Grants	Other sources(specify)	Recurring including salaries	Non Recurring	Special Projects/Anyother, specify	Expenditure per student
8171.35	0	46.43	197.47	6935.31	1679.42	640.17	2.72

Table 3 - CFYm2 2020-2021

Total Income 6614.02				Actual expenditure(till...): 6691.01			Total No. Of Students 3390
Fee	Govt.	Grants	Other sources(specify)	Recurring including salaries	Non Recurring	Special Projects/Anyother, specify	Expenditure per student
5945.59	0	158.01	510.42	5762.98	320.30	607.73	1.97

Table 4 - CFYm3 2019-2020

Total Income 11478.75				Actual expenditure(till...): 10339.09			Total No. Of Students 3649
Fee	Govt.	Grants	Other sources(specify)	Recurring including salaries	Non Recurring	Special Projects/Anyother, specify	Expenditure per student
9982.28	814.34	73.34	608.79	8717.52	1013.39	608.18	2.83

Items	Budgeted in 2022-2023	Actual Expenses in 2022-2023 till	Budgeted in 2021-2022	Actual Expenses in 2021-2022 till	Budgeted in 2020-2021	Actual Expenses in 2020-2021 till	Budgeted in 2019-2020	Actual Expenses in 2019-2020 till
Infrastructure Built-Up	697	568.82	1458.50	1449.02	345.25	169.80	847	762.62

10.2.1 Adequacy of budget allocation (5)

Institute Marks : 5.00

The Annual budget for the Year is prepared in accordance with the needs & requirements of the concerned Departments after taking into consideration the annual student's intake, lab and infrastructure upgradation, faculty and staff requirements and adopting the latest technologies. The Budget so prepared by each Department shall be reviewed in HOD's meeting with the participation of Vice Chancellor, Registrar, Finance Controller and Principal.

After making required amendments and corrections based on the discussions and deliberations a formal budget shall be prepared and the same is being sent to the Principal. The Registrar shall present the draft budget to the Board of Management for approval and sanctioned in the Finance Committee. The budget allocation and utilization for the last three years is found to be adequate.

Table 10.28. Adequacy of budget allocation for the last four financial years

S.No	Financial Year	Budget Allocation (Rs in Lakhs INR)	Actual Expenditure (Rs in Lakhs INR)	Adequacy Status
1	CFY = 2022-23	9614.00	8871.27	Yes
2	CFY m1 = 2021-22	9586.00	9254.90	Yes
3	CFY m2 = 2020-21	7807.00	6691.01	Yes
4	CFY m3 = 2019-20	10640.80	10339.09	Yes

10.2.2 Utilization of allocated funds (5)

Institute Marks : 5.00

Process of Utilisation of Funds started with allocation from Management to the University. Department Heads / Section-in-charges are intimated of the extent of funds allocated against their budget proposals.

Major works like Construction, Up-gradation of Existing Infrastructure, Procurement and Maintenance of Common utilities, House-keeping, Procurement of furniture etc. are controlled directly by the Finance Controller. Actions for procurement of lab equipment, up-gradation of existing lab facilities, purchase of consumables etc. are initiated from the respective Departments through Purchase Department and the funds are released on a case to case basis by the Finance and Accounts section of the University after obtaining approval from the Management.

University forms Technical Committee and Purchase Committee for all the expenditure worth beyond Rs 50000.

During the last three years, the budget was utilized to meet expenses such as salary for Staff, Infrastructure development, Equipment purchases, Expenses towards consumables and contingencies, travel etc. Almost 98% of the allocated budget provided by the Management was effectively utilized by the University in the last four financial years.

Table 10.29. Details of Funds Utilization percentage (%) in the last four financial years

S.No	Financial Year	Budget Allocation (Rs in Lakhs)	Actual Allocation (Rs in Lakhs)	Utilisation (%)
1	CFY = 2022-23	9614.00	8871.27	92.27 %
2	CFY m1 = 2021-22	9586.00	9254.90	96.55 %
3	CFY m2 = 2020-21	7807.00	6691.01	85.71 %
4	CFY m3 = 2019-20	10640.80	10339.09	97.16 %

10.2.3 Availability of the audited statements on the institute's website (5)

Institute Marks : 5.00

AMET Deemed to be University conducts financial audits both by internal and external auditors. Besides, during Academic and Administrative Audits and ISO Audits, account statements are audited.

The audited statement of accounts certified by the qualified chartered accountants are made available publicly in the University website. These statements are also submitted to the Statutory and Regulatory Authorities of the University along with various reports.

Hence, the University is engaging internal and external audits for financial statements which are also publicly displayed.

AMET Deemed to be University also made provision for online Audited statement through website.
(https://www.ametuniv.ac.in/attachment/audited_statementsof_accounts.pdf)

https://www.ametuniv.ac.in/attachment/audited_statementsof_accounts.pdf

Total Income at Institute level: For CFY,CFYm1,CFYm2 & CFYm3

CFY: (Current Financial Year),

CFYm1 : (Current Financial Year minus 1),

CFYm2 : (Current Financial Year minus 2) and

CFYm3 : (Current Financial Year minus 3)

Table 1 :: CFY 2022-2023

Total Budget 215.10		Actual expenditure (till...): 199.19		Total No. Of Students 137
Non Recurring	Recurring	Non Recurring	Recurring	Expenditure per student
42.77	172.33	39.37	159.82	1.45

Table 2 :: CFYm1 2021-2022

Total Budget 228.24		Actual expenditure (till...): 221.68		Total No. Of Students 176
Non Recurring	Recurring	Non Recurring	Recurring	Expenditure per student
37.96	190.28	36.05	185.63	1.26

Table 3 :: CFYm2 2020-2021

Total Budget 209.75		Actual expenditure (till...): 204.13		Total No. Of Students 204
Non Recurring	Recurring	Non Recurring	Recurring	Expenditure per student
40.91	168.84	39.46	164.67	1.00

Table 4 :: CFYm3 2019-2020

Total Budget 334.48		Actual expenditure (till...): 321.45		Total No. Of Students 263
Non Recurring	Recurring	Non Recurring	Recurring	Expenditure per student
43.24	291.24	43.83	277.62	1.22

Items	Budgeted in 2022-2023	Actual Expenses in 2022-2023 till	Budgeted in 2021-2022	Actual Expenses in 2021-2022 till	Budgeted in 2020-2021	Actual Expenses in 2020-2021 till	Budgeted in 2019-2020	Actual Expenses in 2019-2020 till
Laboratory equipment	5	4.89	0	0	0	0	0	0
Software	14	12	3	2.89	3	2.89	0	0
Laboratory consumable	1.90	1.78	2.07	1.88	1.50	2.20	6.49	5.35
Maintenance and spares	38.29	34.79	45.06	44.91	34.18	33.69	66.31	64.64
R & D	11.32	9.54	13.80	17.16	12.74	10.65	18.96	19.53
Training and Travel	103.89	98.23	110.95	105.95	93.34	94.57	158.77	148.68
Miscellaneous Expenses*	40.69	37.97	53.35	48.88	64.99	60.13	83.97	83.26
Total	0	12	0	0	0	0	0	0

10.3.1 Adequacy of budget allocation (10)

Institute Marks : 10.00

The allocated budget was used to meet the new facilities for Equipment, Replacement of outdated Equipment and setting up of new labs due to Revision in syllabi. Budget requirements under recurring and non-recurring heads are collected from each department and section before the commencement of the financial year. Allocations are made as per the availability of funds are monitored by the Finance and Accounts section of the Institute. The University carefully monitors the expenses so that the requirements are met for smooth working of the Departments. The Management has been very efficiently doing this over the past several years and the institution never had any serious Budget Crunch that affected the function of the University.

The details of adequacy of budget allocation for the current financial year and for the last three years for the Programme of Naval Architecture and Offshore Engineering is provided in the following table.

Table 10.31 Adequacy of Budget Allocation - BE-Naval Architecture and Offshore Engineering

S.No	Financial Year	Budget Allocation (Rs in Lakhs)	Actual Allocation (Rs in Lakhs)	Adequacy Status
1	CFY = 2022 - 23	215.09	199.20	Yes
2	CFY m1= 2021 - 22	228.23	221.67	Yes
3	CFY m2 = 2020 - 21	209.75	204.13	Yes
4	CFY m3 = 2019 -20	334.50	321.46	Yes

10.3.2 Utilization of allocated funds (20)

Institute Marks : 20.00

Funds are allocated by the Management of the University. Department Heads are intimated of the extent of funds allocated against their budget proposals. Actions for procurement of lab equipment, up-gradation of existing lab facilities, purchase of consumables, etc. are initiated from the department through Purchase Department and the funds are released on case to case basis from the accounts office of the college on approval by the Management. During the last three years, the budget was utilized to meet expenses like purchase of equipment, expenses towards consumables and contingencies, etc.

The percentage of funds utilization for the current financial year and also for the last three years with respect to the Programme of Naval Architecture and Offshore Engineering is provided in the following table.

Table 10.32 Utilisation of Allocated Funds (Percentage) - (BE-Naval Architecture and Offshore Engineering)

S.No	Financial Year	Budget Allocation (Rs in Lakhs)	Actual Allocation (Rs in Lakhs)	Utilisation (%)
1	CFY = 2022 - 23	215.09	199.20	92.61 %
2	CFY m1= 2021 - 22	228.23	221.67	97.12 %
3	CFY m2 = 2020 - 21	209.75	204.13	97.32 %
4	CFY m3 = 2019 -20	334.50	321.46	96.09 %

10.4 Library and Internet (20)

Total Marks 20.00

AMET Deemed to be University has adequate library and internet facilities. In all the previous assessment years, the University has recorded nil deficiency with reference to library resources and internet facilities.

Quality of learning resources

Prof VBS Rajan Library is the Central Library of the University which is named after Prof.V.B.S. Rajan and is functioning in a newly built Library Block inaugurated by Mr. Efthimios E. Mitropoulos, Secretary General Emeritus, and International Maritime Organization (IMO) on 4th September, 2014.

The University has attained Deemed to be University status in the year 2007. Since then, the Central Library of the University (Prof VBS Rajan Library) has purchased and made access for a huge volume of books, journals, magazines, e books, e journals, databases, software, technical reports, dissertations and Theses, rare books etc. The Library has a Digital Knowledge Centre in which research and publication databases, software etc. are available and updated every year. The Library is fully automated with RFID and Barcoding system. The Library Committee of the University is providing necessary inputs for the purchase of library resources.

Table 10.32. Members of the Library Committee

Sl.No	Name of the Members	Designation
1	Dr. M .Jayaprakeshvel, Registrar	Chairman
2	Capt. Ramakrishnan Venkat Associate Professor Department of Nautical Science	Member
3	Dr. S. Prabhakaran, Professor Department of Marine Engineering	Member
4	Dr. V. Leela Vinodhan, Associate Professor Department of Mechanical Engineering	Member
5	Dr. S.Krishna, Assistant Professor Department of English	Member
6	Dr.Prakash I.N, Librarian, Alliance University, Bengaluru, Karnataka-India	Member
7	Dr.P.Pannerselvam, Librarian Central library, B.S Abdur Rahman Crescent Institute of Science and Technology Vandalur, Chennai	Member
8	Mr. R.. Sundar, Assistant Professor Department of Marine Engineering	Member
9	Dr.K.Sekar, Librarian	Member Secretary

University Library has given equal importance for both hard and soft copy of different category of learning resources. Besides, Prof VBS Rajan Library (Central Library), all the University Departments have Department Library with more specialized books, journals, magazines and technical reports both in electronic and conventional forms.

Relevance of available learning resources including e-resources

The University is having a Library Committee which recommends the up keeping and strengthening of the learning resources both in hard and soft copy formats. The Library consists of large collection of books with updated editions in various titles and volumes. The book collections are available in the field of Engineering, Science and Technology and Management including Marine Engineering, Nautical Science, Management Studies, Marine Biotechnology, Naval Architecture, Petroleum Engineering, Electrical and Electronics Engineering, Mechanical Engineering, Food Processing & Technology and Mining Engineering, Mathematics, Physics, Chemistry, English, Career Information & Guidance, Literature, Books in Regional Languages and other General Areas of study. The general details of physical structure and functional areas of the Prof VBS Rajan Library are given in Table 10.33.

Table 10.33. Brief about the Library Facilities

S.No.	Category	Availability/Value
1	Carpet Area of Library (in m2)	2160.52(2342 Plinth Area)
2	Reading Space (in m2)	1171 Total 2342
3	Number of Seats in Reading Space	First Floor: 200 Second Floor: 200 Digital Knowledge Centre: 40 Audio Visual Room: 40 Total: 480

4	Average No. of Users (Issue book) per Day	152
5	Average No of users (Reading space) Per Day	520
6	Working Hours in Timings:	Working Days 09 am to 09 pm; During Examination 9 am to 10 pm; Holiday 09 am to 05 pm
7	Number of Library Staff	4
8	Number of Library Staff with a Degree in Library Science	3
9	Library Management	Yes – AUTOLIB with RFID
10	Computerization for Search, Indexing	Yes
11	Issue/Return Records Bar Coding used	Yes
12	Library Services on Internet/Intranet INFLIBNET, SHODHGANGA,	Yes
	DELNET or other similar membership archives	
13	Remote Access (INFED)	Yes
14	Dissertation/Thesis Section	Yes
15	Rare Books Collection	Yes

Prof VBS Rajan Library is having over 102738 books, more than 60000 E-books (stored version besides access to several lakhs number of e books) and 123 print journals. The overall collection of library resources is provided in Table 10.34.

Table 10.34. Library Resource Collection

Resources	Numbers
Books	103535
E - Books	206784
E- Journals	40934
Current Journals on Subscription	110
Book Bank	1224
Thesis/Project	2657
CDs & Videos	2841
Bound Volumes of Journals	2461

Reports/Standards/Proceeding	149
NPTEL Video and web courses	36802
Number of Newspaper	13
Total Collection	397510

Besides the above library collection, Prof VBS Rajan Library is equipped with

- Swayam Prabha- 32 DTH Channels of High Quality Educational Programmes
- Well protected with fire alarms and CCTV security systems.
- Separate Audio-Visual Section
- Rare Books Collection
- Wi-Fi enabled with library building
- Equipped with reading halls in two floors with reference section
- Organization of Book Review / Book Talk Sessions
- Organization of Book Exhibitions/Displays of new books.
- Proper system of obtaining, analysis and action pertaining to feedback from the Users of library services.

Availability of E-Journals at Prof VBS Rajan Library

Prof VBS Rajan Library at AMET has subscribed e-journals from various service providers or journal aggregators such as IEEE, Elsevier, Scopus, Science Direct, ASME, ASCE, J-Gate (Engineering and Management), EBSCO, DELNET, INFLIBNET etc. Details are provided in the following tables.

Table 10.35. Details of E-Journals available for the Student and Faculty access at Prof VBS Rajan Library

Sl. No.	Journal Database	Subject Coverage	No. of Journals
1	IEEE	Engineering and Technology (Access from 1998-current issues)	205
2	Elsevier Including SCOPUS and Science Direct	Science, Engineering, Technology, Management, Arts, Social Sciences etc.	340
3	ASME-The American Society of Mechanical Engineers	Mechanical Engineering	30
4	ProQuest E-journals	Engineering and technology	206784
5	J-GATE Engineering	All the Disciplines of Engineering and Technology	35550
6	EBSCO	Science, Engineering, Technology, Management, Arts, Social Sciences etc.	1079
7	J-GATE Management and Social Science	Management and Social Science	4538
8	DELNET	Science, Engineering, Technology, Management, Arts, Social Sciences etc.	4100
Total			252626

Table 10.36. Details of E-Journals Access Links available for the Student and Faculty

Sl.No	DATABASE	LINKS
1	EBSCO (E-Journals)	https://web.p.ebscohost.com/ehost/search/basic?vid=0&sid=7e849e66-(https://web.p.ebscohost.com/ehost/search/basic?vid=0&sid=7e849e66-2ac4-4a80-b0ee-d5db2b1c82e8%40redis) 2ac4-4a80-b0ee-d5db2b1c82e8%40redis
2	IEEE (E-Journals)	https://ieeexplore.ieee.org/Xplore/home.jsp (https://ieeexplore.ieee.org/Xplore/home.jsp)
3	ProQuest (E-Books &E-Journal)	https://www.proquest.com/?parentSessionId=Eo4gj3faZb%2FG (https://www.proquest.com/?parentSessionId=Eo4gj3faZb%2FGYoU5iJ%2BFi7CLfVhxPoZMkv7GZQjQ6%2F0%3D) YoU5iJ%2BFi7CLfVhxPoZMkv7GZQjQ6%2F0%3D)
4	DELNET (E-Books &E-Journal)	http://delnet.in/ (http://delnet.in/) or http://164.100.247.26/ (http://164.100.247.26/)
5	ASME(E-Journals)	Database accessed with in the AMET campus only
6	J-Gate (E-Journals)	Database accessed with in the AMET campus only
7	RFID	For 10,000 Books
8	UGC e-resources developed by UGC	http://ugceresources.in/ (http://ugceresources.in/)
9	NATIONAL DIGITAL LIBRARY (NDL) E-Book and E-Journals	https://ndl.iitkgp.ac.in/ (https://ndl.iitkgp.ac.in/)
10	SHODHGANGA (E-THESIS)	https://shodhganga.inflibnet.ac.in/ (https://shodhganga.inflibnet.ac.in/)
11	The Networked Digital Library of Thesis and Dissertations (NDLTD)	https://ndltd.org/ (https://ndltd.org/)
12	e-ShodhSindhu: Consortium	http://ess.inflibnet.ac.in/ (http://ess.inflibnet.ac.in/)
13	VIDWAN	https://vidwan.inflibnet.ac.in/ (https://vidwan.inflibnet.ac.in/)
14	ShodhShuddhi	https://shodhshuddhi.inflibnet.ac.in/ (https://shodhshuddhi.inflibnet.ac.in/)

Incremental increase in Library Resources (Hard and Soft Copies)

Since the University keeps on expanding the number of programmes offered and increase in intake, Prof VBS Rajan Library has adequately enhanced the Library resources both physical and digital forms. The details of year wise incremental increase in the Library Resources, increase in print and online journals and details of library expenditure on books, magazines/journals, and miscellaneous content are given in Tables 10.37, 10.38 and 10.39, respectively.

Table 10.37. Year wise incremental increase in the Library Resources

Year	Total No. of new volumes added	Total No. of New titles added
2016 – 2017	4334	1215
2017 – 2018	5921	1462

2018 – 2019	7714	1601
2019 – 2020	13162	3326
2020 – 2021	78	78
2021 – 2022	641	512

Table 10.38 Details of E Journals and Print Journals available at the Library over the past six years

Details		2016-17	2017-18	2018-19	2019-20	2020-21	2021-22
Engineering, Technology and Management	As soft copy (E-Journal)	169	10149	10494	340	40965	40935
	As hard copy (Print Journals)	340	212	290	134	121	110

Table 10.39 Details of library expenditure on books, magazines/journals, and miscellaneous count

Year / Category of Resource	Expenditure (Rs)			
	Books	Magazines/Print Journals (for hard copy subscription)	E-Journals (for soft copy subscription)	Miscellaneous contents
2016 – 2017	5551666	1131981	476800	11000
2017 – 2018	7992615	1404562	2602084	29100
2018 – 2019	6171751	1284968	1077798	24800
2019 – 2020	6310033	441577	2331254	506500
2020 – 2021	58614	597308	2332698	171800
2021 – 2022	645401	534077	1758891	296700

Support Services at Prof VBS Rajan Library

a. Thesis/Dissertations Facility

The Library holds all UG, PG Dissertations and Doctoral thesis, of various academic departments of AMET to which they were submitted. 2018 hard copies of the Thesis / Dissertations are available in the Central Library.

b. Selective Dissemination of Information Service (SDI)

User profile and document profile are prepared. User profile contains name of the user, class / department, area of interest and topic of the research. Document profile includes bibliographic information about the books and journals required by the user. User profile and document profile are matched and the required resources will be sent to the individuals either in the form of soft copy or print.

c. Current Awareness Service (CAS)

Current awareness system alerts the user community about each and every new arrivals of the library. When new publication of any format reaches the library, the user community will be intimated about the new arrivals instantly, periodically or upon request. List of latest additions of books are displayed in the New Arrival Rack and subsequently the intimation will be displayed in the Notice Board so as to enable the user about the current awareness.

d. Article Alert Service

Contents of the latest relevant journal articles on request are scanned and sent through e-mail to the concerned faculty members.

e. News Paper-Clippings Service

Information related to Higher Education, research oriented information clippings are preserved and kept intact for ready reference.

f. Reprography Services and Scanning Facilities

Photocopier is provided for users in the Central Library. Students can make photocopy of restricted number of pages not violating the copyright rules from the book, articles from periodicals and newspapers and question papers. Library strictly follows copyright laws while providing this service. A scanner, photo copier cum printer and a printer are available in the library for the users.

g. Library as a space for Knowledge Events

The Audio Visual Hall of Prof VBS Rajan Library (fully air conditioned, audio and video projection facility including smart board; having a seating capacity of 50) is open to all library users (students and Faculty) to organize knowledge promotion events such as Book Talks, Quiz, Student Contests, debates etc.

Details of Digital Resources at Prof VBS Rajan Library

In order to meet the contemporary needs of the library users, the Prof VBS Rajan Library offers the following facilities and value-added services through conventional methods and by using Information and Communication Technologies (ICT).

a. Digital Knowledge Centre

Library has a separate section for digital library. This facility helps the users to explore the electronic databases such as E-Journals, E-books, databases and software subscribed by the Central Library. Access of Internet in Digital Knowledge Centre is enabled through 50 computer systems with an internet speed of 1 GBPS from National Knowledge Network and 100 Mbps from Tata Communications. The computer configurations are as follows

- THIN CLIENT DELL THIN CLIENT WYSE 3030 4GB FLASH/2GB RAM;
- DELL OPTIPLEX 3010 I3-3220 3.30GHZ/2GB/500GB/18.5LCD;
- LENOVA THINK CENTER/i3-4170/3.70GHZ/4GB/500GB/19 LED LENOVA;
- DELL OPTIPLEX330/CORE2DUO 2.53GHZ/2GB/160GH HDD/18.5 LCD DELL

b. E Journals from IEEE

The Institute of Electrical and Electronics Engineers (IEEE) is the world's largest technical professional organization dedicated to advancing technology for the benefit of humanity. It has highly cited publications of Journals to support Engineering professionals. The subscribed IEEE all Society Periodical Package consists of 183 e-Journals related to all the Engineering disciplines with access to back documents since 1998.

c. Elsevier Package: SCOPUS and SCIENCE DIRECT

SCOPUS is the largest abstract and citation database of peer-reviewed literature: scientific journals, books and conference proceedings. The subscribed package of Elsevier SCOPUS – Science Direct consists of 340 e-journals with enormous full-text articles related to various Science, Technology, Engineering and Management studies.

d. E Journals from ASME

The American Society of Mechanical Engineers (ASME) is the premier professional membership organization for Mechanical Engineering. AMET has subscribed ASME with access to 31 most prestigious electronic journals in the area of Mechanical Engineering with an access to back documents since 1960 along with available conference proceedings.

e. E Journals from ASCE

The American Society of Civil Engineers (ASCE) is a leading provider of electronic material to Civil Engineering and related disciplines. AMET has subscribed 31 e-journals from ASCE

f. E-Journals from J-GATE

J-Gate is an important electronic gateway to global e-journal literature which has seamless access to millions of online journal articles offered by a variety of Publishers. AMET has subscribed the two modules from ASME such as Engineering and Technology (with 4285 e- journals) and Management and Social Sciences (with 4538 e-journals).

g. E-Journals and electronic resources from EBSCO

AMET has subscription to EBSCO Business Source Elite, a database, which consists of 1079 full text electronic journals related to all the branches of Management studies. The database consists of the contributions from world renowned publishers in the form of peer reviewed academic journal articles, magazines, abstracts and trade publications.

h. E-Journals and electronic resources from DELNET

The Central Library of AMET has continuous annual institutional membership subscription with the Developing Library Network (DELNET) DELNET, through which access to more than 5000 e-journals and enormous e-books and Thesis/Dissertations are made available. Apart from electronic database, the Central Library of AMET is privileged to access the Union Catalogue of Books to explore Inter-Library Loan facility. DELNET E-resources can be accessed both in Library and also from remotely (from anywhere anytime), through the Remote Access Facility powered by INFED of INFLIBNET.

i. E Books from ProQuest

The subscription package of AMET with ProQuest consists of 40758 e-books covering various disciplines. ProQuest E-resources can be accessed both in Library and also from remotely (from anywhere anytime), through the Remote Access Facility powered by INFED of INFLIBNET.

j. Software

The Library is having following licenced software besides several free and open access software for the student usage

1. URKUND-Plagiarism Detection Software
2. SPSS-Statistical Software

Accessibility to students

a. Access Cards

AMET provides Identity card with Barcode Technology, for all students and staff members, which can be utilized for registering GATE Entry as well as for borrowing the library resources. In addition, the issue of books and return of books are electronically recorded using the OPAC system of the Library through the Library Cards.

b.Remote Access

AMET has also made the Library Resources to be accessed remotely through INFED system of the INFLIBNET. The NDLIBNET Access Management Federation (INFED) has adopted Shibboleth, a standard-based open source software, for authenticating authorized users from institutions and provide them seamless access to e-resources from anywhere, anytime.

c. BARCODE and RFID Systems

The University has adopted barcoding system for recording of issue and return of books. The University has also implemented the Radio-frequency identification (RFID) system to enhance the fidelity in the Library Access process.

d.Library Web portal

Library Portal is an interface to the Library resources and services to access e- question bank. Open source e-journals and magazines, open source e-books, e-newspapers etc., are also made available. Link :<https://ametlibrary.wordpress.com/> (<https://ametlibrary.wordpress.com/>)

e.Institutional Membership & Inter Library Loan (ILL)

To enhance the Library Access coverage, Prof VBS Rajan Library has made collaborations with to explore Inter library Loan (ILL) to provide articles, books, and other materials to faculty members, researchers, students, through Institutional Membership. Prof VBS Rajan Library has the Institutional Members with the following institutions from where AMET students can borrow books and access the library resources.

- Anna University
- MALIBNET
- American Council Library
- British Council Library
- DELNET
- National digital Library (NDL)

f. Library Automation for enhanced Access Experience

The library uses the popular “AUTOLIB” software on windows environment capable of handling of records. It facilitates the students and Faculty to have trouble free access with online Public Access Catalogue (OPAC) and also consists of various modules such as Gate Entry, Acquisition Cataloguing, Circulation, Serial Control and possibility of generating various reports.

g.Web OPAC (Online Public Access Catalogue)

The in-house resource list can be accessed through website of AMET. Central Library has online access facility to search the availability of the required resources for the user community. The search option through OPAC system is unique and can be made through Author-wise, Title-wise, Subject-wise indexes using keywords and can be accessed from on and off the campus. The advanced search provides with more specific search for a title with Boolean operators.

h. Reference services and referral services

Library provides right information to the right person at the right time through Reference Service. Reference sources such as dictionaries, encyclopaedias, almanacs, atlases, etc. are research tools that can help for writing a project. Reference sources provide answers to specific questions, such as brief facts, statistics, and technical instructions and also provide background information. Referral Service includes guiding the users to approach the right institution for the information required.

i. Grievances online

Users can submit their suggestions/grievances through Online Grievance Portal of AMET or through Prof VBS Rajan Library Web Portal which are forwarded to the Library Committee for necessary action. Users can also submit their suggestions/grievances through general interaction, making an application, through e-mail.

Support to students for self-learning activities

The Digital Knowledge Centre of Prof VBS Rajan Library and other digital resources as explained above are supporting the students for their self-learning activities.

a.List of digital resources available for students to enhance their self-learning activities

1. IEEE
2. Elsevier: Including SCOPUS and Science Direct
3. ASME-The American Society of Mechanical Engineers
4. ProQuest E-journals
5. J-GATE Engineering

6. EBSCO
7. J-GATE Management and Social Science
8. DELNET
9. URKUND –Plagiarism Detection Software
10. SPSS-Statistical Software

b. Audio Visual Section (AV Hall)

Audio-Visual Section of the Library is equipped with a computer system well supported by multimedia accessories for enhancing the self-learning experiences by the following ways

1. Effective and efficient utilization of National Programme on Technology Enhanced Learning (NPTEL) courses with the available technical videos, CD ROMS etc. The AV Hall of the Prof VBS Rajan Library is having more than 40000 video lectures (in Engineering, Science and Humanities and Management streams) stored in portable storage device (hard disc) and it is also loaded in the Library Portal Served.
2. Prof VBS Rajan Library has also a collection of locally stored version of more than 50000 books in CD format (more than 3200 CDs). Students can access these materials in the AV Hall to enhance their self-learning
3. Besides, students are using the AV Hall for their self-learning activities such as Club Meetings, Student Organized Weekly Seminars and Research Meetings at the AV Hall.

c. MOOC- Online certification courses

The University has made it mandatory that every student shall undergo minimum one Massive Open Online Course in MOOC platforms such as Coursera, Edx, SWAYAM and NPTEL, etc. per semester. In this regard, students in small groups can access the MOOC courses in AV Hall with state of the art projection, audio and video facility. This enables them to enrich their self-learning experience with current trends, and also to equip themselves with inter-domain subjects.

d. Swayam Prabha

Swayam Prabha is an initiative of Ministry of Human Resource Development (MHRD). It is facilitating access to high quality educational programmes of 32 DTH Channels on 24x7 basis. Everyday, there will be new content for at least four hours which would be repeated 5 times a day, allowing the students to choose the time of their convenience and learn quality online education at their own pace.

e. Institutional Repository

Prof VBS Rajan Library supports the students for self-learning activities through Institutional Repository which is an archives, where each and every administrative and research valuable records in any format can be preserved and accessed anytime. DSpace

Institutional Repository software has been installed in AMET and the repository can be accessed through library web portal. DSpace is open source software for building digital repositories. DSpace preserves and enables easy and open access to all types of digital content including text, images, moving images, mpegs and data sets. Born digital records can be catalogued and uploaded into the DSpace Repository to retrieve the right information at the right time to the user community.

AMET has a total number of Computers 973 for student usage and 251 for Faculty usage. All these Computers/Laptops are provided with broad band internet connectivity. The campus and hostels are fully covered with internet connectivity through Wi-Fi. AMET has 110 Wi-Fi Hot Spots using Dlink in campus and 100 Hot Spots using Ruckus Networks in AMET hostels.

Table 10.40 Details of Internet Provider, Available Bandwidth, Wi-Fi Facility etc.

Particular	Response
Name of the Internet provider	BSNL and TIC FIBER
Available bandwidth	• LAN of 1 GBPS (from BSNL) connectivity under National Knowledge Network (NKN) is provided throughout the campus • TIC Fiber (Thamizhagha Internet Communications)– 100 MBPS • TATA – 100 MBPS (This is the default connection for Administration and Management) • In case of one line failure, each line can be switched to other line at optimized speed to have continuity in connection
Wi- Fi availability	The campus and hostels are fully (100%) covered with internet connectivity through Wi-Fi. AMET has 110 Wi-Fi Hot Spots using Dlink in campus and 100 Hot Spots using Ruckus Networks in AMET hostels.
Access speed	Average 75 Mbps for workstation and desktops
Availability of Internet in Labs	YES. • All the computers in the Labs are provided with Internet Connectivity. • All the Labs are having cat6 Cabling and D link Giga byte switches. • Moreover, all the labs are connected with WiFi also.
Availability of Internet in computing lab	YES • The Computing Labs are provided with Internet Connectivity. • All the Labs are having cat6 Cabling and D link Giga byte switches. • Moreover, all the labs are connected with WiFi also.
Availability in Departments and other units	YES • All the Departments and other functional Units are provided with Internet Connectivity. • They are having cat6 Cabling and D link Giga byte switches. • Moreover, all the Departments and other Units are connected with WiFi also.

Availability in Faculty rooms	YES • All the Faculty Rooms Units are provided with Internet Connectivity. • They are having cat6 Cabling and D link Giga byte switches. • Moreover, all the Faculty Rooms are connected with WiFi also.
Institute own e-mail facility to faculty/students	YES AMET has domain specific mail ID to the stakeholders (username@ametuniv.ac.in)
Security/privacy of e-mail/internet users	YES • Separate firewall for Campus and Hostel are installed with Gateway Content Filtering and IPS/IDS Protection. • Gate way Security –Firewall – Palo Alto 810 for Campus and Class rooms • Antivirus software is installed in all the Desktops and Laptops being used in the campus • Sonic firewall is installed for hostel internet connectivity. • The University is having exclusive IT Support Team which takes care of the security and privacy issues if any.

Annexure I
(A) PROGRAM OUTCOME (POs)

Engineering Graduates will be able to:

- 1. Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- 2. Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- 3. Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- 4. Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- 5. Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- 6. The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- 7. Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- 8. Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- 9. Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- 10. Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- 11. Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- 12. Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

(B) PROGRAM SPECIFIC OUTCOME (PSOs)
Program should specify 2-4 program specific outcomes.

PSO1	Develop innovative and contemporary designs of ships and offshore structures through advanced analysis and experiments to meet the challenges of marine operations in the shipbuilding, offshore, and other maritime industries.
PSO2	Implement engineering design and construction concepts to produce solutions in the field of Naval Architecture and Offshore Engineering.
PSO3	Apply research based knowledge and research methods to analyse and interpret data using modern tools towards the solution of complex problems in the area of Naval Architecture and Offshore Engineering.

Declaration

The head of the institution needs to make a declaration as per the format given -

- I undertake that, the institution is well aware about the provisions in the NBA's accreditation manual concerned for this application, rules, regulations, notifications and NBA expert visit guidelines in force as on date and the institution shall fully abide by them.
- It is submitted that information provided in this Self Assessment Report is factually correct.
- I understand and agree that an appropriate disciplinary action against the Institution will be initiated by the NBA. In case, any false statement/information is observed during pre-visit, visit, postvisit and subsequent to grant of accreditation.

Head of the Institute

Name : Dr. T. Sasilatha

Designation : Principal

Signature :



Seal of The Institution :



Place : Chennai

Date : 13-03-2023 10:15:49